

APM Body of Knowledge



Because when projects
succeed, society benefits

Body of Knowledge

8th edition

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Association for Project Management

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Foreword

The *APM Body of Knowledge* is an essential resource for project professionals seeking to adapt and thrive in today's world. This updated edition emphasises innovation, sustainability and societal impact, offering valuable insights into the principles, frameworks and methods of contemporary project, programme and portfolio management.

Since the previous edition was published in 2019, the world has seen significant change, increased complexity and a heightened focus on sustainability. The project profession is increasingly recognised as a strategic driver of progress across every sector, fostering innovation, change and long-term value, and project professionals bear an increased responsibility to deliver outcomes that positively impact society, the environment and the economy.

As a dynamic and vibrant profession, we thrive when we push ourselves to be creative. The *APM Body of Knowledge* is designed to inspire innovation, encouraging professionals to reflect on their practice and find ways of bringing greater success to their work, through the adoption of new technologies such as artificial intelligence or broadening perspectives through systems thinking. By serving as both a compass and catalyst for change, this text plays a fundamental role in shaping the future of a profession that has become increasingly vital in addressing some of the most pressing challenges of our time.

Whether you're an experienced professional, a career changer, or just starting your journey, this comprehensive resource provides valuable insights and knowledge to help you navigate the rapidly-evolving landscape and to contribute meaningfully towards creating a positive impact on the world around us.

Milla Mazilu

APM Chair 2025

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Introduction

The *APM Body of Knowledge* provides the concepts, functions and activities that make up professional project management. It reflects the developing profession, recognising project-based working at every level, and across all sectors.

The eighth edition builds on the good work of the previous editions. The guiding philosophy, based on consultation with APM members and interested project professionals, is one of evolution rather than revolution.

To this end, the overriding principles of the seventh edition, published in 2019, remain intact.

The term ‘project-based working’ (or the ‘management of projects’) is used to refer collectively to projects, programmes and portfolios, although not all aspects of project-based working apply to projects, programmes and portfolios equally. The term ‘project professional’ is used to refer to anyone working in a defined role within a project, programme or portfolio.

The ability to choose between multiple forms of life cycle is recognised – from one designed to guide the management of deliberate change in a linear fashion to one designed to guide the shaping of emergent change in an incremental, iterative or evolutionary way. We avoid references to ‘waterfall vs agile’, preferring instead to outline the choices that leaders can make, and to acknowledge the reality that many project-based endeavours now adopt some sort of hybrid linear/iterative life cycle approach.

The structure of this edition, too, remains largely true to that of its predecessor, but is bolstered by the addition of two new chapters at the front and back. The chapter running order is as follows:

- **1. Implementing change** is written as an introductory note, outlining the important role that projects and project professionals play in delivering on the wishes of strategists and policy makers.
- **2. Setting up for success** is written primarily for those leaders in organisations who have decisions to make about the role of projects, programmes and portfolios in implementing strategy. Leaders may be in the ‘client’ or investing organisation, or in a supplier organisation that exists to deliver project-based work for clients. The ideas in this chapter apply in both scenarios.
- **3. Preparing for change** is written primarily for those people charged with leading any project, programme or portfolio, of any size and complexity. It addresses early life cycle shaping and late life cycle transition into use for projects, programmes and portfolios, along with matters of assurance, learning and maturity.
- **4. People and behaviours** is written for anyone involved in projects, programmes and portfolios. Influencing and engaging stakeholders, forming, building and leading teams, and the generic skills and responsibilities of being a project professional are addressed with the objective of making it clear that all project-based work relies fundamentally on the ability of people to work together.

- **5. Planning and managing deployment** is written primarily for those involved in the end-to-end process of delivering a project, whether a standalone project or one that is part of a programme and/or portfolio, and regardless of the life cycle approach taken. Although the professional domain has expanded, the detailed matters associated with defining outputs, integrated planning and controlling deployment remain.
- **6. Data analytics and AI in project management** is written for those with one eye on future developments and reflects the growing influence – and importance – of new technologies on the profession.

Perhaps the biggest innovation, though, is alignment between this *Body of Knowledge* and the *APM Competence Framework*. Users can now expand on the knowledge areas included in the *Competence Framework*. To help with navigation, there are section-level headings that mirror the core competence areas. Below these, each section has discrete topics (117 in all) that explore in more detail each competence and provide recommendations for further reading.

As a foundational resource, written by the profession for the profession, we hope you find the revised content in the *APM Body of Knowledge 8th edition* informative and useful in guiding your endeavours to deliver beneficial change through the management of projects, programmes and portfolios.

1 Implementing change

All organisations need to periodically renew and evolve, especially in an increasingly dynamic world. In recent years, increasing interconnectedness has affected the speed of response required from organisations, but now concerns about climate change are rapidly moving sustainability up the agenda, and new technologies, such as artificial intelligence (AI), are changing the world we live in too. Projects and programmes, or portfolios of projects and programmes, are the way organisations implement change so they can continue to survive and thrive. This chapter therefore introduces, at a high level, the key concepts of how projects fit within the wider organisational setting, attributes of projects, and how they are affected by, and create, the organisational culture and change. Each of these subjects is dealt with in greater detail later in this *Body of Knowledge*.

This chapter starts by setting the scene: the overarching need to govern and lead projects and distinguish between project leadership and project management.

The chapter then goes on to outline the key attributes of projects – their phases, the different types of project and broad approaches to delivering projects. In this section of the chapter, we go on to discuss the realisation of benefits and the need for projects to be evaluated, so lessons can be learned.

The last section in this chapter focuses on culture and change. We start by outlining the importance of organisational culture and how the project professional needs to understand the prevailing culture when embarking on change. We discuss the process of change and change management, and creating effective transformation. The chapter concludes by introducing systems thinking as a valuable tool to support the implementation of change.

Delivering change isn't an easy thing to do. Strategists and policy makers can sometimes have great ideas that fail. This is sometimes because the ideas can't be implemented, sometimes because there isn't sufficient time or resource available, sometimes because various stakeholders resist the change, sometimes because the implementation of the change is poorly executed and sometimes because the whole endeavour is overtaken by other events. But, based on skills, knowledge and expertise, project professionals can significantly improve the chances of project success, by creating the governance, leadership and management structures supporting project delivery. Project professionals will also challenge the intent and requirements of what is to be delivered, so that there is clarity and a clear understanding of the resources required to implement the change and the risks involved.

The chapter is composed of three sections:

- 1.1 Projects and organisations**
- 1.2 Projects**
- 1.3 Culture and transformation**

1.1

Projects and organisations

Organisations operate in a dynamic context, full of uncertainty, novelty and turbulence. Projects, programmes and portfolios are introduced with the clear intent of enhancing performance, bringing about change and enabling organisations to adapt, improve and grow. Project work therefore represents intentional investment in development, enhancement and improvement.

The organisation's purpose and strategic intent will drive the need to improve and change. Project work is all about guiding and managing the strategic investments that enable assets, structures, systems, activities and capabilities to be formed, maintained or enhanced so that the organisational plans and ambitions can be realised. In the public sector, this strategic intent may be discussed in terms of policy and policy implementation.

Organisational change is introduced through projects, programmes and portfolios to deliver value to the business or organisation. This value is accrued through realising the benefits that result from delivering successful projects. Managing benefits is part of ensuring that investments are made to deliver value to the organisation. This applies even when the project is being carried out by a supplier or contracting organisation, or if the work is needed to maintain current capability or to conform to new regulations or directives so that smooth business operations can proceed.

The successful deployment of change, the support of new behaviours and the utilisation of new capability, resulting in the realisation of benefits, involves engaging with, promoting and working with diverse communities and groups. To ensure that value is created and sustained, organisations need to consider and address the full investment life cycle to ensure that forecast benefits materialise.

Delivering strategy is enabled through projects, programmes and portfolios. Portfolios structure investments in line with strategic objectives, while balancing, aligning and scrutinising capacity and resources. Programmes combine business as usual (BAU) with projects and steady-state activity dictated by strategic priorities. Projects are transient endeavours that bring about change and achieve planned objectives. Together, they combine to deliver the beneficial change required to implement, enable and satisfy the strategic intent of the organisation.

Having clarified the nature of project work and the differences between portfolios, programmes and projects, this section addresses the key overarching elements influencing project delivery. The section includes:

- 1.1.1 Projects in context:** Embracing and managing uncertainty
- 1.1.2 Governance:** Providing direction on goals and objectives
- 1.1.3 Project leadership:** Creating the mission and vision around the project
- 1.1.4 Project management:** The processes that deliver the project

1.1.1 Projects in context

Embracing and managing uncertainty

In many contexts, project work engages with novelty and uncertainty, extending into an unknown future. Projects and programmes therefore entail management under uncertain conditions. Yet, it is widely recognised that traditional business models, which focus on efficiency, top-down control and desired predictability, address only a small proportion of a rather complex, uncertain and interconnected landscape.

Uncertainty arises from many sources (Figure 1.1.1). Leaders increasingly face new challenges, such as the emergence of new markets, interconnected global competition, new sources of innovation, rising customer expectations, disruptive technologies, the growing gig economy and the increasing diversification of the workforce. Stable and predictable contexts are hard to find, and the models and approaches used for managing need to be updated to reflect a world characterised by uncertainty, turbulence, novelty, ambiguity and complexity. Moreover, the combination of economic unknowns, with political, social and environmental concerns relating to the proposed actions and their longer-term implications, requires ways of managing this uncertainty.

The US military coined the term 'VUCA' to reflect the 'volatility, uncertainty, complexity and ambiguity' of general conditions and situations associated with a multilateral world following the end of the Cold War. The term has been widely adopted to represent increasingly vulnerable and unpredictable contexts. The key implication of VUCA conditions is that there is an inherent uncertainty that makes it difficult to predict and plan with great accuracy.

The rigidity that comes from expecting full and perfect knowledge is unsustainable and unattainable in turbulent contexts. Uncertainty defies anticipation and detailed planning. Enforcing detailed planning and fixed-price contracting on an uncertain future can be counterproductive and damaging to business. Change is natural and ongoing as managers learn more about the context they are operating in, enabling them to identify emerging opportunities, respond to new conditions, and address shortfalls and differences in outcomes. Embracing and managing uncertainty lies at the heart of good project management, and an insistence on certainty may unwittingly result in mismatches between plans, models and reality, resulting in poor project performance.

Addressing uncertainty entails developing organisational capabilities for dealing with change; this fosters readiness (1) to exploit new opportunities, and (2) to respond and adapt. This is frequently translated into strategic flexibility, corporate resilience or organisational agility (see 1.2.3). Flexible plans, iterations and prototyping offer vehicles for the experimentation and adaptation that are needed to inform, adjust to and exploit uncertain contexts. These can complement open collaboration approaches and enhanced abilities to innovate and move rapidly and flexibly in order to shape opportunities, change strategic directions and build on adversity in uncertain settings. Organisations that invest in flexible planning, options evaluation and scenario planning are already better prepared to respond to emergent conditions – and such preparation needs to be reflected in project work. Having suitably qualified and experienced project managers is a critical element of the required organisational capability. Equally important is having an open culture, the psychological safety and being able to admit mistakes, which allows wider engagement and better decision-making.

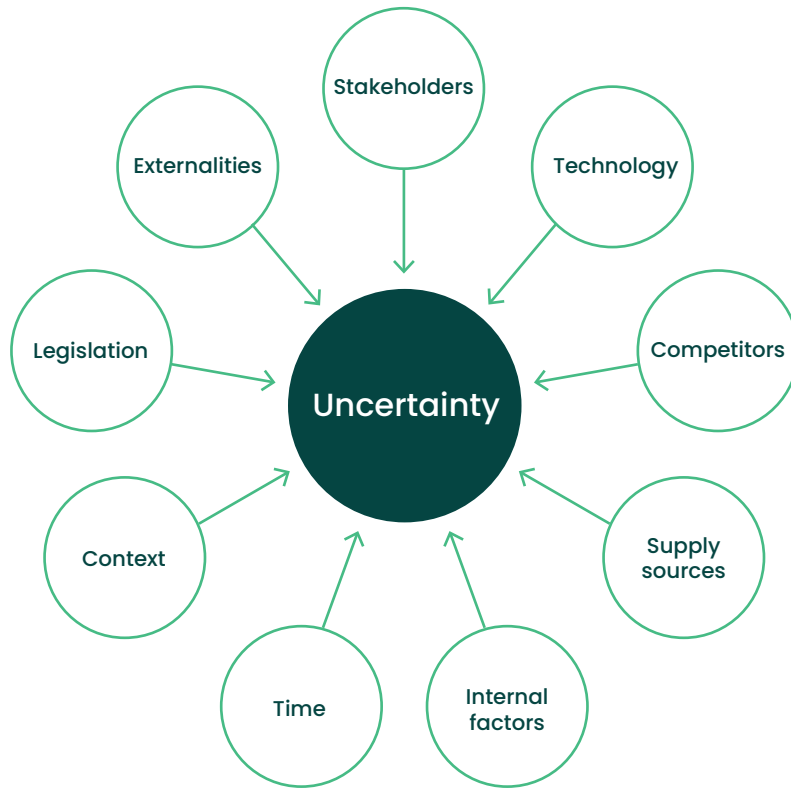


Figure 1.1.1 Sources of uncertainty

Recommended reading

- *Managing the Unknown* (2006) offers a way of looking at managing projects in novel and unknown environments. The authors propose a combination of trial-and-error learning, using multiple independent trials to identify the best options in novel projects.
- *Managing Project Uncertainty* (2016) focuses on the impacts of novelty in the uncertain world that projects inhabit, providing ways of identifying the symptoms of uncertainty and developing strategies to deal with it. The book offers senior managers ways to improve project and programme strategy by exposing new ideas and concepts that can be harnessed to tackle uncertainty in its many guises.
- *Managing in a VUCA World* (2015) is an edited collection focused on defining VUCA and uncovering the wider impacts on management. It makes the case for broader knowledge and the application of new concepts and frameworks to deal with unpredictable and rapidly changing situations.

1.1.2 Governance

Providing direction on goals and objectives

Governance is the system by which the organisation and its portfolio, programme and projects are directed and controlled. The objective of governance is to ensure appropriate management of activities. Governance therefore must facilitate entrepreneurial activities while balancing these with the judicious use of resources in the furtherance of delivering the longer-term success of the project, programme or portfolio, and the parent or commissioning organisation.

Governance is about *who* is allowed to decide *what*. Not all decisions on a project can be taken by a single individual; therefore the governance system has to allow for the delegation of decision-making within parameters. In many projects, there are two distinct roles: first, the project director, who is responsible for the project, and, second, the project sponsor, or project owner, who is responsible for receiving the outcomes of the project and making them work in the wider organisation. These roles may be held by individuals but, often in larger projects, boards are created to oversee these activities. However, the use of a board must not dilute the duties to be delivered or delay decision-making.

As the governance system enables delegation, the system must clearly communicate what is to be achieved and, if necessary, how this will be achieved. The detail of the 'how' will vary from level to level, with broader guidance at the higher level and more detailed guidance, when necessary, at the coalface of project delivery. Although the objective of the project may be clear, how this is to be achieved is rarely fully understood at a senior level, as those closer to the actual delivery have more recent and detailed knowledge of circumstances that need to be taken into account. Therefore one general principle of governance is to delegate decision-making to the lowest level possible, while still maintaining control of direction and oversight of progress, including what has actually been done.

In principle, there are two levels of governance control that should be considered:

- control of direction
- control of implementation

At the organisation level, control of direction refers to the decision-making process around the actual content of the strategy – what the organisation is trying to achieve. At the project level, it refers to managing the process that determines the problem to be solved, and defines the outcomes and benefits to be achieved. In practice, this control should also assure that the project's goals are aligned with the organisational strategy. Strategy and direction need to be periodically reviewed, so control of direction also covers external monitoring of the environment and oversight of the strategy review process.

At the organisation level, control of implementation refers to oversight of the strategy implementation process. For projects, this requires managing the alignment of the project delivery with the outcomes that have been set, and oversight of the key milestones and deliverables, monitoring progress at a high level and taking corrective action as and when required. Governance of strategy implementation should also involve periodically reviewing the project's delivery, and expected outputs and outcomes, to ensure that they will continue to deliver the benefits expected.

Expanding on this, governance needs to be applied appropriately at each level of an organisation, including at the portfolio, programme and project levels. Taking a viable systems model approach, governance needs to be fractal – that is, the same mechanisms are used at each level. Governance must inform the decision-making process, but the decision-making process must be informed by the direction set from above, including the goals and objectives passed down, the information gathered from continually monitoring the environment and the feedback information flowing up from subsidiary operations and processes (see Figure 1.1.2). In the governance system, each decision-making level being governed also governs the level below and provides feedback to the level above. In addition to those directional and informational flows captured in Figure 1.1.2, each level should periodically validate what is happening at the level below through an audit or other appropriate mechanism, to ensure that what is being requested and reported is actually happening.

It is a responsibility of leaders to support governance, but governance must also constrain leaders by requiring them to work within a prescribed decision-making process.

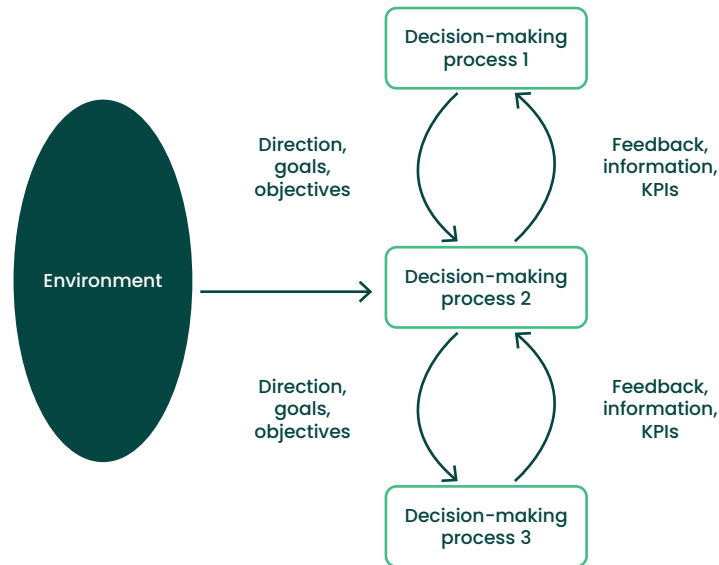


Figure 1.1.2 A framework of governance between levels in an organisation, portfolio, programme or project

Recommended reading

- *Directing Change* (2018), *Governance of Co-owned Projects* (2017) and *Sponsoring Change* (2018) by APM's Governance Specific Interest Group (SIG) cover different aspects of project governance.
- *The Fractal Organization: Creating Sustainable Organizations with the Viable System Model* (2009) offers practical guidance on how to achieve joined-up thinking at different levels in an organisation.

1.1.3 Project leadership

Creating the mission and vision around the project

There is a significant and established leadership literature, but there are subtle and important differences between general leadership and project leadership. First, project leaders lead temporary 'organisations', in that projects are planned to have a start and a finish. Second, project leaders are managing change, rather than BAU. Third, project leaders rarely have line management responsibilities outside the project and within the organisation where the project is being delivered, so they are required to have significant influencing skills. Finally, project leaders hand over their project to others, who ultimately deliver the benefits and outcomes through managing the resulting BAU situation created by the project.

Therefore, while there is a significant overlap between general leadership and project leadership, these four differences create a context which necessitates a different skill set.

The difference between project management and project leadership is more widely understood than the difference between leadership and project leadership. Project management is the application of processes, methods, skills, knowledge and experience to achieve specific project objectives according to the project acceptance criteria within agreed parameters. It has final deliverables that are constrained to a finite timescale and budget. On the other hand, project leadership is providing vision, direction, feedback and support, so that people can do their best work, with a focus on goals, team and stakeholders.

Process knowledge and technical project management skills are not enough to deliver a successful project. The project needs to be led. Key elements of this leadership include:

- creating the mission and vision around the project and what it is to achieve
- building and sustaining collaboration with and between stakeholders in the furtherance of the project
- creating and building the team to plan, design, deliver and hand over the project
- creating and maintaining the project culture to enable the team and individuals to thrive and work to the best of their abilities
- leading the governance of the project to ensure that it remains aligned with the original purpose and that the tools are used appropriately to empower and constrain
- developing the expertise and capability of the team

Put simply, the project leader's role covers:

- understanding the aim of the project, its scope and constraints, and approaches to making it tangible
- putting in place a team to fulfil these requirements
- creating the environment for that team to work to the best of their ability
- communicating with all stakeholders to ensure they contribute and understand process and progress
- monitoring the environment for changes that may affect the outcome
- ensuring that the project is handed over effectively to the client

Each project is different, from megaprojects involving the construction of infrastructure that will take years, to smaller-scale IT projects that may last for a few months, but the underlying principles are the same, as are the skills and abilities required of the project leader to fulfil their role effectively.

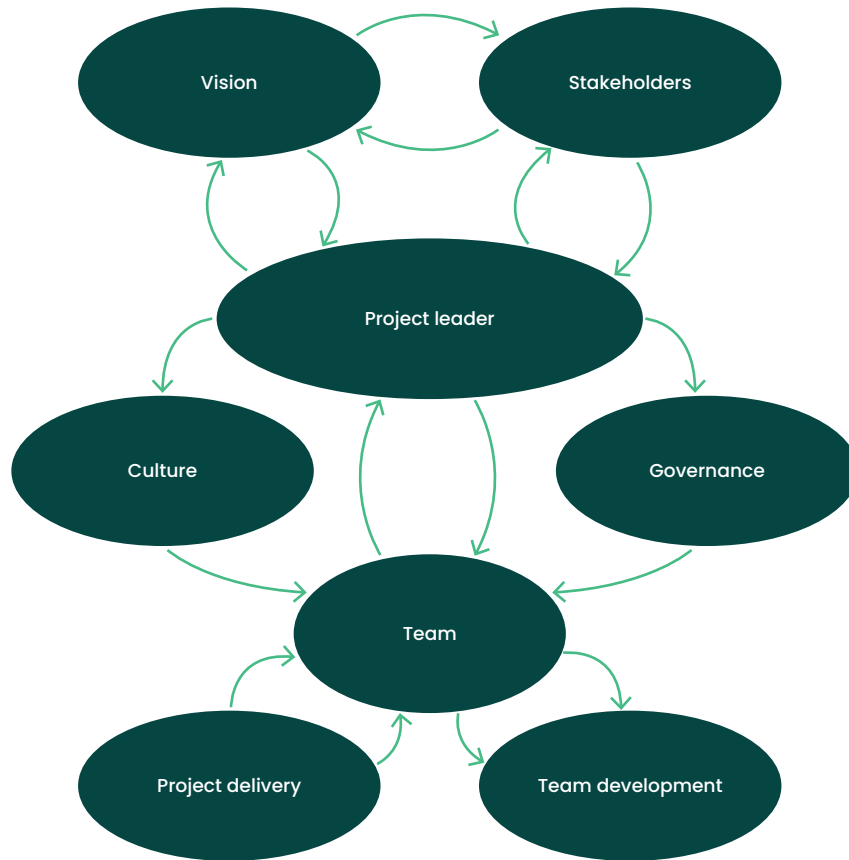


Figure 1.1.3 Key roles of a project leader

Recommended reading

- *Evolving Project Leadership* (2021), published by APM, looks at how project leaders can encourage greater agility and collaboration in their teams.
- *Project Leadership: Skills, Behaviours, Knowledge and Values* (2018) is an APM research report that identifies the competences needed to deliver major, complex projects successfully.
- The APM research paper *Leadership: Responding to Complexity* (2019) provides a framework for responding to complexity.

1.1.4 Project management

The processes that deliver the project

Given the roles of the project leader set out in 1.1.3, the role of project management is to develop, implement and control the processes that deliver the project. As outlined in 1.1.3, project management is the application of processes, methods, skills, knowledge and experience to achieve specific project objectives according to the project acceptance criteria within agreed parameters. Project management has final deliverables that are constrained to a finite timescale and budget (see Figure 1.1.4). This will include:

- evaluating ideas and options
- creating the business case and establishing the outcomes and benefits of the project
- managing the project through the business case approval and stage-gate processes
- managing the stakeholder engagement process, both to establish needs and to keep stakeholders informed of progress and developments
- validating and evaluating potential supplier bids and contributions
- managing the change management process in the wider organisation
- recruiting and managing the team to deliver the project
- monitoring and developing team capability to ensure the team members have the necessary skills, or develop the necessary skills, to deliver the project
- acquiring the resources to deliver the project
- managing the financial processes so that expenditure is controlled in line with authorisation requirements
- managing the financial reporting process by tracking both spend and progress
- managing the risk management process
- ensuring the implementation of first-line assurance (see 3.4.2)
- managing the management information system so that appropriate data is captured and information flows to appropriate decision makers and governance organisations
- ensuring accurate and appropriately prompt reporting, including of key performance indicators
- managing the integrated planning process, including the requirements of key dependencies
- managing the contracting process to comply with regulations, organisational requirements and good practice
- managing oversight of the supplier base to ensure good working relationships and the continuing capability to deliver
- managing the supplier delivery process so that deliveries are timely and cost effective, and quality standards are met
- managing the scope of the project and its change controls
- managing the configuration of the final product or service to be handed over to those running BAU
- managing the definition of verification or testing procedures prior to and during the handover phases

- managing the benefits realisation process, including defining and tracking the realisation of benefits over time
- managing the post-project evaluation to identify lessons to be learned and to contribute to the organisation's knowledge base

Projects require leadership, but they also require the discipline of project management processes to ensure that focus is maintained, resources are used efficiently and effectively, and information is regularly provided to support the decision-making processes.

The triple constraints in Figure 1.1.4 constitute the Barnes Triangle¹ (sometimes referred to as the Iron Triangle). However, the word quality is often misunderstood. For Barnes, quality refers to the whole of the requirement to be delivered to the customer or end user, and not just the quality of the project work as the word implies.

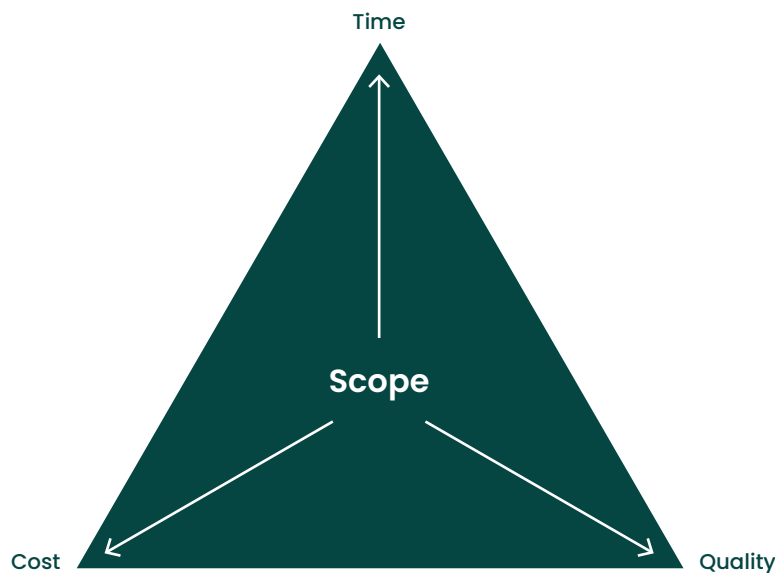


Figure 1.1.4 Trading the triple constraints of time, cost and quality

Recommended reading

- *Project Controls in the 21st Century* (2025), published by APM, covers all aspects of the project control processes, which are fundamental to good project management.

¹ Dr Martin Barnes CBE, a founding member of APM, is credited with the invention of the classic Time/Cost/Quality triangle.

1.2

Projects

Successful organisations must constantly evolve to keep up with changes in their environment. Alongside tactical improvements made as part of business-as-usual operations, most organisations invest in projects, either standalone or as part of larger programmes or portfolios, to bring about major changes that contribute to their strategic objectives.

At any one time, large organisations may be running hundreds of projects, each with its own objectives, challenges, stakeholders and interdependencies. To plan, coordinate and govern all their changes, many organisations use a top-down hierarchy of portfolios, programmes and projects to align all of their efforts with the organisation's objectives:

- **Portfolio:** A collection of projects and/or programmes used to structure and manage investments at an organisational or functional level (covered in section 2.1).
- **Programme:** A strategic endeavour, incorporating a group of related projects and BAU activities, undertaken to achieve a broader beneficial change.
- **Project:** A unique, transient endeavour undertaken to bring about change and to achieve specific planned objectives.

Working back up the hierarchy, projects use expenditure and resources to create tangible outputs, such as acquiring, upgrading, maintaining or replacing assets, products, services or capabilities. These outputs (e.g. implementing a new software tool) help generate improved service outcomes (e.g. increased user satisfaction), which, in turn, create benefits that contribute to strategic objectives (e.g. greater system use and less expensive phone support).

Whether it's building a skyscraper, improving a helpdesk process or restructuring a business department, projects take many forms. But, as projects are the underpinning vehicle that delivers change, understanding what they are, their core characteristics and how to approach them is essential knowledge for every project professional, regardless of their industry, organisation or background.

This section is for all project professionals tasked with supporting, leading or overseeing projects. Specifically, it will cover:

- 1.2.1 Project phases:** Breaking the project journey into distinct phases
- 1.2.2 Project types:** The 'what' and the 'how' of different projects
- 1.2.3 Project approaches:** Agreeing values, principles and behaviours for project delivery
- 1.2.4 Project benefits:** Justifying investment with a positive, measurable change
- 1.2.5 Project evaluation:** Determining a project's success

1.2.1 Project phases

Breaking the project journey into distinct phases

Managing a project is a journey. What starts as a new and exciting idea develops and grows over time, eventually producing an output that becomes part of an organisation's DNA. But turning an idea into a reality is often a long and complex process, requiring different types of work, skills, tools and techniques at different times to make it a success.

To help project professionals manage this journey, it can be useful to break projects down into distinct, manageable phases. Structuring projects in this way has many benefits, including the following:

- **Better grouping of activities:** Whether it's analysis, planning or building, identifying phases helps to group activities into different types. In turn, this helps project professionals assign the right resources to the right type of work at the right times.
- **More accurate plans and estimates:** While big-picture thinking has its place, it is much easier to plan and estimate work when it's broken up into smaller chunks. This ultimately leads to greater accuracy and increased confidence in project plans and estimates.
- **Helping to track and monitor progress:** Phases create clear break points that can be used to monitor progress against agreed checkpoints and milestones. This makes it easier to determine when things are going well or if support might be needed.

Later sections of this book explore the concept of project life cycles (see 2.5), including the practical frameworks, tools and techniques that can be used to manage a project in an orderly and efficient manner. But, regardless of the life cycle that's chosen, all projects go through the same high-level phases throughout their journey (Figure 1.2.1):

- **Concept:** Every project starts as an idea, often with the goal of exploiting an opportunity or solving a problem. This idea is analysed to determine whether it is viable and aligns with the organisation's objectives.
- **Definition:** If the idea is deemed a good one, the way to turn it into a reality is planned. This includes more detailed analysis, estimation and planning, culminating in approval to proceed from the project sponsor.
- **Deployment:** If approved, the plans to create a new output are implemented. Through testing and assurance, the project's performance is continually monitored to ensure the intended outcomes and benefits remain on track.
- **Transition:** The outputs are now a reality. Once they are accepted, they are transitioned into operational use, the benefits are tracked and the project is closed (see 3.6).

Understanding the project journey, and the phases and activities within it, helps project professionals quickly adapt to new and in-flight projects, no matter the industry, organisation or context they operate within.

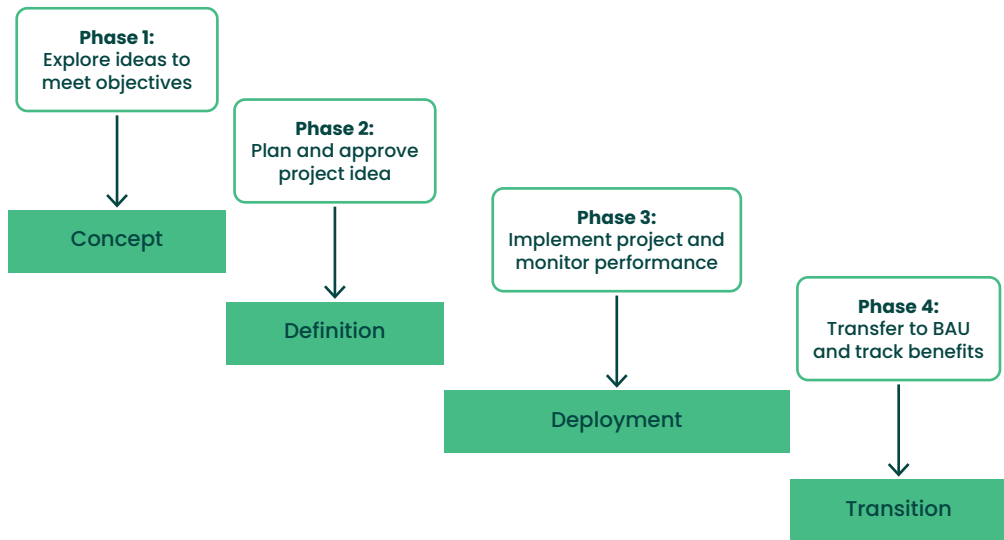


Figure 1.2.1 Project phases

Recommended reading

- *The Project Workout* (2019) provides a guide to structuring projects, combining useful advice with practical techniques. The resource identifies many of the key activities required to utilise the 'staged' framework, with an extended focus on governance, monitoring and control, information management and the relevant standards.
- *How Big Things Get Done* (2024) provides a modern, case-study-driven look at how some of the world's biggest projects gained success by starting small and following a phase-by-phase approach.

1.2.2 Project types

The 'what' and the 'how' of different projects

Despite their many similarities, all projects and project teams start from a different point of understanding. In a bid to help project professionals succeed, over many years academics have worked to classify different 'types' of projects and how best to approach analysing, planning and managing them.

The common consensus is that project teams should start by understanding the 'what' and the 'how' of their projects. Specifically, project teams start by asking themselves:

- How well do we know what we are doing?
- How well do we know how we will do it?

But this is not always possible, so Turner and Cochrane (1993) created a 'goals-and-methods matrix' as an approach to dealing with projects having ill-defined goals or methods of achieving them (see Figure 1.2.2). In very simple terms, there are four types of projects.

Type 1: These are projects with well-defined goals and methods, e.g. large engineering projects. Also known as earth projects, as they tend to have solid foundations.

Type 2: These are projects that have clearly defined goals but poorly defined methods of delivery, e.g. product development projects. Also known as water projects, as they flow with a sense of purpose but in a haphazard way.

Type 3: These are projects with poorly defined goals but a clear delivery method, e.g. software development projects. Also known as fire projects because they generate heat in the definition stage of the work, but they can burn with no apparent purpose.

Type 4: These are projects with poorly defined goals and methods, e.g. organisational development projects. Also known as air projects because they are difficult to catch hold of and deliver 'blue-sky' research objectives.

Once the type of project is understood, project professionals can tailor their planning and management approach accordingly.

In 1995 Obeng further developed this idea, giving the four types of project names: 'Walking in the fog', 'Making a movie', 'Going on a quest' and 'Painting by numbers'.

Where uncertainty is high (i.e. 'Walking in the fog'), projects should spend longer in the concept and definition phases to gain more information and build confidence in their delivery. Teams on 'Making a movie' projects should focus their early efforts on the 'what' by developing a clear business case (see 2.4). 'Going on a quest' projects often speed through the concept phase and spend time between definition and deployment as team members experiment with different 'hows'. Where certainty is high across the board (i.e. 'Painting by numbers'), the concept and planning phases can be brief, with teams aiming to deliver the agreed outputs as quickly as possible.

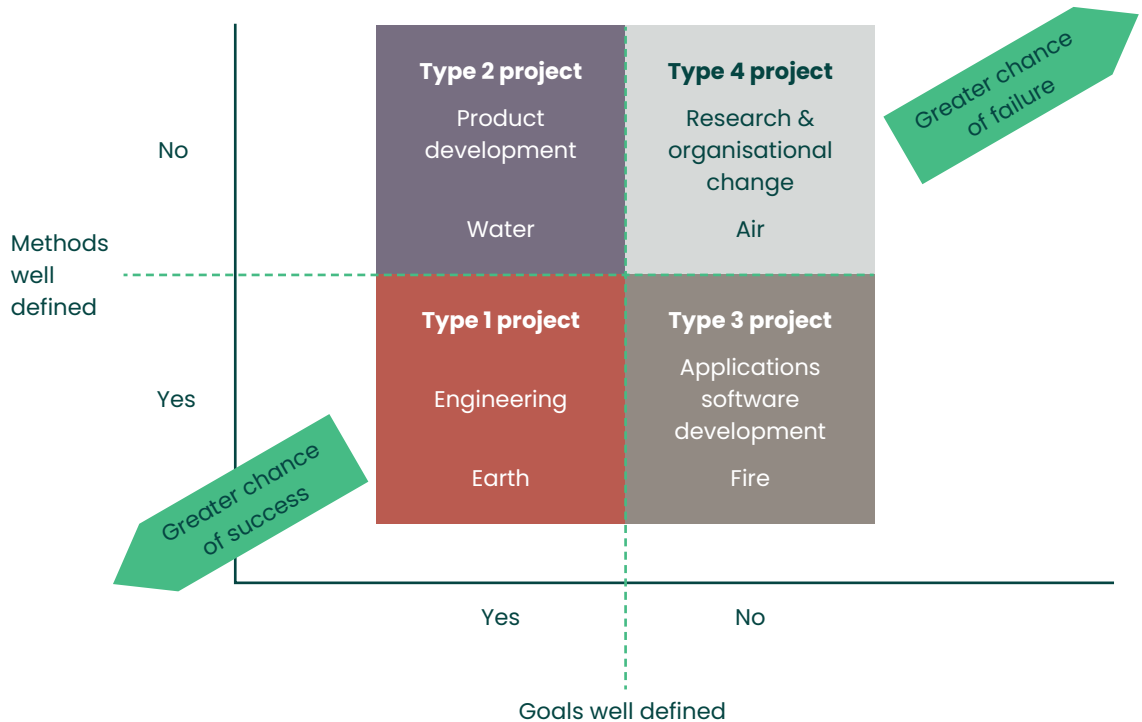


Figure 1.2.2 The four types of projects

Source: Turner and Cochrane (1993)

Recommended reading

- *Goals-and-methods matrix* (1993) is a paper in the *International Journal of Project Management* that defines four project types.
- *All Change!* (1995) by Eddie Obeng explores the different types of project scenarios which project professionals may take and how different management approaches better suit each one.

1.2.3 Project approaches

Agreeing values, principles and behaviours for project delivery

By their very nature, projects are complex undertakings, sometimes taking months or years to complete. During this journey, conflicts will arise, trade-offs will need to be made and best-laid plans will inevitably need redrawing. To help guide teams through this complexity, all project teams should agree on a high-level project approach.

A project approach refers to the overarching values, principles and behaviours the project works to throughout its delivery (Figure 1.2.3). Agreeing a project approach sets guiding principles that the team will use when planning, making decisions and solving complex problems, rather than jumping straight into choosing project management frameworks, tools or techniques.

While it's common to see projects being labelled as taking either an 'agile' or a 'waterfall' approach, in reality, a project's approach isn't black and white, and often moves along a spectrum of agility, risk taking and autonomy as it progresses. When agreeing their project approach, teams should consider factors such as the following:

- **Risk appetite:** How much risk is the project willing to take? This often aligns to the broader organisational culture and risk appetite, which can vary by industry and domain, too.
- **Cost of change:** What is the project's cost of change? This refers to how easy it is for a project to pivot and change mid-deployment. For example, it is much easier to rewrite software code than it is to demolish and rebuild a football stadium.
- **Business agility:** How much agility exists in the organisation? A flexible, resilient and adaptable organisation will have different expectations of a project to a more rigid, vulnerable or bureaucratic organisation.
- **Decision-making autonomy:** Is the team empowered to make decisions? A project team that is empowered approaches conflicts, decisions and trade-offs differently to a team that requires layers of approvals before acting.

Projects with a higher risk appetite, a low cost of change, team autonomy and a strong business agility are more likely to prioritise fast progress and learning through trial and error rather than spending a lot of time on detailed analysis, governance and planning.

On the other hand, where the risk appetite, business agility and team autonomy are low, and the cost of change is high, project teams simply can't afford to make mistakes. As such, they are more likely to adopt a more controlled, cautious and methodical approach.

The chosen project approach often manifests itself into other areas, such as in a project charter or mission statement, the mix of life cycles, tools and techniques used, and the team management style (see 4.4). As projects progress, they frequently move in and out of periods of stability, uncertainty and risk and, as such, their approach should be adapted in line with the changing environment and the wants and needs of their stakeholders.

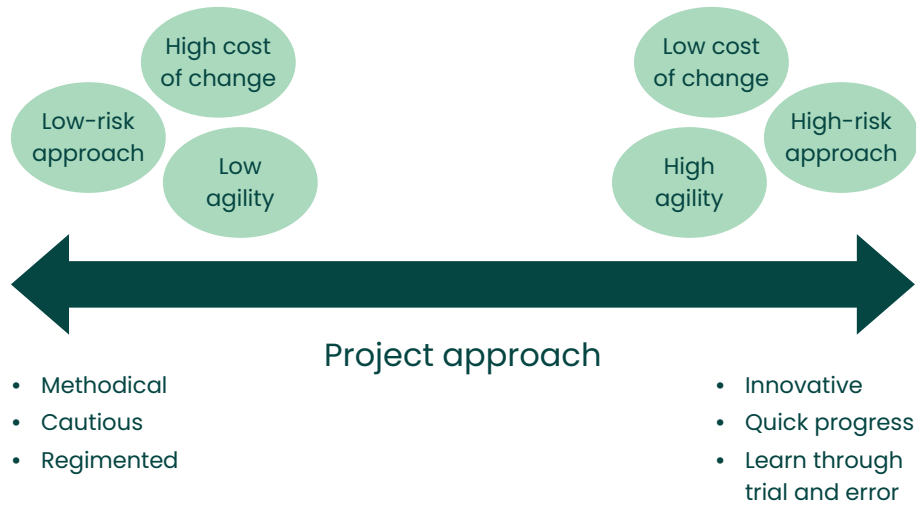


Figure 1.2.3 Different project approaches

Recommended reading

- *Understanding Agile in Project Management* (2022) is an APM research paper that makes steps towards an enhanced understanding of agile in project management.
- *Agile Beyond IT* (2022) explores how agility can be applied to all aspects of project management – from leadership through to planning and implementation.
- *Can agile be scaled?* (2017) is a research paper that looks at the extent to which scaled agile tools, techniques and roles are in place in corporate organisations.

1.2.4 Project benefits

Justifying project investment with a positive, measurable change

Delivering benefits is the primary reason why organisations undertake change. Given that no organisations have unlimited resources, before starting any project there's a need to clearly understand how the investment will generate benefits for the organisation. After all, what is the point of spending money on a project if it doesn't provide a return?

A benefit is a positive and measurable impact of change. In most organisations, benefits positively impact revenue, profit, opportunities, relationships, expertise or market position. These benefits can be tangible (e.g. money saved, jobs created) or intangible (e.g. reputation gained). They may or may not also be quantifiable in cash terms (e.g. reduced costs or greater customer satisfaction).

All project benefits should be derived from and aligned to the organisation's strategic objectives. In some cases, there may also be unavoidable declines or negative impacts arising from a change. If accepted, these are called disbenefits and should be identified, managed and tracked in the same way as benefits (see 3.1).

Governance boards and sponsors have a crucial role in assessing and approving projects, thus accepting the balance between a project's investment, benefits and disbenefits.

As is explored in section 3.1, benefits require proactive management throughout the entire project life cycle. An organisation identifies the benefits it needs and provides investment to achieve those benefits. Then, projects and programmes are implemented to create new outputs, capabilities and outcomes which ultimately lead to those benefits being generated.

During the project, the team must monitor performance indicators that can reliably predict whether the project's benefits remain viable and achievable. If they are not, it's the responsibility of project professionals to raise this with their stakeholders. Ultimately, if a project no longer represents value for money, the best decision may be to stop it.

In certain instances, to ensure that value is created and sustained from change initiatives, benefits have to be managed throughout the investment life cycle, including after the project or programme has closed. Ultimately, the sponsor owns the business case and is accountable for the realisation of the benefits. To help with this, it is therefore important to engender a shift from a culture of delivery towards an ethos of long-term value (Figure 1.2.4).

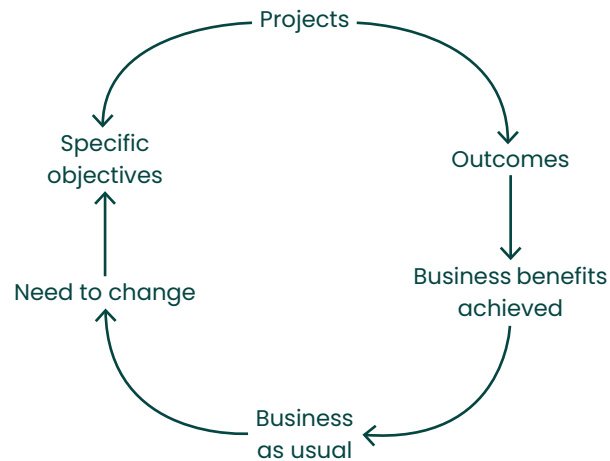


Figure 1.2.4 Projects creating benefits

Recommended reading

- *Guide for Effective Benefits Management in Major Projects* (2017) pulls together insights and lessons from significant major projects. The report synthesises the key principles and activities required to successfully deliver benefits in such projects and offers advice and guidance on extending the life cycle.
- *A Guide to Using a Benefits Management Framework* (2019) was compiled by the APM Benefits Management Specific Interest Group. The guide develops an understanding of the need for benefits realisation, offering a framework for addressing contextual factors and developing the capability to realise benefits.
- *The Information Paradox* (2007) is a classic book, offering the first detailed treatment of benefits. First published in 1998, it still provides fresh ideas, concrete advice and thoughtful reflection on the role of benefits in project work and the wider issues regarding a shift towards value-based focus.

1.2.5 Project evaluation

Determining a project's success

Despite projects sharing many similar characteristics, the unique challenges, complexities, dependencies and environment of each one makes it difficult to measure them all with the same criteria. That said, evaluating project performance is important for the growth and credibility of the project profession, so project teams should use a range of measures to determine the success of their project deliveries.

Learning lessons from what went well, and what can be improved, is also a crucial part of the project management profession at large. Capturing and sharing lessons learned helps project professionals to improve the accuracy, confidence and success rate of future projects, ultimately maximising the positive impact that projects have on society.

Project success can be determined through many different lenses, each with its own level of quantitative or qualitative measure (Figure 1.2.5). These include the following:

- **Technical performance:** The traditional measures of scope, time, cost and quality are a common, objective way of evaluating a project's performance and, ultimately, its success.
- **Benefits realisation:** All projects set out to deliver a return on their investment, so measuring benefits is another common way to determine project success, especially when utilising an extended project life cycle.
- **Stakeholder perception:** Especially for client-facing projects, success can also be determined through stakeholder satisfaction. Even when quantitative measures look good, if the client isn't happy, the project may not be deemed a success.
- **Business capabilities:** Has the project left the organisation in a better state than it was before? While some projects may not deliver large tangible benefits, they may enable the organisation to achieve future objectives.
- **Personal growth:** Have the knowledge, skills and competences of the team improved throughout the project? If so, this professional maturity represents a benefit to the organisation and could enhance the performance of future projects.

It is best practice for project professionals to agree success measures with stakeholders at the beginning of any project. These measures often form part of the project's requirements, objectives and success criteria (see 5.1) and make it easier for the entire project team to measure performance throughout, and at the end of, the project.

But it's also important to remember that projects operate in volatile environments. Changes in the business environment, especially those outside of the project's control, can both help and hinder a project's delivery. For this reason, it's important to use a mix of different measures to properly evaluate project performance, and to help identify learnings that can be applied in future projects.

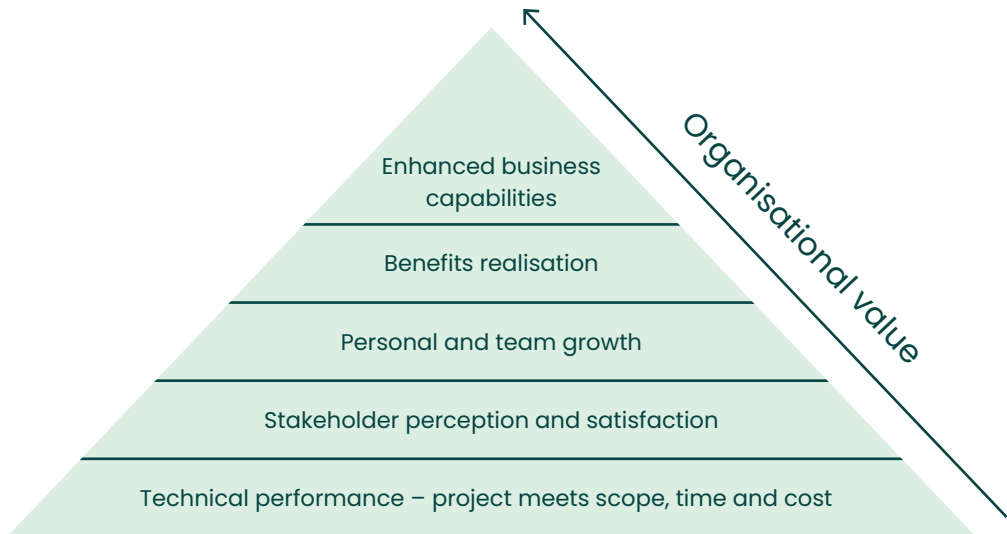


Figure 1.2.5 Project success factors

Recommended reading

- APM's *Dynamic Conditions for Project Success* report (2021) seeks to identify the core factors which lead to the successful delivery of projects, programmes and portfolios.

1.3

Culture and transformation

The business of project professionals is change. They create things that have consequences in the world. Their stakeholders live in societies and work in organisations. The changes project professionals create are affected by and will impact on the cultures of those societies and organisations, so the subjects of culture, change and transformation are fundamental to project-based working.

Organisational culture is the unwritten set of rules that influences individual and group behaviour and attitudes. Project professionals need to understand these rules if they wish to create successful change. This means appreciating the elements of that culture and the way they manifest within an organisation or its parts – and remembering that, of course, project-based working creates organisational subcultures of its own.

Delivering change is what project professionals do as a response to the never-ending flow of external change that organisations must navigate. Those organisations chart their strategy, and project-based working creates the processes, assets, products and changes that deliver it. To be successful, projects and programmes must often change attitudes, behaviours and culture.

The work of creating these changes is change management. Change management is a systematic process to move people from current to future behaviours and culture. It is the part of project-based working that focuses on people: their capabilities, attitudes, behaviours and emotions. It requires an understanding of psychology and an enthusiasm to work closely with people, who are often under pressure.

Transformation is a fundamental, strategic shift in an organisation's core structure and capabilities. It is a large and disruptive set of changes that go deeply into and broadly across an organisation. It requires project professionals to marshal a broad range of skills and experience, and they need to operate at the top levels of the sponsoring organisation.

Delivering change or transformation within an organisation is demanding for many reasons, not least the complexity of the systems, processes and cultures that need to be addressed. So an important toolset in these contexts is systems thinking, which focuses on the interconnections between the parts of a system, to understand them as a unified whole. It is a body of knowledge that more project professionals could usefully incorporate into their practice.

This section is written for project professionals who recognise the importance of change on the delivery of their projects and programmes. The section includes:

- 1.3.1 Organisational culture:** Habit, tradition and personality
- 1.3.2 Change:** The need to stay relevant
- 1.3.3 Change management:** Helping people to thrive through the changes
- 1.3.4 Transformation:** Creating a fundamental and lasting change
- 1.3.5 Systems thinking and change:** Seeing everything as a complex whole

1.3.1 Organisational culture

Habit, tradition and personality

Organisational culture is the unwritten set of rules that influences individual and group behaviour and attitudes. It applies at different levels of an organisation and may vary across a large organisation. This may be due to differences in the local cultural context of regional offices, or it may reflect the distinct team cultures of functions, departments or projects.

Culture can be deeply or lightly embedded in an organisation. Business theorist and psychologist Edgar Schein (2004) suggested it exists at three levels (see Figure 1.3.1):

- **Artefacts:** Elements of culture that are easily visible to outsiders, although not always easy to understand.
- **Values:** These determine norms of behaviour and how people think things 'should be done'. They drive relationships, decision-making and, sometimes, public statements like straplines.
- **Shared assumptions:** These are the beliefs that people don't question. They underpin and explain values and artefacts. Outsiders are rarely aware of them.

There are many different types of organisational culture. A single model can rarely give a satisfactory description of a complex real culture. However, Kim Cameron and Robert Quinn (2011) developed a widely used model, based on two pairs of competing values. These are stability versus flexibility, and inward- versus outward-looking. These are the four cultures they give:

- **Clan culture (inward/flexible):** A collaborative, nurturing culture with light-touch leadership and a values-driven, family feel. Project professionals who favour a collegiate, coaching leadership style encourage this culture.
- **Adhocracy culture (outward/flexible):** An energetic, entrepreneurial culture where innovation and risk taking are welcome. Iterative projects often have a strong element of adhocracy culture.
- **Hierarchy culture (inward/stable):** A rigid culture with a strong sense of order and control. Consistency is valued over creativity. Projects subject to strict governance may be suited to this culture.
- **Market culture (outward/stable):** A results-orientated culture that can be highly competitive. Operationally, this manifests in a focus on profit and market share. In projects, it creates rivalry with other 'competing' project teams.

To encourage a deeper understanding of the range of organisational cultures, Geert Hofstede (1991) proposed more cultural dimensions or competing values. In project-based working, the most relevant is process versus results orientation. Gerry Johnson and Kevan Scholes (1999) suggested that a web of seven factors impact on organisational culture.

Project professionals need to understand the prevailing culture of an organisation to help with both leading their teams and engaging with stakeholders. Often, project teams develop a subculture that overlaps with the main organisational culture but differs in some ways. Multi-organisation teams will develop a distinct culture of their own. It is a role of leadership to guide this development.

Projects and programmes can change the organisation's culture as either a deliberate goal or a consequence of the changes they create. In general, change has a small impact on culture; transformation can have a large impact. Topics 1.3.2 and 1.3.4 look at change and transformation respectively.

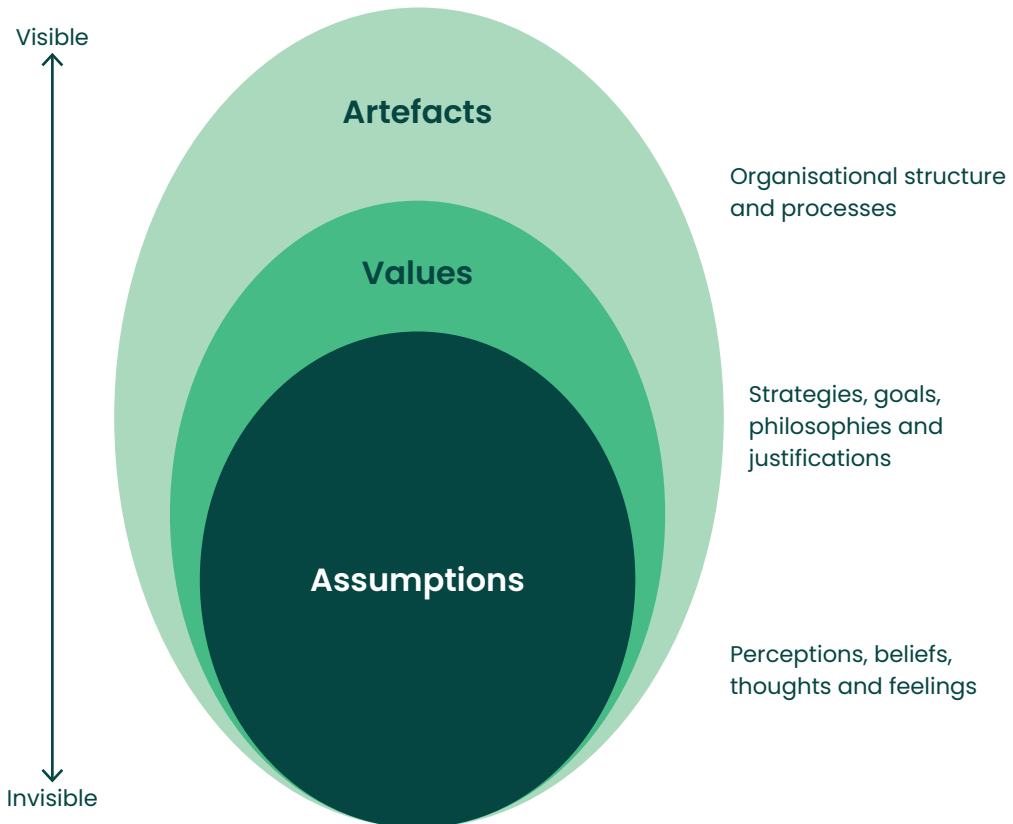


Figure 1.3.1 Three levels of organisational culture

Source: Adapted from Schein (2004)

Recommended reading

- *Organizational Culture and Leadership* (2004), an established classic, analyses how culture begins, thrives or dies with leadership, how to manage cultural change effectively, and the leader's role in managing disparate groups.
- *How the Way We Talk Can Change the Way We Work* (2001) is a practical and effective guide that provides change practitioners with the tools to fill gaps between what we intend and what we are able to accomplish.
- *Understanding Organizations* (1993) argues that the key to successful organisations lies in a better understanding of the needs and motivations of the people within them. The fourth edition of the classic management text offers an extended dictionary of the key concepts of culture and motivations in organisations.

1.3.2 Change

The need to stay relevant

Change is a constant of life. Organisations must respond to all kinds of pressures, including political, economic, societal, technological, legal and environmental (known as PESTLE), and commercial changes. If they don't respond, they will not thrive – and they may not survive.

To stay relevant, organisations need new approaches, capabilities and strategies. There must be a constant process of renewal and update. Strategic intent (see 2.1.1) drives organisations to maintain their competitive advantage – or seek a new one. Organisational strategies lead to the development of a portfolio of change initiatives. Each initiative develops new assets, capabilities, processes and resources that the organisation needs.

Project professionals design and implement projects and programmes to deliver these initiatives. But delivering assets, capabilities, processes and resources does not deliver change. For an organisation to change, the people within it need to change their attitudes and behaviours. Without these changes, the organisation will get little or no benefit from its investment.

Behaviour change needs a level of commitment (or 'buy-in'), but the default human response to imposed change is resistance. Change management is the process of:

- engaging with people
- educating and training them
- handling their resistance
- supporting them through the change

While change management and project management share many similarities, it is important not to see them as the same thing. They are two distinct disciplines with distinct roles and responsibilities. For example, those working in change management are often known as 'change managers' or 'change leaders'. It is important that project professionals understand the basic skills and toolsets of these change professionals.

Project management creates the deliverables that organisational strategy specifies. Change management ensures that people adopt and use these deliverables as the strategy requires. Combining change management with the management of project work embeds change, and this enables the planned benefits.

Successful change requires changes to behaviours, so there is a significant focus on people, culture and behavioural norms. Change management needs a range of resources to deliver this change. It draws on specialists, alongside project or programme teams and operational resources from within the organisation.

Organisational change demands new ways of doing things and new behaviours. Project work delivers the material changes: assets, structures, processes and systems. Change management makes it possible to bring people along, prepare them intellectually and emotionally, and embed the changes. A change programme combines these disciplines. Together, they allow the organisation to get the benefits that meet the business case for change.

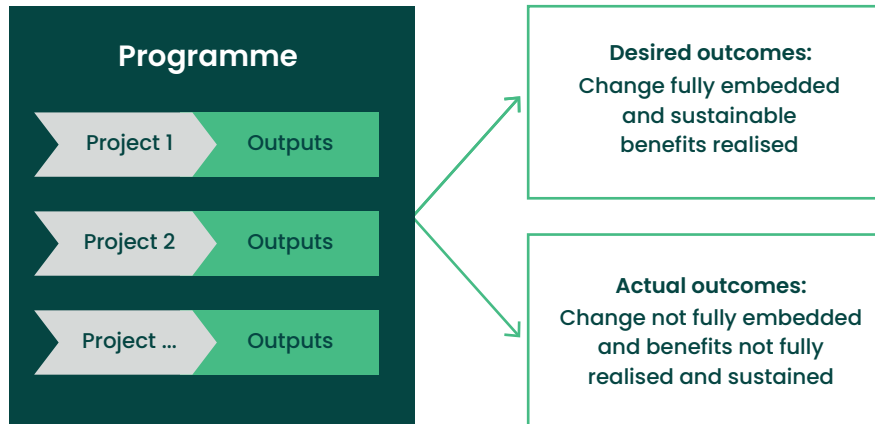


Figure 1.3.2 Outcomes of change efforts

Recommended reading

- *Introduction to Managing Change* (2017), developed by the APM Enabling Change Specific Interest Group, offers a dedicated resource focused on the management of change. The book provides a rationale for change management, a list of key factors for successful change, a further distillation of key factors in selecting a change approach, and a summary of many of the key approaches to change management.
- *Exploring Strategic Change* (2015) engages with the process of delivering strategic and organisational change, offering detailed guidelines, specific strategies, theoretical insights and practical solutions.
- *Managing Projects in a World of People, Strategy and Change* (2019) proposes a rethinking of some of the methods used in initiating, managing and governing projects, programmes and change initiatives. The book positions change as an opportunity to reflect and offers new perspectives on ethics, strategic initiatives, governance and change, along with new ideas focused on antifragility, commercial management and ethics.

1.3.3 Change management

Helping people to thrive through the changes

Change management is a systematic process to move people from current to future behaviours and culture. It has a body of knowledge, methods and tools to address the challenges people and groups must deal with in the face of external change.

Those challenges arise from human psychology. Change feels uncomfortable and people tend to resist, so change management must be able to:

- establish the need for change
- communicate a vision effectively and with respect
- draw together advocates for the change
- manage the development of new knowledge and processes
- inform, educate and train people to thrive in the future
- engage with and handle resistance to change
- deal with dissent, conflict and demotivation
- support people emotionally
- embed a new culture
- understand human psychology

There are many models for the process of managing change. All of them have the following elements:

- **Demonstrate the need for change:** This is the basis for the business case.
- **Craft a compelling vision for the change:** Having created a push away from the past, this creates a pull to the future.
- **Build a team to deliver the change:** This needs specialists in change management and project delivery, plus advocates from within the organisation. Change must receive positive support from organisational leaders, and active engagement at all levels.
- **Create a delivery plan:** This must coordinate project management and change management elements. Communication will play a vital role.
- **Handle the inevitable resistance:** People will resist change. Address their concerns with patience and respect. Engage them with pilots and prototypes, so they can help shape and influence the change.
- **Deliver the products and changes:** Active project implementation will combine the work of the product delivery and change management teams. An important part of maintaining motivation is celebrating successes along the way.
- **Follow through, beyond delivery:** Successful change will need to establish new behavioural norms and a shift in workplace culture. Work on these extends beyond the life cycle of a delivery project.

Some things are vital for success. Clearly, understanding human psychology and treating each individual with respect is fundamental. Other key points include the need for conspicuous sponsorship of the change, which demands leadership from the top tiers of the organisation. But perhaps the most common cause of failure is excess. Trying to deliver too much change in too little time leads to burnout, demotivation and change fatigue. Change takes time, and too much change delivers little more than chaos.

Change management is a discipline that project professionals need to understand. It is the soft, people dimension that balances the hard, process aspects of the role. Practitioners may not specialise in it, but change management is no less important than risk, cost or procurement management, for example.



Figure 1.3.3 The process of managing change

Recommended reading

- *Beyond the Wall of Resistance* (1995) gives advice on how to overcome opposition and build support for change.
- *Leading Change* (2012) outlines the process every organisation must go through to achieve successful change.
- The APM Enabling Change Specific Interest Group's *Introduction to Managing Change* (2017) discusses the importance of managing change effectively to deliver the benefits of projects and programmes.
- *Creating an Effective Public Sector* (2022) provides a comprehensive overview focusing on delivering change and performance in the public sector.

1.3.4 Transformation

Creating a fundamental and lasting change

Transformation is a fundamental strategic shift in an organisation's core structure and capabilities. It goes beyond incremental improvements. It typically involves comprehensive changes across multiple dimensions, including the business model, operations, technology, processes, culture and strategy.

Transformation is not just optimising existing processes. It is reinventing the organisation to achieve a disruptive and permanent change in its performance, market position and long-term viability.

By contrast, change is doing things in different ways. It can be at any scale, from small, incremental improvements to massive overhauls. It can happen at any level, from a single team to the whole organisation. There are no formal definitions of change and transformation in the context of project-based work, but several ideas build a distinction:

- **Scale:** Change is incremental. Transformation offers a radical shift.
- **Strategic nature:** Change is about operations and tactical advantage. Transformation is about vision and strategy.
- **Multidimensional:** Change can start anywhere and affects part of the organisation. Transformation starts at the top and affects everything.
- **Depth:** Change is shallow and affects methods, tools and processes. Transformation is profound and affects values and culture too.
- **Ambition:** Change is doing things differently and better: 'What we do now is not good enough.' Transformation is becoming a different organisation: 'What we do now is the wrong thing.'

There is no hard boundary between change and transformation. As a result, the project professionals who deliver change and transformation have overlapping but different skill sets.

Change managers work with people to help them navigate and thrive in their new roles, or to do tasks in different ways. As well as mastering a wide range of tools, they need to develop a deep understanding of the psychological and emotional consequences of change.

Transformation leaders need to understand the skills of change management, but they don't need as much depth of knowledge. Theirs is a more strategic role. They coordinate a portfolio of strategic initiatives and provide visionary leadership.

The primary challenges that transformation leaders face include the:

- scale and duration of transformation initiatives
- need for constant adaptation over the long term
- complex interplays of environment, strategy and organisational politics
- breadth of knowledge and skills needed to deliver transformation
- depth of culture change that is often needed for success
- influence and gravitas to operate at board level

These challenges make transformation leadership a demanding role for experienced project professionals. It offers an opportunity to work at board level and across a whole organisation. Transformation leaders deal with the intersection of strategy, implementation and politics.

Figure 1.3.4 shows the nine pillars of organisational transformation. These are the key themes of the transformation journey, grouped into three overarching processes.

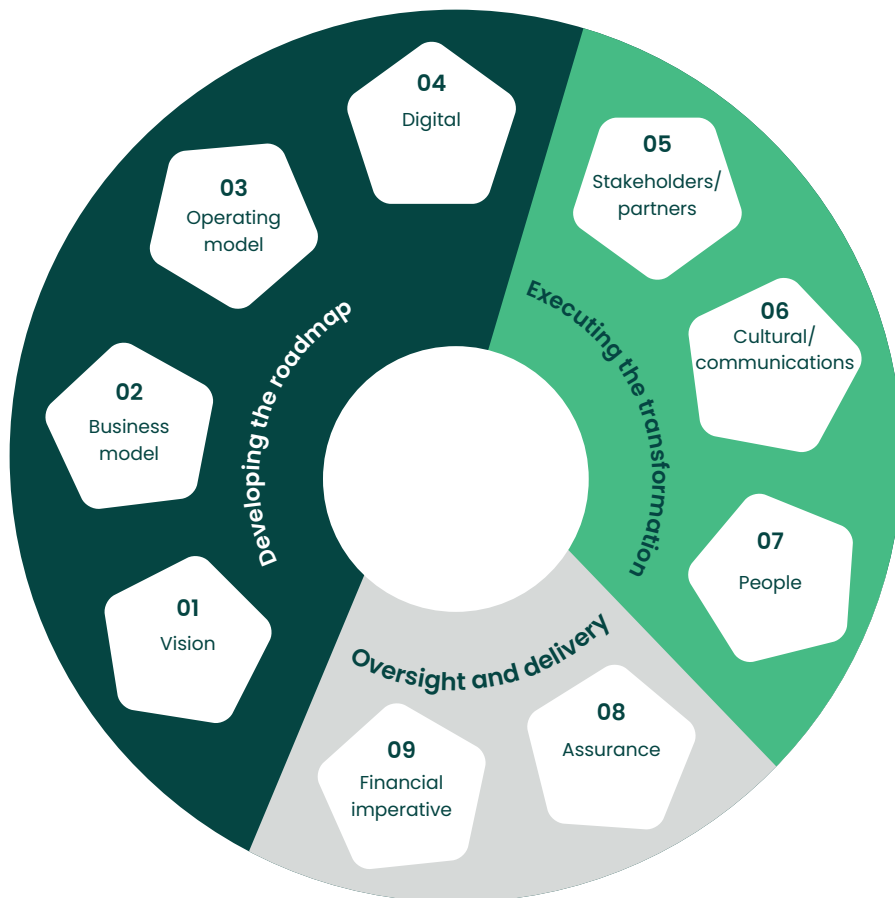


Figure 1.3.4 The TLBoK (Transformation Leaders Body of Knowledge) framework

Source: Lockwood (2024)

Recommended reading

- *Transformation Leaders Body of Knowledge* (2024) is a comprehensive guide to leading organisational change and transformation.
- *A Transformation Lens* (2023) provides practical advice on how to approach and navigate the transformational journey.
- APM's *Introduction to Managing Change* (2017) offers a dedicated resource focused on the management of change.

1.3.5 Systems thinking and change

Seeing everything as a complex whole

Projects, programmes and portfolios are made up of multiple, interacting systems containing connected elements that form patterns, relationships and rules.

Systems thinking contrasts with linear thinking (see Figure 1.3.5), which is suitable for systems where cause and effect are predictable. A complex system is more than the sum of its parts; systems thinking focuses on the interconnections between elements. It recognises that we cannot predict a whole system's behaviour, but we can analyse its elements and focus on patterns of change.

Project-based work is often complex, balancing multiple objectives and constraints. System thinking views a project as a system, rather than as individual elements. It can result in better-informed decisions and effective risk management, resulting in sustainable and innovative outcomes.

System thinking recognises interactions with other systems, including the business-as-usual systems it is often designed to change. It identifies the interconnections and dependencies that teams might otherwise overlook, helping to identify root causes rather than symptoms, leading to effective and lasting solutions. Treating projects, programmes and portfolios as dynamic and interconnected systems increases the chances of success.

Using this approach, project professionals can anticipate consequences of change by:

- confirming that problems are properly understood
- evaluating solutions and their impact on other systems before implementation
- exploring the project's context and scope, and the consequences of alternative scopes on implementation, outcomes and benefits
- understanding how changes in context will affect the project
- identifying interdependencies, constraints and the keys to success
- planning, resource optimisation and simulation

A holistic approach ensures change initiatives remain aligned to long-term organisational goals. This enhances overall system performance and value delivery. Understanding the complex interplay of the different elements in an organisation is also essential in understanding organisational change and business transformation. Its application to change management can:

- reveal conflicts between stakeholders and sources of resistance to change
- identify and anticipate potential impacts and consequences
- determine communication strategies
- guide and adapt to culture shifts

Systems thinking helps transformation leaders appreciate broader dynamics and highlight interdependencies, feedback loops and leverage points. With this knowledge, project professionals can spot unintended consequences of actions, and respond proactively to enhance overall system performance and value delivery.

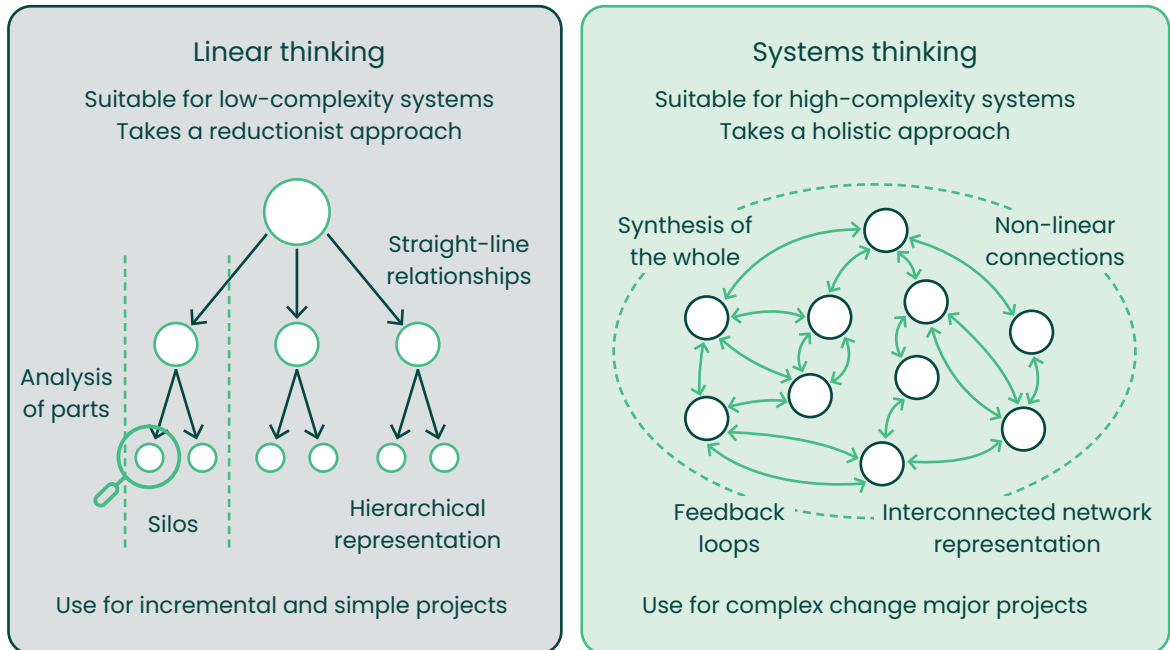


Figure 1.3.5 The natures of linear and systems thinking

Recommended reading:

- The APM Systems Thinking Interest Network's publication *Doing the Right Project: Using a Systems Thinking Approach to Selecting Successful Projects* (2025) includes systems thinking tools and techniques that project professionals can use to support their projects.
- The APM research report *Systems Thinking: How Is It Used in Project Management?* (2018) looks at the application of systems thinking in project management.

2

Setting up for success

Projects, programmes and portfolios need to be aligned to deliver what the organisation is trying to achieve. This overarching direction of travel is often referred to as the strategic intent of the organisation. The organisation then creates strategies aimed at delivering the strategic intent, and the projects, programmes and portfolios are the way this strategy is delivered.

However, we live in a VUCA world, so although the strategic intent remains constant over a period of time, the strategy for achieving the intent will change as a result of an evolving environment and as a result of the organisation's direct experience of implementing the strategy. These developments require project professionals to regularly review their portfolios, programmes and projects to ensure they remain fit for purpose and continue to align with the organisation's ambition. This chapter starts with a section that discusses these issues and provides guidance on maintaining alignment.

Projects, programmes and portfolios have to be financially viable, but increasingly they have to reflect the need for organisations to be sustainable and, in some cases, deliver social value. This chapter therefore contains sections on sustainability, financial management and business cases, drawing these different elements together.

However, as stated above, the world is in constant flux, and project professionals have to work within this ever-changing environment. This means that the life cycle of a project may not be linear. The path of the project may need to change in flight, so there is an important section covering different approaches and life cycles.

While projects are being delivered and change is happening, there needs to be oversight ensuring that alignment to the strategic intent is maintained and that project work is delivered efficiently and effectively. The governance practices of the organisation need to be fit for purpose, so that timely and appropriate corrections can be made; hence the last section in this chapter.

This chapter is composed of six sections:

- 2.1 Portfolio shaping**
- 2.2 Sustainability**
- 2.3 Financial management**
- 2.4 Business cases**
- 2.5 Project life cycles**
- 2.6 Governance arrangements**

2.1

Portfolio shaping

A portfolio is a collection of projects and programmes used to structure and manage investments at an organisational or functional level. Portfolio shaping is setting up portfolios for the efficient delivery of strategic objectives. It prioritises, balances and groups change initiatives, taking into account factors such as resource, risk, short- and long-term gains, benefits and priority of delivery. The goal of portfolio management is to deliver change initiatives that optimise the value delivered from investment. Yet there must be a balance with business-as-usual (BAU) activities.

A portfolio is different from a project or programme. Projects and programmes focus on delivery of outputs, outcomes and benefits. Portfolios are coordinating structures to support their deployment.

Portfolios can also exist at lower levels in an organisation. For example, they can coordinate the deployment of multiple projects in one department or business area. They help manage the prioritisation and allocation of scarce resources. Funding for such portfolios can be on a rolling annual basis, as part of a departmental budget.

Governance is always important because of the strategic nature of a portfolio. It will usually align with corporate governance. Where this is not the case, it is vital to establish a solid process that the executive team understands and buys into.

In a portfolio, project and programme sponsors must be prepared to compromise on their initiatives' individual priorities if necessary. What matters is the health of the wider portfolio, so the organisation must align culture, incentives and behaviour with its strategic intent.

Building a balanced portfolio means understanding the strategic intent. This creates a direction towards which the organisation can align its initiatives. Misaligned projects or programmes are wasteful and do not contribute to strategic objectives.

As soon as portfolio delivery starts, it needs constant evaluation. Organisations have two concerns: first, the performance of individual initiatives, and second, the composition and timing of the portfolio as a whole. Changing organisational priorities and external business environment mean portfolios cannot remain static. A well-managed process for regular review is key to good governance.

This section is written for project professionals who work at portfolio level or on projects or programmes within a portfolio. The section includes:

2.1.1 Strategic implementation: Making strategy happen

2.1.2 Building the portfolio: Crafting a balanced portfolio that aligns with strategic intent

2.1.3 Assessing ongoing viability: Maintaining an optimum portfolio

2.1.1 Strategic implementation

Making strategy happen

Strategic intent is the driver to achieve an important outcome. It encapsulates the broad direction of travel for the organisation and, ideally, stays constant over a reasonable period, while ways of achieving the strategic intent may change. Organisations use it as a benchmark to measure their progress over this sustained period of time.

Strategic intent provides the basis for aligning the portfolio to changes to the organisation's ambition, and drives strategy (Figure 2.1.1). While the strategy drives the alignment between ambition and resources, it can become a constraint on growth. The future is uncertain, so it is useful to develop a more dynamic portfolio selection process. Emerging opportunities and threats should cause ambitions and strategies to evolve, while the strategic intent remains the guiding overarching goal.

Investment in change is one way of implementing strategy. Projects, programmes and portfolios tend to flow out of strategic decisions made by the organisation. They are strategic investments that develop processes, technology and capabilities. Project work implements the organisational strategy to realise its benefits and accrue value.

Senior leaders in the organisation need to demonstrate how each project or programme they fund contributes to the overall strategy and delivers the strategic intent. Where projects or programmes do not align with the organisational strategy, executives must question their relevance.

Project work may also be an enabler to:

- create capabilities or assets needed for later change initiatives
- maintain existing capabilities or assets
- ensure compliance with new legislation or regulations
- satisfy professional, ethical or social imperatives

Many executives see strategy implementation as the hardest part of the strategy process. This is the domain where project professionals work, delivering it by executing strategic projects and programmes to realise targeted benefits. Project work is an essential part of making strategic investment successful.

The focus of project working is creating value. Deliberate and emergent strategies determine what outcomes will hold the most value. This will be set out in the business case (section 2.4), so project professionals must attend to the realisation of the benefits that justify the investment (section 3.1). The choice of life cycle plays a role in ensuring that an initiative can deliver the intended benefits and value. Different elements of a portfolio will follow their own life cycle model.

Supplier organisations, like consultants and contractors, have different motivations. They manage projects for their clients where there is a commercial return for the work they do and the risks they incur.

Corporate leaders are accountable for delivering corporate objectives, for example on service delivery, profitability or social value. Project work is a critical part of delivering this. Portfolio management plays an important role in balancing these initiatives by maintaining the alignment between project work and strategic objectives, and maintaining the infrastructure that enables the realisation of the benefits of change.

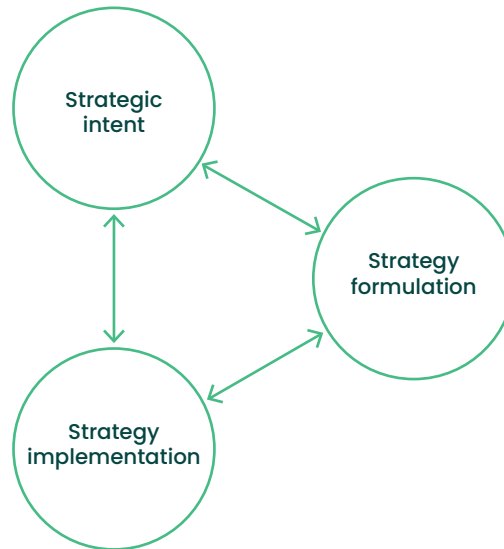


Figure 2.1.1 Hierarchy of strategic intent

Recommended reading

- *Strategic intent* (2005) revisits and updates the original concept of strategic intent established by the same team of authors 16 years previously. In this contribution, the authors compare Western companies to Japanese corporations, encouraging a rethinking of strategy and a repositioning of strategic intent.
- *Introduction to Managing Change* (2017), developed by the APM Enabling Change Specific Interest Group, introduces the importance of managing change and sponsoring such efforts. It makes the case for aligning change projects to the organisational business strategy, and supporting the strategy by articulating well-defined benefits.
- *The Evolution of Project Management Practice* (2018) is an edited volume that makes the case for shifting the focus from the staged delivery of artefacts towards consideration of stakeholders, benefits, value and complexity. The authors offer new perspectives on planning, business cases, benefits, collaboration, sponsorship, strategy execution and overall performance.

2.1.2 Building the portfolio

Crafting a balanced portfolio that aligns with strategic intent

A portfolio is a collection of projects and programmes. It offers a balanced investment at an organisational or functional level. Building an effective portfolio is about optimising strategic benefits or operational efficiency.

A portfolio will therefore be strongly influenced by the strategic intent of the organisation. This maps out strategic objectives and outcomes, and the benefits the strategic plan envisages. It sets the destination for the organisation and its functions (Figure 2.1.2). The portfolio will contain projects and programmes which will move the organisation in this direction.

Organisations need a governance body to assess each project or programme candidate. As well as their individual merits, documented in a business case, they must fit well as components of the wider portfolio. The primary requirement is that they align with the direction defined by the strategic intent, because misaligned initiatives risk using scarce time and resources for a goal that serves the organisation poorly.

It is often the case that there are more suitable initiatives available than the organisation can manage, resource and fund. Decision makers need to prioritise according to appropriate considerations. That decision process must be transparent and robust. The main prioritisation criteria organisations use include:

- **Dependencies:** Some initiatives are enablers for further projects or programmes.
- **Value:** Whatever measure the organisation adopts.
- **Imperative:** Some changes are necessary results of legislation, regulation or social responsibility.
- **Strategic impact:** How much the initiative delivers the organisation's strategic intent.

There are many measures of value (see 2.3.1). What matters is that the organisation selects with care and applies the methodology consistently.

Decision makers should also aim to achieve a balanced portfolio. This means balancing factors like:

- risk
- resource utilisation, particularly scarce resources
- short-term versus long-term impacts
- different types of initiative
- the commitment needed from different parts of the business

Once the organisation has identified potential projects and programmes, project professionals become more involved. There will be a period of planning, capacity evaluation and trial resource allocation. Initiatives must be scheduled with care – seek to avoid overlapping calls on resources, clashing delivery dates or interference with critical BAU activities. Initiatives often fail because they exceed the capacity of the organisation to either deliver or absorb change. Project professionals must liaise with operational teams to understand capacity and schedule constraints.

Portfolios need rigorous and ongoing review. Quite literally, the change portfolio holds the future of the organisation at stake. A regular, formal review should consider emerging threats and opportunities in the business environment.

Some initiatives may no longer yield the value that their business case promised. For some, this means adapting. For others, it means cancellation. Changes may highlight gaps in the portfolio, in which case the organisation will need to commission new projects and programmes to address the situation.

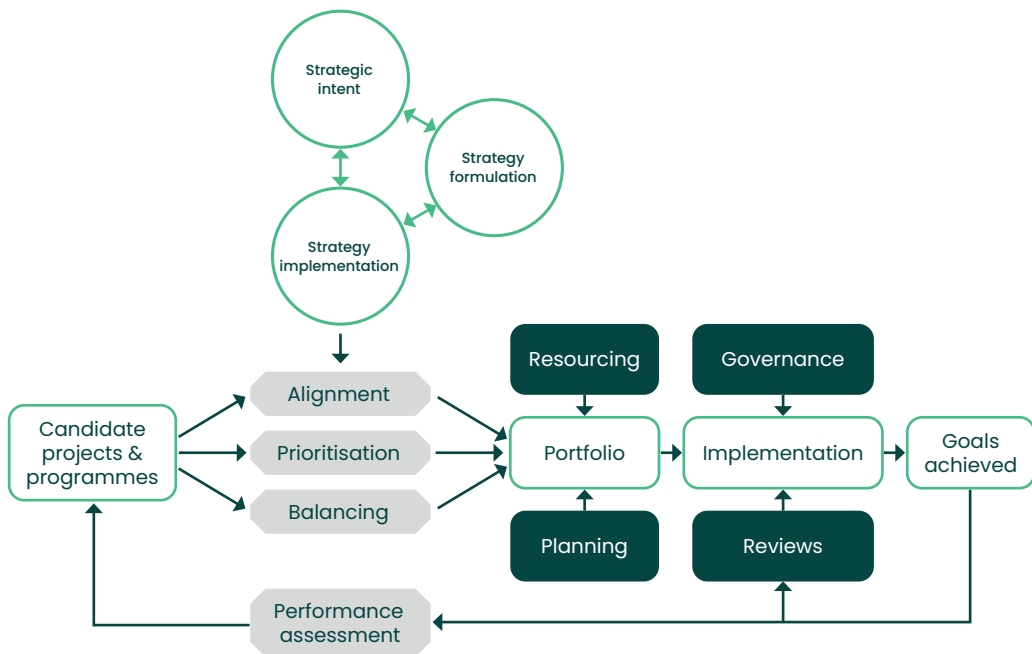


Figure 2.1.2 Key elements of portfolio management

Recommended reading

- The APM Portfolio Management Specific Interest Group's *Portfolio Management: A Practical Guide* (2019) includes many example templates and tools that the project professional can use to manage the challenging task of aligning the deployment of projects and programmes with organisational strategy.
- *Portfolio and Programme Management Demystified* (2013) is a useful guide that compares the management of multiple projects through portfolios and programmes. It brings together complex topics into one place and discusses them in an approachable and entertaining way.
- *The Handbook of Project Portfolio Management* (2019) is a large, edited collection of contributions, addressing many aspects related to the use of portfolio management in practice. Topics range from different types of portfolios and portfolio components to exploring the use of portfolios in different sectors, and covering the prerequisites for using portfolios, managing portfolios and developing the relevant capabilities.

2.1.3 Assessing ongoing viability

Maintaining an optimum portfolio

Portfolios should never be static. As projects and programmes proceed, the organisation learns from the experience and the external business environment also changes. A well-optimised portfolio may cease to represent either best value or the best route to delivering the strategic intent. The solution is a regular review cycle that assesses ongoing viability.

Refining a portfolio for optimum performance (Figure 2.1.3) starts with understanding the pressures for change. These occur:

- within the portfolio, as projects and programmes evolve; some will perform well, others less so
- within the organisation, as changes impact on processes, priorities and culture
- outside the organisation, as the social, political and commercial environment shifts

As a result, individual initiatives may not continue to offer either good value for money or outcomes that deliver the future that the organisation now needs. Each initiative needs a proper evaluation, with the options to:

- maintain it as is
- modify it to improve performance
- advance it, perhaps by applying extra resources
- delay it or slow it down
- close it down to avoid wasted effort

There will also be the option to bring forward other initiatives that have been on hold. In extreme cases, the organisation may need to launch a new project or programme to address the changed situation.

The process needs to follow all the principles of good governance. Any decision is likely to have profound impacts on the organisation's future, which means there needs to be a formal governance body or portfolio board to make the decisions. This needs to have robust and up-to-date evidence and data to consider. Facilitating this whole process will be the responsibility of the portfolio management team.

The portfolio board needs to meet according to a schedule, frequently enough to address changes before too much resource is committed to out-of-date priorities. Critically, its decision-making process needs to be rigorous and evidence-based. The evidence it considers, and its decisions, need to be on record for full internal transparency. Often, these decisions need to go to the corporate board for endorsement.

This whole process has a large number of stakeholders, including people within and outside of the organisation. Portfolio management teams must consider what information each group will need, and how best to communicate with them. Some stakeholders will also have a legitimate reason to seek to influence decision-making, so portfolio governance must include this within the wider decision process.

Portfolios can persist and evolve over many years. With an effective portfolio review process, organisations can keep them fresh, focused and relevant.

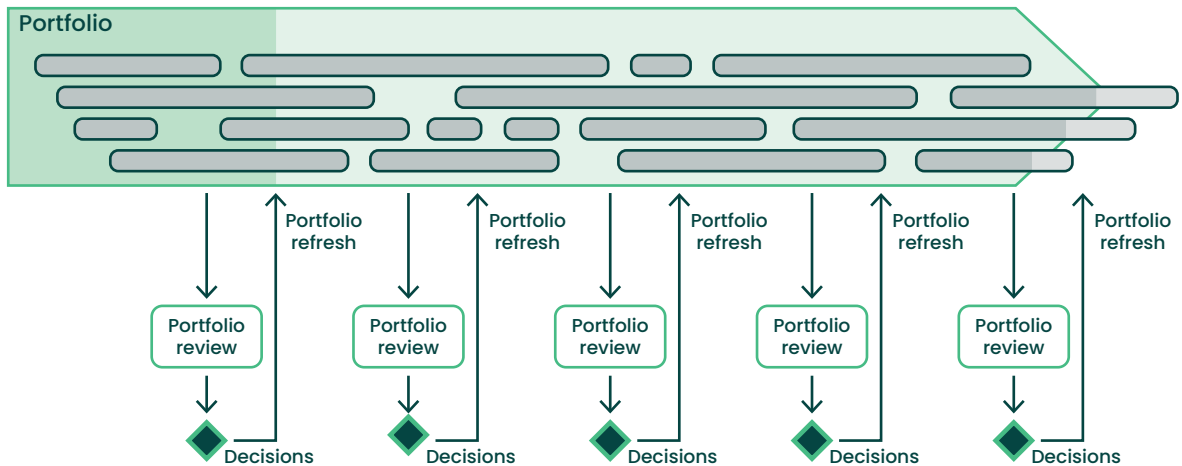


Figure 2.1.3 Refining a portfolio for optimum performance

Recommended reading

- The APM Portfolio Management Specific Interest Group's *Portfolio Management: A Practical Guide* (2019) includes many example templates and tools that the project professional can use to manage the challenging task of aligning the deployment of projects and programmes with organisational strategy.
- *Portfolio and Programme Management Demystified* (2013) is a useful guide that compares the management of multiple projects through portfolios and programmes. It brings together complex topics into one place and discusses them in an approachable and entertaining way.
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2.2 Sustainability

Sustainability integrates economic, environmental, social, human and governance factors throughout the project life cycle. This impacts a project's objectives and how it is delivered. The aim is to deliver sustainable outputs and outcomes that regenerate the environment and communities in the longer term.

The Paris Agreement (United Nations, 2015) aims to limit global warming to below 2°C above pre-industrial levels, aiming for 1.5°C. Businesses are committing to formal carbon reduction targets and prioritising reducing carbon emissions within projects and project management processes.

There are many interlinked issues to consider. While reducing carbon emissions is vital, the United Nations' Sustainable Development Goals (*The 17 goals, 2024*) address a wide range of challenges, including social inequalities, climate change and biodiversity loss. These goals provide a comprehensive sustainability framework for countries and organisations. There is an increasing expectation that projects reflect this holistic approach to sustainability.

This is sometimes referred to as responsible project management: "The concept of managing projects with specific attention to the intended and unintended impacts of the project and its outcomes, in both the short and long term, thereby delivering economic, social and environmental impact." (Thompson, 2020).

Combined, these dimensions are known as the triple bottom line, expanding traditional financial reporting to include social and environmental performance. This is often summarised as the three Ps: people, planet and profit.

This framework can be used to evaluate a company's performance through its environmental impact, social responsibility and governance practices, which are integrated into investment decisions and corporate decision-making. Sustainable project management serves as a bridge between an organisation's sustainability strategy and the achievement of sustainability goals through robust project management practices.

The section includes:

2.2.1 The importance of embedding sustainability into project management:

Understanding sustainable principles and requirements

2.2.2 Integrating sustainability processes into the project life cycle: Incorporating sustainability at every stage of the project

2.2.3 Sustainable products: How project professionals can influence product development and delivery

2.2.4 Sustainability assessments: Using assessments to inform decision-making

2.2.1 The importance of embedding sustainability into project management

Understanding sustainable principles and requirements

In our changing world, sustainability in projects is more important than ever. The shift towards sustainable projects expands the role of a project manager significantly beyond meeting traditional time, cost and quality criteria (see 1.1.4) to consider a wider return on investment – what do stakeholders get back for what investment?

Project professionals must ensure they have a good understanding of sustainability and regenerative principles and requirements. In addition, they need the ability to implement practices that drive innovation, enhance stakeholder satisfaction and future-proof projects against emerging challenges.

It is a fundamental competence vital for improving and facilitating effective project, programme and portfolio management. To understand what project professionals need to know and deliver for a project to meet sustainability standards, see the *APM Competence Framework*.

The importance and benefits of integrating sustainability into project management (Figure 2.2.1) include the following:

- **Long-term value:** Enhancing project outcomes and stakeholder satisfaction. A sustainable and regenerative approach is increasingly recognised as a value generator rather than a cost centre.
- **Innovation:** Driving innovation by fostering new low-carbon solutions and technologies that can lead to competitive advantages, operational efficiencies and solutions that promote job creation, mental wellbeing and community building.
- **Cost savings:** Adopting sustainable practices can lead to cost reductions through efficiency improvements and resource conservation (e.g. using less energy, using fewer materials, producing less waste).
- **Regulatory compliance:** Ensuring projects align with environmental laws, policies and standards, and have a 'licence to operate'.
- **Future-proofing:** Adopting future-proofing strategies to anticipate and adapt to upcoming regulatory changes, technological advancements, environmental issues and societal expectations, to ensure that projects and products remain relevant and resilient in our rapidly changing world.
- **Risk management:** Proactively identifying and mitigating sustainability-related risks, such as resource scarcity, rising energy costs and market shifts. It is important to build resilience against environmental and social disruptions, thereby safeguarding project continuity and performance.
- **Meeting stakeholder expectations:** Ensuring products and services meet stakeholders' social and environmental concerns, gaining quicker approvals and receiving positive rather than negative feedback. Rather than merely 'managing' expectations, projects are now expected to proactively secure and maintain engagement from and between stakeholders. Enhancing transparency through clear sustainability performance reporting can improve credibility with stakeholders.

- **Market and brand value:** Adopting sustainable approaches can enhance market competitiveness, brand reputation and customer loyalty.
- **Partnership opportunities:** Collaborating with suppliers, partners and communities on sustainability initiatives can create shared value and strengthen relationships.

Leadership for delivering long-term sustainable and regenerative outcomes, by integrating it into governance, decision-making, project management processes, and the organisation's cultural and behavioural change, optimises the likelihood of a successful project.

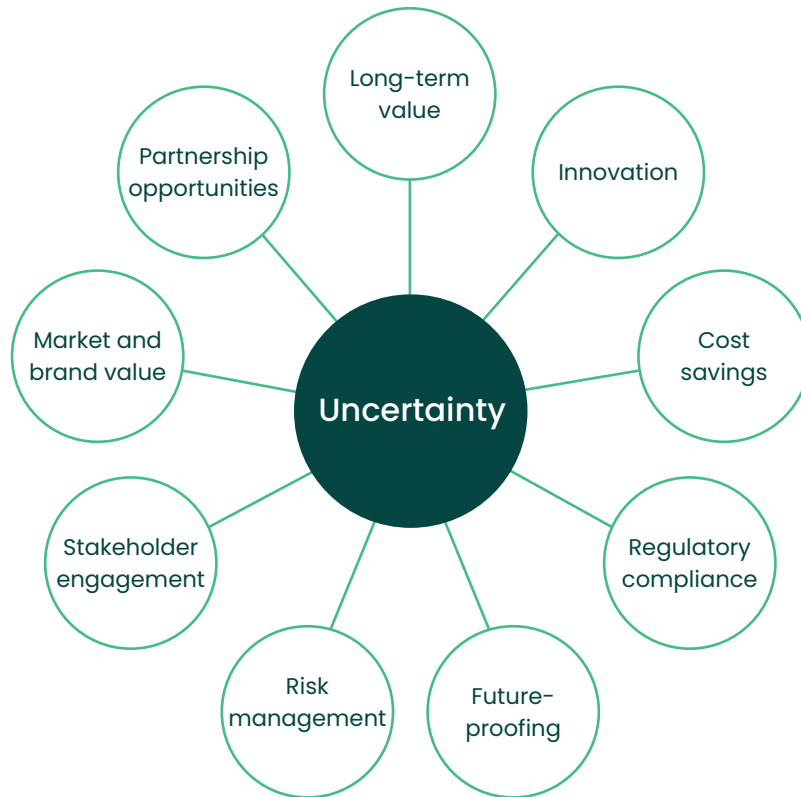


Figure: 2.2.1 The importance and benefits of integrating sustainability into project management

Recommended reading

- The *APM Competence Framework* (2022) lists the competences and practices that project professionals need to deliver projects in a sustainable way.
- Professor Peter Morris's report *Climate Change and What the Project Management Profession Should Be Doing About It: A UK Perspective* (2017) sets out the argument for managing climate change as a project or programme.
- The APM research paper *Are We Ready for Net Zero in Project Management?* (2024) investigates the preparedness of the project management profession in the UK to tackle the challenges of climate change.

2.2.2 Integrating sustainability processes into the project life cycle

Incorporating sustainability at every stage of the project

Project professionals play a fundamental role in incorporating sustainability throughout the project life cycle.

At concept stage, it is important to understand the evolving policies, legislation, best practice, frameworks, opportunities and risks that make up the project's context.

The purpose(s) of the project needs to consider sustainable (reduce, reuse, recycle) and regenerative (enhance, improve) approaches. These should align with organisational sustainability strategies and, where relevant, international frameworks.

This needs to be embedded in the project's business case with environmental impacts and sustainability factored into any options analysis. Frameworks such as the Five Case Model – natural, human, social, manufactured and financial – or the UK Government's *Green Book* can be referenced in developing business cases (see 2.4.1). Projects can also adopt a natural capital approach, where decision-making considers the value of the natural environment for people and the economy; creating social value should also be considered.

At the definition stage, benefits management approaches can help define how sustainability will be achieved and measured and included in related key performance indicators (KPIs). A project team should have suitable skills and experience, including specialist skills where required. They should be challenged to 'do better' if the brief is not ambitious enough.

Relevant sustainability assessments should be commissioned, and sustainability and regenerative approaches embedded into project governance, decision-making and processes.

During the deployment phase, the sustainability credentials, experience and solutions of suppliers need to be aligned to the project's goals. Opportunities and risks need to be managed, and teams must be prepared to change direction or extend timescales if a new, more sustainable option becomes available.

Throughout this phase, the project must maintain engagement with stakeholders, and include the emergent and changing benefits of sustainable and regenerative approaches when deciding how to overcome obstacles. Environmental and social performance should be monitored against KPIs throughout.

During transition, a sustainability handover plan ensures users and beneficiaries benefit from the intended environmental and social outcomes. Performance should be evaluated with the aim of obtaining sustainability certifications, which can enhance the project's credibility and market value. These achievements should be reported to stakeholders.

Monitoring post-project sustainability and regenerative performance should be maintained throughout the adoption phase to track the ongoing impact of the project. Lessons learned and opportunities for continuous improvement should be documented and shared for future projects.

All this should aim to achieve long-term outcomes and benefits. These achievements should be documented and communicated. The outcomes of the project and the project itself should contribute to a sustainable future.

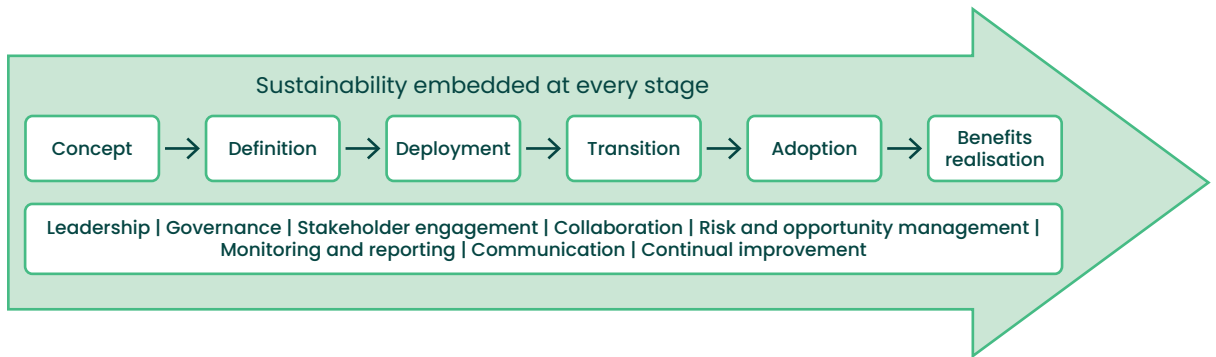


Figure 2.2.2 Embedding sustainability at every stage of the project

Recommended reading

- The *Green Book* (2022) is UK Government guidance on how appraise projects and programmes, both before and after implementation.
- Supplementary guidance to the *Green Book* above, *Enabling a Natural Capital Approach* (2023), which advises on how project appraisals can incorporate natural capital that benefits the environment, economy and society, and *International Guide to Developing the Project Business Case* (2018), which includes the Five Case Model.
- The APM research paper *Sustainability: Inclusive Storytelling to Aid Sustainable Development Goals* (2022) shares the findings from 60 interviews with project professionals and how they are addressing the sustainability agenda.

2.2.3 Sustainable products

How project professionals can influence product development and delivery

Sustainable products are goods and services designed to minimise environmental impact, regenerate biodiversity, enhance social outcomes and lead to commercial success. Robust project management processes are required to design, develop and deliver sustainable products.

There is an increasing demand for sustainable products from businesses and individual consumers. Evolving consumer preferences for sustainability are driving market changes and influencing product development, resulting in significant innovation programmes in supply chains. Regulatory changes also impact the development and delivery of sustainable products, such as the Producer Responsibility Obligations (Packaging Waste) Regulations 2007 and Extended Producer Responsibility (EPR) for packaging in the UK.

There is a difference in scale of the environmental impacts of producing physical products/goods, and services – as described in Figure 2.2.3.

Projects to produce physical products/goods have a higher impact (for example, constructing buildings and producing manufactured products such as laptops or electrical equipment). They require raw material extraction, transportation, and processing, assembly or construction, which uses energy and produces carbon emissions, as well as having wider environmental impacts. Impacts from these stages in a product life cycle are often termed ‘embodied impacts’ or ‘embodied carbon’. They also have operational (‘use phase’) impacts as well as resulting in waste at the end of their useful life. These types of projects often have extensive supply chains that need to be proactively managed from an environmental and ethical perspective, to achieve the project sustainability strategy. Approaches such as low-carbon design, design for the circular economy, and sustainable manufacturing processes can be adopted for these types of physical products.

Digital products, or projects for service-based organisations, generally have a lower environmental impact than those involving physical products. Examples include implementing a new customer relationship management (CRM) system or developing a mobile app. While these projects usually occur in office environments and don’t rely on extensive materials or supply chains, there are still ways to reduce carbon emissions. For instance, data centres – vital to digital operations – consume significant energy. Similarly, data usage, artificial intelligence (AI) and blockchain solutions can carry high carbon footprints. On the positive side, digital projects can be designed to enhance inclusivity, such as creating accessible multilingual apps, and they can replace paper-based systems or reduce travel, offering clear sustainability benefits.

Project professionals have more influence than we might think to prioritise sustainability in product development and delivery processes. They are in a unique position to effect change in projects and to help produce more sustainable products.

Project professionals need to understand the key environmental, societal and wellbeing risks and opportunities, integrate sustainability into the project life cycle, and ensure that products have legitimate environmental claims – this is crucial for consumer trust and brand reputation.

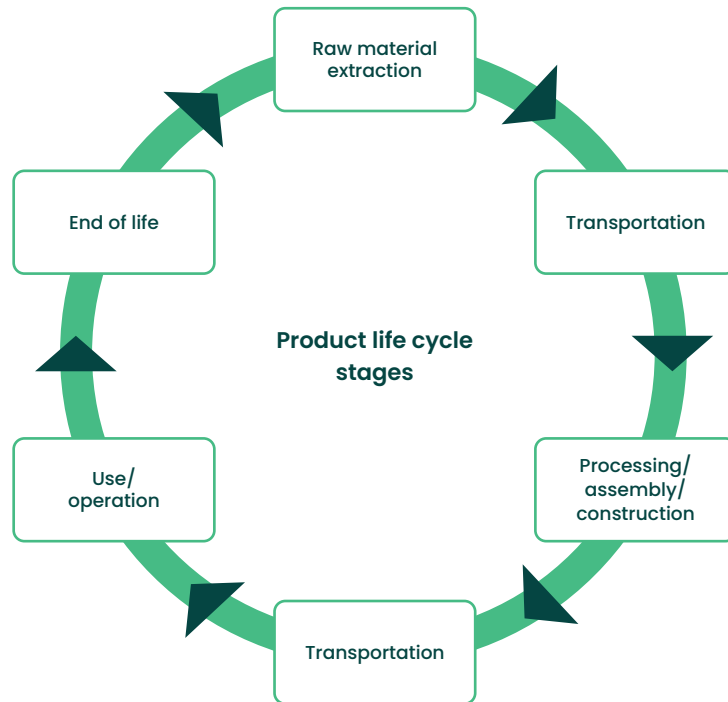


Figure: 2.2.3 The influence of the project professional on the product life cycle

Recommended reading

- *Cradle to Cradle* (2009) presents an alternative manufacturing model where materials and components can be repurposed – or recycled indefinitely.
- *How Bad Are Bananas? The Carbon Footprint of Everything* (2020) sets out the challenge of addressing climate change and the need to become energy- and carbon-literate to help inform the decisions we make.
- *The Ellen MacArthur Foundation* has a range of resources that explore the principles of the circular economy that aims to eliminate waste and reduce environmental impact. See ellenmacarthurfoundation.org.

2.2.4 Sustainability assessments

Using assessments to inform decision-making

Sustainability assessments are systematic processes used to evaluate the environmental, social and economic impacts of projects or products.

Sustainability assessments serve several useful purposes. They:

- help identify potential environmental and social risks, and which issues are the most important. This is the concept of 'materiality' – the threshold of importance to both project viability and organisational impact
- help us to understand and enhance sustainability and regenerative performance
- guide decision-making towards more sustainable outcomes
- quantify the benefits of sustainable approaches
- help communicate the sustainability credentials of a project or product externally

They can be undertaken informally and internally but are most credible when aligned with a particular process, standard or certification scheme. Expert sustainability consultants are often appointed to carry out this service.

Below are some common types of sustainability assessments:

- **Materiality assessment:** Identifies and prioritises the sustainability issues that are most significant to the business and stakeholders. It helps organisations to determine which topics are critical for their strategy, operations, products and reporting.
- **Life cycle assessment (LCA):** Evaluates the environmental impacts of a product throughout its life cycle, from raw material extraction to disposal. The ISO 14040 series standards cover LCAs.
- **Environmental product declaration (EPD):** Quantifies environmental information about the life cycle of a product, enabling comparisons between products fulfilling the same function.
- **Environmental impact assessment (EIA):** Assesses the environmental consequences of plans, policies or projects prior to the decision to proceed. An EIA is commonly used for construction projects.
- **Carbon footprint:** ISO 14067:2018 specifies principles, requirements and guidelines for the quantification and reporting of the carbon footprint of a product.
- **BREEAM and LEED:** Certification schemes for assessing the sustainability of buildings (among other schemes).

Project professionals have a role to plan for sustainability assessments, appoint expert consultants, coordinate stakeholders, develop plans to act on the results, communicate the outcomes and foster continuous improvement. By overseeing the sustainability assessment process, project professionals can drive sustainable outcomes and positive impacts.

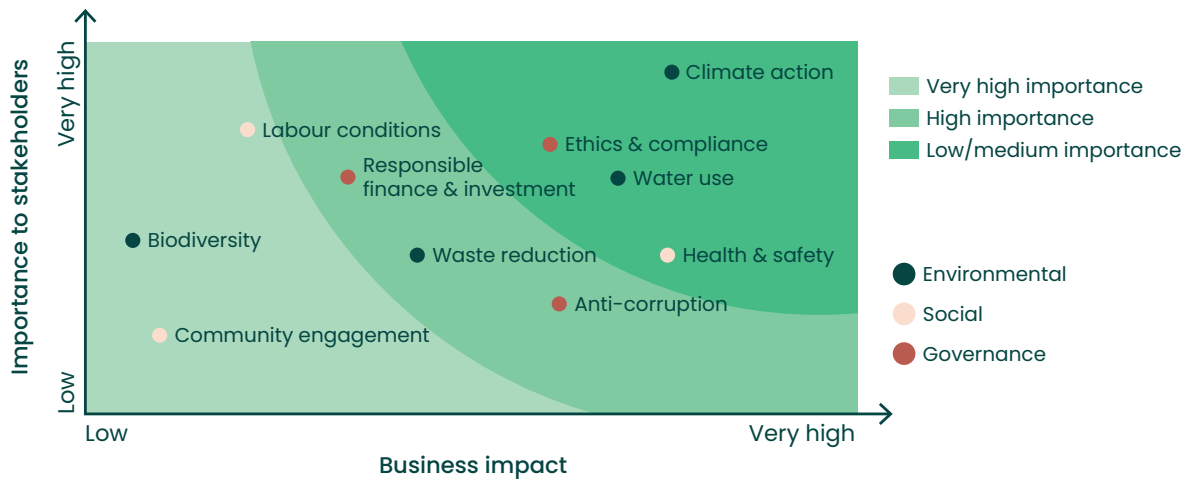


Figure 2.2.4 A sample materiality assessment

Source: Envoria

Recommended reading

- *The 17 goals* (2024) is an online resource that provides an overview of each of the 17 United Nations' Sustainable Development Goals (SDGs), including targets and indicators for successful implementation.

2.3

Financial management

Projects, programmes and portfolios need funding to deliver their objectives. Financial management is the planning, spending, monitoring and controlling of this money. Solid financial management allows project professionals to provide reliable reporting, to inform decisions.

Financial management is a specialised process that senior project professionals need to understand. Project working requires funds, which can come from within or outside the sponsoring organisation. This expertise includes knowledge of capital and project finance.

Financial management starts with financial appraisal, which establishes the potential value of a change initiative and leads to an investment decision. Following that, a project needs to have a budget as the basis to secure funding. Once the project starts, it will draw down that funding. Project professionals must monitor and control project expenditure. Finally, good governance requires reliable financial reporting.

Investment decisions must balance a wide range of factors, both financial and non-financial, but they rest upon a robust justification for committing limited resources. There are many tools available for assessing the financial value that the sponsoring organisation can achieve from a given initiative.

Once a decision has been made to invest in a change initiative, funding needs to be secured. This can come from many sources, inside or outside the organisation. The source of funding will introduce a group of stakeholders with a strong interest in the performance of the project, programme or portfolio. Usually, funding is not released in one tranche at the start of work. Instead, there will be a number of releases that the project professional needs to draw down throughout the life cycle. There will often be conditions to satisfy before this can happen.

The financial stakeholders will want to know about the performance of the initiative that spends the money they have allocated, given or lent, so financial reporting is both a governance and a communications imperative. Project professionals need to create reports that assess both status and forecasts. These form the basis of financial decision making by funders, sponsors and governing boards.

This section is written for project professionals who need to understand and carry out financial management for project-based working. The section includes:

2.3.1 Investment decisions: Evaluating the return on investment

2.3.2 Securing funding: Getting the money to deliver the changes

2.3.3 Financial reporting: Informing stakeholders about financial performance

2.3.1 Investment decisions

Evaluating the return on investment

Investment decisions are part of governance. Therefore they must rest upon a robust justification for committing limited resources. This is true in the public, private and non-profit sectors. There is always a choice about how to invest funds and capabilities to deliver value. Sponsors make the case that a possible initiative can realise best value for the capital, operational expenditure and staff time.

Investment decisions balance many elements, including the following:

- **Strategic alignment:** Will the outcomes align with the organisation's strategic goals? Are they consistent with policy, regulation and legislation?
- **Affordability:** Are the available funds sufficient to manage the risks and deliver the benefits?
- **Value delivered:** Is the investment likely to deliver a suitable financial return or other measurable value? This must account for capital and operational costs, and the benefits, over the economic life of the product. Is this the best achievable impact from these funds?
- **Portfolio effect:** How does the investment fit among the full set of operational and change activities?

Organisations have a choice of tools to evaluate investment decisions within an investment appraisal. These include simple measures like value ratio, net benefit and return on investment (ROI).

There are also sophisticated measures that account for the time value of money. Net present value (NPV), net present social value (NPSV) and internal rate of return (IRR) derive from discounted cash flows (DCF). These compare with measures of other ways the organisation can use its funds, such as the weighted average cost of capital (WACC). Organisations typically define hurdle rates, which represent the minimum target return on their investments. These concepts are part of capital budgeting within corporate finance (Figure 2.3.1).

For commercial reasons, private organisations usually base their investment decisions on financial considerations. Funding comes from debt, investment or reserves. Public and non-profit organisations are more often concerned with value for money, affordability and service. Funding often derives from government, grants or donations.

Decision makers also consider non-financial factors. These include the following:

- **Practicality:** Can the project be delivered technically, with appropriate risk, meeting regulatory standards and organisational values?
- **Maturity of definition:** Are scope and requirements defined in enough detail to give confidence that all costs and benefits have been captured?
- **Decision bias:** Are estimates fair or are they biased by agendas or prejudices? Are psychological biases or blind spots influencing decision-making?

For each investment, the sponsor brings together financial and non-financial considerations, with a risk analysis. This gives decision makers confidence in the projected returns and a prudent financial contingency. But the investment decision must not be

made in isolation. It should take place in the context of the full portfolio of initiatives (see section 2.1).

Assurance of the governance process is important. Independent review and scrutiny of the investment case would recognise a range of possible outcomes, not a single case, especially early in the life cycle.

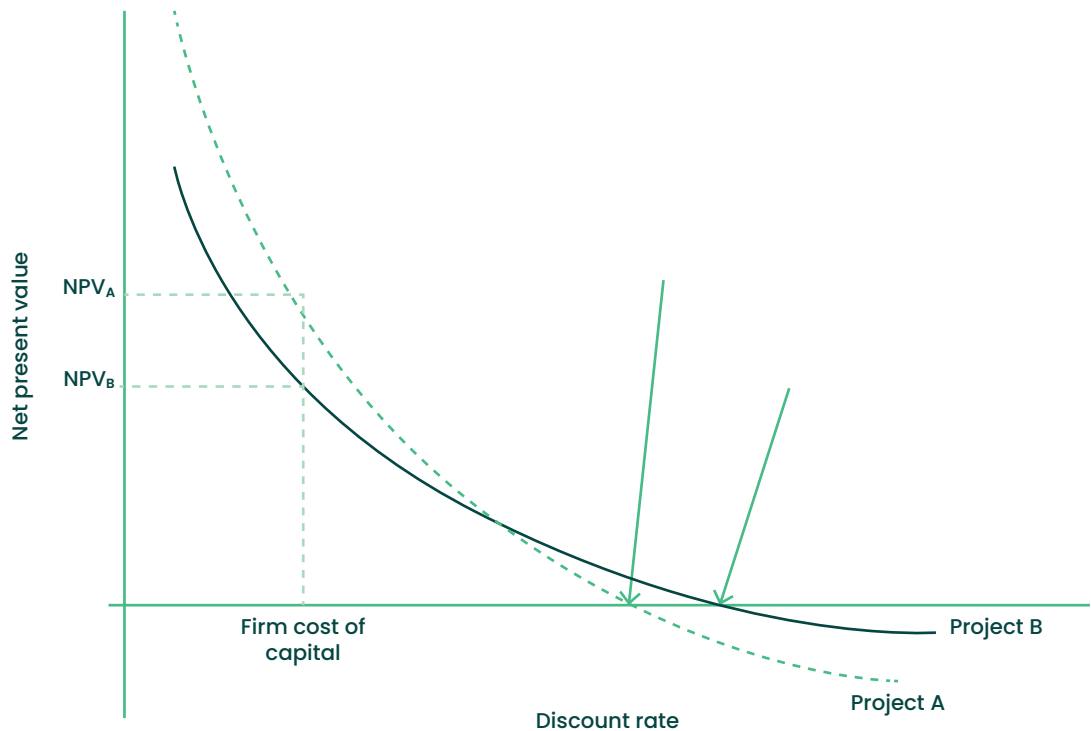


Figure 2.3.1 Comparing options to invest capital

Recommended reading

- *Net Present Value and Risk Modelling for Projects* (2016) explores NPV in detail, suggesting how it can be used during the early stages of project work to improve forecasts when uncertainty is at its highest and the opportunities to influence are at their greatest.
- *Delusions of success* (2003) examines how executives fall victim to what psychologists term 'planning fallacy' when forecasting the outcomes of risky projects. Executives spin scenarios of success while overlooking the potential for mistakes and miscalculations. As a result, managers pursue initiatives that are unlikely to deliver the expected returns.
- *The hidden traps in decision making* (2006) proclaims that, although making decisions is the most important job that executives undertake, it is also the toughest and the riskiest. Bad decisions can be traced back to the way decisions were conceived and considered – and sometimes the fault lies not in the decision-making process but rather in the mind of the decision-maker.

2.3.2 Securing funding

Getting the money to deliver the changes

Funding is the means by which the money required to undertake a project, programme or portfolio is secured and then made available as required. It can come from one or more of many sources – both within and outside the organisation. These include:

- current revenue for operational expenditure (opex) or capital reserves for capital expenditure (capex)
- external investment, debt financing or overdrafts
- government funding or grants
- charitable grants or donations

Internal funding can come from different sources. Smaller projects, with a narrower scope, draw funding from delegated departmental budgets. In this case, the budget holder may act as sponsor.

Funding for larger projects and programmes comes from the organisational level. Typically, the executive board oversees this. So, timing of funding decisions ties into the organisation's strategic and business planning cycles. Financing projects from the organisation's own balance sheet is called 'corporate finance'.

When the funding comes from external sources, this introduces important stakeholders, who may be investors, lenders, grant-making bodies or shareholders. They will have a strong interest in how project professionals are using their money, which raises the need for reporting and excellent stakeholder communication. Financing projects from outside the organisation (off-balance-sheet finance) is called 'project finance'.

It usually falls upon the sponsor to secure funding. At the very least, they would expect to lead this process. In many circumstances, they will also be the person responsible for managing the funding. However, for large programmes and portfolios, they may have a finance expert to support them.

External funding often comes with costs, which the investment appraisal needs to account for (see 2.3.1). Project professionals need to consider the costs of arrangement fees, borrowing costs (interest or capital charges), foreign exchange costs and administration. For long projects and programmes, interest and exchange rate variations pose significant risks. Managing these is part of the project finance discipline.

Another concern for project professionals is the timing and management of the release of funds. Often, projects and programmes can only draw down funding at specific times or events. These include:

- at decision gates or other milestones
- on delivery or handover of products
- due to procurement commitments or other contractual dates
- on scheduled drawdown dates

It is important to avoid unnecessary delays or issues. This means understanding the procedures in place within the organisation and the funders. Project professionals need to manage accountabilities, reporting and completion of documentation. If issues or variances occur, they will need to consider impacts on their funding needs, using mechanisms such as change control and contract variation orders to secure changes in the amounts and timings of funding drawdown.

	Capital Funding for capital expenditure (capex)	Revenue Funding for operational expenditure (opex)
From within the organisation (capital finance)	Capital reserves Investment funds	Current revenue Operational budgets
From outside the organisation (project finance)	Government grants Charitable grants Development funds Rights or bond issues Third-party investment Debt	Working capital finance Overdraft Government grants Charitable donations

Figure 2.3.2 Examples of sources of funding for projects, programmes and portfolios

Recommended reading

- *Principles of Project Finance*, 2nd edition (2013) features concepts and techniques, making it essential reading for those who want to succeed in financing large projects.
- *Project Finance for Business Development* (2018) shows how different elements of project finance come together to structure viable and financeable projects.
- *Project Finance in Theory and Practice* (2023) includes case studies and insights from project finance experts.

2.3.3 Financial reporting

Informing stakeholders about financial performance

Change initiatives are inherently risky. They undertake work that is novel to the organisation, involving substantial resources and cost, and the future of the organisation can depend on their outcomes. Good governance is therefore critical. Funders whose money is at risk will have an acute interest in the financial performance of a project, programme or portfolio.

This makes financial reporting an essential responsibility for project professionals. Often, it will be overseen by a sponsor, a project board, a project management office (PMO) or the corporate finance function.

But financial reporting can be important to stakeholders beyond the sponsoring organisation. These might, for example, be lenders, investors, grant-giving bodies or shareholders. They will want to know that the money they have lent, invested or given is being used wisely. Is it delivering the promised results? Will there be any unforeseen calls for extra funding?

In financial reports, these stakeholders will want to see backward-looking status reporting and forward-looking forecasts. Status reports show expenditure against budget, actual cash flow against plan and delivery progress against schedule. Earned value analysis (EVA) is a powerful tool that project professionals can use to compare rates of delivery and expenditure against plan.

Stakeholders also want expenditure forecasts and risk assessments – particularly financial threats. If there are risks, they will want to know about mitigations. Fundamentally, they want to know that the delivery team has the project or programme under control, so the frequency and timing of reporting will be important.

Ideally, systems allow detailed expenditure tracking in real time, with precise cost allocation. Where possible, they will be able to produce both reports and forecasts, with the content, detail and format tailored to the needs of each stakeholder group.

Information should be available as close to real time as the systems allow. Often, financial reports will be part of corporate-level reporting. Indeed, for substantial programmes and portfolios, they may also appear in statutory reporting and annual reports to shareholders and analysts. When this is the case, project professionals should expect oversight from the finance function.

Financial reporting for projects that follow an iterative life cycle can be more straightforward than for linear projects. Each iteration will have a large fixed-cost element and progress is easy to measure. However, project professionals often choose an iterative life cycle approach when there is a high degree of uncertainty about the specification of the completed product or the development process. This makes it harder to assess funding requirements at the outset. It also makes financial forecasting a challenge, unless the project works with a fixed total budget.

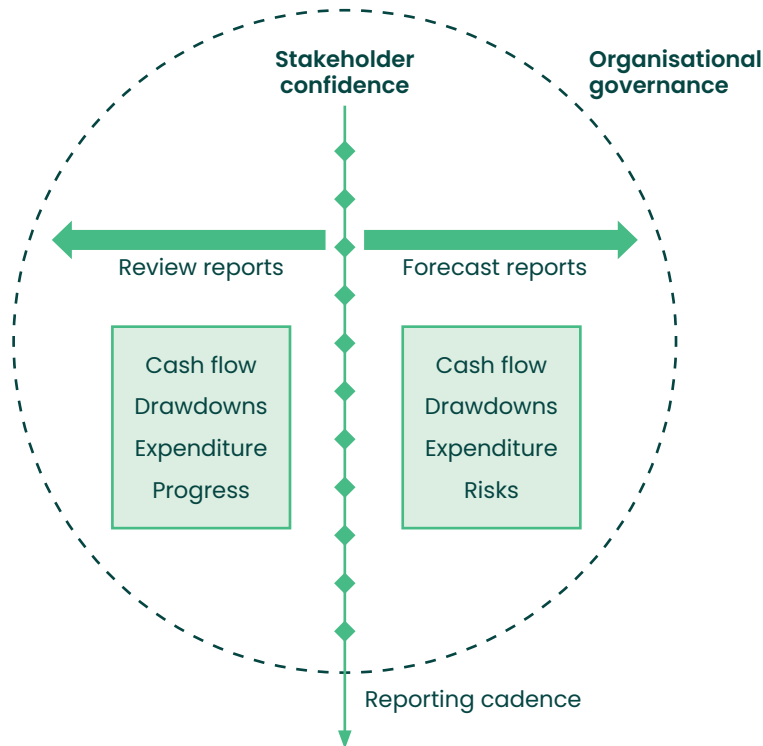


Figure 2.3.3 Financial reporting serves organisational governance and stakeholder confidence

Recommended reading

- *Cost and Value Management in Projects* (2023) addresses cost from a strategic perspective, examining project management decision areas that have the potential to enhance value and providing an integrated framework for managing cost.
- The APM Earned Value Management Specific Interest Group's *Earned Value Management Handbook* (2013) provides guidance on earned value tools and techniques that enable users to report on the level of expenditure against the budgeted cost of work.

2.4

Business cases

A business case provides the justification for undertaking a particular project, programme or portfolio. Once a business need is identified, project teams analyse the time, cost, risks and benefits of different options, before providing a recommendation of the best solution to achieve the desired outcome. Once approved, the business case is regularly reviewed to ensure the initiative continues to offer the organisation value for money.

Even though the project sponsor is accountable for the business case, in reality, the work to prepare and manage it is often delegated to project professionals. This process begins in the early stages of the life cycle, where project professionals work with subject-matter experts to create the initial business case. A strong business case will show how the initiative meets the following five dimensions:

- **Strategic case:** Showing how the initiative aligns to the organisation's objectives.
- **Economic case:** Demonstrating positive financial, societal and environmental benefits.
- **Commercial case:** Evidencing how the project will be paid for.
- **Financial case:** Proving the organisation can afford to do the initiative now.
- **Management case:** Detailing the project management approach to be taken.

As different options to deliver the initiative are analysed, project professionals and the project sponsor work together to put forward a recommendation for approval. If decision makers believe the initiative aligns with the organisation's strategic objectives, represents value for money and has a high chance of success, approval to proceed will be granted.

But the role of a business case doesn't end there. Project professionals must use it as a living document throughout the project life cycle. As changes in the project environment occur and new information comes to light, the business case should be updated to reflect the latest situation, justifying the project's continuation on the grounds that it will deliver value for money.

Where changes have a significant impact on the time frame, cost, risk or benefits of the initiative, the project sponsor may need to seek reapproval from decision makers to determine whether the project should continue or not.

In extended project life cycles, the business case is also used as a reference to measure whether the anticipated benefits of the project have been realised. Where variances occur, project professionals should share lessons learned to ensure greater accuracy for future business cases.

This section covers the following topics:

2.4.1 Purpose of a business case: Justifying the investment in a project, programme or portfolio

2.4.2 Creating a business case: Creating, reviewing and approving a business case

2.4.3 Maintaining business cases: Keeping the business case updated to enable ongoing justification

2.4.1 Purpose of a business case

Justifying the investment in a project, programme or portfolio

All projects, programmes and portfolios cost money. Because of this, when sponsoring stakeholders identify a need to change, they must demonstrate that the associated costs, effort and benefit represent good value for money for their organisation.

A business case provides the justification for undertaking a particular project, programme or portfolio. It evaluates the time, cost, risks and benefit of different options, before providing a recommendation of the best solution to achieve the desired outcome.

The information presented in the business case results from work conducted by project professionals in the early phases of the project life cycle (see 2.4.2). While all organisations will have their own business case format, all business cases typically include five key dimensions or cases (Figure 2.4.1):

- **Strategic case:** The compelling case for the project, programme or portfolio and how it aligns with other initiatives to support the organisation's strategic objectives.
- **Economic case:** The project's return on investment, comparing the time, cost and resources required with the expected organisational, societal and environmental benefits.
- **Commercial case:** How the project will be paid for and how the resources will be sourced (e.g. the procurement approach).
- **Financial case:** The project's affordability to the organisation in the timescales proposed (i.e. can we afford this project now, or should we wait?).
- **Management case:** The way the project will be run, including team roles, governance, the choice of life cycle and the ways of working.

Combining these dimensions creates a well-rounded view of an initiative, providing decision makers with the information they need to make the best decisions. Without a clear business case, decision makers risk investing money in unsound endeavours, leading to consequences such as financial losses, reduced productivity, detrimental environmental impacts and damaged stakeholder confidence.

But most organisations aren't short of problems to solve or opportunities to exploit. So, decision makers regularly have to make trade-offs when deciding which initiatives to start and stop, often making those decisions based on the current environment and the organisation's priorities.

Because of this, project professionals should remember that business cases are ultimately storytelling documents. The best business cases combine the technicalities of an investment appraisal with an evidence-based narrative of how the project team will work together to create success. Striking this balance, and telling a compelling story, will help project professionals convince decision makers that the initiative is worth progressing over other viable alternatives.

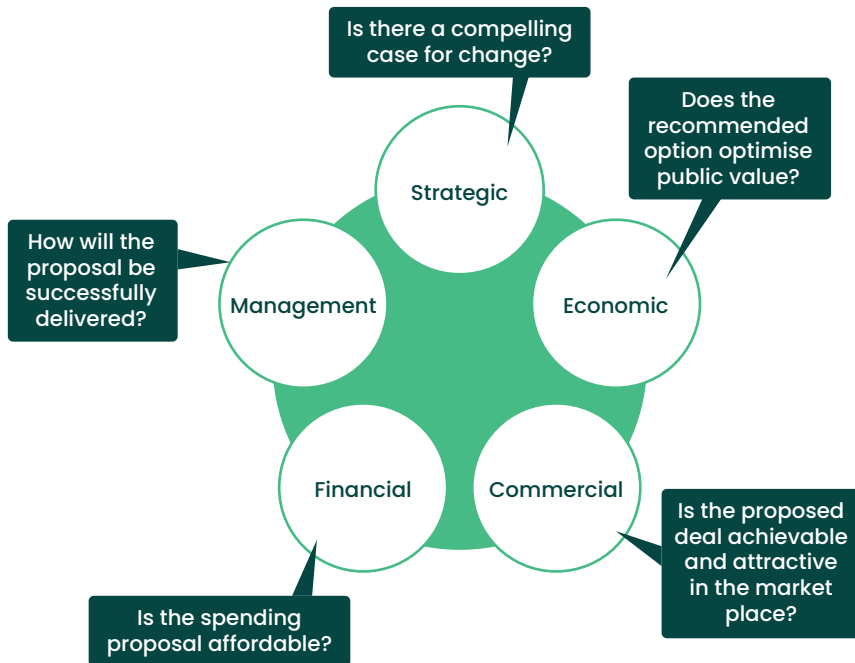


Figure 2.4.1 Five dimensions included in a business case

Source: Adapted from HM Treasury (2018)

Recommended reading

- The APM Benefits Management Specific Interest Group has produced *A Guide to Using a Benefits Management Framework* (2019). The resource includes an insight into the investment decision from the perspective of benefits realisation.
- The UK Government's *International Guide to Developing the Project Business Case* (2018) provides a detailed look at the Five Case Model, and step-by-step support on creating an outline or full business case for projects or programmes.
- The *Green Book* (2022) contains advice from HM Treasury on public sector investments. The advice has many universal concepts that apply equally across the private and charitable sectors.

2.4.2 Creating a business case

Creating, reviewing and approving a business case

Creating a well-rounded business case is a team effort, requiring inputs, expertise, challenge and assurance from many different stakeholders. While all organisations will have their own defined process for creating, reviewing and approving a business case, project professionals should familiarise themselves with the common activities to give them the best chance of success.

Even though the project sponsor is accountable for the business case, in reality, preparing the contents of the business case is often delegated to project professionals. This active involvement is actually good practice, as it brings them closer to the shaping process, facilitating knowledge sharing and developing a common understanding.

Creating a business case starts with clearly defining the business need. Here, project professionals work with the sponsor to define the problem to be solved or the opportunity to be exploited, linking either one back to its impact on the organisation's strategy.

Once a business need has been established, much of the work to create a business case now goes into analysing different solutions to meet the need. Project professionals should work with a range of subject-matter experts to determine the feasibility of different options, gathering high-level estimates of:

- the work required and how long it will take
- the resources and costs required
- the risks associated
- the societal, environmental and sustainability impacts of the change

Given that the business case is created in the early stages of the project life cycle, there will be uncertainty and a lack of detailed information. To overcome this, project professionals should use proven estimating techniques (see 5.4.1), risk-based contingency planning (see 5.8.3) and third-party assurance (3.4) to improve estimating accuracy and build confidence in their assumptions.

Once the options have been analysed, project professionals and the project sponsor should then review them together, before agreeing on a recommendation to put forward for approval. Within this recommendation, they should provide confidence in how the work will be achieved, including examples of the project's governance approach, the chosen life cycle and how factors such as environmental sustainability will be managed (see 2.2).

Once the business case is ready, the project sponsor will submit the business case and seek approval for the project to proceed. In most organisations, business cases are reviewed in defined governance forums, such as executive boards or investment committees, with approval resulting in the formal start-up of the project.

The level of rigour and analysis required when creating a business case often depends on the organisational culture and the chosen project life cycle. For example, projects with a linear life cycle are more likely to include a detailed analysis up front, with iterative approaches opting for a lighter business case, as detail will be gained when the project progresses.

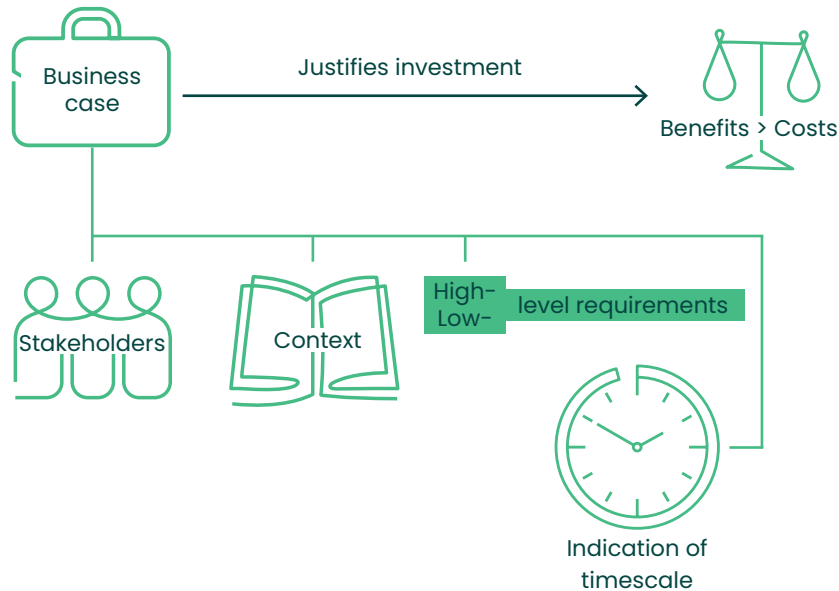


Figure 2.4.2 Contents of the business case

Recommended reading

- *Making the Business Case* (2009) is a straightforward guide to writing effective business cases. It offers practical examples and reflective exercises, with advice covering the journey from strategy to options consideration, and detailed content related to identifying and defining the benefits, costs and achievability.
- *International Guide to Developing the Project Business Case – Better Business Cases: For Better Outcomes* (2018) is government guidance on developing and appraising spending proposals to deliver best value for money.
- APM's short guide on *How to Improve Business Case Estimates* (2024) gives practical advice on how to overcome the challenges to creating accurate business case estimates.

2.4.3 Maintaining business cases

Keeping the business case updated to enable ongoing justification

Projects operate in complex environments, with internal and external changes happening daily. Even though most business cases are created to justify starting a new initiative, project professionals must remember that a business case is a living document, designed to provide ongoing evidence that the project is still delivering value for money.

Numerous events can impact the viability of a project's business case. Examples include:

- **new stakeholder requirements:** creating additional work and cost
- **economic instability:** changing the cost and availability of resources
- **technological advances:** reducing the benefit of the proposed solution
- **organisational restructuring:** creating additional risk and uncertainty
- **project team changes:** causing delays, additional cost and changes in approach

But while it is important to stay on top of changes, project professionals can't spend all their time reviewing their project's business case. Instead, it's best practice to use other project governance and control points as opportunities to reassess the viability of the project, including:

- **change control:** where the impacts of requested changes are assessed against the project's baseline
- **project reviews:** where project teams review progress, comparing the performance of the project's schedule, resources, scope and budget against the forecast
- **phase/stage-gate reviews:** where project teams and stakeholders approve the project to proceed to the next phase of the life cycle

Where new information significantly impacts the project's business case, the project professional should bring this to the attention of the project sponsor.

If the impact is considered to be significant, the sponsor may need to update, or seek renewed approval from, decision makers to decide whether the project should continue or not. Where multiple initiatives are vying for common resources, a project may be asked to pause, even if its business case may still deliver benefits, if other initiatives are deemed to be better value for money.

During extended project life cycles, project professionals will also use the business case as a reference to measure whether the anticipated benefits of the project have been realised. This crosses over into the concept of benefits management (see 3.1), with over- or under performance reviewed, in-conjunction with the project sponsor, and the business case updated accordingly.

Given the ever-changing project environment, it is inevitable that the business case of a particular initiative will change as new information emerges. As changes are managed, project professionals should also seek to share lessons learned with each other, to improve the accuracy and assumptions of future business cases.

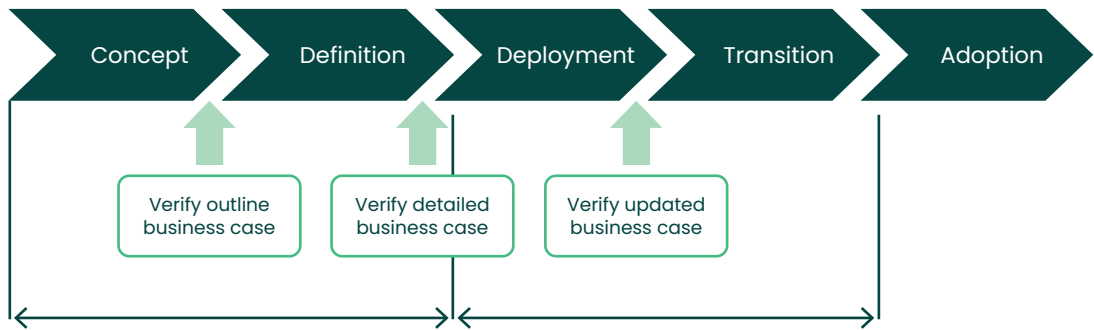


Figure 2.4.3 Stage-gate reviews to assess the viability of the business case

Source: Adapted from prince2.wiki/management-products/business-case/ by Frank Turley

Recommended reading

- *APM – ACostE Estimating Guide* (2019) is created by project professionals with real-life experiences. It helps readers to understand and diligently apply the core values of estimating cost to improve the clarity and robustness of an estimate for better decision-making.
- *HBR Guide to Building Your Business Case* (2015) provides guidance and tools to help maintain a strong business case, including how to align with strategic goals and how to present the case to stakeholders.
- *How to Learn Lessons Effectively* (2024), a short guide from APM, covers the process and techniques to help identify, capture and share lessons imaginatively and consistently.

2.5

Project life cycles

Projects are complex undertakings, requiring careful structuring and shaping to maximise their chances of success. One of the most important early decisions a project professional will have to make is about the most appropriate life cycle to help them manage the work.

A life cycle is a framework comprising a series of distinct phases required to transform an idea into reality in an orderly and efficient manner. It offers a systematic and organised way to undertake project-based work and can be viewed as the foundation underpinning the entire initiative.

At a high level, there are two fundamental life cycle approaches:

- **Linear:** This takes the project through a set of distinct, sequential phases in a single pass, only progressing from phase to phase once all the work on the current phase is completed.
- **Iterative:** This takes the project through a set of repeatable phases to create an initial output, which is then built upon with further deliveries of incremental value.

Each life cycle has its own strengths and weaknesses, making each one more or less suitable for particular initiatives. Project professionals must consider several factors when choosing the right life cycle, including their project's environment, the amount of information available to them, their appetite for risk, and the skills, competences and experience of the project team.

In many instances, the complex nature of business environments means that project professionals adopt a hybrid approach. They blend the philosophies, principles, tools and techniques of both linear and iterative life cycles to create a way of working that offers their project the best chance of overall success.

While most project life cycles end once the project has deployed the agreed outputs, extended project life cycles include additional phases to manage business adoption and the realisation of benefits. Here, project professionals are given the accountability and additional investment to manage these activities, helping the project to turn outputs into outcomes and increase the chances of the anticipated benefits being realised.

This chapter is for project professionals tasked with supporting, leading or overseeing project life cycles. Specifically, it will cover:

2.5.1 Linear life cycles: Following a sequential, phased approach

2.5.2 Iterative life cycles: Building project detail with an incremental approach

2.5.3 Hybrid life cycles: Combining linear and iterative life cycle techniques

2.5.4 Extended life cycles: Managing business adoption and benefits realisation

2.5.1 Linear life cycles

Following a sequential, phased approach

Linear life cycles, often referred to as 'waterfall' life cycles, divide the project into a set of distinct, sequential phases (Figure 2.5.1). These phases help the project develop from an initial concept through to the deployment of a final output in a single pass, only progressing from phase to phase once all the work has been completed.

By working in this sequenced way, linear life cycles enable project professionals to tightly control the project's progression, providing a clear framework for processes such as scheduling, budgeting, allocating resources and risk management. Linear life cycles also provide clear gateways between phases, which are helpful for arranging milestones, setting governance and conducting project reviews.

Because of this structured approach, linear life cycles work particularly well for clearly defined or high-risk projects, where project professionals can adjust the time, cost and risk to achieve the agreed scope and quality. They also work well for projects with a high cost of change, because they provide additional rigour to the design and planning of deliverables.

On the other hand, linear life cycles assume a lot of knowledge up front about the project, making them less suitable for developing or highly changeable environments. The linear sequencing of the phases also means end users must wait longer to receive value, with limited mechanisms in place to make major changes after the initial delivery is complete.

Most linear life cycles include variations of the following phases:

- **Concept:** An initial idea is developed through conducting research studies, gathering high-level requirements or carrying out a viability assessment, often culminating in an outline business case.
- **Definition:** The project is defined in greater detail, plans are created and a fuller statement of requirements is gathered, which provide a full justification for the work and outputs.
- **Deployment:** The defined plans are then implemented, with project professionals using testing and assurance to measure the project's performance against the intended outputs.
- **Transition:** Project sponsors and end users accept the outputs, resulting in a handover and a formal project closure.

The exact phasing, structure and naming of the adopted linear life cycle often varies according to an organisation's nature, purpose, environment and governance culture. For example, smaller projects may adopt a shorter cycle, with other projects incorporating an additional benefits realisation phase as part of an extended project life cycle (see 2.5.4).

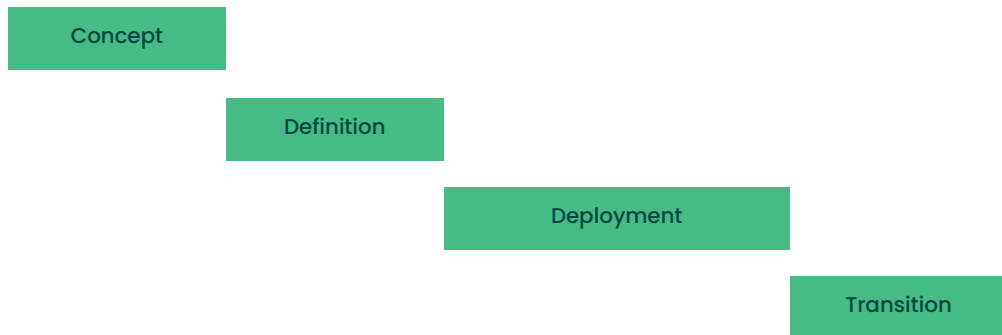


Figure 2.5.1 Linear project life cycle

Recommended reading

- *The Project Workout* (2019) provides a guide to structuring projects, combining useful advice with practical techniques. The resource identifies many of the key activities required to utilise the 'staged' framework with an extended focus on governance, monitoring and control, information management and the relevant standards.
- *APM Introduction to Programme Management* (2016) offers guidelines on the application of life cycles in the context of programmes. It provides detailed guidance on the key activities related to each of the major stages and on the governance of programme life cycles.
- *Guide to Life Cycles and Life Cycle Models* (2017) covers a plethora of life cycles and approaches identified by a joint task force involving members from the International Council on Systems Engineering (INCOSE) and APM. The coverage includes project and product life cycles with variations from many sectors and disciplines.

2.5.2 Iterative life cycles

Building project detail with an incremental approach

Iterative life cycles follow a set of repeatable phases to create an initial output which is then built upon by further deliveries of incremental value. Iterative life cycles are based on the concepts of continuous improvement and evolutionary development, where the solution evolves as learning takes place and new information emerges.

Recognising that change is fluid, iterative life cycles begin with developing a high-level vision, with the finer detail of the outputs defined and built during each iteration cycle. By allowing the specification and design to run in parallel, the deployment is 'fast-tracked' to deliver value earlier to the end users. This rapid deployment enables project professionals to gain rapid feedback, so that they can better understand the users' needs to make adaptations and improvements moving forwards.

For this reason, iterative life cycles work well for projects that are exploring new and innovative ideas without having the luxury of detailed up-front knowledge. They're also useful for developing new concepts, such as prototypes or pioneering technologies, and work best when there is a low cost of change, so that they can pivot and adapt quickly if required.

On the other hand, iterative life cycles are grounded in principles of collaboration and co-creation, relying on engagement and regular feedback from end users to progress. Iterative projects often work best when teams are co-located, as it helps break down communication barriers and enables the team to stay closer to the end user.

Much like linear life cycles, there are many variations to the structure, naming and phasing of iterative life cycles. One of the most common varieties stems from the dynamic systems development method (DSDM; see Figure 2.5.2), which includes the following phases:

- **Pre-project and feasibility:** The initial idea for the project is established, ensuring it aligns with organisational objectives, is cost effective, is technically possible and has a clear and measurable goal.
- **Foundations:** The project moves into planning and design, with the creation of a high-level delivery plan and a technical design.
- **Evolutionary development and deployment:** The project enters repeatable, iterative cycles of detailed design, development, testing and deployment. Early iterations create an initial output before adding further value as learning takes place and new information emerges.
- **Realisation:** Once the work has been completed, the final step is to embed the project outputs and evolve the next iteration.

Many iterative life cycles apply the agile concept of sprints, a timeboxed iteration with a fixed end date. Often, this culminates in the rapid delivery of an initial 'minimum viable product', with additional scope items being assessed, prioritised and planned for in future iterations. During this process, a daily scrum – or meet-up – is used to facilitate team collaboration and synchronise work.

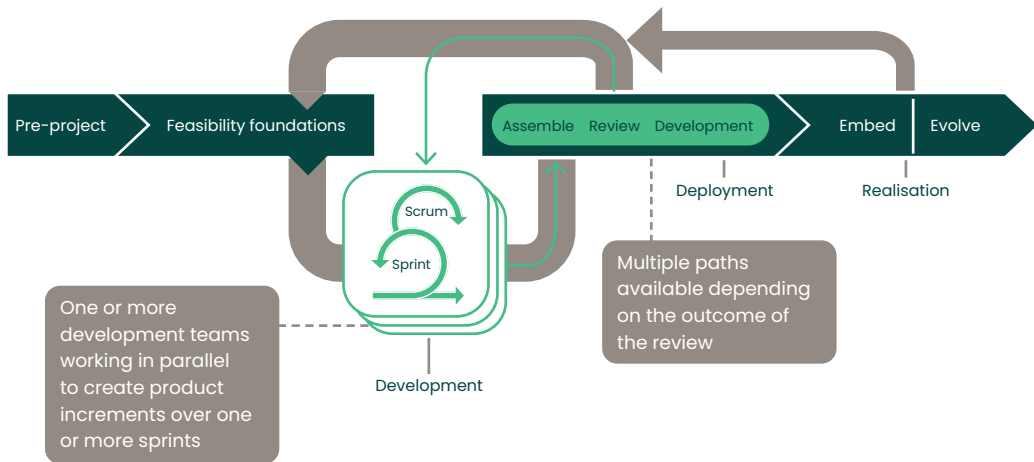


Figure 2.5.2 Iterative development in a dynamic, agile context

Source: Agile Business Consortium (2024)

Recommended reading

- *The Project Workout* (2019) covers concurrent engineering and other life cycle variations in the context of projects. It also offers some guidance on the relationship between agile deployment and project management.
- Guidelines on life cycles related to the management of programmes is covered in *APM Introduction to Programme Management* (2016). This touches on the idea of incremental development and deployment supported through a set of tranches. It also considers the relationship between projects and programmes, as implied by the deployment structure. This resource offers useful guidelines related to the use of increments and timeboxes in agile settings.
- *AgilePM, Vol. 3* (2024) describes the project life cycle in an agile setting. The text explores agile philosophy, highlighting how and where project management can be represented throughout an agile or iterative life cycle.

2.5.3 Hybrid life cycles

Combining linear and iterative life cycle techniques

Every project is unique, meaning there's no one-size-fits-all life cycle approach. The choice of which life cycle to use often hinges on what the project is trying to achieve, alongside the culture, governance and risk appetite of the organisation. In many instances, this means taking the best parts of different life cycles and bringing them together to maximise the project's chance of success.

Hybrid life cycles enable a pragmatic mix of life cycle elements to create an approach that's right for a particular project. Typically, hybrid life cycles fuse together elements of linear and iterative life cycles, creating an approach that caters to a range of factors, including:

- the amount of known information at the beginning of a project
- stakeholders' expectations of the project's time, cost, scope and quality
- the project team's skills, competences and experience
- the project's risk appetite

A common example of a hybrid life cycle is shown in Figure 2.5.3, where an iterative approach is used in the concept and definition phases to help overcome initial uncertainty and the lack of up-front information. Once clarity has been gained, the project moves to a linear approach for the deployment and transition phases. This provides stakeholders with certainty about the project's scope, while benefiting from a greater structure to help control the project's schedule, budget, quality and risks.

Given the complexity and uncertainty of many business environments, project professionals should remember that there is no need to apply one approach for the duration of a project. Instead, many of the most successful projects adapt their approach as the project progresses, utilising a mix of linear and iterative techniques to achieve the best outcomes.

When changing approach, the key to success is aligning the elements of a project's hybrid life cycle with the organisation's broader governance, such as project reporting and change control. This ensures that stakeholders are aligned, project team members are clear about their roles and responsibilities, and any dependencies on other change initiatives are fully understood.

Blending linear and iterative life cycles, principles, tools and techniques is equally beneficial for portfolios and programmes. For example, agile portfolio estimating techniques can be extremely useful for organisations that need greater flexibility when operating in uncertain times. Equally, programme benefits can be realised earlier through iterative deployments thanks to the ability to gain rapid feedback from end users.

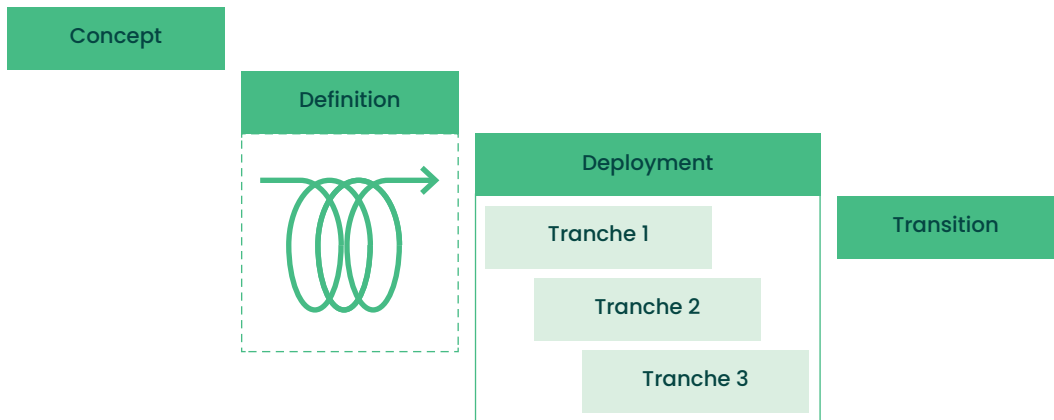


Figure 2.5.3 Hybrid programme life cycle

Recommended reading

- *A Guide to Assurance of Agile Delivery* (2017) was developed by the APM Assurance Specific Interest Group (SIG) to add assurance considerations to projects characterised by agile deployment forms. The guide incorporates aspects related to agile projects into the assurance activities, offering some specific guidance for hybrid projects, and reminding assurance specialists to consider the different constituent parts of the project.
- *Directing Agile Change* (2016) aims to provide guidance in relation to overseeing the deployment of agile projects. The resource was created by the APM Governance SIG. It identifies the distinctions between linear and iterative modes of deployment and acknowledges the implications of utilising hybrid combinations that draw on both perspectives.
- *Agile-stage-gate hybrids: The next stage for product development* (2016) was written by a product development and innovation expert. The article makes the case for integrating elements of agile product development into traditional gating processes, leading to faster physical product releases, better responses to changing requirements and improved team communication.

2.5.4 Extended life cycles

Managing business adoption and benefits realisation

The expectations of projects can differ and, as a result, the scope of a project's life cycle can take various forms. When they are part of a larger programme, projects need only to be concerned with delivering their outputs, but standalone projects may be expected to help turn their outputs into improved business outcomes by managing business adoption and benefits realisation activities before they close.

Extended life cycles add adoption and benefits realisation phases to the end of a standard project life cycle. This provides project professionals with the accountability and additional investment to ensure that changes are fully embedded into business operations and the anticipated benefits of the project are realised. Specifically, this includes:

- **adoption:** activities such as communications, training, knowledge sharing and change management to ensure operational teams correctly utilise the project's outputs
- **benefits realisation:** activities to measure business performance, ensuring that adoption leads to the desired outcomes and the realisation of the anticipated benefits, for example, tracking improvements to business expenses, sales revenue or staff morale

As adoption of the project's outputs increases (e.g. the use of a new sales tool), the organisation should begin to see improved business outcomes (e.g. increased customer engagement), thus realising the anticipated benefits of the project (e.g. higher sales revenue). For this reason, the adoption and benefits realisation phases often run in parallel (Figure 2.5.4).

Most programmes include adoption and benefits realisation phases as standard but, for standalone projects, a decision should be made about whether to extend the project's life cycle. On the one hand, extended project life cycles typically lead to higher adoption rates and a greater chance of realising the anticipated benefits – and these are often the key elements in determining stakeholders' views on whether the project has ultimately been a success. On the other hand, the organisation will incur additional costs for project professionals' time and effort from supporting these activities, versus asking operational teams to manage them themselves.

Where the decision is made to adopt an extended project life cycle, project professionals must plan for this in the early phases of the life cycle, considering the requirements, work, resources and risks of the additional phases. This crosses over into the concept of benefits management (see 3.1), where the identification, definition, planning, execution and tracking of benefits are all required as key activities within the broader project management plan.

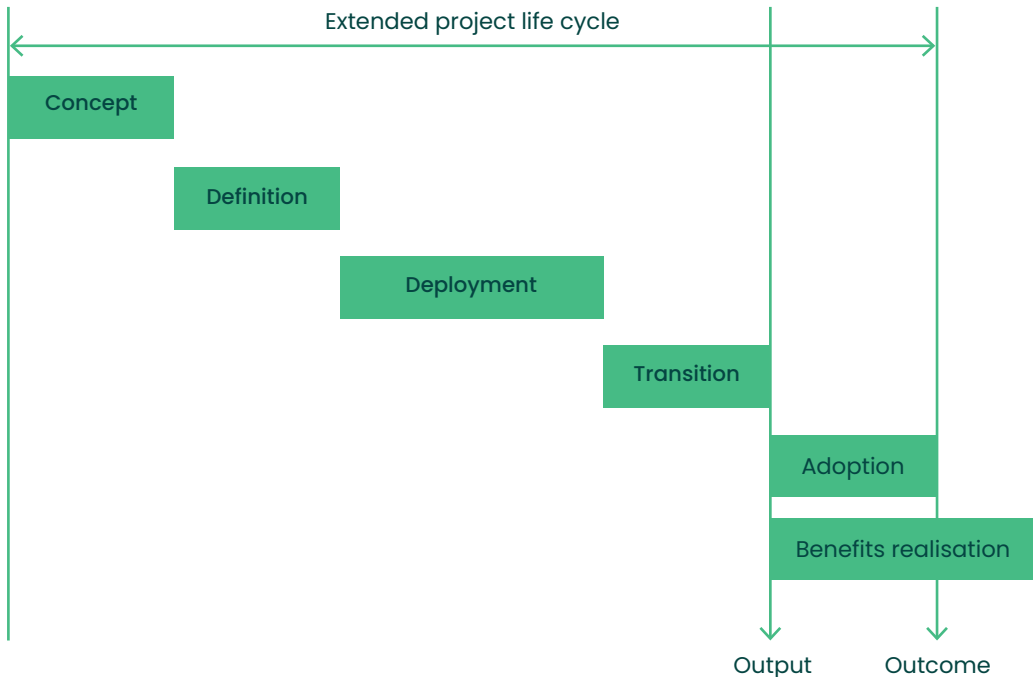


Figure 2.5.4 Extended life cycle

Recommended reading

- *Guide for Effective Benefits Management in Major Projects* (2017) pulls together insights and lessons from significant major projects. The report synthesises the key principles and activities required to successfully deliver benefits in such projects, and offers advice and guidance on extending the life cycle.
- *A Guide to Using a Benefits Management Framework* (2019) was compiled by the APM Benefits Management Specific Interest Group. The guide develops an understanding of the need for benefits realisation, offering a framework for addressing contextual factors and developing the capability to realise benefits.
- *Benefit Realisation Management* (2010) is a practical guide to implementing benefits realisation in organisations. The book explains the processes required to support benefits realisation practice, identifying the key additions required to supplement projects and programmes, and embed benefits realisation practice as a measure of success.

2.6

Governance arrangements

Governance is a framework of authority and accountability. It defines and controls the outputs, outcomes and benefits from projects, programmes and portfolios. It is how the investing organisation exerts oversight and control over the work and the realisation of value.

It is the role of governance to ensure that projects, programmes and portfolios:

- **do the right things** to serve their investing organisations...
- **... in the right way** by complying with good practices, ethical standards, policies and procedures, regulations and legislation

Securing the most business value depends on sound governance and assurance (see 3.4). Governance and assurance ensure delivery is consistent with the business case and that resources are used wisely. Successful governance depends on the sponsor and governing board. They need both the right terms of reference and the capabilities to carry out their responsibilities.

The governance principles for a project, programme or portfolio should integrate with the wider corporate governance. They must also recognise the nature of the initiative: its scale, complexity and impact, and the life cycle model it follows. Good governance needs good and timely information and fully engaged people.

Leading the governance process will be the sponsor. This individual is responsible for ensuring the work meets its objectives and the organisation's needs. They are an advocate, a supporter, a decision-maker and an overseer of the initiative. Sponsorship is a big and important role.

Project, programme and portfolio organisations are temporary structures within a wider organisation. As with governance arrangements, these structures should reflect the organisation's permanent structure and the needs of the initiative. Project professionals need to balance competing pressures from the permanent and temporary organisations.

Governance boards can be responsible for project, programme or portfolio governance, or they can be a supporting structure for sponsors who have full responsibility. They need clear terms of reference and members who have the right skills and expertise to fulfil their board roles.

The section includes:

- 2.6.1 Governance principles:** Establishing control of deployment of projects, programmes and portfolios
- 2.6.2 Sponsorship:** Championing the work to ensure it delivers the intended benefits and value
- 2.6.3 Temporary structures:** Aligning and balancing temporary and permanent organisational structures
- 2.6.4 Governance boards:** Putting governance principles into practice

2.6.1 Governance principles

Establishing control of deployment of projects, programmes and portfolios

Governance of project-based work overlaps with corporate and operational governance (Figure 2.6.1). When it is effective, it provides confidence to the board of directors, trustees or members that investments in projects, programmes and portfolios are being well managed. Project professionals design these governance structures. This needs an understanding of corporate culture and structures, and of the complexity and risks of the change.

Good governance needs a clear definition of the roles and responsibilities of the team and governance tiers. A responsibility assignment matrix can be useful here. It clarifies which roles:

- are accountable and responsible for activities and decisions
- must be consulted during decision-making
- need to be informed of outcomes

Creating and maintaining clear responsibilities throughout the life cycle lets governance fulfil its functions. These include efficient, effective and accountable:

- consultation with key stakeholders and experts
- decision-making and direction setting
- oversight and review
- provision of resources to maintain the right capacity and capability

Governance empowers project professionals while also supporting them. It defines delegated limits of authority and establishes a clear escalation route for issues, decisions and change requests.

Governance records underpin the internal and external auditing of a project, and maintaining accurate, well-organised and accessible records is essential.

Project professionals select a life cycle model for each initiative (see section 2.5). Governance arrangements need to fit with this choice. The structures, standards and levels of rigour should also be appropriate to the work. Ensuring all the requirements of a preceding phase are met before work progresses is important in all life cycle models.

For linear projects, decision points between life cycle phases are known as decision gates (see 3.5.3). They rely on assurance of the work carried out (see 3.4) and integrated plans for the work to come (see 5.4).

For iterative life cycles, there is a formal review of completed work at the end of each iteration. Work is drawn down for the next cycle only if the value proposition remains positive.

Good governance relies on robust reporting and good-quality information. Sponsors, boards and other senior stakeholders need this to understand the work and make informed decisions. Project-based work often has multiple owners, so governance structures need to give each stakeholder an appropriate level of influence, and a route to resolve any disputes. This will depend on their stake, status and expertise.

Adopting these governance principles ensures that project professionals:

- deliver the outcomes and value the investing organisation commits to
- have the escalation and decision support they need to do their job
- adhere to legal, regulatory, corporate, ethical and professional standards

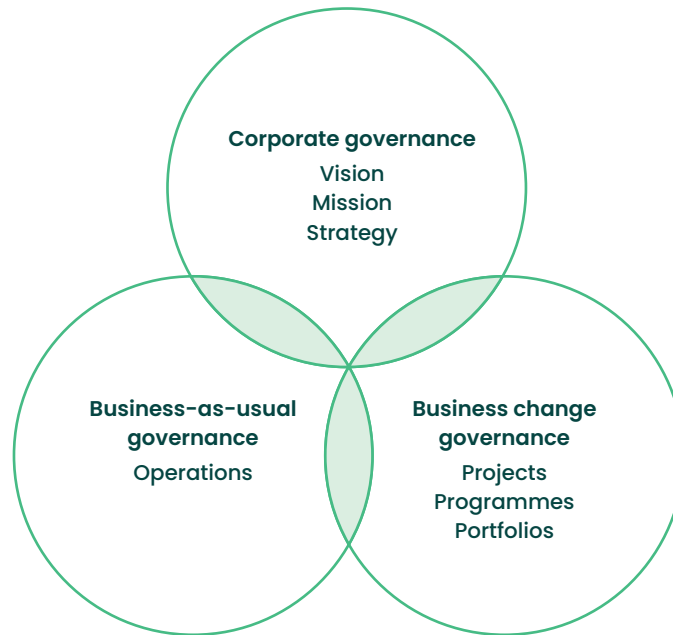


Figure 2.6.1 Interaction of different levels of governance

Recommended reading

The APM Governance Interest Network (formerly APM Governance Specific Interest Group) has published a comprehensive suite of guides on a range of governance principles:

- *Developing the Practice of Governance* (2019) is an APM report making recommendations for different governance approaches dependent on the different project life cycles.
- *Directing Change* (2018) provides advice to those with corporate governance roles on how to adopt solid practices for the governance of complex change. It also links change with BAU and corporate governance activities.
- *Directing Agile Change* (2016) explains how good governance of projects that adopt iterative/agile approaches is enabled and suggests collaborative behaviours that can be applied at any level of leadership in the organisation, from the main board and chief executive level downwards.
- *Governance of Co-owned Projects* (2017) offers a succinct guide for boards and their advisors, providing a set of underlying principles that can be assessed and applied when there are multiple organisations investing in a project.

2.6.2 Sponsorship

Championing the work to ensure it delivers the intended benefits and value

The sponsor (sometimes known as the senior responsible owner) is accountable for good governance and ensuring the work meets its objectives and the organisation's needs (Figure 2.6.2).

Sponsors are business leaders. They play a key role in promoting, advocating and shaping project work. The sponsor oversees the conduct of a project or programme, remaining accountable for the realisation of the benefits during and after delivery.

Sponsorship is a crucial role, so sponsors need to have the status and authority to influence the deployment of the project or programme. Like other project professionals, they need clarity about their role, authority and responsibilities. They should seek formal terms of reference from the investing organisation.

The sponsor's involvement continues throughout the life cycle of the change initiative. However, the extent of their engagement will fluctuate, tending to peak at the initial and final stages of the project.

At the start, the sponsor oversees the work that sets up the project or programme. This includes identifying needs and requirements, establishing the business case and securing funding.

During the main part of the project, they lead the day-to-day governance activities. These focus on oversight of the work, making decisions, and interfacing between the organisation and the project or programme.

In the closing stage, the sponsor ensures the project team has properly closed out the work. They oversee the handover of deliverables to operations and ensure the realisation of planned benefits.

The sponsor's role in the life cycle usually starts before and continues after that of the project or programme manager. As part of this, they may select and appoint the project professional who will lead delivery.

Sponsors play a central role in governance. They:

- ensure the continuing validity of the business case
- lead decision-making processes, particularly in setting the direction of the project
- secure funding and top-level contingency (management reserves)
- act as the first point of escalation for the project or programme manager

In addition to the governance role, effective sponsors support the project professional. This is a leadership, management, communications and coaching role. They may also help to support the wider team, for example by motivating them through difficult times. They also have a role in stakeholder engagement, particularly with influential and senior stakeholders. Through these working relationships, the sponsor can add great value.

A successful sponsor is:

- a leader and decision-maker who can work across boundaries within the organisation
- a stakeholder influencer with authority to act on behalf of the investing organisation
- an enthusiastic advocate of the work and the change it brings about
- prepared to commit sufficient time and support to undertake the role
- aware of the practices that underpin project-based working. They can make informed decisions about whether the work is being managed effectively, responsibly and sustainably. They can assess performance against plan and challenge project professionals appropriately
- aware of their own shortcomings, and a facilitator bringing in the appropriate technical expertise, industry experience and influencers when and where necessary.

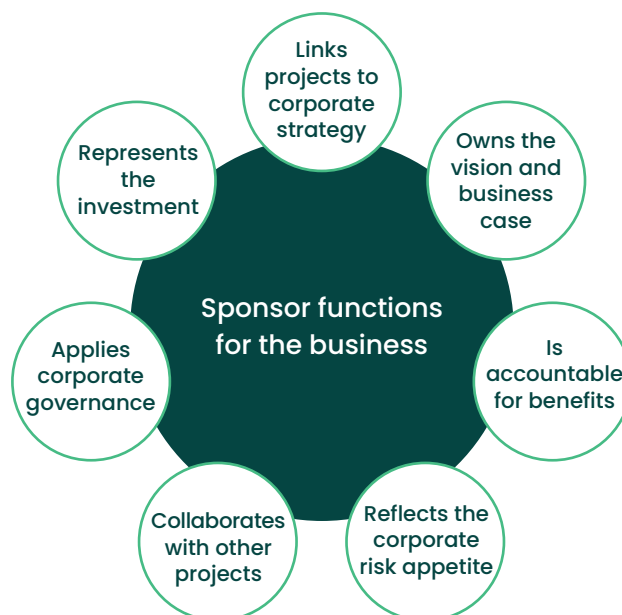


Figure 2.6.2 The scope of project sponsorship

Recommended reading

- *Building Sponsors: Future Project Leadership* (2018) is an APM report based on outputs from a sponsors' summit exploring the key themes of the sponsor's role. The event and the report focused on real-world experiences from a wide range of sectors and laid out key themes relevant to a broad consensus.
- *Sponsoring Change* (2018), by the APM Governance Specific Interest Group, covers areas such as why projects need sponsors, how the board's support is important for sponsor and project success, the attributes of effective sponsors, what sponsors do for the business and what sponsors do for the project manager, and gives some useful pointers on choosing and selecting sponsors.

2.6.3 Temporary structures

Aligning and balancing temporary and permanent organisational structures

Organisational structures are about power and information. They define who holds roles, responsibilities and power, and influence how information flows between different levels of management. This means they allow some people to control the actions and decisions that achieve (or frustrate) strategic objectives.

For day-to-day work, there is a 'permanent' organisational structure that provides a relatively stable environment for decision-making and information flow. There are many structures that organisations can adopt. At one end of the range is a federal structure with autonomous units and at the other is a matrix structure. Individual operational units are subject to competing direction from different managers, which creates the challenge of reconciling competing priorities.

Projects and programmes are temporary endeavours with temporary organisational structures. Project professionals design them to manage activities and resources in order to deliver specific objectives within predetermined time frames. These do not typically match the structure of the permanent organisation but they do need to be coordinated.

Organisations are made up of functional departments, for example finance, information technology, product development and human resources. These functions provide a structure for holding the resources, processes and technologies that perform work.

Project professionals work across both permanent and temporary structures. How easy or hard this is depends on how well the organisation's structures align with each other.

Organisations may also have a function that manages a portfolio of change initiatives. They might call this a portfolio management office, a value management office or similar. Within it, portfolio managers work with senior stakeholders across the organisation to:

- identify and define change initiatives
- prioritise them and ensure they align with strategy
- track the intended benefits they deliver
- fulfil other functions the organisation chooses for them

An organisational resourcing strategy identifies and allocates staff to serve the business. The resourcing strategy balances permanent staff with contractors or other third parties and links to talent management within the organisation. Its role is to select people for each role and give opportunities for professional development and progression.

Temporary organisation structures often include temporary staff; for example, a complex project may require a highly experienced project professional, but this person may not be part of the permanent staff, so could be brought in for the project. Alternatively, the organisation may choose to appoint a less experienced candidate. They would support them with effective mentoring, coaching and learning opportunities.

Project-based working often spans functions, so it is vital to have clear accountability within both permanent and temporary structures. The sponsor plays a key role in managing the interface between permanent and temporary structures, to:

- negotiate to ensure projects have access to suitably skilled and experienced people. Drawing these from the permanent organisation can affect operational performance
- manage tensions across the boundary between the temporary and permanent organisations
- ease the transition from project delivery to operational implementation

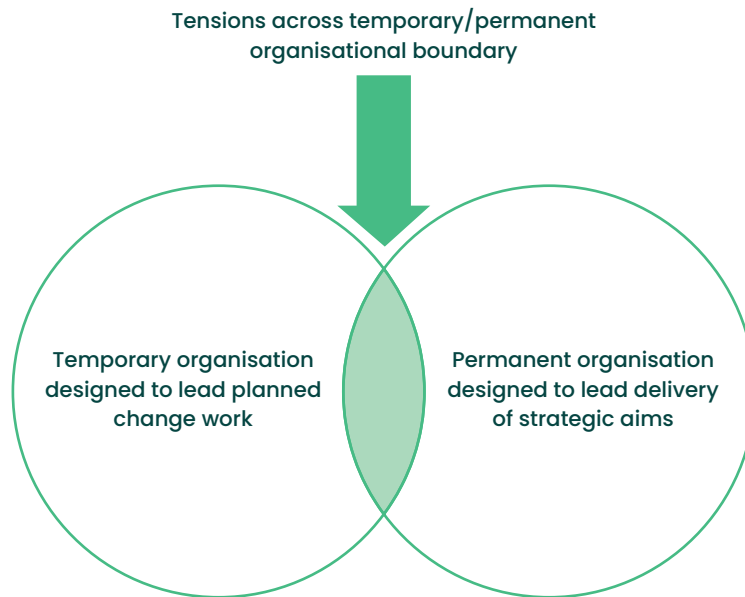


Figure 2.6.3 Relationship between the permanent and temporary organisations

Recommended reading

- *Managing and Working in Project Society* (2015) offers an examination of the challenges associated with temporary project organisations in a permanent society. The authors make a strong case for renewing institutions to ensure that they are able to meet the imperative for, and challenges of, increasing and accelerating projectification.
- *Advancing Research on Projects and Temporary Organizations* (2014) is an edited volume, bringing together multiple contributions to focus on the relationship between modern organisations and project management. The book provides an excellent research-orientated overview of temporary organisations and the importance of understanding project work as a social process.
- *Temporary organizing* (2016) is the introduction to a special edition of *Organization Studies*, a leading management journal, and discusses key challenges and opportunities in the study of temporary organising, including methodological issues, how to theorise time and how to relate the temporary to the more permanent.

2.6.4 Governance boards

Putting governance principles into practice

Governance boards oversee projects, programmes and portfolios. They review performance and make decisions in a way that is consistent with the size and complexity of the work. The board will represent financial, provider and user interests. It may include representatives of the functions and departments investing in, or affected by, the changes. Common terms for governance boards include project or programme board, or steering group or committee.

The responsibilities and authority of a governance board are set out in its terms of reference or charter. It is usually the sponsor who develops this. Terms of reference may set out specific roles for members and escalation routes for decisions beyond the board's authority.

In most organisations, governance boards follow procedures that are consistent with corporate governance, including submitting papers in advance and taking down minutes.

It is important that the members have the right skills to perform the board's functions. Often, members will include experts able to assess specific elements of the initiative. Where projects are co-owned, each owner will be represented in a way that reflects their stake and interests.

Wider stakeholder representation on the board may be beneficial, as it can promote engagement and the management of expectations. The extent of their participation is a matter for the board to decide.

The sponsor plays a key role in establishing the board's structure, culture and working practices. Sponsors often chair governance boards (Figure 2.6.4). Two models are common:

- The board is the primary governance body. It has delegated authority from the sponsoring organisation. The sponsor is one member of the board.
- The sponsor is the primary source of governance. They have delegated authority from the sponsoring organisation. The board's role is to advise the sponsor.

The governance board discharges its responsibility for the performance of the project, programme or portfolio by:

- overseeing progress and reviewing alignment with the organisation's strategy and strategic intent
- overseeing the realisation of benefits
- overseeing risk
- monitoring progress through project reports, assurance and audit
- scrutinising progress using objective metrics
- overseeing remedial measures for deviations from planned or forecast performance and agreeing major departures from plan
- reviewing and determining change requests (see 5.11.1). They may escalate them to a higher authority where necessary

- ensuring policies, regulations, statutory obligations, and professional and ethical standards are followed
- making formal go/no-go decisions at defined decision gates (see 3.5.1)

In some instances, there is a need for many experts or representatives, which can create an unwieldy governance board. In these cases, there may be a small governance board that makes decisions, but it will be supported by a wider steering group, with an advisory role only.

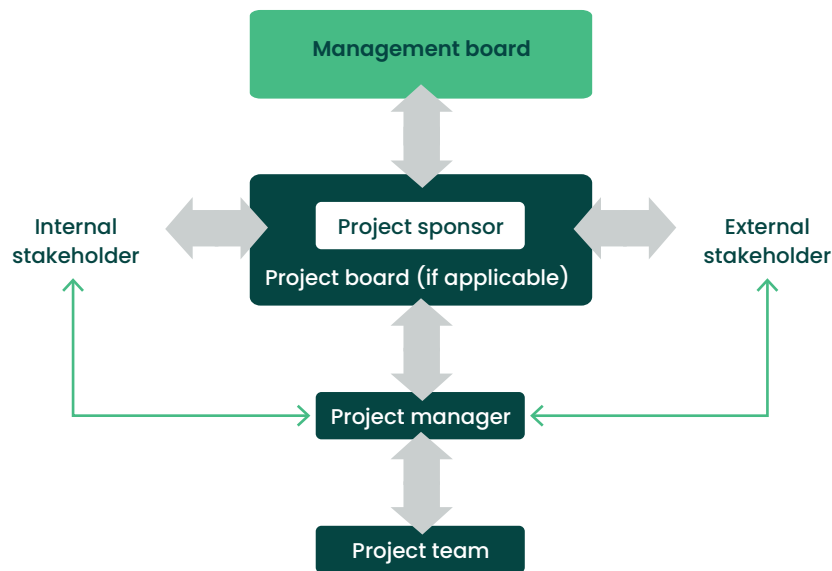


Figure 2.6.4 Generic project governance structure

Recommended reading

- *Project Governance* (2009) focuses on the structures required for effective governance. The project-steering group and its role are explored in detail, allowing readers to engage with two main approaches to governance (namely, transaction and agency perspectives).
- *Project Governance: A Practical Guide to Effective Project Decision Making* (2009) introduces the principles of effective project governance and detailed guidance on a project-governance model.
- *The Handbook of Board Governance* (2016) is an excellent resource for members of all governance boards. The edited collection offers comprehensive insights, addressing many critical aspects relevant to projects and some of the issues that need to be addressed.

3 Preparing for change

By preparing for change before, during and after the project has been completed, organisations can ensure project success and develop a culture that consistently applies best practice.

The chapter begins with benefits management, the essential process of defining, planning and tracking a project's benefits to ensure they deliver their intended organisational and social outcomes. This process establishes and monitors a roadmap for achieving intended benefits.

Capability development focuses on the continuous improvement of project management skills, competences and practices within an organisational context. Developing and retaining talented project professionals, combined with the creation of communities of practice and the effective management of knowledge, can foster a culture of continuous learning and improvement. These elements ensure that organisations are prepared to manage projects, programmes and portfolios consistently and effectively.

The role of procurement is explored to understand its contribution to project success. An effective procurement strategy with a structured process, alongside the optimal mix of internal capability and external specialist suppliers, will ensure that resources align with the project's needs and goals, mitigating risk, driving efficiency and delivering the desired outcomes effectively.

Assurance provides confidence that a project or programme will deliver its objectives and intended value. Through effective implementation of assurance principles, alongside robust audit processes, organisations can ensure their projects remain aligned with strategic objectives and meet defined standards.

Continuing the theme of ensuring success throughout the project, the role of reviews is explored through effective decision gates and project reviews. The section also features the strategic role of the project management office (PMO) and the important role information management plays in creating and promoting a structured environment for overseeing project progress.

Finally, the chapter concludes with transition management, discussing the processes for transitioning project outputs into operational use, handling unexpected project endings, and closing projects and programmes effectively. These activities ensure that organisations can seamlessly integrate project outcomes, maintain continuity and realise the full value of investments.

This chapter is composed of six sections:

- 3.1 Benefits management**
- 3.2 Capability development**
- 3.3 Procurement**
- 3.4 Assurance**
- 3.5 Reviews**
- 3.6 Transition into use**

3.1

Benefits management

A benefit is a positive and measurable impact arising from a change initiative. All projects should deliver benefits, but they don't just happen on their own. Instead, project professionals must carefully manage benefits, ensuring that projects and programmes deliver the right outputs and outcomes to create a positive impact and justify the investment.

As a concept, benefits management includes processes for the identification, definition, planning, tracking and realisation of benefits. Once portfolio-level stakeholders have aligned change initiatives with the strategic objectives of the business, project and programme teams manage the entire benefits journey, covering everything from identification through to realisation.

In most instances, project professionals do this through a range of defined processes, which come together to form the overall benefits management strategy. These processes include the following:

- **Benefits identification:** Benefits are identified, quantified and agreed with the project sponsor and other expert stakeholders. Each benefit is documented, including what it is, the target to be achieved and how improvements will be measured.
- **Benefits planning:** A benefits realisation plan is created, detailing how benefits will be generated, and the granular activities and resources required to create and adopt the underpinning outputs and outcomes.
- **Benefits tracking:** As projects progress, project professionals continually confirm that expected benefits are still viable, as well as tracking progress towards them. As changes occur, impacts to the benefits are reviewed, with material impacts escalated to sponsoring stakeholders, if required.
- **Adoption and benefits realisation:** Outputs and outcomes don't automatically create benefits. Instead, business users need support to adopt an initiative's deliverables. As this adoption increases and new outcomes are generated, benefits should be measured to determine whether the original targets are now being met. If not, action should be taken to improve adoption or agree new targets.

While it is not covered in this section, at a portfolio level, many organisations operate a broader benefits management framework. This framework provides a structure for categorising benefits, while providing visibility on their contribution to organisational strategy and helping to inform the prioritisation of initiatives and investment decisions.

This chapter is for project professionals tasked with supporting, leading or overseeing benefits management. Specifically, it will cover:

3.1.1 Benefits planning: Defining and planning organisational benefits

3.1.2 Benefits tracking: Ensuring benefits remain achievable as the project progresses

3.1.3 Adoption and benefits realisation: Maximising adoption to increase benefits realisation

3.1.1 Benefits planning

Defining and planning organisational benefits

All projects must deliver benefits to justify their investment, but benefits don't just appear out of thin air. In fact, like many things in project management, benefits management starts with creating a plan. A benefits management plan clearly defines an project's benefits, who is responsible for them, and how they will be delivered and measured.

Identifying and planning benefits is an integral part of an project's early shaping, forming the foundation of the business case and helping define success criteria, scope and requirements.

While the project sponsor will often have their own view of the benefits, project professionals should bring together a range of subject-matter experts to agree a common understanding of the benefits on offer from a particular project.

During this identification process, it is wise to remember that benefits take many forms. Financial benefits (e.g. revenue increases or expense savings) are often easier to define, while the equally important non-financial benefits (e.g. regulatory compliance or customer satisfaction) can be harder to quantify but should not be overlooked.

As each benefit is identified, quantified and agreed, a benefit profile should be created. This should include the following information as a minimum:

- the benefit's description
- the benefit category (e.g. financial versus non-financial)
- the current baseline (e.g. the revenue today)
- the future target (e.g. future revenue)
- how the benefit will be measured
- how the benefit links to the organisation's strategic objectives (see Figure 3.1.1)
- a benefit owner

Defining a benefit's baseline, target and measurement is especially important, as otherwise it's difficult to determine whether the benefit has been achieved and whether it has had a positive impact on the organisation's objectives. While the project sponsor takes overall accountability for benefits being delivered, benefit owners should be nominated to ensure that the business is ready to accept the project's outputs, improve outcomes and, ultimately, realise the benefits.

With the benefits defined, a plan can be created to show how they will come to life. At an organisational or portfolio level, a high-level benefits management plan will identify the project or programme responsible for delivery, along with the timeline, level of risk and key stakeholders involved.

At the project or programme level, a detailed benefits realisation plan will form part of the integrated project management plan. This should highlight the outputs and outcomes that will help generate the benefits, and the granular activities and resources required to deliver and adopt them.

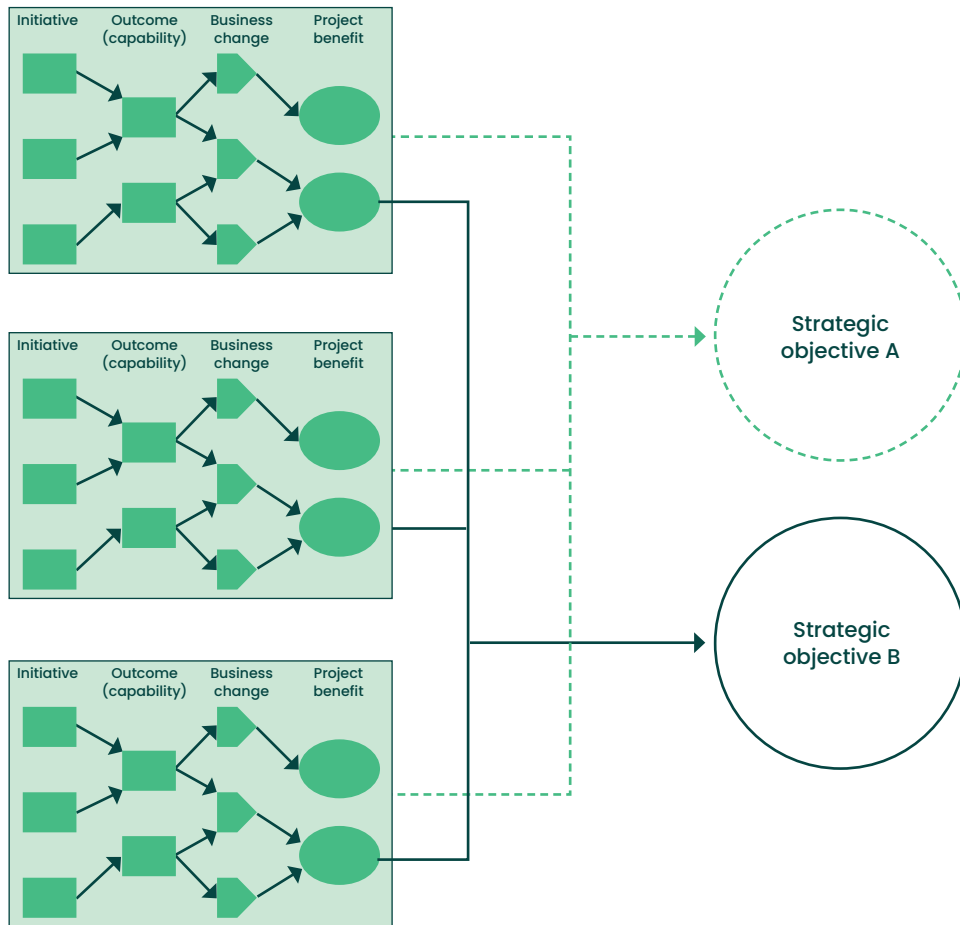


Figure 3.1.1 Project benefits linked to an organisation's strategic objectives

Recommended reading

- APM's *A Guide to Using a Benefits Management Framework* (2019) provides a comprehensive overview of benefits management frameworks. Specifically for this chapter, further information relating to benefits planning can be found in Chapter 3.
- *The Importance of Conventions: A Critical Evaluation of Current Practice in Social Cost-Benefit Analysis* (2017) is an APM research paper that provides a critical evaluation of methods used by project planners and evaluators in the public and third sectors to quantify social benefits and costs.

3.1.2 Benefits tracking

Ensuring benefits remain achievable as the project progresses

Given that projects are uncertain endeavours, sponsoring stakeholders like to keep up to date with the progress of project benefits. This means that project professionals must monitor and track benefits closely, working hard to create visibility while also ensuring the impact of any changes is managed to provide confidence that the benefit forecast is on track.

Benefits tracking is the process of continuously monitoring the viability of forecast benefits, as well as the project or programme's progress towards realising them. It is best practice that benefits tracking forms a core part of an initiative's reporting cycle, with many organisations adopting a consistent benefits reporting approach that all projects or programmes must follow (e.g. a benefit dashboard).

In reality, it is highly unlikely that a project or programme's benefits will materialise exactly as planned. While project professionals work hard to accurately estimate benefits, as initiatives progress, changes occur that will inevitably impact the benefit target.

There are many events that may positively or negatively impact forecast benefits. Examples include:

- **economic instability:** changes in costs of resources or materials
- **business performance:** reduction or increase in market share or brand loyalty
- **organisational restructuring:** creating additional risk and uncertainty
- **timeline changes:** delaying or bringing forward output/outcome delivery

As these changes occur, project professionals should quantify the impact on the benefit target and plan any appropriate actions. Where the impact on the business case is material, project professionals should raise this with the project sponsor immediately. Depending on the organisation's governance, sponsors may then need to update, or seek renewed approval from, decision makers to decide whether the project or programme should continue.

As well as formal changes, project professionals should take a similar benefits-focused approach when assessing the project's progress, risks and issues. All these factors can influence the benefits realisation timeline, so a proactive culture of benefits management across the team will ultimately increase the chances of success.

This ongoing benefits-tracking process crosses over into other project management monitoring techniques, such as business case management (see 2.4.2) and change control (see 5.11). It is best practice for project professionals to also use governance points, such as stage-gate reviews, to complete a benefit 'health check' (Figure 3.1.2), as this offers a great opportunity to provide stakeholders with confidence in the broader benefits management plan.

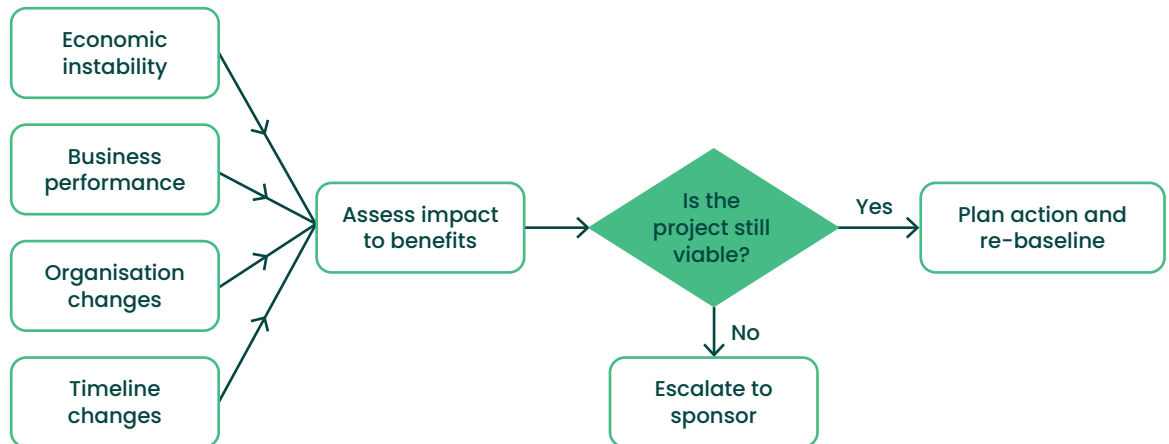


Figure 3.1.2 A simple benefit health check

Recommended reading

- APM's *A Guide to Using a Benefits Management Framework* (2019) provides a comprehensive overview of benefits management frameworks. Specifically for this chapter, further information relating to benefits tracking can be found in Chapter 3.

3.1.3 Adoption and benefits realisation

Maximising adoption to increase benefits realisation

Once a project delivers its outputs, it doesn't automatically generate business benefit. Instead, the adoption and use of outputs (e.g. a new software tool) creates improved business outcomes (e.g. increased customer usage), with those outcomes going on to generate the anticipated business benefit (e.g. lower service delivery costs).

Benefits realisation is the process of ensuring that benefits are derived from delivered outputs and outcomes (Figure 3.1.3). To succeed with benefits management, organisations have to put adequate effort, control and measurement into business adoption, to ensure that the work of the project has the best chance of delivering value to the organisation. The groundwork for business adoption should have already been set in the benefits planning phase (see 3.1.1). Specifically, project professionals will have already planned the activities and resources required for outputs to be transitioned into use, including agreeing testing schedules, acceptance criteria and quality assurance thresholds. Benefit owners and change managers work throughout the project to ensure the business is ready to accept the outputs, facilitating activities such as training, communications and readiness. These readiness activities are covered in greater detail in section 3.6, 'Transition into use'.

As output adoption increases, the benefit measures agreed in the planning phase should then be used to determine whether the benefit targets are being met (e.g. is the weekly report showing that usage is increasing as planned?). Measurement techniques will differ from benefit to benefit, with operational improvements and financial benefits often using quantitative techniques, whereas non-financial benefits often need to adopt a more qualitative approach (e.g. user feedback).

The project sponsor is ultimately accountable for ensuring the measurement process is robust, often using formal benefits realisation reviews to discuss measurements, align on progress and determine whether the benefits are impacting the organisation's objectives. Where benefits aren't being realised as planned, actions may be taken to increase adoption or re-baseline the benefit target.

The timing of benefits realisation can vary, depending on the project's life cycle. In most linear life cycles, benefits are realised at the end of the project once all outputs are delivered, whereas most iterative projects incrementally realise benefits as they progress.

While most projects end at the point of transition, project professionals may actively support business adoption and benefits realisation activities if the project decides to use an extended life cycle (see 2.5.4). Even though this often leads to higher adoption rates and, in turn, greater benefits realisation, organisations have to balance this against the additional costs incurred for project professionals to support adoption activities. Finally, there is also the advantage of the project professional learning from how their endeavours land within the wider organisation and deliver the benefits.

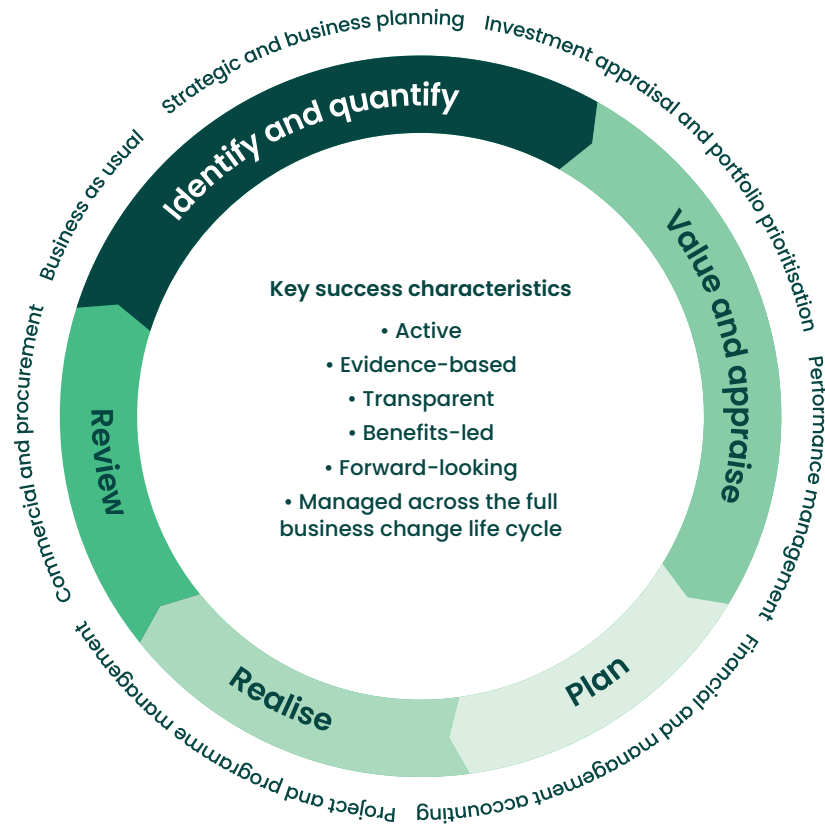


Figure 3.1.3 The benefits realisation cycle

Source: Adapted from Jenner (2012)

Recommended reading

- APM's *A Guide to Using a Benefits Management Framework* (2019) includes a chapter on embedding and making benefits business as usual.
- *Managing Benefits* (2012) guides readers on how to manage the delivery of the benefits used to justify investment in change. Suitable for all those involved in successful change delivery from senior responsible owners and directors through to portfolio, programme and project managers.

3.2

Capability development

Capability development addresses the continuous improvement of competences within an organisation. It invests in people and knowledge, improving the predictability of delivering results, and this creates the correct context for teams to perform.

Project professionals need a constant focus on their own professional development but, at the same time, they must also take some responsibility for the professional development of the people who work for them. Project-based working is a strong platform for developing new skills. Project professionals need to master a huge range of knowledge areas, and the environment they work in changes constantly, which makes capability development a career-long endeavour.

One objective of capability development is to increase the maturity level of project-based working. The usual measure of this is a five-level framework that assesses capabilities, processes and behaviours. These range from unpredictable and reactive to stable and subject to continuous improvement.

It is people who deliver projects, programmes and portfolios. Organisations seek to attract, nurture and grow the best people available. The term for this is 'talent management'. Project professionals benefit from this in two ways: first, as the subject of it and, second, from the skills of the people who join their teams. Project professionals have a responsibility to support talent management within their organisation.

Project professionals benefit significantly from the expansion, codification and sharing of knowledge. Joining a community of practice allows them to share their own knowledge. These communities of practice develop the knowledge, methods and tools that project professionals use every day. This is one of the contributions that professional bodies, such as the Association for Project Management (APM), make to the profession. APM's communities of practice are known as APM Interest Networks.

Gathering this knowledge, ordering it, storing it and making it available is knowledge management. There are plenty of software tools available to support this work, but the degree of success of knowledge management within an organisation is, to a far greater degree, a product of the culture and the behaviours it drives.

This section is written for project professionals who take an interest in the processes of developing their own capabilities and those of colleagues. The section includes:

3.2.1 Maturity of practice: Investing to make delivery more predictable

3.2.2 Talent management: Attracting, deploying, supporting and retaining talented people

3.2.3 Communities of practice: Investing in developing knowledge

3.2.4 Knowledge management: Connecting people to curate, share and use knowledge to improve outcomes

3.2.1 Maturity of practice

Investing to make delivery more predictable

Organisations that invest in project-based working want their investments to succeed. The reality is that some projects fail to meet some or all of their objectives and planned benefits. Better-developed project practices can increase the chances of successful delivery, so many organisations set out to increase the maturity of their project-based working.

The first step is understanding current capabilities, processes and behaviours. The second is identifying a structured improvement path to increase efficiency, effectiveness and predictability.

Organisations assess maturity against a maturity model. The Capability Maturity Model (CMM®) is the basis of most of them. It was created to assess supplier capability for a software project but is now used to model the maturity of a wide range of processes. It has five maturity levels (terminology can vary):

- **Level 1 – Initial:** The deployment of capability is reactive, unpredictable and sometimes chaotic. Few processes are defined and success depends on individual effort.
- **Level 2 – Repeatable:** Basic monitoring and control processes are used for each project, programme or portfolio. Discipline is in place to repeat earlier successes.
- **Level 3 – Defined:** Projects, programme and portfolio processes are documented and standardised. There is proactive tailoring of these processes.
- **Level 4 – Managed:** Performance metrics are used to control performance.
- **Level 5 – Optimising:** Stable processes are subject to continuous improvement. New ideas and technologies are used to respond to quantitative feedback data.

An organisation can define maturity separately for projects, programmes and portfolios. For example, management of projects may be at Level 3 but management of programmes may only be at Level 2.

A richer assessment of maturity comes from dividing practices into key process areas. Examples include risk management, benefits realisation, stakeholder engagement, project culture and scheduling. Each perspective will have its own attributes that indicate a particular maturity level.

Focusing on addressing specific practices can help drive improvement in a manageable way, for example, by moving risk management from a bureaucratic ‘tick-box’ exercise to one that contributes to decision-making. The changes that will have the biggest impact on an organisation are often changes to the attitudes of the people who work there.

Organisations that want to deliver effective and efficient project-based working usually aim to achieve and sustain Level 3 as a minimum. Reaching higher maturity levels requires significant investment in time, resources and process development, so a formal investment decision is appropriate in most cases. This would evaluate the trade-off of costs and risks against the benefits of higher levels of maturity. Often, competitive pressures drive the decision to invest in increasing maturity. Many commercially available maturity models enable benchmarking across sectors, but organisations can equally develop their own, to measure progress.

Continuous improvement of maturity levels requires sponsorship and commitment from senior leaders. It is perhaps best managed as a project or programme in its own right.

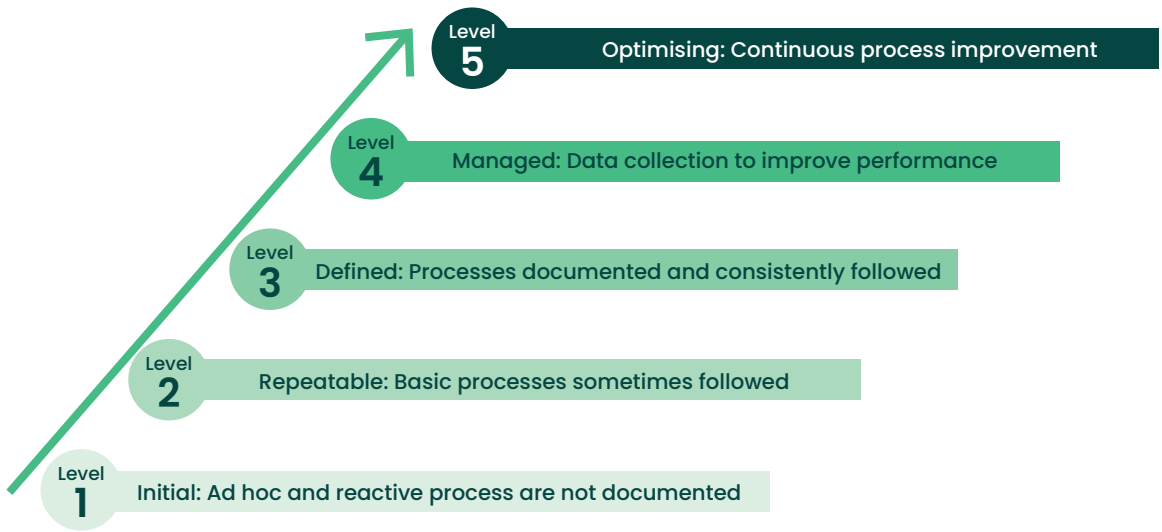


Figure 3.2.1 Five levels of a capability maturity model

Source: Based on the Capability Maturity Model, adapted from Paulk (1995)

Recommended reading

- *CMMI® Development, V2.0 Driving Performance through Capability* (2018) describes practices that help organisations to improve their processes and business capability. This technical report and maturity model developed by the Software Engineering Institute (SEI) based at Carnegie Mellon University provides a comprehensive integrated set of good practices required to improve capabilities, products and processes in order to drive business results and optimise business performance in an increasingly demanding business setting.
- Chapter 38 of the *Gower Handbook of Programme Management* (2016) is dedicated to assessing and improving programme management maturity. It offers a six-level indication of maturity across different aspects of programme management including strategic alignment, sponsor's and project manager's competence, benefits, stakeholders and governance.

3.2.2 Talent management

Attracting, deploying, supporting and retaining talented people

Organisations need good people to deliver their strategic goals and operational objectives. Talent management is the way organisations attract, motivate, develop and retain high-quality people.

Successful organisations manage the relationship between their strategies and the resources available to deliver them. These resources include technology, finance, information and, of course, people. Proactive management of talent has a real effect on the success of project-based work.

The uncertainty and change in project environments mean that priorities are always shifting. This makes effective resource management hard. Because people are critical and costly resources in projects, effective talent management is vital.

Talent management requires developing the knowledge and capabilities of people to prepare them for future challenges. It is an investment in the organisation's core competences. Talent management processes include talent planning, resourcing strategies, training needs analysis, and learning and development activities.

Deliberate leadership attention to talent management can deliver benefits such as:

- gaining project skills, knowledge and practices
- encouraging cross-functional working and strengthening wider relationships
- generating greater business awareness
- increasing diversity, and broadening thinking and approaches to problem solving
- supporting continuous improvement and development
- transferring newly acquired skills and knowledge back into operational roles
- enabling succession planning and future talent pools
- building greater process maturity, individual capabilities and staffing resilience

Project professionals understand the work programme, role profiles, team structures and competences required to deliver the project. This is increasingly complex for programmes and portfolios. Career paths must motivate and retain talent for the organisation's long-term success, as should the culture of the project team and wider organisation.

Project professionals may be involved in many talent management practices (Figure 3.2.2):

- **Talent audit:** Identifying needs and understanding any availability gaps.
- **Attract and secure:** Identifying ideal candidates and bringing them into the project.
- **Inclusivity audits:** Ensuring that the organisation's brand, recruitment and retention processes, and culture attract the widest talent.
- **Resource project teams:** Deploying people with the relevant competences into team structures.
- **Build maturity of project-based working:** Developing systems and processes to free up talent for creative and strategic working.
- **Performance management:** Observing, feeding back and appraising talent.
- **Recognition and reward:** Recognising efforts and offering incentives, including salary and advancement.

- **Succession planning:** Building maturity, resilience and expansion capacity to give people a reason to stay.
- **Learning and development:** Transferring knowledge and developing skills to nurture talent and build capacity.

Learning and development is a critical part of talent management, where project professionals can play a large part by offering opportunities to take on stretching roles and providing on-the-job learning. This supplements both formal and informal training through seminars, conferences, courses and more.

Finally, project professionals can be excellent coaches and mentors. As coaches, they support learners to think through solutions to workplace and career challenges. As mentors, they offer the benefit of their own experience and wisdom.



Figure 3.2.2 Steps to building and retaining project management talent

Recommended reading

- *Resourcing and Talent Management* (2014) offers a good depth of coverage of resourcing and talent management issues for organisations.
- *People Resourcing and Talent Planning* (2010) is engaging and provides good examples of resourcing and talent-planning issues in organisations. This book also considers the challenges facing organisations in implementing talent management strategies.
- *Strategic Human Resource Management* (2007) is a heavyweight textbook that provides an in-depth analysis of strategic talent management issues from a global perspective. This is particularly important to global project teams collaborating on a range of project portfolios.

3.2.3 Communities of practice

Investing in developing knowledge

Communities of practice are a type of learning network, used within and between organisations to maintain, develop and share knowledge. They create and support the use of knowledge, and so are an effective knowledge management method.

Four characteristics distinguish communities of practice from work teams, networks and other types of community. In a community of practice:

- There is a focus on knowledge in a specific domain, for example, risk management or thin-film solar panels.
- Members form a community through regular interaction. This builds relationships that provide a social basis for creating and sharing knowledge and learning.
- Members are active practitioners in the domain. They have shared practices with shared experiences, language, stories and approaches to solving problems.

Communities of practice differ from communities of interest. In communities of interest, members are not necessarily active practitioners. Regular interaction and strong relationships distinguish communities of practice from looser networks.

The members of communities of practice typically cross organisational boundaries, hierarchies and other divisions. The communities can be formal or informal, emergent or planned, and of any size. Some communities of practice have sponsors, objectives and targets; others set their own agenda.

In most communities of practice, some members adopt leadership, management, coordination and technical roles. Facilitation and leadership are key to communities of practice success. Collaboration can be face to face or online.

In project-based working, communities of practice bring continuity and stability across the profession. This doesn't mean that knowledge or membership is stable: both are dynamic. They develop ideas and methods, and help teams to increase the maturity of their practice (see 3.2.1). Communities of practice are a resource for project professionals to draw from.

Examples of how communities of practice can benefit project professionals are:

- as a steward of domain knowledge
- as a hub for sharing knowledge
- to prevent knowledge loss when teams disperse or people leave the organisation
- as a source of advice and guidance for project and programme teams
- to solve problems, including troubleshooting and resolving issues
- to manage knowledge-related risks, such as knowledge gaps
- as a source of continuing professional development (CPD – see 4.6.1)
- to develop and improve project management processes, products and tools
- to provide motivation, recognition and professional satisfaction

Communities of practice develop tacit and explicit knowledge. Explicit knowledge is easy to document and share. People hold tacit knowledge in their heads, as experience and intuition. Part of the role of a community of practice is to codify tacit knowledge into explicit knowledge.

Supporting and encouraging communities of practice is an investment in the success and maturity of project-based working (Figure 3.2.3). Organisations that allow staff to take part in them get a lot in return, as their people can access the knowledge and resources to deliver improved performance.

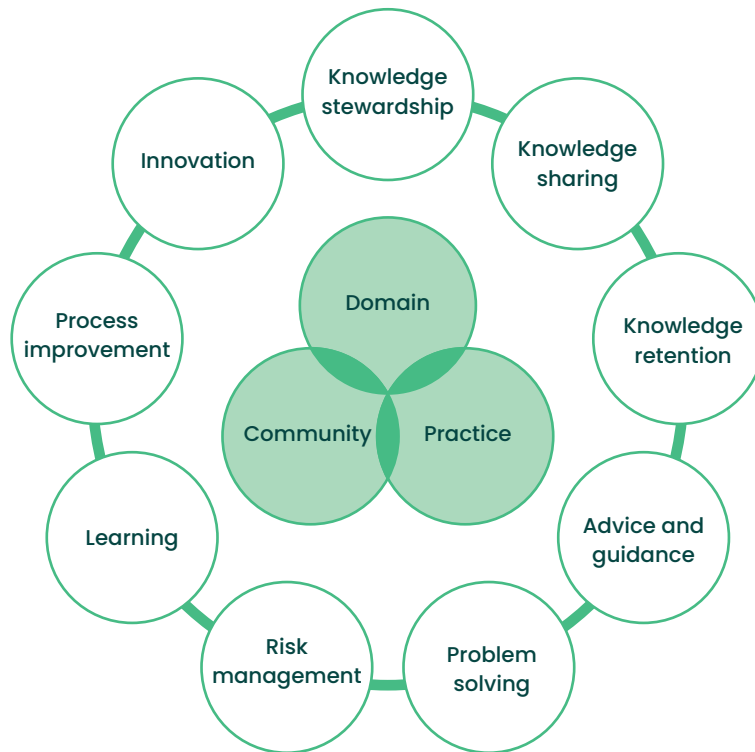


Figure 3.2.3 Single community of practice showing potential benefits/functions

Source: Created by Judy Payne for the APM Body of Knowledge

Recommended reading

- *Introduction to Communities of Practice* (2015) is a comprehensive and accessible introduction co-authored by Etienne Wenger-Trayner, the originator of the term 'communities of practice'. It includes examples, answers to frequently asked questions, links to additional resources and a further reading list of Wenger-Trayner's highly respected books and articles.
- *Harnessing your staff's informal networks* (2010) summarises the way communities of practice have developed from informal, unstructured groups into a formal mechanism for coordinating work across organisational boundaries. A good, quick read, with examples and practical tips.
- *Buzzing Communities* (2012) is a detailed, practical guide based on experience and research. Although the book is about managing online communities of all types, many of the principles and concepts also apply to face-to-face communities. It is especially good for community managers.
- APM's communities of practice have many resources: see apm.org.uk/community.

3.2.4 Knowledge management

Connecting people to curate, share and use knowledge to improve outcomes

Knowledge management is the practice of gathering, creating, organising, storing and sharing knowledge within an organisation. It happens within and between projects, programmes, portfolios and organisations, and it spans the whole life cycle and beyond.

Project professionals can work with knowledge to add value by, for example:

- anticipating, understanding and responding to changing conditions
- avoiding repetition of mistakes
- generating options and solutions
- supporting decision-making processes
- enabling benefits realisation

A wide range of software tools is available to support knowledge management. These can host knowledge within or outside the organisation. However, more important than the raw capabilities of these tools is the uptake by users. Adoption of knowledge management relies on attitudes to drive behaviours, so implementing knowledge management is more a culture change programme than a technical project.

Project professionals aim to use knowledge to improve outcomes (Figure 3.2.4). There are some principles that should be borne in mind:

- Knowledge is intangible and complex. It encompasses much more than data, information or documents (explicit knowledge). It exists in people and can be difficult to express (tacit knowledge).
- Because knowledge is intangible, knowledge management helps people articulate what they know. This allows them to contribute ideas and share them with others.
- Knowledge management can have multiple objectives. Its focus may be on creating new knowledge or using existing knowledge. Both involve sharing knowledge but each needs a different approach.
- Knowledge creation requires an environment of shared purpose, trust and autonomy.
- Using existing knowledge is easiest in an ordered and controlled environment. Codifying knowledge is rarely enough to create shared understanding.
- There is no one-size-fits-all knowledge management solution. It should be tailored to the context, culture and needs of users. Each project, programme, portfolio and organisation must find its own solution.
- Knowledge management needs to be part of the planning for the project or programme life cycle, but processes will need to adapt, driven by feedback and evolving needs.
- Technology can support and enhance knowledge management, but passive collection of lessons learned rarely has a significant impact. Active use of technology can enrich workplace interactions.
- Knowledge management needs a supportive culture that values knowledge and learning. Leadership, diversity, structure and governance all affect this culture. Knowledge management benefits from explicit and tangible leadership support. It is most effective when the sponsor is an active champion.

- There is great value in a supportive infrastructure. This is a role that a PMO (project , programme or portfolio management office) can usefully play, across all project-based working in an organisation (see 3.5.4).

Knowledge management relies on connecting people. Systems and documents are little more than enablers. There are many ways to harness people's knowledge, including peer-assisted learning, facilitated workshops, mentorship, ideas generation sessions and communities of practice (see 3.2.3). These support knowledge curation, creation and transfer.

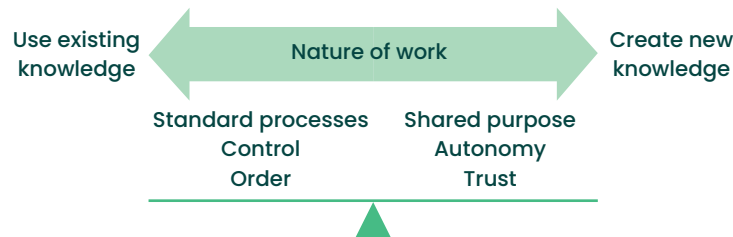


Figure 3.2.4 Nature of work and the working environment

Source: Payne, Roden and Simister (2019)

Recommended reading

- *ISO 30401:2018 Knowledge management systems: Requirements* (2018) is the international standard for knowledge management. It sets requirements and provides guidelines for establishing, implementing, maintaining, reviewing and improving an effective management system for knowledge.
- *Managing Knowledge in Project Environments* (2019) is a definitive short guide to knowledge management in projects, programmes and portfolios. The book presents knowledge management as a series of principles, choices and contextual factors, providing readers with a framework for understanding and thinking about what knowledge management means for their context, projects and working environment.
- *Managing Knowledge Work and Innovation* (2009) is a comprehensive, fairly academic guide to knowledge work. It connects knowledge and knowledge management theories to work practices, using multiple perspectives and case studies. It is good for readers who want to understand the complexity and theoretical underpinning of knowledge management.

3.3

Procurement

All projects need resources to succeed, and in many instances these resources come from outside the organisation. Procurement is the process by which goods and services are acquired from an external provider, with project professionals working with procurement experts to form collaborative relationships with suppliers that help meet the project's needs.

Most organisations have established procurement policies and processes which project professionals need to align with. At a high level, the standard procurement process includes stages for:

- **strategy:** breaking the project into packages and deciding on an approach
- **contract set-up:** choosing the type of contract that will best achieve success
- **tender and award:** assessing suppliers and awarding a contract
- **management and closure:** managing the supplier until the work is complete

Procurement strategy and strategic sourcing topics are covered further in this section, with contract types, contract awarding and supplier management covered in greater detail in section 5.9.

Working from the project's business case, procurement strategy is defined by breaking the project down into packages using a package breakdown structure (PaBS). From here, project teams complete an initial market consultation before considering factors such as business finances, existing knowledge and delivery capability to decide whether to make or buy each package.

Then project professionals can agree a procurement strategy for each significant package, with an associated procurement management plan presented to the project sponsor for approval.

With a high-level strategy defined, the strategic sourcing approach should then be agreed by considering the buying strengths and weaknesses of the organisation and, by extension, the project. Put simply, project teams weigh up the benefits and financial potential against the supply chain risk to decide on whether they pursue a simple, transactional contract or a long-term partnering arrangement.

Procurement experts can help with these decisions, facilitating analysis of whether marginal cost savings are outweighed by the advantage of working with a supplier who can embed themselves into the organisation and thus better meet the project's needs.

This chapter is for project professionals tasked with supporting, leading or overseeing procurement. Specifically, it will cover:

3.3.1 The procurement process: Acquiring goods and services from external providers

3.3.2 Procurement strategy: Matching supply chain engagement to needs

3.3.3 Strategic sourcing: Choosing strategies to obtain best value from suppliers

3.3.1 The procurement process

Acquiring goods and services from external providers

Procurement is the process by which goods and services are acquired from an external provider. While many projects, programmes and portfolios need to acquire goods and services, procurement isn't a discipline that's exclusive to the project profession. In fact, many organisations have procurement functions that operate in their own right, often with established governance, policies and processes already in place.

Because of this, project professionals typically need to align their initiatives to their organisation's procurement process. While all organisations manage procurement slightly differently, a typical project procurement process (shown in Figure 3.3.1) has stages for:

- **setting an overarching procurement strategy:** breaking the project down into packages, testing market viability and determining make-or-buy criteria
- **defining an approach to strategic sourcing:** understanding the buying strengths and weaknesses of the organisation to determine how best to work with suppliers
- **agreeing the type and contents of a supplier contract:** choosing the type of contractual arrangement that best enables the project to succeed
- **completing a competitive tender:** assessing suppliers on their ability to support the project's objectives
- **awarding a contract:** selecting the best supplier(s) and agreeing a contract for the delivery of goods and/or services
- **managing the supplier:** working with the supplier to ensure the contract is delivered to the agreed level of quality and performance
- **contract closure, handover and support:** closing the contract down or handing it over to operational teams once the project ends

The early stages of the procurement process are strategic in nature, as project professionals must work to establish a firm understanding of their requirements to select the right approach. In procurement, it is all too easy to slip into specifying functional requirements; therefore the early strategic focus must consider lifetime cost of ownership, fitness for purpose, durability, availability, reliability, maintainability and efficiency. Selecting for lifetime cost of ownership can result in a higher cost of purchase in many instances, but better outcomes in the longer term. Typically, these decisions will be made at the beginning of the project life cycle with project sponsors, with clear accountability for a procurement management plan as part of the broader integrated project management plan. Procurement strategy and strategic sourcing topics are covered further in topics 3.3.2 and 3.3.3.

As the procurement process progresses, it becomes more tactical in nature, with project professionals working closely with subject-matter experts to evaluate suppliers, draw up contracts and oversee performance. Contract types, contract awarding and supplier management topics are covered in greater detail in section 5.9.

Procurement is a specialist discipline, requiring project professionals to work closely with procurement experts throughout the entire process. To achieve the best results as the procurement process develops and contracts are drawn up, it is best practice to also utilise the knowledge of other experts in areas such as finance, legal, risk and compliance.

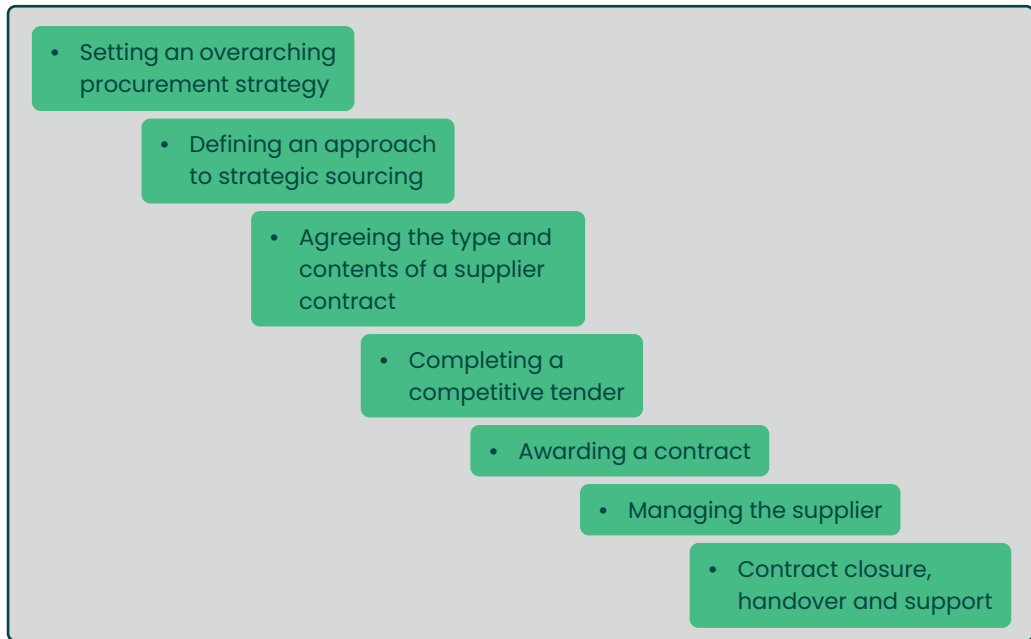


Figure 3.3.1 The procurement process

Recommended reading

- The *APM Guide to Contracts and Procurement* (2017) is an accessible and comprehensive manual for all procurement activity. It provides greater detail into the entire procurement process, including the key inputs, outputs and activities for each stage.

3.3.2 Procurement strategy

Matching supply chain engagement to needs

The procurement strategy sets out the high-level approach for securing the goods and services required from external suppliers to meet the project, programme or portfolio needs (Figure 3.3.2). It is a collaborative exercise performed alongside procurement professionals and leads to the development of the procurement element of an integrated project management plan (PMP).

To define the appropriate procurement strategy, the project team must work together to:

- create a PaBS to define which parts of the project scope may require procurement from external suppliers
- complete an initial market consultation to confirm that appropriate goods or services in each package exist
- establish 'make-or-buy' criteria to validate and confirm the best procurement strategy for each package

Working from the project's business case, developing a PaBS is an iterative process whereby project professionals and subject-matter experts identify individual requirements for purchasing. Much like other breakdown structures, the level of detail will vary between projects, but all PaBS structures should ensure goods and services are grouped in a way that enables economies of scale to be achieved.

Once a PaBS is established, it is best practice to complete an initial, high-level market consultation to confirm that the goods and services of each package exist and are viable for purchase. While market consultation is essential to validate the strategy, project professionals should ensure they are not prejudiced against, and do not unfairly advantage, any suppliers ahead of a subsequent appraisal or tender activity (see 5.9.2).

If viable goods and service exist, the strategy for each package is determined against a common set of make-or-buy decision criteria. Common make-or-buy criteria themes include:

- business circumstances (e.g. are there any financial restrictions?)
- knowledge (e.g. does the organisation have the knowledge required?)
- delivery (e.g. does the organisation have the capabilities required to deliver?)
- service (e.g. once implemented, can the organisation meet the required service level?)
- risk (e.g. does contracting create risk that cannot be managed?)

For each significant package, project professionals should define and agree a procurement strategy, and create an associated procurement management plan. In line with other project governance, the procurement management plan should be reviewed and signed off by the project sponsor before proceeding to further stages of the procurement process.

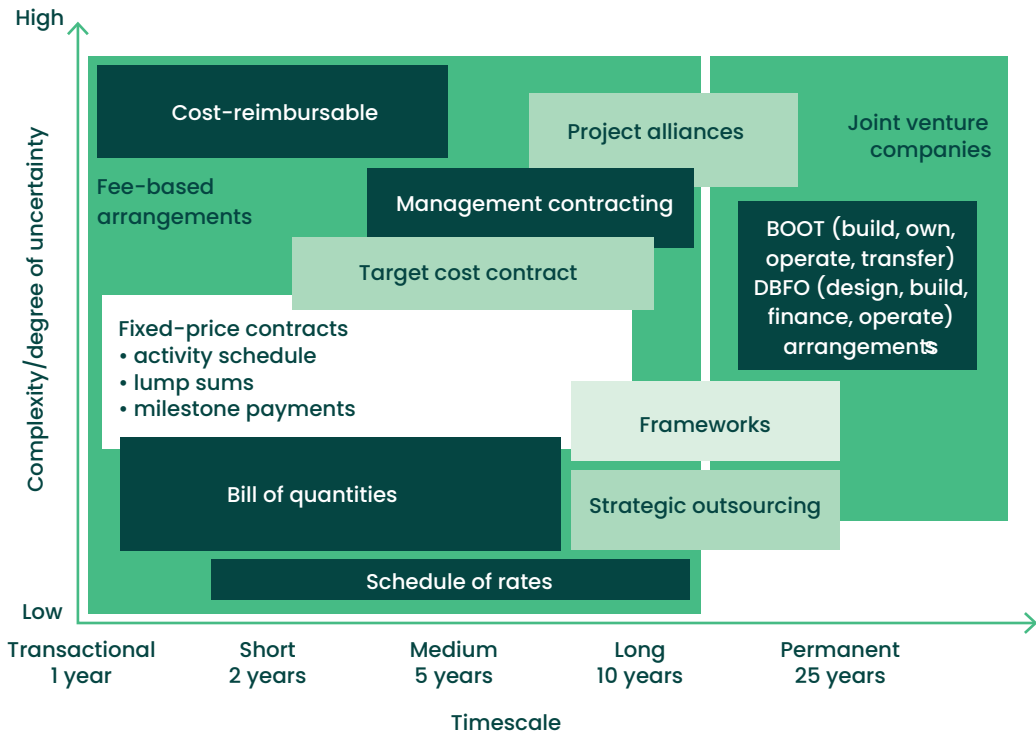


Figure 3.3.2 Choice of contracting strategy

Recommended reading

- The *APM Guide to Contracts and Procurement* (2017) is an accessible and comprehensive manual for all procurement activity. Chapter 3, 'Project procurement strategy', offers step-by-step guidance on developing a procurement strategy.
- *Project 13 Commercial Handbook* (2018) is a guide for developing a commercial strategy. It seeks to develop a new business model – based on an enterprise, not on traditional transactional arrangements – to boost certainty and productivity in deployment, improve whole-life outcomes in operation, and support a more sustainable, innovative, highly skilled industry. Although targeted at the construction industry, much of the content has universal procurement relevance.
- *Procuring Successful Mega-Projects* (2015) is a mentor's guide for project directors that distils practical advice on how to set up, develop and negotiate effective major government contracts.

3.3.3 Strategic sourcing

Choosing strategies to obtain best value from suppliers

With the procurement strategy for each significant package defined, but before engaging with the market to acquire goods or services, analysis is needed to understand the buying strengths and weaknesses of the organisation and, by extension, the project. This enables project teams to maximise their value for money while remaining adequately prepared for any risks in the supply chain.

To do this, project professionals and procurement experts need to understand the:

- financial impact potential of each package
- risk of supply chain disruption of each package

This analysis results in an understanding of both components, carefully balancing the 'upside' of suppliers that drive greater profitability with the 'downside' of those that may cause risk to the organisation. The Kraljic matrix (see Figure 3.3.3) is a widely adopted tool used to assist in this process, allowing projects to easily 'plot' their position on the matrix.

Based on the outcome of the analysis, the sourcing strategy for a particular package might vary from a simple transactional, 'over-the-counter'-type arrangement for routine commodities to long-term partnering or alliancing arrangements for high-risk items such as niche technologies or specialist design services.

In all strategic sourcing decisions, there is a balance to strike between multiple criteria, including the long-term sustainability of the supply chain and its ability to support business operations once the project is complete. This is particularly true where the organisation is fast-moving or is subject to regularly changing requirements.

Project professionals and business stakeholders must also establish how closely they want to work with multiple suppliers. Many organisations choose to engage a principal contractor, consultant or lead supplier that they can work with on a one-to-one basis, who will manage other suppliers on their behalf. Suppliers who are managed by these lead suppliers are often referred to as second- or third-tier suppliers.

This decision also impacts the need for skilled internal supplier management resources. The advantages of hiring capable lead contractors should be balanced with the ability of the organisation to grow its own talent. At programme or portfolio level, this decision process would include considering the option of an external partner to provide a fluctuating level of support throughout the life of the work.

Project teams, typically supported by procurement professionals, should also balance the cost advantage gained from frequent tendering against the benefits of long-term relationships. A marginal cost saving may be easily outweighed by the advantage of working with a supplier who knows the organisation and its culture, and can thus better meet the project's needs. However, in public bodies, wider ethical and legal considerations on tendering have to be borne in mind.

Project professionals should familiarise themselves with the organisation's corporate procurement policies and procedures, as they may be able to take advantage of existing frameworks or, conversely, be constrained by preferential supplier lists. In the public sector, there are often frameworks that support procurement practices too.

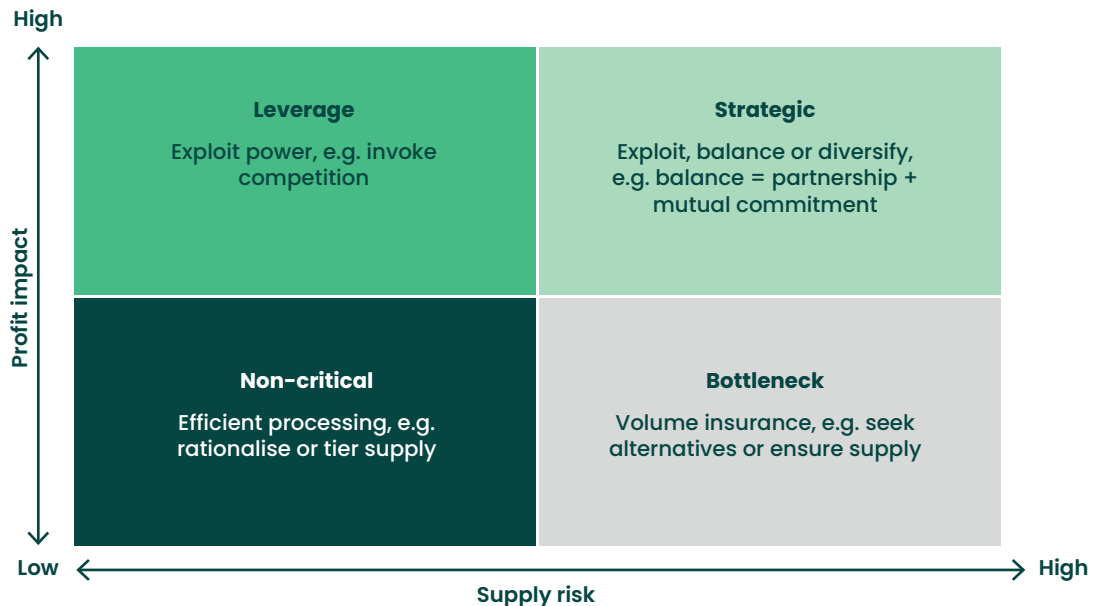


Figure 3.3.3 Supplier-based segmentation using a matrix

Source: Based on the Kraljic Matrix (1983)

Recommended reading

- The *APM Guide to Contracts and Procurement* (2017) is an accessible and comprehensive manual for all procurement activity. Chapter 4, 'Package contracting strategy', is particularly relevant to this topic.
- *Project Routemap: Setting up Projects for Success* (2022) is the UK Government's support tool for novel or complex major projects. It provides useful checklists to help clients with selecting the most appropriate supplier to meet project objectives.
- *Strategic Sourcing Management* (2016) examines procurement and supply management in detail, covering the three dimensions of competitiveness, effectiveness and efficiency. The book is organised in four parts: strategic decisions, operational management of procurement and related supply chain, management of human resources, and dedicated information systems to support procurement.

3.4

Assurance

The purpose of assurance is to give governance boards and stakeholders confidence in their projects, programmes and portfolios. It assesses the likelihood that initiatives achieve their objectives and realise their benefits, and it identifies weaknesses and remedial actions to improve the chances of success.

Assurance is an essential part of the governance of both project-based working and the wider organisation. It is not an instant solution to the problem of project and programme failures, but it is an essential part of minimising the risks. However, it can only ever be as good as the level of support, capacity and expertise that organisations give it.

Assurance too can fail. When it does, it is usually for one of two reasons. With poor process and under-resourcing, assurance can miss important findings. This is 'uninsightful assurance'. The other common cause of failure arises when organisations fail to act on findings and recommendations. This happens when the 'voice of assurance' is not powerful enough or, equally, when the organisation chooses to not listen.

Unlike project reviews (see 3.5.2), assurance is independent of the project. But, like project reviews, teams should welcome assurance as a supportive, rather than adversarial, process. Assurance strategies may span a programme or portfolio, or may serve a single project. Whichever is the case, however, reviews need to fit with the organisation's wider governance principles.

There may be several layers of assurance, carried out by different teams. 'Integrated assurance' is the term for coordinating and streamlining these layers. Project-based working has adapted the 'three lines of defence' model that is common in the world of internal audit. This integrates assurance of internal controls, compliance with those controls and independent scrutiny of the effectiveness of governance and control. These report to different layers of management: line 1 reports to the project or programme team, line 2 to the sponsor and governance board, and line 3 to the organisational executive board.

The third line of defence is an independent audit. This can be in the hands of a PMO, the organisation's internal audit or compliance function, or an external organisation. It is the final level of assurance to the people responsible for the organisation's operation, value delivery and ethics. Where integrated assurance works well, the independent audit would expect to endorse the findings of the first two levels of assurance.

This section is written for project professionals who want to understand assurance. They will be subject to assurance oversight throughout their careers and many will, at some stage, undertake assurance or audit work. Some may even choose to specialise in this area of work. The section includes:

3.4.1 Assurance principles: Providing confidence to stakeholders

3.4.2 Integrated assurance: Protection with three lines of defence

3.4.3 Audits and assurance: The third line of defence

3.4.1 Assurance principles

Providing confidence to stakeholders

Assurance is a process that supports delivery of successful change initiatives with a peer review of the team's approach. This gives the governance board and stakeholders confidence that the project, programme or portfolio is on track to deliver its intended value.

Effective assurance is objective and independent. It comes from outside the change initiative and works together with governance and risk management. Governance relies on assurance to support sound decision-making and oversight. Risk analysis guides assurance towards the greatest vulnerabilities (see 5.10.3).

Evidence from assurance improves risk assessment and informs the decision-gate process (see 3.5.3). This increases the probability of projects, programmes and portfolios meeting their objectives. Governance, assurance and risk management work together; weakness in one will compromise the others.

The primary responsibility for setting up assurance lies with the sponsor. They agree an approach with other stakeholders and the team. This may be part of a formal assurance strategy that applies across a programme, portfolio or organisation.

Where a project is independent, the sponsor develops an assurance approach from first principles. They agree with principal stakeholders on what will provide the necessary objectivity and confidence. From the agreed approach, an assurance team will develop its plan.

In developing an assurance approach, the main questions to address are:

- Who are the stakeholders, and what are their needs from, and roles in providing, assurance?
- What are the levels of risk, value and criticality of the initiative?
- Are there any current concerns or critical delivery issues?
- What are the characteristics of the initiative and the life cycle model?
- What are the governance arrangements of the organisation, portfolio, programme and project?
- What is the availability of specialist skills and capacity?
- How will the process ensure appropriate levels of independence?

To be effective, an assurance plan must meet these criteria:

- It demonstrates independence, objectivity and proportionality to stakeholders' needs.
- It is targeted towards the greatest significance and risk.
- There is clear ownership and accountabilities for assurance arrangements, activities and outputs.
- Assurance activities are planned, coordinated and funded.
- There is a clear governance route for reporting outcomes and resolving issues.

In a large project or programme, there may be several layers of assurance, with different providers. Examples include internal audit, consultant-led reviews and external decision-gate scrutiny. In these cases, the sponsor coordinates the activities of the separate

providers to ensure there are no gaps in the coverage. They also work with the project leader to secure the team's cooperation.

Many different stakeholders may carry out a broad range of assurance activities. 'Integrated assurance' is the term for coordinating and streamlining them.

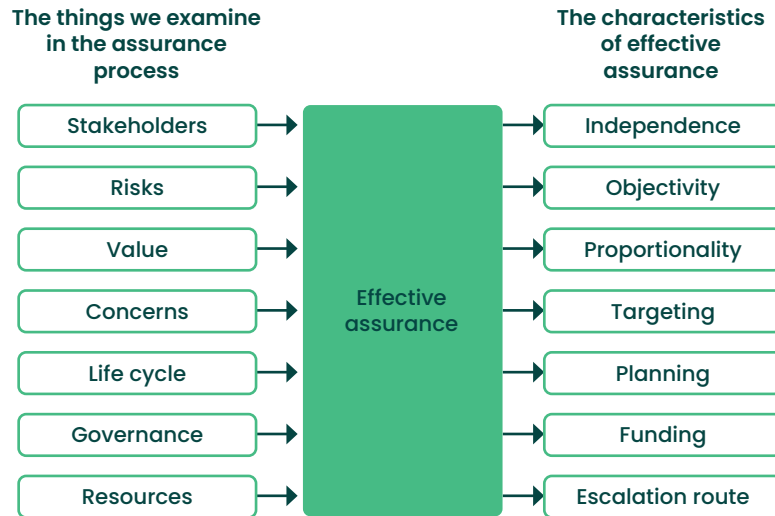


Figure 3.4.1 Effective assurance

Recommended reading

- The APM Assurance Specific Interest Group has published a comprehensive suite of guides on assurance which offer practical advice to providers and receivers of assurance in any industry sector:
 - *A Guide to Project Auditing* (2018) is intended for the use of auditors in developing an approach to review and assurance.
 - *A Guide to Assurance of Agile Delivery* (2017) addresses the different approaches required for agile approaches, where observation, as opposed to documentation, plays a larger role in assurance.
 - *A Guide to Integrated Assurance* (2014) explains the benefits of the integrated approach and provides guidance on how to implement an integrated assurance model.

3.4.2 Integrated assurance

Protection with three lines of defence

Integrated assurance coordinates different assurance activities. Different reviewers carry these activities out independently of each other, and at different tiers. By integrating them, the different approaches work together to benefit the project or programme, the sponsoring organisation and its stakeholders.

The three lines of defence model organises different levels of assurance (Figure 3.4.2). It originated as a way to manage operational risk in the UK financial services sector, but the principles translate well into project-based working.

In this model, three tiers of assurance support the management and governance of projects, programmes and portfolios. These are the three lines of defence:

- **First line:** The systems and processes within the project-based working have controls to manage delivery and the risks that flow from it. This assurance takes place within the initiative. It is the responsibility of the project, programme or portfolio manager. It includes internal review findings that are reported to the project manager and sponsor.
- **Second line:** Assurance reviews check the compliance and effectiveness of the systems in place. The reviewers are independent of the initiative and may come from, for example, a PMO (see 3.5.4) or the organisation's own compliance or risk management functions. Separate reviews may cover things like health and safety, quality, procurement or environmental performance. They report to senior management: the sponsor and governance board.
- **Third line:** This is independent assurance of governance as well as delivery processes. It considers the effectiveness of the first and second lines of assurance. This reports at organisational level to the leadership team and board. They may also share findings with external stakeholders. Reviewers are responsible to the organisation. They are either an internal audit or assurance function, or external to the organisation. In the public sector there are multiple organisations responsible for this role; in the UK, the National Audit Office and the Government's Internal Audit Agency are two examples.

Integrated assurance needs to follow some core principles, which include:

- **accountability** to the sponsoring organisation and its stakeholders
- **independence** and objectivity of third-line assurance
- **taking action** to correct any failings that the reviews discover
- **collective** working between the three lines

The three lines of defence model is not prescriptive. Organisations adapt the model to their own needs, considering factors like risks, acceptable tolerances, context and culture. It is good practice to document the principal areas of risk and tabulate which aspects of each line of defence will address each risk.

Applying the three lines of defence model is no guarantee of project, programme or portfolio success, but it can clarify the effectiveness of project processes and controls, and help improve their effectiveness. It is a valuable contribution to good governance and risk management.

Project professionals should welcome integrated assurance as an aid to their work. Organisations should endorse it and supply enough resource to assurance capabilities. Its success depends largely on the support it receives and its standing within the organisation.

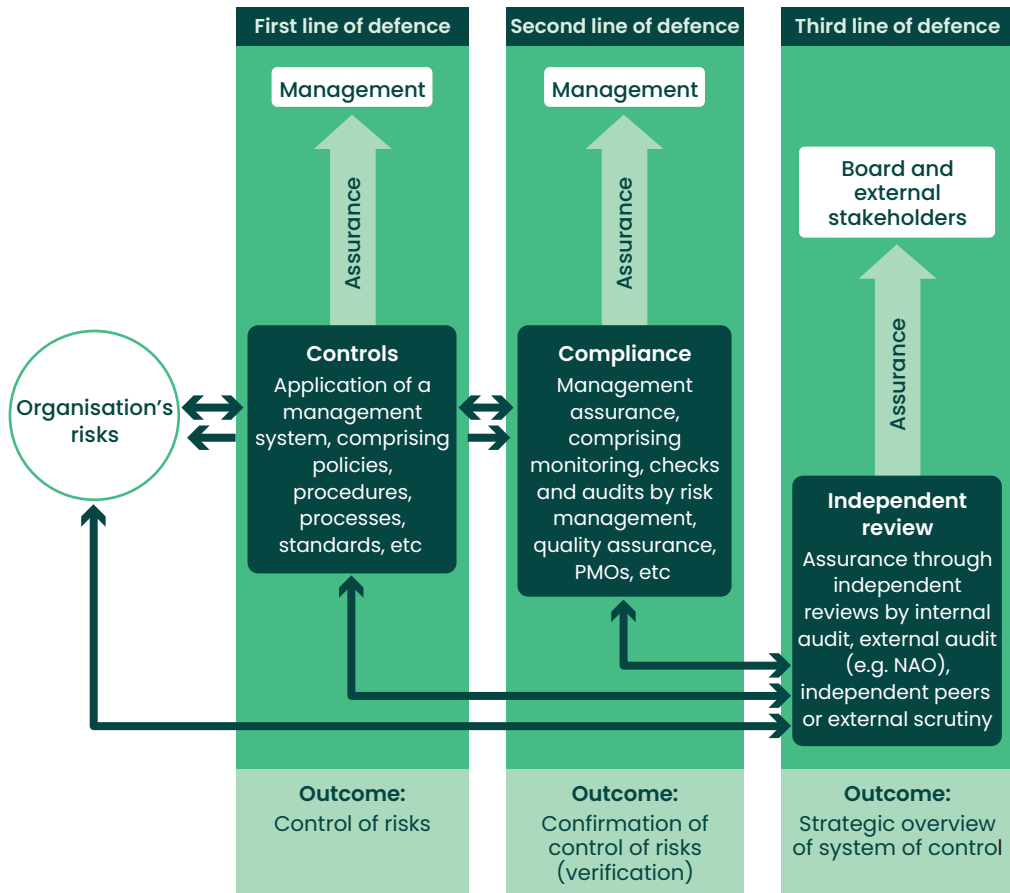


Figure 3.4.2 The three lines of defence model for assurance

Source: APM Assurance Specific Interest Group (2014). Used with permission of Roy Millard

Recommended reading

- The APM Assurance Specific Interest Group's *A Guide to Integrated Assurance* (2014) explains the benefits of the integrated approach and provides guidance on how to implement an integrated assurance model.

3.4.3 Audits and assurance

The third line of defence

Integrated assurance covers a range of assurance activities. Their purpose is to provide confidence to stakeholders that projects, programmes and portfolios will achieve their objectives and realise their benefits.

Assurance focuses on ensuring that the governance, processes and controls are fit for purpose. It assesses the procedures and their implementation. Assurance complements disciplines like quality and risk management (see 5.3 and 5.10).

An important tool in the assurance third line of defence is audit. Audit is used as part of an assurance process to provide an objective evaluation of compliance with governance practices, organisational process, project controls and standards leading to insights on the likelihood that the initiatives will realise their expected benefits. They are independent of the project or programme and its team. The design focuses on gathering objective data. Audits should be:

- independent, objective and accountable
- supported by the organisational board
- planned and coordinated with the organisation's governance processes
- proportionate to the value and risk potential of the initiative, and the assurance needs of its stakeholders
- risk-based, against an independent risk evaluation
- reported in full, with the intent of helping the organisation address weaknesses

Audit teams can come from:

- a project, programme or portfolio management office (PMO – see 3.5.4)
- an organisational internal audit function
- a third-party provider (typically a consultancy, auditor or accreditation body)

It is critical to agree the purpose and scope of the audit with relevant stakeholders before work starts. The auditee is the person who will be responsible for responding to audit findings. The audit process can be summarised in four stages:

- **Planning:** Auditors gather contextual information, set terms of reference and assemble a team.
- **Fieldwork:**
 - Auditors assess the effectiveness of governance
 - Auditors assess the design, implementation and operation of key management controls.
 - Auditors assess the relationship with stakeholders – external stakeholders and internal including the project team
- **Evaluation and reporting:** Auditors develop and document their insights and recommendations to produce a final report.
- **Implementation:** Auditors follow up to confirm that the sponsor and project team have taken agreed actions.

Audit findings may include adequacy of systems, deviations from plan or procedural irregularities. Auditors may make any observations they believe will help the auditees to perform more effectively.

Typically, organisations grade the severity of their concerns. Traffic-light reporting systems are common. For example, they might rate major concerns as red, intermediate concerns as amber, minor concerns as green and, sometimes, observations as blue.

Audit findings need a response from the auditee within an agreed timescale. Assurance and audit are not decision-making functions, but governance tiers base decisions on their findings.

Actions may include replanning, developing new controls or adjusting the team. Follow-up audits will confirm that actions have been completed before the audit is closed out.

Assurance tends to be forward-looking. The first and second lines of defence focus on what is in place. Audit, as the third line of defence, puts more emphasis on examining past decisions and actions, with the intention of finding ways to make improvements for the future.

Risk description	First line			Second line				Third line		
	Management actions and reporting	Programme boards and reviews (incl. 'the plan')	The management system	Risk management	Quality assurance	HSE assurance	Project assurance	Internal audit	External scrutiny	External audits
Risk 1	✓	✓	✓	✓			✓	✓		✓
Risk 2	✓	✓		✓				✓		
Risk 3	✓	✓	✓	✓				✓		
Risk 4	✓	✓		✓				✓		
Risk 5	✓			✓				✓		
Risk 6	✓	✓		✓				✓		

Figure 3.4.3 Adopting a risk-based approach to planning assurance

Recommended reading

- *A Guide to Auditing Programmes and Projects* (2023) outlines why audits are needed and the steps involved in the audit process, including planning, fieldwork, reporting and the implementation of recommendations.
- *Measures for Assuring Projects* (2016) is a toolkit developed by the APM Assurance Specific Interest Group. It is designed as a reference for assurance practitioners. It provides a generic basis for the assurance of projects, programmes and portfolios.

3.5

Reviews

Reviews gather information about a change initiative, to provide an internal assessment of its status and ongoing viability. Reviews can be used to give advice, and to provide confidence and a sense of authority to projects. But their primary function is to be part of the good governance of projects, programmes and portfolios. They are a means of oversight and the basis for informed decision-making.

Unlike assurance, reviews are internal to the change initiative. They can involve a range of stakeholders, such as customers and users, as well as the governance tiers of the project.

The principles of project, programme and portfolio reviews are the same for both linear and iterative life cycles.

Linear life cycles tend to have reviews at key milestone events, such as the delivery of products, boundaries between stages or when issues emerge. Reviews involve the collection and analysis of data, and an examination of deviations from plan. The essential purpose is to understand issues in order to determine appropriate remedial actions.

Iterative life cycles have given project professionals a renewed focus on the principles of project reviews and new ways to conduct them using a broader range of tools.

Decision gates are formal review points. People outside the change initiative can conduct them as part of project assurance, but, more often, they are internal to the project and are the responsibility of a governance board. In these cases, they are a form of review. However, the stakes are higher than for other reviews. Decision gates typically, but not always, make a go/no-go determination for the future of a project.

Outside of the decision gate process, reviews are critically important to ensure the project remains on track and relevant. Reviews should always check that the assumptions made about outcomes and benefits are still valid and that the project remains aligned with current policy and strategy.

Reviews are not tick-box exercises. To be effective, they require ongoing support and personnel. A PMO can help with this.

Finally, reviews take in and consider a large amount of information. This needs to be managed: that is, collected, stored, curated, disseminated, archived and, at some point, destroyed. Information can take many forms, including data sets, documents and images. This is another discipline that may become the responsibility of a PMO.

This section is written for project professionals who want to use reviews to improve outcomes, to see a review as a positive support to their work, and to learn from the insights it can offer them. The section includes:

3.5.1 Project reviews: Learning lessons from a live project

3.5.2 Reviews of the iterative life cycle: Iterative approaches have regular, rigorous reviews

3.5.3 Decision gates: Managed progression through the life cycle

3.5.4 PMO: Support structures for projects, programmes and portfolios

3.5.5 Information management: Capturing evidence to support buy-in, learning and assurance

3.5.1 Project reviews

Learning lessons from a live project

Project reviews are a way to gather information about a change initiative. They provide an internal assessment of its status and ongoing viability. They are also a source of advice and guidance to the project professionals leading the work. Project reviews differ from project assurance activities. People external to the project, programme or portfolio lead them, but reviews may include an element of peer assessment.

When planning reviews, it is important to consider what to assess, when to conduct the review and what information to gather (Figure 3.5.1). This leads to a choice of review format, which must align with the governance structures in place and any legal or regulatory requirements. Often the team will consult stakeholders.

Timing should link to the life cycle of the initiative and major milestones. One important form of review is a decision-gate review (see 3.5.3), which happens at the end of a project or programme stage. Most reviews are less formal, but the style and frequency depend on the risk, complexity and value of the initiative.

An important consideration is what data to gather and how to validate, analyse and present it. Reviewers can make the most objective decisions when they can access raw data. Each layer of analysis and interpretation can introduce bias.

Stakeholders are an important part of a project review. They are a source of information and have an interest in the review's outcomes. These findings may inform their own decisions. They will certainly want to understand the status and consider their attitude towards any proposed remedial actions. Project professionals should liaise with stakeholders to learn what information they need and how they prefer the team to present it.

Project reviews are a key part of governance. They identify and record deviations from plan, budget, scope or quality, so it is vital to know the tolerance levels that will determine the level of concern. Project control documents such as the master plan and business case set the benchmark.

As well as identifying and documenting deviations, the team must determine the reasons for them, but the real value is in the remedial actions they identify. This means governance structures must scrutinise response plans. Depending on the severity of deviations, there may be stages of escalation.

After governance approves the responses, the team implements them. This can involve diverting resources from the activities in the baseline plan. Governance tiers will usually expect monitoring and reporting of any remediation.

The final part of a project review is identifying lessons from it. Good practice is to conduct a lessons-learned review and to document the conclusions. Where the organisation has a knowledge management process (see 3.2.4), this will assimilate into the lessons-learned document and make it widely available. If there is a PMO (see 3.5.4), it may play a role in getting other projects, programmes and portfolios to apply these lessons.

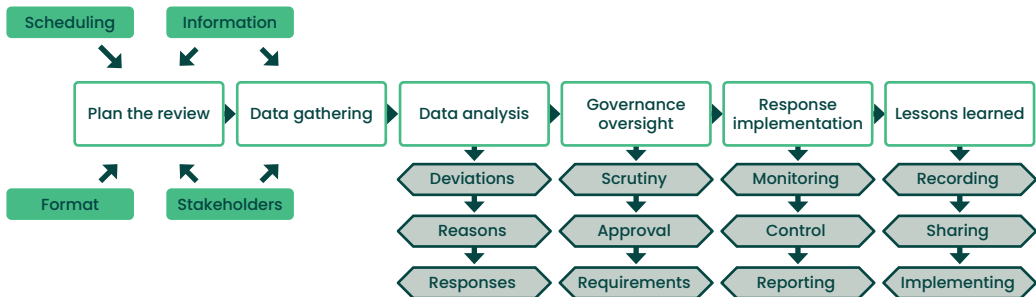


Figure 3.5.1 Schematic of the project review process

Recommended reading

- *Project Controls in the 21st Century* (2025), published by APM, covers all aspects of project reviews, including stakeholder reviews and feedback loops.

3.5.2 Reviews of the iterative life cycle

Iterative approaches have regular, rigorous reviews

In an iterative life cycle, project reviews are structured evaluations of the project's progress, performance and outcomes. They take place at regular points throughout the life cycle to ensure the project is on track to meet its objectives.

Typically, reviews take place at the start or end of each iteration or sprint. This differs from linear life cycles, where they usually happen at key milestones or at the ends of stages. They provide continuous feedback to ensure the project adapts to changing needs.

Reviews of iterative projects follow principles that can apply equally to linear projects (Figure 3.5.2). They include:

- **user, customer or stakeholder collaboration:** greater stakeholder involvement to generate more and better feedback
- **data-driven:** use of metrics and performance indicators to assess progress and identify issues
- **focus on learning:** the goal is to learn and improve rather than assign blame
- **continuous improvement:** following the principles of kaizen, 'change for the better'
- **transparency:** encouraging open discussions and wide communication of status
- **risk mitigation:** identifying and addressing potential risks as soon as possible
- **adaptation and responsiveness:** adjusting the plan to address issues
- **actionable recommendations:** reviews must make a difference to outcomes
- **progress over perfection:** later iterations can focus on optimisation
- **regularity:** following the pattern of iterations

Within the wide range of iterative approaches and frameworks, there are many approaches to project reviews. Here too, there is much that project professionals from a background of linear projects can learn. These include:

- **iteration (or sprint) reviews:** the team demonstrates completed work at the end of each iteration, to get feedback
- **demos and showcases:** demonstrate work in progress, for feedback and course correction
- **continuous integration and testing:** provides feedback about the quality and stability of the product in a production environment
- **retrospectives:** the development team reviews its performance and processes to optimise how it works together
- **ad hoc reviews:** enable the flexibility to address issues or changes as they arise
- **backlog grooming:** reviews the backlog to simplify drawdown for later iterations
- **adaptive planning:** reviews the plan based on new information, changed priorities or feedback from previous iterations
- **stakeholder feedback or check-ins:** get insights from stakeholders about their priorities, perceptions and suggestions
- **risk and opportunity reviews:** assess possible risks and opportunities, allowing proactive management and adapting the project strategy

- **continuous improvement workshops:** provide a forum to identify process improvements based on learning from previous iterations
- **work in progress (WIP) reviews:** identify and resolve bottlenecks in iterative projects

Project reviews in iterative life cycles are a vital way to ensure alignment, quality and continuous improvement. They not only provide checkpoints to assess progress but also foster a culture of ongoing learning that helps teams manage and adapt to change, and so drive project success.

This approach contrasts with linear project management reviews. These are less frequent, more formal and less suited to projects in dynamic or uncertain environments.

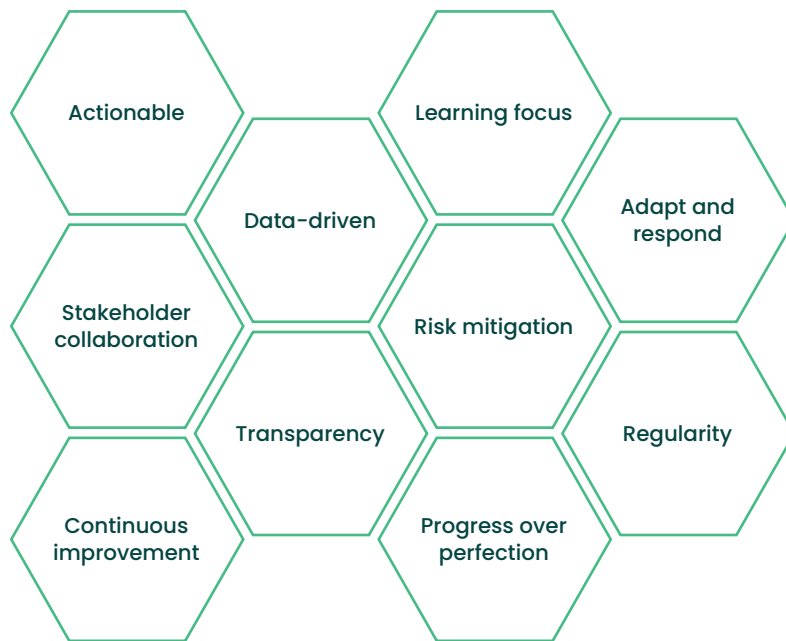


Figure 3.5.2 Principles of iterative life cycle project reviews

Recommended reading

- APM's guide *Project Controls in the 21st Century* (2025) provides an overview of iterative or agile working practices, including iteration reviews held at the end of a sprint or significant milestone.

3.5.3 Decision gates

Managed progression through the life cycle

Decision gates are points in the life cycle where there is a review to confirm the viability of the work against the business case. Alternative terms are 'stage gates', 'gateways', 'boundary gates' and just 'gates'.

In linear life cycles, decision gates are event-driven – they happen at the end of a phase of work (see Figure 3.5.3). In iterative life cycles, they fall in line with the fixed cycles. Hybrid life cycles combine main decision gates, at the end of major phases of work, with interim review points that reflect the iterative nature of the development. In all cases, the sponsor and the governance board are accountable for the decision about whether to continue the work.

Reviews at decision gates ask backward- and forward-looking questions:

- **Backward-looking:** Has the initiative achieved everything it should before moving on to the next stage?
- **Forward-looking:** Is the business case still viable? That is, can the project achieve the desired benefits for an acceptable level of cost and risk?

Within standalone projects, decision gates consider only the viability of that project. In programmes and portfolios, decisions include whether to rephase or terminate existing, or initiate new, projects.

Decision gates are a key governance process. They can be internal to the project, with scrutiny a role of the governance board, or they can be external, with the review team drawn from the wider organisation or beyond. Thus, decision gates can be part of a project review or project assurance (see 3.4).

Project professionals should not view decision gates as adversarial. The review team's role is to be a critical friend, as their scrutiny should help the project team to identify and resolve issues. Decision gates are also an opportunity to obtain authority for matters such as procurement, drawdown of finance or changes to scope. Between decision gates, the sponsor is accountable for overseeing the work and making big decisions (see 2.6.2).

Decision gates also provide a chance to highlight achievements, influence stakeholders, seek support for risk responses or issue resolution, and renew support by gaining attention from senior stakeholders.

Decision gates are 'go/no-go' decision points, but, in some cases, the decision will be a 'conditional go' subject to a set of conditions that need to be met within a set time frame. The timing of the next gate is usually the last decision for review and confirmation.

The project, programme or portfolio manager is responsible for being ready for a decision gate. This includes making sure relevant documentation is available on time.

Understandably, in preparing for decision gates, the focus is on obtaining a 'go' decision – but the responsibility of the review is to make the right decision for the organisation. This may be to pause, replan or terminate the work. This is how organisations avoid the sunk cost trap.

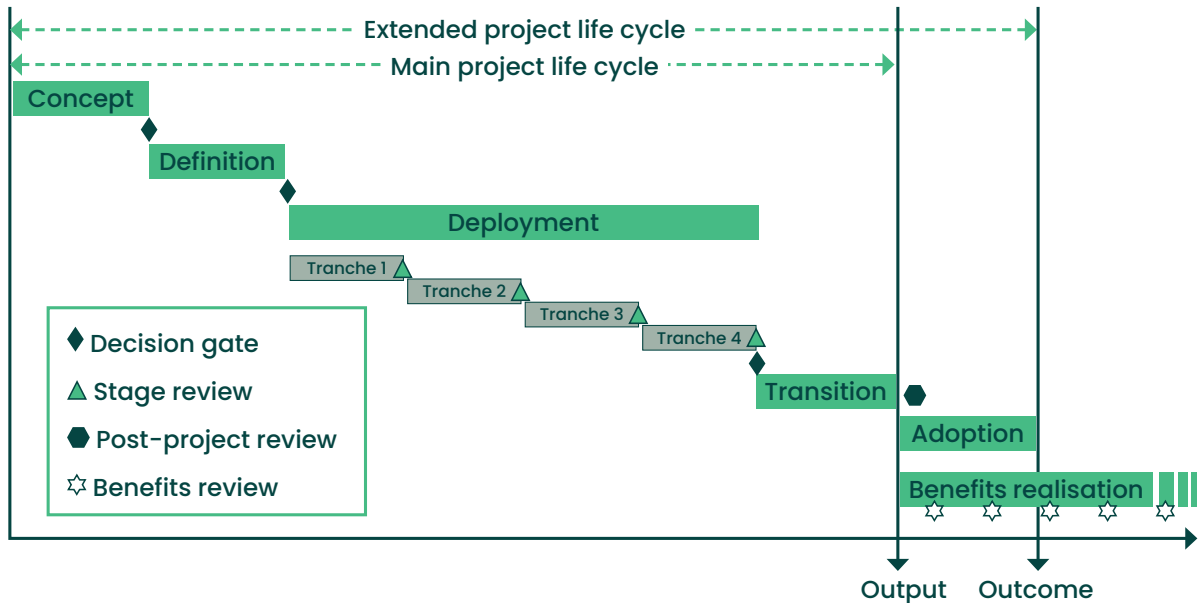


Figure 3.5.3 Decision gates in a project life cycle

Recommended reading

- *Gate Review Process* (2021) is anchored to the Five Case Model for business cases, and examines programmes and projects at key decision points in their life cycle to provide assurance that they can progress successfully to the next stage.

3.5.4 PMO

Support, guidance and frameworks for projects, programmes and portfolios

An organisational structure offering frameworks, guidance and support to its project management community is a PMO. Its role and approach vary between organisations, so it can be known as a project, programme or portfolio management office.

PMOs feature four delivery areas:

- **Delivery support:** Providing benefits that cannot be resourced on a project-by-project basis. This enables delivery consistency and visibility through reporting and analysis that enables effective decision-making.
- **Governance frameworks:** Providing appropriate controls such as assurance to ensure a repeatable approach, meaning projects are delivered in the right way to give confidence to sponsors.
- **Prioritisation and alignment to organisational strategy:** Supporting capacity planning and enabling a balanced portfolio of change along with insights to achieve organisational objectives.
- **Capability development and knowledge management:** Developing a learning environment and a culture of continuous improvement in project management practice.

There are four typical PMO models (Figure 3.5.4), driven by organisational need, project management maturity, team nature and size, and organisational structure:

- **Centralised or embedded PMO:** Most PMO functions that include project delivery are controlled by a central project, programme or portfolio manager. Processes and procedures are owned by the PMO, with organisational processes defined at a higher level and passed down for the PMO to manage.
- **Hub and spoke:** A hub-and-spoke PMO has a central structure linked to satellite PMOs. Project delivery is part of the PMO, but likely to operate within individual organisation functions. Mandatory processes and procedures are centrally owned, but some functional autonomy is allowed.
- **Central:** A central structure provides services and support to projects and programmes across all business units. Processes and procedures are owned centrally and provided to each delivery team to use as they see fit, typically to a defined minimum standard.
- **Governance and delivery split:** A strategy-and-governance-focused PMO, with no links to delivery, encourages collaboration for process design and adherence. A community of practice develops as a driver for engagement and partnership between delivery teams and the PMO.

The organisational context will determine which approach is best; each has its own benefits and risks. PMOs can be temporary or permanent structures, so approaches can be tailored to fit the organisational reality.

A PMO needs a range of skills including data analysis, facilitation, problem solving and collaboration. Training and coaching skills help teams develop, and assurance and audit competence are needed to ensure compliance, provide challenge and drive improvement.

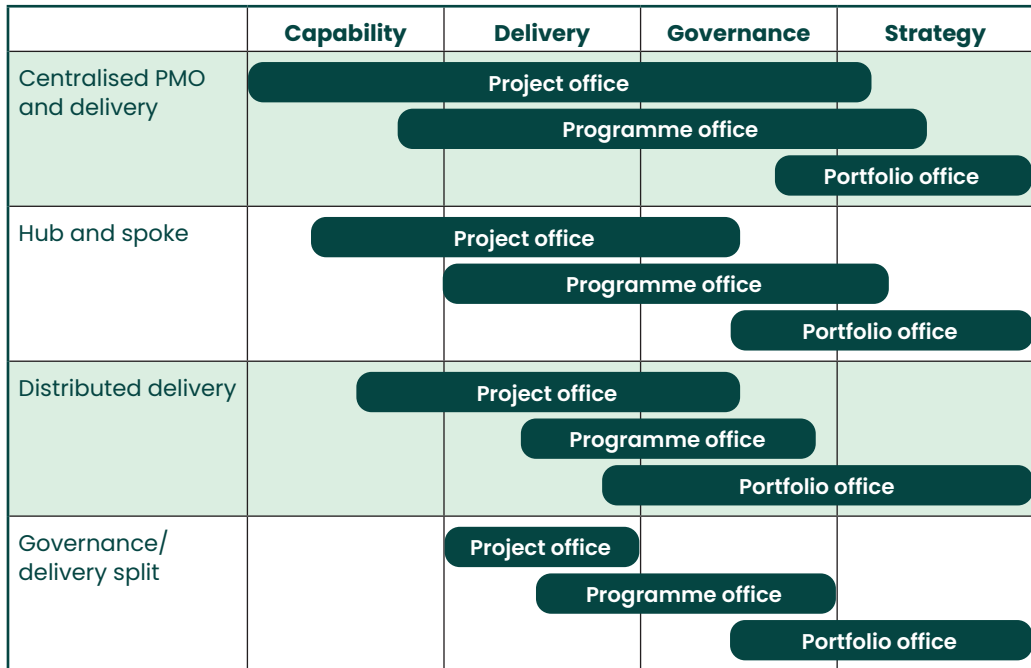


Figure 3.5.4 Different forms of PMO

Source: Reproduced by kind permission of Emma-Ruth Arnaz-Pemberton, Wellington Ltd

Recommended reading

- The *PMO Competency Framework* (2021) is a guide for PMO professionals interested in assessing and developing organisation-wide, team and personal competences within a PMO environment.
- The *Complete Project Management Office Handbook* (2014) explains how to use the PMO to influence project outcomes in a way that serves both project and business management interests.

3.5.5 Information management

Capturing evidence to support buy-in, learning and assurance

Information management is the collection, storage, curation, dissemination, archiving and destruction of documents, images, drawings and other sources of information (summarised in Figure 3.5.5). Project-based working relies on accurate and timely information, which teams and stakeholders need to make informed decisions and fulfil their roles.

Projects, programmes and portfolios use and generate a lot of documentation. Project professionals need plans, controls and reports. They use information to communicate with their teams and wider stakeholders, and to provide documentary evidence for assurance.

Defining information management processes and responsibilities is a key set-up activity. Many organisations have standard processes and tools, often supported by a PMO (see 3.5.4). Standard templates are useful to help document creators to cover all the key points. They also make it easier for reviewers and those involved in assurance, audit and process improvement.

Another key aspect of information management is the quality and integrity of data and information. This means paying attention to their accuracy, completeness, consistency, timeliness and formatting. Alongside this is the need for good data governance policies and procedures.

Project professionals can adapt standard ways of working, but they must document these approaches in the information management section of the project management plan. This must include data security. The sponsor or governance body should approve these processes.

In project-based working, documents and other information will change, so rigorous version control will avoid wasted time and expensive errors which can arise from people working with out-of-date versions. Information management processes also:

- establish ways to communicate to relevant stakeholders any changes to documents. Where there is no standard system for document control, project professionals create their own
- design information storage and retrieval, with accessibility in mind. Information that cannot be accessed is of no value. Establishing access rights for all project information helps people access information efficiently and it is critical for meeting data protection requirements
- archive information when it is no longer needed. Archived information provides an audit trail of changes. At some point, organisations will need to destroy information and documents. This must be subject to statutory requirements and organisational policy

The information an organisation requires a project, programme or portfolio to produce varies. Factors like the purpose and complexity of the work, standard processes and policies or the type of life cycle will determine the rules. Iterative projects place more emphasis on discovering and recording emergent information rather than pre-approved plans, so they are likely to capture information in more dynamic ways.

Projects sitting within a programme or portfolio may enjoy a reduced administrative burden. Information management may be the responsibility of the coordinating framework or its PMO.

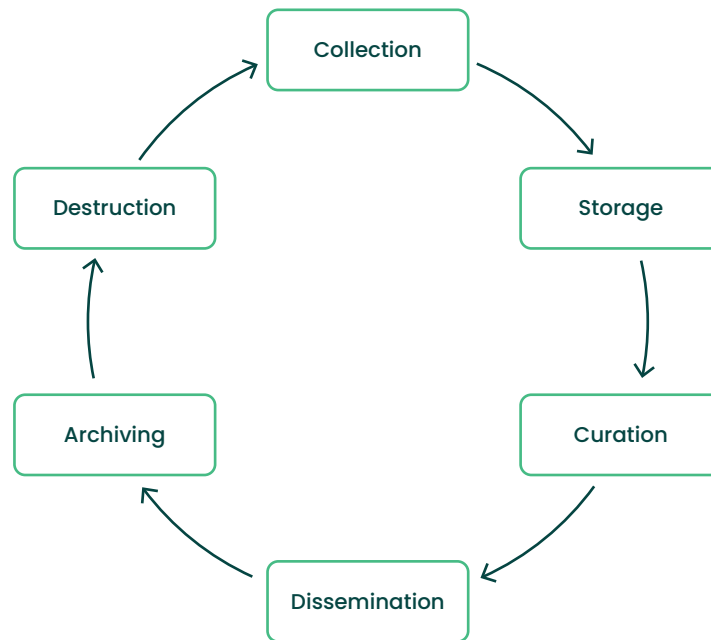


Figure 3.5.5 Scope of information management

Recommended reading

- *Essentials of Management Information Systems* (2016) is a book that sets out why information systems are one of the major tools available to project professionals for achieving operational excellence, developing new products and services, improving decision-making and achieving competitive advantage.
- *Ethical Data and Information Management* (2018) is concerned with how information is managed, processed and governed in a sensitive and ethical fashion. The book offers practical, actionable methods and tools for implementing information management within organisations. The ethical perspective which frames the concepts and methods on offer is both timely and unique.
- *Records and Information Management* (2018) is an extensive guide to using records and information management (RIM). The key principles advocate a focus on articulating the useful life of information and getting rid of it as soon as there is no longer a legitimate reason to hold on to it.

3.6

Transition into use

Transition management is multifaceted. Its purpose is to facilitate changed capability and bed in new processes, practices, tools and techniques.

It includes organisational change management and will direct benefits realisation management.

The overarching aim of projects is to produce outputs which can be transformed into outcomes and benefits for an organisation. The transition of outputs is therefore an important process which requires careful planning. A transition plan needs to identify priorities and potential disruptions, and assign ownership of outputs.

The approach adopted for transition is closely linked to the chosen life cycle. Many variants are possible. In all cases, the temporary change team should:

- identify and engage with the recipients of the change
- inform them of the new processes, products, systems or ways of working
- champion positive factors which would come from adoption
- identify and address any barriers which would impact adoption

People from within the change initiative or the wider business should work throughout the life cycle to prepare for successful handover of project outputs, and adoption of those outputs to produce outcomes and benefits.

Sometimes, project work doesn't end as planned. Early closure should not be considered a bad outcome. It shows good organisational decision-making, as it prevents further investment into projects which are unlikely to yield benefits.

In some sectors, this is a vital and a positive organisational capability. Projects and programmes are transient endeavours. Therefore their managed closure – whether as planned or earlier – is necessary to bring the investment to a tidy close.

This section will be of particular interest to project, programme and portfolio leaders who are thinking about how to manage the end of change initiatives. It will address the transition from the temporary team to the permanent organisation. It includes:

3.6.1 Transition of project outputs: Ensuring that outputs enable the intended benefits

3.6.2 Unplanned project endings: Knowing when closure of the original project is the right business decision

3.6.3 Administrative closure of projects: Shutdown of all the deployment activity and corporate acceptance of completion

3.6.4 Closing programmes and portfolios: Retiring coordinating frameworks for projects when they cease to add value

3.6.1 Transition of project outputs

Ensuring that outputs enable the intended benefits

The transition of project outputs happens when the project team has delivered what has been agreed. The project sponsor and users must be able to accept the deliverables (see Figure 3.6.1).

In a linear life cycle, this is typically in the final phase. In an iterative life cycle, it will be when the sponsor or product owner accepts the outputs. In all cases, transition triggers the beginning of adoption, use and benefits realisation (see 3.1.3).

Transition planning starts at the beginning of a project. The investing organisation needs to specify what needs to be 'handed back' at the transition point. This could include user manuals, operating procedures, asset registers and drawings, but, by the transition point, these may have changed as a result of the project.

As part of the transition process, the deliverables, together with assurance evidence (see 3.4), are prepared for passing to the sponsor and the user. This will involve testing component deliverables in a safe, non-operational mode. This may include tests of non-physical items, for example, software.

Behavioural changes also need to be considered: for example, coaching the business-as-usual team to help them to adopt new ways of working.

Following successful offline testing, the complete set of deliverables will be tested in operational mode. This will often include representatives of the users. If the deliverables meet the acceptance criteria, they will be signed off by the sponsor.

There are several transition strategies that can be adopted. The one chosen will depend on the:

- situational context – 'why we are doing it'
- strategic priority
- capacity of the organisation to accept the transition
- complexity of the transition

For example, a phased adoption may be preferred if the system required:

- significant training
- knowledge transfer
- changes to job roles

In all cases, plans for acceptance and transition are prepared and agreed in the project management plan.

The transition process may also include:

- acceptance of the relevant documentation
- acceptance certificates
- transfer of responsibility from the project team to the sponsor/users
- formal transfer of ownership

On high-risk projects, contingency plans may be developed in case the initial transition does not proceed as planned. This may form part of post-implementation support or it could be a warranty arrangement between the project and the investing organisation.

Once the outputs have been handed over successfully, the benefits realisation process can start.

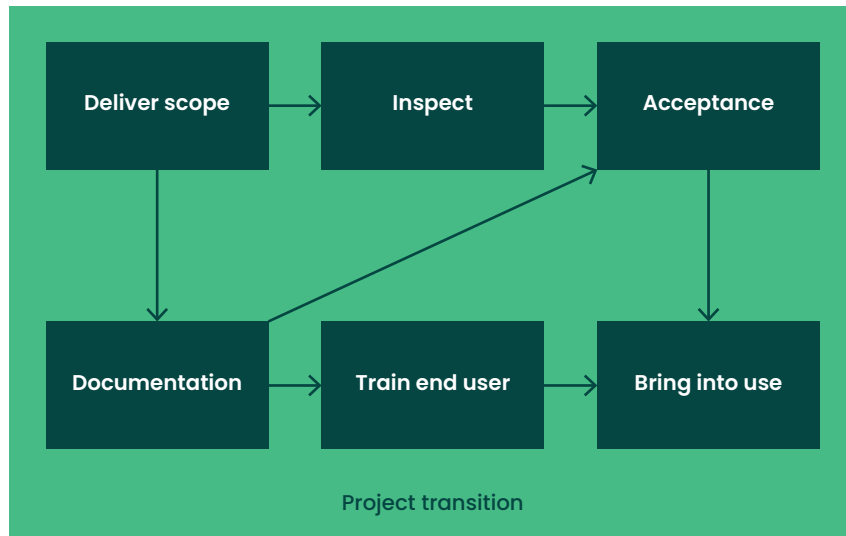


Figure 3.6.1 Activities involved in project transition

Recommended reading

- The APM Governance Specific Interest Group guide *Sponsoring Change* (2018) explains the importance of the role of the sponsor and their accountability for the realisation of desired outcomes and benefits from any investment.
- The APM Specific Interest Group for Planning, Monitoring and Control guide on *Planning, Scheduling, Monitoring and Control* (2015) contains a specific section on the transition of projects. It offers practical guidance on planning transitions based on best practices.
- The APM Earned Value Management Specific Interest Group's *Earned Value Management Handbook* (2013) describes some of the issues in the management of transition of project deliverables.

3.6.2 Unplanned project endings

Knowing when closure of the original project is the right business decision

Projects exist to produce planned objectives which:

- satisfy a business need, and/or
- bring about a business benefit

Over time, circumstances may change. External and/or internal events may influence the business strategy. Unacceptable risks may be identified. The original objectives may no longer be relevant or desirable in the light of these changes.

In such a situation, it is logical to close the project early. This means that the investment can be directed towards a more useful opportunity (see Figure 3.6.2).

The use of decision gates can facilitate this (see 3.5.3). This governance process ensures that investment does not continue if there is no longer a viable business case. The culture of the organisation needs to support this governance, so early project closure should be seen as a positive decision and not a failure.

Many sectors understand this. The concept of failing fast is built into the planning and decision-making processes. For example, in drug development, many promising compounds will enter the portfolio, but those that will not make it through to full clinical trials or come to market are stopped as soon as possible.

Many projects may look promising but cease to be so when more information is known. It is not wrong to start something when the outcome is uncertain, but it is wrong to continue something when the evidence suggests that sufficient value cannot be created.

The governance approach should provide the framework and guidance that helps the right decision to be made.

The organisation can support early closure by:

- creating incentives and 'safety nets' for the team
- championing the sharing of knowledge
- looking for ways for other teams to use resources that become unexpectedly available
- providing support for stakeholders who are affected
- promoting internal communications, explaining what has happened and why

In a programme or portfolio, closing projects needs to be viewed in the wider context. It should not be seen as a failure but as an opportunity for better utilisation of resource. It may result in a reprioritisation of projects that better serves the organisation's needs. It is also a significant learning opportunity if the reasons for closure are captured and disseminated across the organisation's project professionals.

A sponsor's decision to close a project for the greater good of the organisation needs to be welcomed and acknowledged.

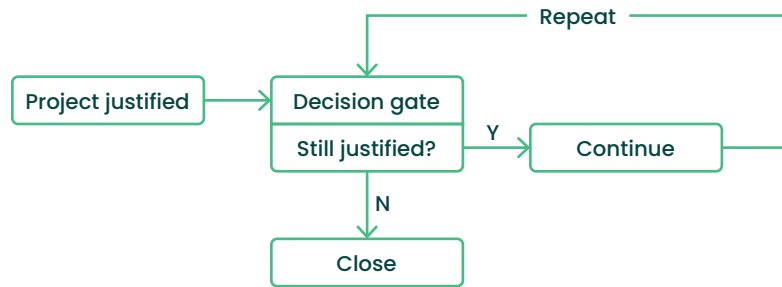


Figure 3.6.2 Closing projects when they are no longer justified is good practice

Recommended reading

- *Project Management* (2013) covers most aspects related to managing projects. Chapter 28 focuses on project closure, including consideration of different reasons and rationales for closing projects, and the need for as-built data records for projects terminated prematurely, in case customers subsequently decide to resurrect them.
- *Managing Project Ending* (2009) is dedicated to exploring the different scenarios for project ending, emphasising that ending entails more than simply closing the effort. While the book is focused on developing a project ending as a strategy and a competence, it explores premature termination, late termination and non-termination, in addition to planned termination.
- *Enforcing strategic fit of project portfolios by project termination* (2012) is a research paper asserting that senior managers should terminate projects that no longer conform to corporate strategy to ensure strategic fit, and indicating that such involvement can affect their new construct of project termination quality.

3.6.3 Administrative closure of projects

Shutdown of all the deployment activity and corporate acceptance of completion

The project team will disperse once all the deployment activity has been completed. It is therefore the responsibility of the sponsor to ensure that the administrative closure happens effectively.

Administrative closure typically takes the form of a project closure report (see Figure 3.6.3). This will be endorsed by governance through a final decision-gate approval. The PMO may produce this, but the sponsor will remain accountable.

A typical project closure report contains:

- evidence that the project has delivered its intended outputs and that they have been accepted
- acknowledgement of any benefits that have already been realised
- the plan for the realisation of any remaining benefits
- evidence that contracts have been completed
- responsibilities for ownership of any residual matters

The requirements for archiving documents are determined by the project management plan, corporate policy and legislation (see 3.5.5). The sponsor ensures that these are followed. Retention periods and disposal requirements must also be specified.

Staff need to be reassigned as the project approaches closure. Team members may have acquired new skills during the project, so these need to be reflected in their personal records. Provision should be made for their learning to be used – there may be opportunities to use their skills in other related projects. The PMO or human resources (HR) function may be best placed to identify development areas for new staff competences.

The advice and support of people professionals (HR) needs to be sought if job roles no longer exist for team members. Their exit from the organisation needs to be planned and implemented. The way staff are managed at the close of a project is critical, as it impacts anyone leaving the organisation as well as those remaining in the organisation. Organisations that regularly can't find jobs for project professionals at the end of a project will, over time, lose talent and find themselves in the situation where people leave projects early so that they continue to be employed. This will adversely impact on current and future project delivery.

The finalisation of all supplier contracts is an essential part of administrative closure. The project professional needs to obtain any post-deployment technical documentation. The supplier's contract will have specified the precise requirements which need to be distributed as appropriate within the host organisation.

Contracts within the supply chain need to be formally closed. This will include confirmation that:

- all the supplier's obligations have been met
- final accounts have been agreed

The legal or commercial team may need to be called upon in the case of any ongoing disputes to ensure that project closure can be effected as quickly as possible.

The sponsor closes the project accounts and returns any surplus funding when all project liabilities have been discharged.

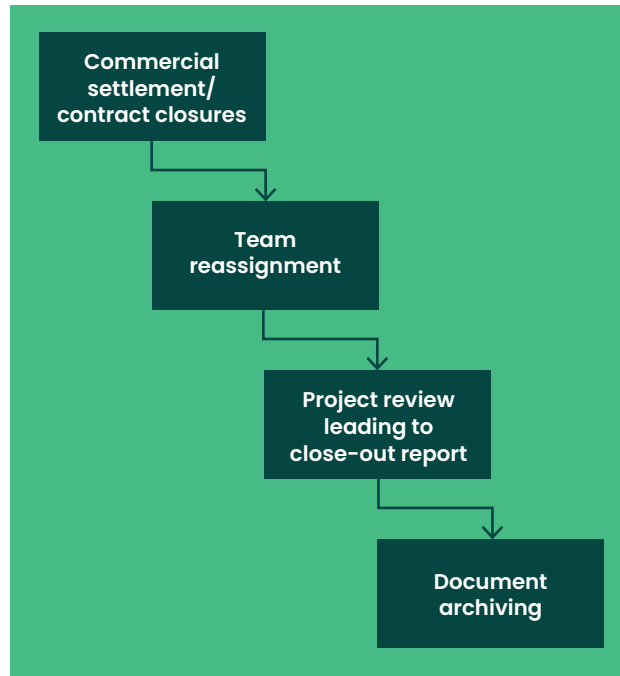


Figure 3.6.3 Key elements of project closure

Recommended reading

- APM's Planning, Monitoring and Control Specific Interest Group guide *Planning, Scheduling, Monitoring and Control* (2015) describes handover and closure activity in Chapter 30 and usefully differentiates between the separate but closely related natures of handover and closure.
- *Project Management* (2013), Chapter 28, offers advice and guidance regarding project closure, including cost cut-off, disposal of surplus materials, authorising of post-project expenditure, project closure documents, final project cost records and other documentation. It also addresses the as-built condition of a multiple manufacturing project, security, and the management of files and archives.
- *How can we hand over projects better?* (2017) is an APM research report that provides advice and guidance to support project professionals in the planning and management of the transition to BAU.

3.6.4 Closing programmes and portfolios

Retiring coordinating frameworks for projects when they cease to add value

Programmes and portfolios have distinct objectives (see 1.2). They are the coordinating frameworks for collections of projects and take into consideration other organisational activities.

Closure of a programme is justified if:

- all the required outcomes have been delivered
- the business case is no longer viable
- residual work would be better delivered as part of another project or programme

The administrative closure of a programme is like the administrative closure of a project (see 3.6.3). If the programme has delivered all its outcomes and fulfilled its intended functions, there should be a:

- formal review of achievements
- review of organisational learning

A different approach may be used if a strategic portfolio is the vehicle for change. It would not be usual to close the portfolio; rather, constituent parts of the portfolio may change over time. However, the whole portfolio may close under certain circumstances, for example, if there is a change in senior leadership or significant and unexpected changes in context.

The ongoing viability of managing programmes and portfolios needs to be periodically reviewed. As projects reach completion, the number of constituent projects will reduce until there comes a point when the administrative and management costs will outweigh the benefits. The structure is abandoned when this point is reached. The decision will be made by the agreed governance body. It will be informed by the advice of the sponsor, and potentially supported by the PMO.

The governance and administrative arrangements of any continuing projects are then revised. Consideration will be given to the following:

- Roles and responsibilities will need to be reassigned to other elements of the business.
- Reconfirmation of sponsorship or alternative sponsorship arrangements should be agreed.
- Continued functional support will need to be provided, for example commercial or finance.
- Briefings with stakeholders should take place to advise them of organisational changes and facilitate engagement.

The above arrangements are made easier if the remaining active projects can be transferred to an existing programme or portfolio. However, how they will integrate needs to be thought through. Consideration needs to be given to the fit from the organisational, technical, process and human perspectives.

These changes can be unsettling at the team and individual levels. Ideally, this should have been thought through and planned well in advance when this is possible.

Those impacted should be comprehensively briefed and given honest advice about the risks and opportunities that the changes represent (Figure 3.6.4 lists these activities). Organisations should also think about what can be done, for example, celebrating and recognising personal and collective achievement, support for future deployment, or reflecting on learning to take into future work. As described in topic 3.6.3, the quality of how you close a project (or programme or portfolio) and support people as they move on is really important, not only for those directly involved but for the organisation's future capability to deliver project work.

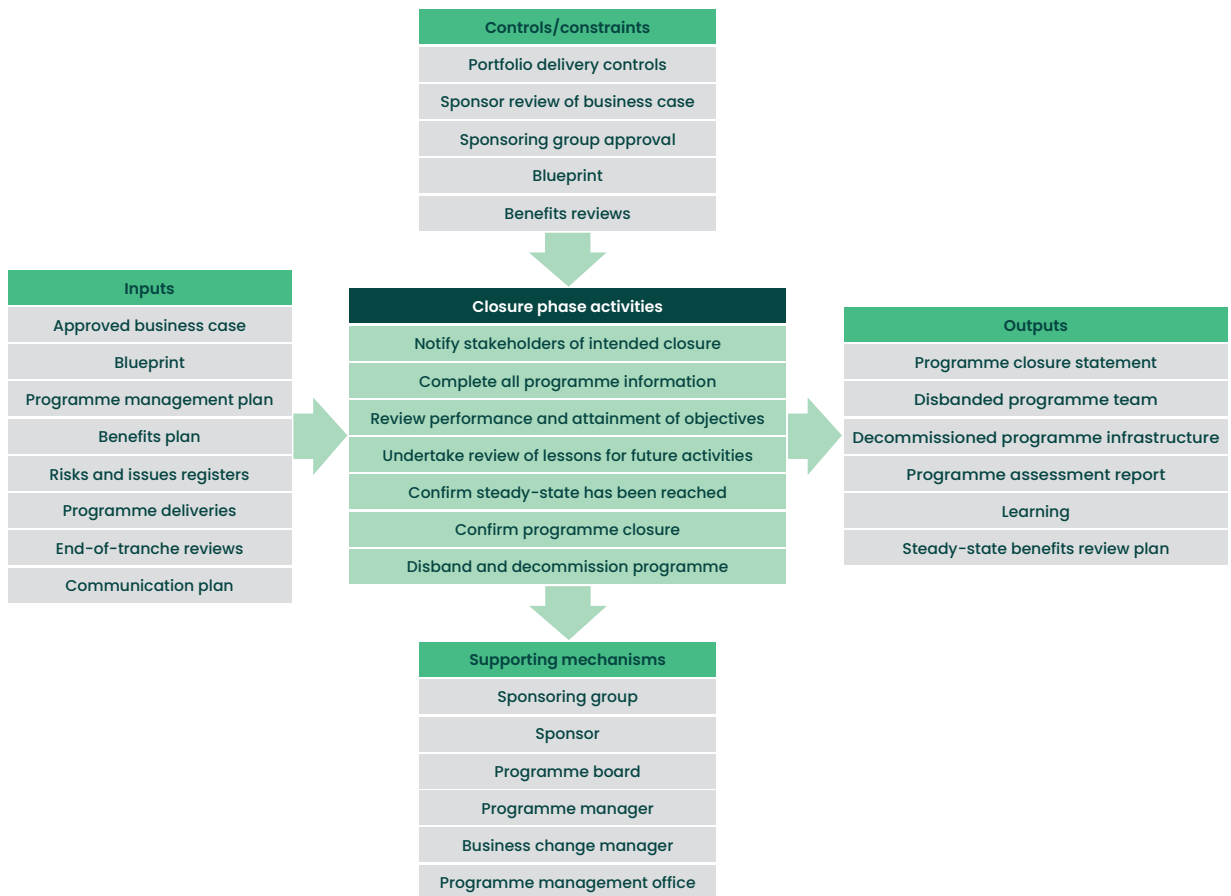


Figure 3.6.4 Programme closure activities

Recommended reading

- *APM Introduction to Programme Management* (2016) is the APM Programme Management Specific Interest Group guide. Section 2.7 covers programme closure and has practical guidance about how to achieve it.
- *Program Management* (2015) includes detailed coverage of programme management activities and processes. Chapter 11 is dedicated to programme closure and includes value realisation assessment; managing programme completion, including the transfer of assets, residuals and resources to the business, and closure report; and finalisation of lessons learned.

4 People and behaviours

Projects are delivered by people. Alongside the technical aspects of project planning and delivery that need to be addressed, the people aspects of projects are critical.

This chapter starts with stakeholders. Projects are conceived to deliver benefits to stakeholders, so the starting point of any project is understanding the stakeholder requirement, engaging with stakeholders and communicating with all concerned.

At some stage of a project, there is invariably some level of conflict. This may be between the different stakeholders, with suppliers or within the team itself. Not all conflict is bad as it allows issues to be voiced. But conflict has to be managed and resolved, often using negotiation techniques.

Projects have to be led and require specific leadership roles. Leaders need to understand the wider context of the project, oversee the engagement with stakeholders and create the project vision. They need to adapt their approach to the different situations and phases of a project. For this to happen, leaders need to be self-aware, of both their strengths and weaknesses. No leader is perfect, so this self-awareness allows the leader to build a team that complements their strengths and, more importantly, overcomes their weaknesses.

Teams need to be managed, so in this chapter there is a focus on team development, on managing a dispersed team and on managing wellbeing. Increasingly, with remote working, all these activities are harder to do, but are essential to the health of a project team.

Team diversity is useful, as it broadens everyone's perspective of the project and how it can be delivered. Inclusion allows the project to make the best use of all the talent available. In this chapter, we talk about the benefits of inclusion and how to create an inclusive team. We discuss bias: being aware of bias and compensating for it.

All projects are delivered within a wider legal, regulatory and ethical environment supported by guidance, standards and codes of practice. Continuing professional development (CPD) is necessary to stay abreast of developments in these areas and to support successful delivery.

This chapter is composed of six sections:

- 4.1 Stakeholder engagement**
- 4.2 Conflict resolution**
- 4.3 Leadership**
- 4.4 Team management**
- 4.5 Diversity and inclusion**
- 4.6 Ethics, compliance and professionalism**

4.1

Stakeholder engagement

Without stakeholders, project-based work would have no purpose. They define what the project needs to do to succeed. They also pose a risk to that success, so understanding and engaging with stakeholders is critical work. A stakeholder is anyone who has an interest or role in a project, programme or portfolio, or is impacted by it. The last edition of this *Body of Knowledge* moved away from the term 'stakeholder management'. Since then, the term 'stakeholder engagement' has become near-universal. What has not changed is the importance of stakeholders to project professionals. If stakeholders can be included in the articulation of the project, its outcomes and benefits, this often leads to better delivery and results. But people only feel included when they have experienced genuine engagement; they can feel cynical if the engagement is seen as either coercive or manipulative. Engagement takes time and it cannot be rushed, so it is important to plan properly for this phase of a project.

This section tackles four topics, listed in bold below, that project professionals need to master.

Engagement planning is the process for engaging with stakeholders. It has different steps to follow – see 4.1.1. Each of these is an expertise in itself. This needs to be a constant and dynamic process that continues throughout the life cycle of the project-based work.

Project-based working takes place in a complex social and political context. This extends beyond the sponsoring organisation and embraces all stakeholders. The interconnected relationships, political positions and power dynamics both complicate and enrich the jobs of project professionals. Yet they often have little formal power of their own, so they need to understand how to use the forms of power and the social skills that they do have.

One of the most valuable skills for project professionals is facilitation. This enables them to get the best contributions from their teams. It also allows them to learn from, influence and broker agreements among stakeholders. It offers an effective leadership style. The techniques of facilitation are many, and can help in a wide range of situations.

Stakeholder engagement takes place through communication. There are many means of communication, each with benefits and disadvantages. What matters most is that project professionals are deliberate in crafting messages, selecting media and planning each communication (using a communications plan).

This section is written for project professionals who recognise the importance of engaging positively with their stakeholders. When they understand and respect their stakeholders, success is more likely. The section includes:

4.1.1 Engagement planning: Creating positive engagement with stakeholders

4.1.2 Social context: Navigating social and political complexity

4.1.3 Facilitation: Making it easy to collaborate and solve problems

4.1.4 Communication: Ensuring the exchange of relevant information

4.1.1 Engagement planning

Creating positive engagement with stakeholders

A stakeholder is anyone who has an interest or role in a project, programme or portfolio. It is the stakeholders who determine the success, or not, of project-based initiatives. They represent key success factors and points of potential risk, so project professionals must engage with stakeholders in a respectful and productive way. Stakeholder engagement follows five steps: identify, analyse, plan, act and review.

Stakeholders exist within and outside the sponsoring organisation. Project professionals should look as widely as possible to **identify** their stakeholders. Some may be close to the project. Indeed, some project professionals see their team as stakeholders. Other stakeholders have decreasing levels of interest and involvement. Some have interests in different aspects of the work and at different times.

After identifying the stakeholders, project teams will **analyse** them. Effective engagement requires project professionals to understand their stakeholders. Of the many factors they may consider, some are more often critical, like:

- power, influence or impact on the initiative
- interests, needs and priorities
- support for, opposition to or undecided attitude towards the initiative

Stakeholder analysis is a dynamic process. Project professionals should revisit it throughout the project life cycle. They will update assumptions and track changes to attitudes and relationships. A simple method of analysing stakeholders is to plot their influence and interest on a standard power-interest graph (see Figure 4.1.1). In complex social contexts, teams will need more advanced approaches to stakeholder analysis. These must account for factors like political positions, alliances and sources of power (see 4.1.2).

Stakeholder analysis forms the basis for planning stakeholder engagement. The first step is to determine the objective of engaging with each stakeholder or stakeholder group. This may be to:

- maintain and use positive support
- minimise or counter opposition
- harness knowledge, expertise or insight
- influence opinion, attitudes or action
- gather requirements (see 5.1)

A stakeholder engagement **plan** sets out how the team will meet these objectives, with contacts and communications. It allocates responsibilities and sets out timings. And it ensures the team tracks the impact and outcomes of stakeholder engagement.

Then, project professionals will **act** on their plan. They coordinate engagement across the project, programme or portfolio to avoid missed or conflicting messages. Where multiple initiatives affect stakeholders, teams must coordinate with each other. Stakeholder engagement at portfolio level often needs support from senior leaders.

For effective engagement, project professionals need:

- situational awareness: ability to choose the right time, place and message
- cultural awareness: understanding of background, perspectives and values
- communication skills: ability to craft clear, compelling and persuasive messages (see 4.1.4)
- influencing skills: ability to shift opinion, reduce opposition and win support
- conflict resolution skills: ability to challenge others and resolve conflict fairly and respectfully (see 4.2)

Stakeholder engagement continues throughout the life cycle of the work. Project professionals should regularly **review** the effectiveness of their work and changes to the stakeholder landscape.

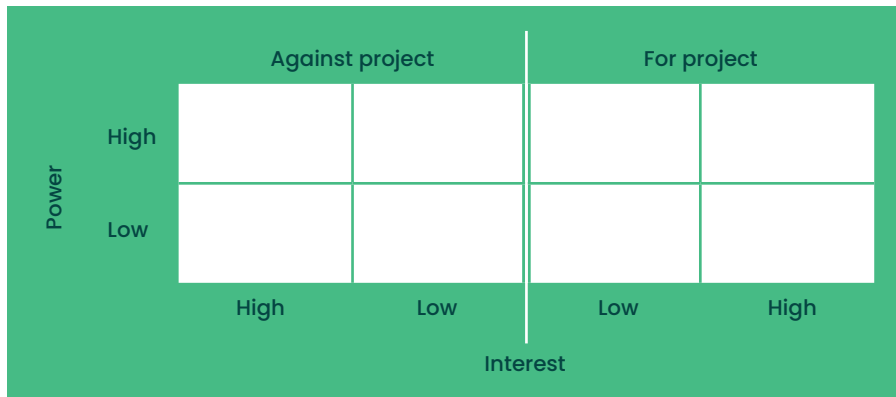


Figure 4.1.1 An approach to capturing analysis of stakeholders

Recommended reading

- *Engaging Stakeholders* (2020), published by APM, seeks to address central questions such as: What does engagement look like? What tools have I got available? How do I best reach, engage and work with stakeholders?
- *Rebel Ideas: The Power of Thinking Differently* (2021) explores the idea of intelligent and unintelligent teams and the skills needed to address the gaps, and how teams can work together constructively.
- *Practical People Engagement: Leading Change through the Power of Relationships* (2013) provides a rich source of practices and techniques that help the reader get better results from the change they are trying to lead. The book distils the principles of people engagement from the observation of high performers. Different forms of engagement are explored, including those that are effective in supporting agile approaches.

4.1.2 Social context

Navigating social and political complexity

Project-based working is all about people. They must work together to deliver solutions and solve problems, often under pressure. Project professionals need to get the best from team members and create a positive project culture. They must navigate organisational politics and influence stakeholders.

Organisations are complex, and project-based working must often deal with additional challenges. These include working with multiple organisations, and encountering conflicting priorities and different cultural norms. But complexity also arises from people's behaviours. These can include power struggles, misaligned communication and conflict.

So, project professionals need to be able to build an understanding of the social systems at play; that is, to map out the network of relationships among stakeholders. It is helpful to visualise who knows whom, and the influences between them (Figure 4.1.2). This produces a social network diagram (also called a soft systems diagram or a sociogram).

Systems of interconnected people and groups produce their own behaviours. They differ from those of individuals or groups acting alone. Systems are more than the sum of their parts.

Power is an important aspect of a social system. Project professionals need to look beyond simple ideas of high or low power to understand the sources of that power. Within the organisation these include:

- position (hierarchical or legitimate) power
- reward and coercive power
- resource and information power
- expert power
- connection, or alliance, power
- personal (referent) power

Beyond the organisation, these include:

- political power
- approval power

In most situations, project professionals do not have positional power over stakeholders. They need to influence through their credibility, personality and character. They deploy expertise, backed up by solid information. And they need to build alliances to support themselves.

The use of rewards and coercion rarely has integrity. It is okay to congratulate and celebrate contributions, or to highlight the risks of certain decisions, but inducements and threats go against the ethical codes that topic 4.6.3 describes.

Organisational politics mixes personal and project objectives. It creates explicit and hidden agendas, alliances and conflicts. Understanding the network of relationships and interests is vital to help craft engagement and communication strategies. It can also be useful to help identify people- and behaviour-based risks.

Understanding the social network of a project is not enough. Project professionals also need to develop social skills and political acumen. Social skills allow us to communicate and interact with each other effectively. Political acumen is the ability to understand and navigate the dynamics of a political landscape.

Political complexity can also influence the choice of implementation strategy. A linear life cycle may not be effective when it is hard to anticipate evolving requirements. Iterative approaches favour stakeholder collaboration, co-creation of solutions and adaptive response to changing priorities. They can better balance the need for pace and progress with resolving the uncertainties of complex social contexts.

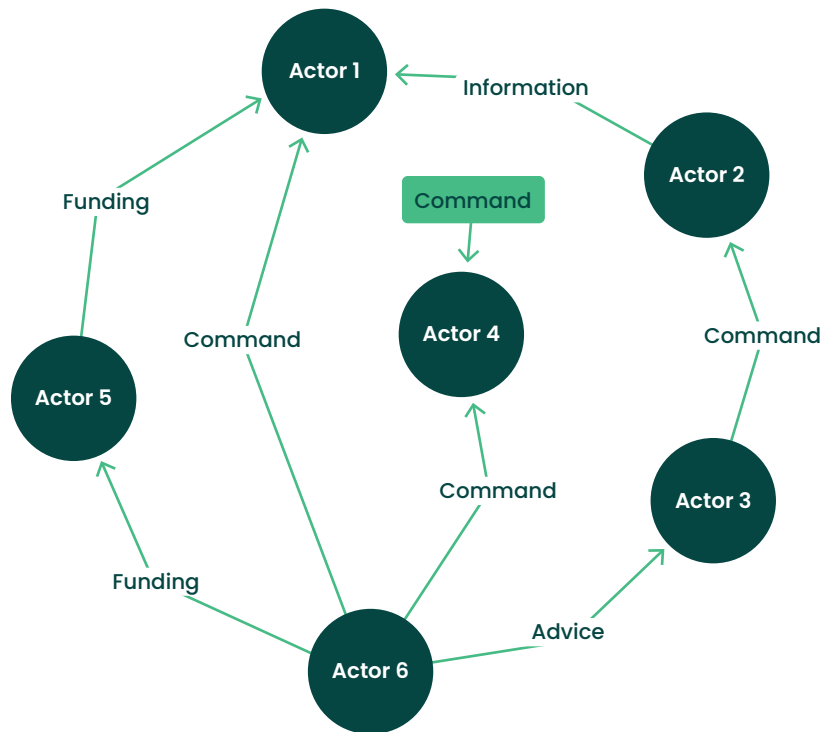


Figure 4.1.2 An example social network diagram

Source: APM/INCOSEUK Systems Thinking Specific Interest Group (2018)

Recommended reading

- *Systems Thinking: How Is It Used in Project Management?* (2018) is the output of an APM Research Fund project conducted by the APM Systems Thinking Specific Interest Group, in collaboration with University College London. It provides broad information about systems thinking in general (not just about social systems) applied specifically to project-based management.
- *Social Network Analysis* (2017) is a classic text, now in its fourth edition, that helps people involved in analysing social contexts to understand how to map and understand relationships between people and groups.

4.1.3 Facilitation

Making it easy to collaborate and solve problems

Project professionals need to engage stakeholders and get the best from their teams. They may sometimes need to be directive, especially in times of crisis. At other times, it makes much more sense to act as facilitators. Facilitation is creating an opportunity for people to participate in a conversation, but it also allows for co-creation of better solutions.

This means remaining neutral and allowing people to collaborate to solve problems or reach agreement. Good facilitation encourages people to contribute and fosters openness, ownership and creativity. The project professional's role is to support and encourage people, helping everyone to do their best work.

Facilitation is a necessary skill that applies in many contexts, including:

- identifying risks, dependencies, constraints or stakeholders
- planning and budgeting
- helping stakeholders reach a decision
- leading a team in solving a problem
- chairing meetings

Facilitation is both a method and a style of leadership. A facilitative leadership style shows trust in your team. It can harness the diverse experiences, ideas and approaches they bring. To be effective, a facilitator must work hard to make sure everyone is able to contribute and be heard. This means stepping back from the content of the discussion and focusing on the processes. The job of a facilitator is to help the group get the results they need.

Sometimes, project professionals can feel too close to the details or have a stake in a decision, so it may not be appropriate for them to facilitate a meeting or workshop. In this case, a neutral facilitator can allow everyone to participate equally.

Facilitation uses a wide range of skills, tools and techniques. These are invaluable for project professionals. Success factors for effective facilitation include:

- pre-planning, to ensure that all those who need to contribute can do so
- creating psychological safety so everyone feels confident to join in. Contracting with participants at the start of a workshop is a good way to do this (see Figure 4.1.3)
- asking good questions and ensuring attentive listening from the group
- drawing out contributions and managing responses
- observing the group, to assess mood, behaviour, frustration, agreement or conflict
- summarising and synthesising themes and ideas

Facilitation is useful at all levels of project-based working:

- **Organisational level:** Spanning organisational boundaries, facilitating cross-boundary discussions.
- **Portfolio level:** Explore how to execute strategy with a workshop to explore risk profiles of different options.

- **Programme level:** Bring together stakeholders with different needs and perspectives, to agree a common vision for the programme.
- **Project level:** Adopt a facilitative style of leadership to motivate and encourage team members.

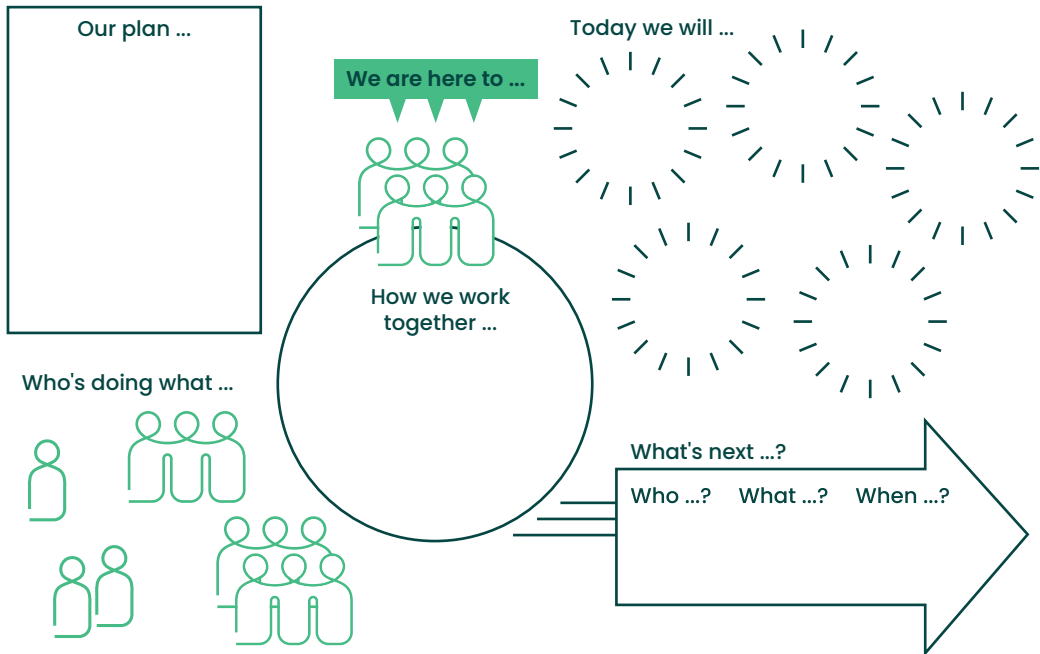


Figure 4.1.3 An approach to contracting with participants of facilitated sessions

Source: Reproduced with kind permission of Penny Pullan

Recommended reading

- *Facilitator's Guide to Participatory Decision-Making* (2014) covers decision-making with groups when aiming at consensus. It explores how to come to a sustainable agreement and introduces the idea of the 'groan zone', which groups often get stuck in before they've reached true consensus.
- *A Short Guide to Facilitating Risk Management: Engaging People to Identify, Own and Manage Risk* (2011) looks at the whole risk management process from the point of view of facilitation. It introduces the concept of a risk facilitator and discusses where to focus at each step of the risk management process.
- *Visual Meetings: How Graphics, Sticky Notes & Idea Mapping Can Transform Group Productivity* (2010) explores visual facilitation techniques that work very well for both in-the-room-together and virtual projects. It discusses how to balance the areas of attention, energy, information and operations when facilitating.

4.1.4 Communication

Ensuring the exchange of relevant information

The ability to communicate is a core skill for project professionals. It's how they share an understanding of everything that matters. This includes requirements, objectives, plans and benefits. It is how they align stakeholders, motivate teams and embed knowledge.

There are many ways to communicate. Effective communicators consider both the message they want to pass on and the method (medium) for sharing the message (Figure 4.1.4). Choices of communication methods must account for the:

- target audience
- urgency of the issue
- complexity and emotional impact of the message
- likelihood and consequences of miscommunication

However, effective communicators must also hear feedback. This includes feedback on progress, uncertainty, lack of progress, emerging issues, risks, disagreements and discontent. Active listening is an extremely important skill, as well as being able to read verbal clues.

Projects, programmes and portfolios use communication plans. These build on a stakeholder analysis to outline the 'what, why, when, who and how' of communication, both within the team and between the team and stakeholders. It is important to measure effectiveness, so the team can adjust the plan for greatest impact. But effective communication plans must also include channels for feedback, both formal and informal. The better these channels are, the earlier issues are raised and the sooner they can be addressed.

Many factors affect the success of communication. These include cultural influences, the method of communication and the language used. Project professionals have choices of media. These combine written words, symbols, voice and non-verbal signals in different ways. Real-world face-to-face communication is the richest. Non-verbal communication includes gesture, expression and voice. It can amplify or undermine the words used. Being able to control non-verbal signals to align with your message is an advanced skill. Video communication brings some, but not all, non-verbal signals to remote communication.

Face-to-face communication may not always be ideal. For example, detailed numerical data is best represented in writing, or as tables or graphs, but the message behind the numbers has to be understood for people to take action. Likewise, a group working through feedback on a document will put more focus on the words and format in a voice call.

Project professionals working with dispersed teams (see 4.4.2) face further challenges, which are greater still when part of the team is co-located because it risks introducing a disparity in the treatment of local and remote team members. Project professionals need to find ways for efficient, effective and inclusive communication.

Project professionals have a wide range of electronic communication and collaboration tools available. However, the choice often takes place at organisational level: project professionals will be expected to communicate via the channels made available to them. It is more often the culture and the practices within the project that determine the effectiveness of communication. The specific tools that teams use to communicate among themselves, and with stakeholders, often have less impact. Good practices deliver good results. Poor behaviours lead to misunderstandings and conflicts.

Many organisations and sectors develop their own protocols and standards for communication. Adhering to these can be important. Examples include procurement, governance oversight and securing approvals. Project professionals must learn the communication norms of the organisations where they work. The alternative will be avoidable delays, errors and conflict.

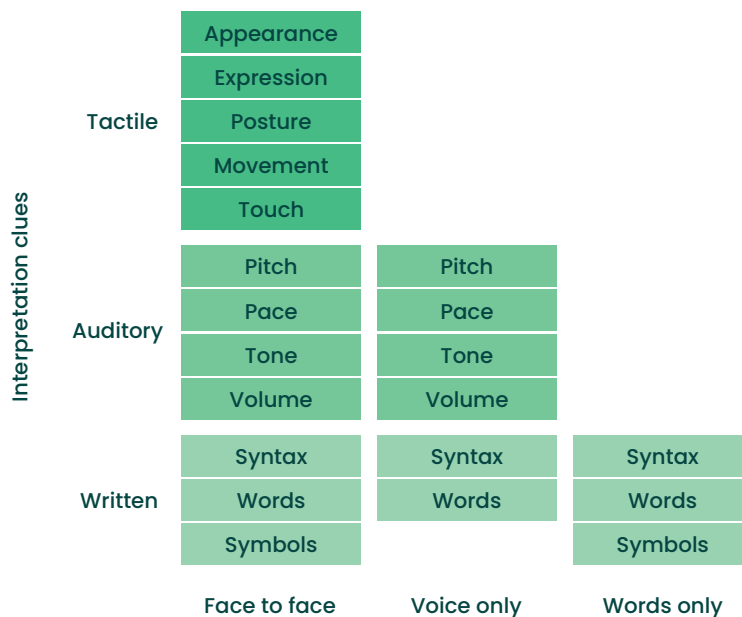


Figure 4.1.4 Considering the medium and the message

Recommended reading

- *Communicating Projects: An End-to-End Guide to Planning, Implementing and Evaluating Effective Communication* (2013) gives project and programme communicators a framework for developing an effective strategy to achieve behaviour change and improve levels of engagement.
- APM's *Communication Planning* guide (2020) outlines the process of creating a communication plan and the different factors that make communication work.
- The APM People Interest Network's *Active Listening Panel: Framework* (2022) describes an engagement tool that focuses on building relationships with stakeholders through understanding what matters to them.

4.2

Conflict resolution

Conflict is a difference in ideas, interests, priorities or perceptions. It is a feature of humanity and takes many forms. In project-based working, there are three forms that are of most interest: types of conflict, managing conflict and how to negotiate to find a suitable resolution.

First is the need to recognise that conflict is not necessarily bad. Positive conflict is a respectful disagreement. It is necessary if project professionals want to find optimal solutions and make robust decisions. With no constructive challenge, there is the risk of falling into groupthink. At best, better outcomes are missed and, at worst, serious risks are created for projects, programmes and portfolios.

But sometimes conflict is not conducted in a constructive way. This is negative conflict and it needs to be managed to find a suitable resolution. If it becomes acrimonious, it is necessary to attend to any relationship damage that occurs, otherwise it can contaminate morale, motivation and trust in the project team and stakeholder group. Project professionals need a series of steps to de-escalate the conflict, and they can support them with a set of strategies they can follow.

Negotiation is another form of positive conflict. It follows a process and is conducted with respect. But it does fit our definition of conflict, and it is a common part of project-based working. So it is helpful to have a negotiation process and, because it is a form of conflict, conflict strategies to assist with this process. Large procurements will almost certainly involve formal negotiations. These will cover the principal conditions of the deal and the detailed contract terms.

This section is written for project professionals who are dealing daily with petty and major conflicts, and with small and big negotiations. For high-stakes conflict and negotiation, there is specialist training. Project professionals may also need to call upon expert support from trained mediators, arbitrators, procurement professionals and lawyers. The section includes:

4.2.1 Conflict: positive and negative: Conflict is not always bad: it can be necessary

4.2.2 Approaches to managing conflict: Facilitating win-win solutions where appropriate

4.2.3 Negotiation: Planning, making and following up on agreements

4.2.1 Conflict: positive and negative

Conflict is not always bad: it can be necessary

As stated previously, conflict is a difference in ideas, interests, priorities or perceptions. In project-based working, it can arise over what to do and how to do it.

We may think 'conflict bad, harmony good'. This is not true. Where someone promotes an idea that is weak or wrong, an alternative should be welcomed. What makes conflict disruptive is how it happens.

A disagreement discussed respectfully is positive and is a necessary way to challenge thinking, find better solutions and make good decisions. Without conflict, teams fall into groupthink. This happens when people find disagreement uncomfortable, so the group seeks consensus before it should. As it approaches consensus, the group rejects dissenting voices or people may self-censor contrary points of view.

When teams fall into groupthink, decisions are based on 'what we all know'. Team members feel they cannot challenge the consensus, so relevant new information and ideas are not fully explored.

When this happens, groups can endorse higher-risk decisions than individuals would. This may result from the degree of confidence due to a lack of dissent. It is called 'risky shift'.

So, lack of conflict is dangerous – 'negative conflict = bad, positive conflict = good':

- Negative conflict is acrimonious, disrespectful and poorly managed.
- Positive conflict is considered, respectful and well facilitated.

Negative conflict arises when emotions overcome reason. This can be because the parties care about the issue at hand, but the situation can easily become bad if it escalates into conflict between people rather than issues. As soon as things are personalised, acrimony starts (see Figure 4.2.1).

Conflict escalation is a series of rising steps moving from accord to disagreement, then to irritation, anger, abuse and even violence. It is possible to spot the signs in body language and voice:

- posture: body angle, leaning in (aggressive) or out (aversive)
- gesture: aggressive or self-soothing
- facial expression: mouth, nostrils, eyes and brow
- physiological changes: breathing, flushing and pupil dilation
- tone of voice
- choice of words

Project professionals need ways to avoid negative conflict and facilitate positive conflict. It starts with creating a positive environment where people challenge each other respectfully. This is an inclusive culture where:

- people respect one another
- finding optimal solutions is more important than being right
- being wrong is not seen as failure

To support this, project professionals need the skills to facilitate discussions effectively. They need to allow:

- everybody to contribute, so that charismatic individuals are not dominating
- the group to assess all evidence and ideas
- constructive challenge of ideas and evidence
- respect for dissenting opinion
- time to agree actionable outcomes

But even with a respectful culture and strong facilitation, negative conflict can still arise, so project professionals also need the skills to spot signs of rising negative conflict, and to defuse it, while encouraging respectful discussion of the important issues.

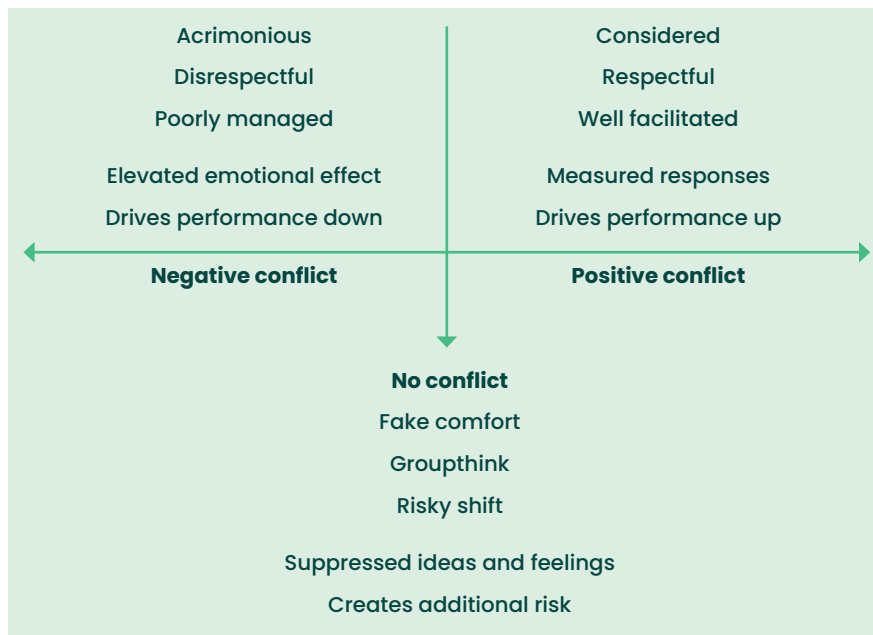


Figure 4.2.1 Impacts of negative, positive and no conflict

Recommended reading

- *The Five Dysfunctions of a Team* (2002) analyses what makes teams work effectively, including the desire to preserve artificial harmony – which in turn, harms productivity.

4.2.2 Approaches to managing conflict

Facilitating win-win solutions where appropriate

Conflict can arise from multiple sources, including differing opinions or opposing interests that matter to the people involved and are hard to reconcile. Conflict may arise from what to do, processes to follow or relationships.

Project professionals should encourage positive conflict. But, when negative conflict arises, they have a responsibility to intervene. Ignoring this can put the project, programme or portfolio at risk. It can damage morale, poison relationships and undermine confidence.

In managing negative conflict, there are two objectives:

- Finding a fair outcome that serves the stakeholders. This may mean one party is less content.
- Addressing relationship damage to the conflicting parties.

Resolution can start when a third party intervenes or a participant becomes aware of a damaging dynamic. It begins with a choice to engage positively. This means demonstrating respect to everyone involved.

Next, the conflicting parties must build rapport. Without it, they will not listen to and understand each other. Only then can they share facts, definitions, perspectives, concerns and feelings.

Then they can agree criteria for a satisfactory resolution. Each sets out their requirements and preferred outcomes. From this, they can explore options for resolution. They continue to explore until they reach agreement.

Figure 4.2.2 illustrates a model of five strategies for reaching agreement. It is based on a diagnostic tool for assessing preferences. It describes conflict in two dimensions:

- Assertiveness: desire to achieve our own objectives. A more assertive strategy arises from giving high priority to the outcome.
- Cooperativeness: desire to support others' objectives. A more cooperative strategy arises from giving high priority to the relationship.

Some conflicts – where neither outcome nor relationship is important – are not worth pursuing. And, where one priority dominates, a strategy of competing or accommodating will be best. This leaves two strategies:

- Compromising is the easier strategy. Both parties give up something to create a fair balance.
- Collaboration takes a lot of work. Both parties work together to put more into the discussion. This creates mutual benefit and enhances the relationship to create a 'win-win' outcome. It needs a lot of facilitation skill. The investment is not always appropriate.

Other skills important in conflict management include:

- assertiveness: standing up for principles, the project, the team and yourself
- listening and understanding
- communicating ideas, emotions and perceptions
- resilience: remaining resourceful under pressure

Sometimes, someone else is needed to resolve a conflict. This may be a sponsor, stakeholder or third-party mediator or arbitrator. A mediator is someone who helps protagonists to communicate. An arbitrator judges the situation and makes a decision.

Projects need clear protocols for escalating conflicts, to project governance, programme or portfolio level. In contractual disputes, the organisation may need litigation or alternative dispute resolution to de-escalate a conflict. It is always better to create an agreed conflict resolution that accommodates stakeholders' requirements wherever possible, but it is probably useful to identify a final arbiter in case this situation arises and a final decision needs to be made.

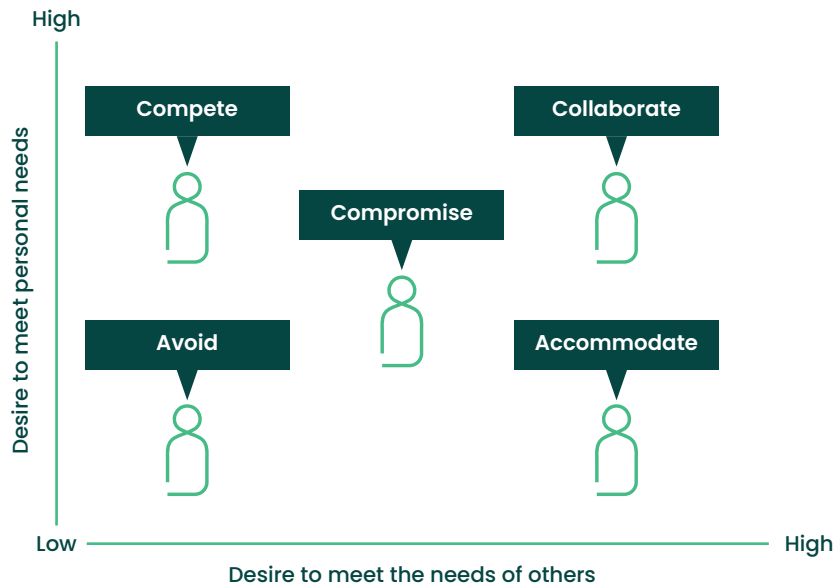


Figure 4.2.2 The five conflict strategies: a common model to consider approaches to dealing with conflict

Source: Adapted from TKI Conflict Model, Kilmann Diagnostics LLC

Recommended reading

- The *Thomas-Kilmann Conflict Mode Instrument™* is a frequently used model to explore the options for management or resolution of a conflict. Kilmann's website provides access to reading materials and the self-diagnosis instrument to help develop skills in dealing with conflict.
- *Everyone Can Win: Responding to Conflict Constructively* (2007) is a practical book that provides the essentials for handling personal and workplace difficulties with emotional intelligence, including handling clashes of values and toxic power issues.
- *A Practical Approach to Alternative Dispute Resolution* (2018) is a comprehensive and digestible commentary on the ways to resolve conflicts out of court. This is very relevant to some projects where conflicts within the organisation, or in the supply chain or between partners, cannot be resolved easily.

4.2.3 Negotiation

Planning, making and following up on agreements

There are many situations in project-based working where 'deals' are needed. They range from agreeing access to project resources, to defining scope, to negotiating contracts. Negotiation principles and skills apply to all of these.

Negotiation is the process of searching for an agreement that satisfies all parties. It is more formal than influencing and more structured than an argument. Negotiation is conflict that follows a process and is conducted with respect.

There are four steps in the negotiation process:

1. Preparation
2. Opening
3. Bargaining
4. Closing

Preparation is vital. Most important is knowing what you want (your goal) and your bottom line – this is the worst deal that is acceptable to you. To understand it, you need to know your best alternative to a negotiated agreement (BATNA). Any deal less favourable than your BATNA represents a loss.

Preparation should include understanding the party you will negotiate with. Consider what their objectives and BATNA might be. As shown in Figure 4.2.3, the BATNAs of each party define a zone of possible agreement (ZOPA).

In many organisations, there is an obligation to achieve value for money or 'best value' in commercial negotiations. This means understanding what contributes to value in the context of the negotiation. It is rarely as simple as cost alone. Some negotiations have implications for the wider organisation, portfolio or programme, so project professionals need to understand what value means at that level. They may need to reject a short-term or 'good' local deal that has adverse wider consequences.

The opening step is where we:

- make a positive impression and build rapport
- agree the basis of the meeting and ground rules
- understand each person's authority to make an agreement
- establish each party's objectives for the negotiation

The five conflict strategies in 4.2.2 are also relevant to the bargaining step negotiation. 'Driving a hard bargain' and tactics that pressure the other party to 'beat the opposition' are the competing mode. This does not build sustainable relationships. It is more usual to seek compromise or a 'win-win' collaboration. This does not mean that competitive prices are not important, but negotiators balance the price with the value to be achieved. Concessions and extra value are the currency of the bargaining step.

When the terms seem to satisfy both parties, one party states this and tries to close the negotiation. If the other agrees, only the formalities remain. If not, the parties need to explore outstanding issues to find a resolution.

Document the agreement to minimise future conflict, and follow up on agreed actions. Where negotiation is between legal entities, there will be a contract. Project professionals need support from procurement and legal specialists to ensure agreements are legally sound as well as best value for the organisation.

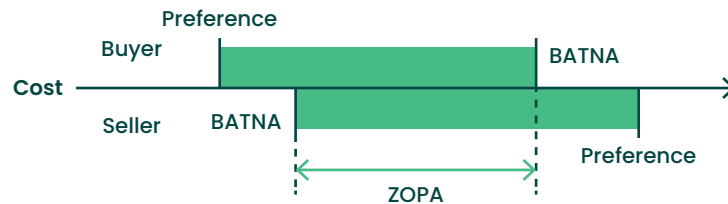


Figure 4.2.3 The concept of a best alternative to a negotiated agreement (BATNA)

Recommended reading

- *Getting to Yes: Negotiating an Agreement Without Giving In* (2011) is a best-selling text that has helped millions of people secure agreements by providing a simple but highly effective framework for negotiation. It is the original source of the idea of BATNA and ZOPA.
- *Successful Contract Negotiation* (1993) is a classic, comprehensive and well-established manual on the negotiation of business contracts. It includes not just the process of negotiation but also the content (prices, payments, warranties, liabilities and claims), looking at the sellers' and buyers' viewpoints.
- *3-D Negotiation: Powerful Tools to Change the Game in Your Most Important Deals* (2006) outlines an approach to align set-up, deal design and tactics to achieve superior results. It contains practical steps and engaging examples.
- The APM book *Navigating Project Negotiations* (2023) covers a simple five-step approach to negotiating in a project-based environment.

4.3

Leadership

Leadership is the ability to establish vision and direction, to influence and align others towards a common purpose, and to empower and inspire people to achieve success.

This means that leadership skills are essential for project professionals. They apply at the project, programme and portfolio levels. And, within each of these tiers, there are leadership roles at all levels.

This section starts with an assessment of the components of project leadership (4.3.1). It looks at the leadership requirements for delivering successful projects. The section then goes on to discuss creating the vision and the role the vision plays in directing programmes and projects (4.3.2). After creating the vision, another key project leadership skill is judgement (4.3.3). Judgement is key to good decision-making.

Topic 4.3.4 discusses three key elements leaders need to focus on to develop their own leadership skills. This starts with being self-aware and understanding how their leadership impacts on others. Leaders need resilience, but they also need to focus on their own wellbeing to be in a position to lead and look after the wellbeing of others. Finally, good leaders reflect and learn. In that way, they continue to develop and become better leaders.

This section ends with a topic on situational leadership (4.3.5). The focus is on the need for leaders to lead in different contexts and highlights the different leadership requirements depending on the maturity of the team.

The perfect project leader doesn't and will never exist. But, by understanding people's strengths and weaknesses, a leader can build a team that complements their own strengths and weaknesses to provide far stronger leadership of a project than can be delivered by any single individual. But this also means that there are multiple leadership roles, and project professionals at all levels need to understand their own skill base and how they can fit into the leadership structure.

Finally, it is very difficult to break leadership down to individual elements, so this section needs to be read and understood as a whole, ideally with wider literature on leadership to help develop a fuller picture.

The section includes:

4.3.1 The challenges of project leadership: The key requirements of project leadership

4.3.2 Creating vision: Knowing where we are going and why

4.3.3 Judgement and decision-making: Using expertise to make good decisions

4.3.4 Leadership of self: Self-awareness and understanding others

4.3.5 Situational leadership: Responding to the environment

4.3.1 The challenges of project leadership

The key requirements of project leadership

Figure 4.3.1 summarises the key requirements of a project leader.

One way of categorising these requirements is to divide them into two groups: hardware and software. The hardware can be seen and includes governance processes, application of project management tools and techniques, a capable project team, and resources, such as finance and equipment. The software is intangible, such as personal qualities of the leader, an appropriate culture and good working relationships. Although the software is less visible, its effects are absolutely tangible.

In the early leadership literature, there was a movement that considered great leaders to have certain traits, including attributes such as above-average intelligence, extraordinary energy, intuition, a particular physical appearance, and particular values and beliefs. The world has moved on and, although energy is still required and values and beliefs are extremely beneficial, our understanding of the effects of physical appearance has changed and there is now a greater emphasis on emotional intelligence.

Many of the practices contained in Figure 4.3.1 are described elsewhere – ‘Stakeholder engagement’ (4.1), ‘Building the team’ (4.4 and 4.5), ‘Governance’ (2.6) and appropriate use of tools and processes (as described in Chapter 5) – so the focus here is on personal skills and abilities, shaping the culture and creating a network of resource.

Project leaders need to be self-aware: they need to understand others and be authentic. This creates trust and enables the leader to better influence others. Being able to deal with ambiguity together with the issues of uncertainty and risk creates confidence in the leader. Developing good judgement is critical too, and often courage is required to challenge and deal with difficult situations. Good leaders develop their leadership of self (see 4.3.4) but, importantly, create the time and space to make good decisions, engage with stakeholders, and lead and develop their team.

In a project situation, the leader should create the culture within the team early. Developing the most appropriate culture is essential for successful project delivery. Each project is different, but certain principles are important:

- **Valuing all employees:** Workers will have no sense of ownership and are unlikely to give of their best if they do not believe their work is appreciated or that they will be treated fairly as individuals.
- **Allowing a level of freedom to operate:** The level of freedom will depend on the nature of the work, but individuals work best when they understand what their role is, are given guidelines and parameters within which to work, and are then allowed to do the job as they see fit.
- **Encouraging freedom of expression:** As no individual has the monopoly on good ideas, encouraging comments and thoughts adds value. Also, allowing people to speak freely about problems helps to prevent pressure building up.
- **Actively promoting learning:** This not only motivates individuals but also enhances their capabilities.

- **Setting standards but tolerating failure:** People need to know what standards are required but, equally, it is important to know when those standards are not being met or when they are not appropriate. If there is what is generally known as a 'blame' culture, few people will be willing to admit to errors or to voice their opinions for fear of sanction.

It is hard for any single project leader or employee to negotiate their way around the myriad of connections, overwhelming wealth of information and sometimes conflicting requirements. This is why it is important for project leaders to build 'networks of resource', which can comprise trusted advisors as well as reliable sources of information and reliable suppliers. Most leaders will be familiar with the need to secure reliable suppliers, but the other two categories are less well considered. These networks help in creating resilience – they can be called upon when needed. Ideally, a network should also include senior and influential people whose support can be called upon and their advice sought.

What matters here is that the sources are trusted and capable in the area in which they are being used. A trusted advisor, for example, may be reliable in their field of expertise but less so in another. The same is true of sources of information. Building good relationships with advisors means they are more likely to be available when they are needed.



Figure 4.3.1 Key requirements of a project leader

Recommended reading

- *Evolving Project Leadership* (2021) establishes a vision of what good project leadership looks like and offers concrete steps to achieving this.

4.3.2 Creating vision

Knowing where we are going and why

A key part of leading a change initiative is the clear articulation of vision, values and objectives. In project-based working, 'vision' describes the future state at the end of the change initiative. This is delivered by selecting projects and programmes that deliver one or more of the strategic objectives of the sponsoring organisation.

These strategic objectives connect, in turn, to the organisation's vision for where it is heading and what it wants to be. The result will be a portfolio of change initiatives. Within that portfolio are individual projects and programmes (Figure 4.3.2).

Project professionals get the best from their teams when they convey both the vision and what it means. People are motivated by understanding the purpose and value of their work. If they don't know why they are doing something, it is hard to maintain commitment. Understanding context makes it easier for team members to do their work, as they can find more appropriate solutions to the challenges of project-based work, and make better decisions.

The vision (or mission) statement encompasses the reason for doing the project – its greater purpose. Project leaders should craft a vivid statement that is short and precise, and uses simple language. Nothing crushes enthusiasm like 'management-speak'! They should gather their team together and set out:

- where the project is going: the 'vision'
- what they need to achieve: the 'goal', 'objective', or 'mission'
- who they are working for: the users, customers, stakeholders or beneficiaries
- why they are doing it: the 'purpose', benefit or value of the work

Once the vision has been understood, it is up to project professionals to share it at every opportunity.

The vision and purpose of the change initiative need to mesh with the team's culture. Project professionals are responsible for leading this. The culture must offer the right levels of:

- physical and psychological safety: everyone is protected from harm and able to flourish
- empowerment and autonomy: people can make choices and do things their way
- inclusiveness and belonging: everyone feels comfortable, respected and valued (see 4.5.1)
- care and resilience: people look out for one another, and systems address the risks from stress (see 4.4.3)

Underpinning this will be the team's values. These set out what things are important. They therefore drive choices about what the team does and how it behaves.

Values can easily become little more than 'good words'. But, when project professionals lead well, values like accountability, integrity, collaboration and respect can become guiding principles for an effective team. The best values are those that the team sets for itself, with the help of a skilled facilitator.

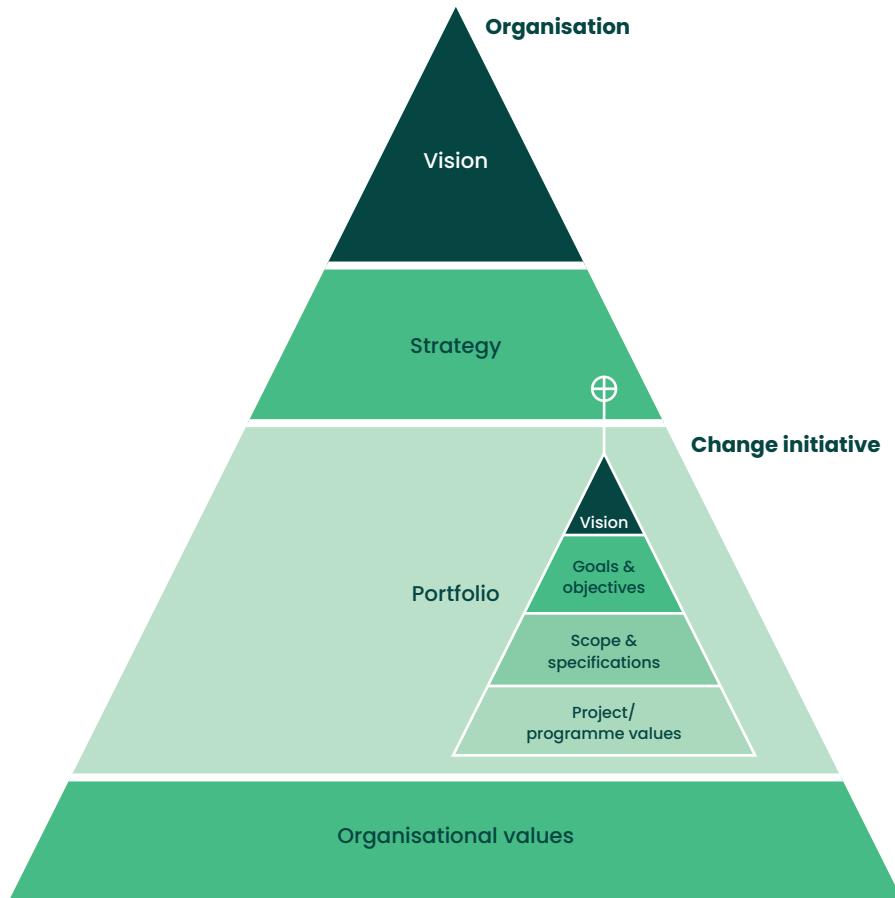


Figure 4.3.2 Vision of change initiative related to organisational strategy

Recommended reading

- *Organizational Culture and Leadership* (2004) provides a deeper understanding of the interrelationship of organisational culture dynamics and leadership.
- *Understanding Organizations* (1993) explores the key concepts of culture, motivations, leadership, role-playing, coordinating and consultation.

4.3.3 Judgement and decision-making

Using expertise to make good decisions

Project-based working calls for good judgement and sound decisions at all levels. Project professionals must be able to both make decisions and support decision-making. Examples from different areas include:

- project and programme selection
- governance decisions that guide and oversee projects, programmes and portfolios
- prioritising anything from user requirements to risks
- selecting a preferred option to resolve an issue or problem
- choosing suppliers in project procurement
- allocating and scheduling resources

At the point that project professionals make a decision, they can never know if theirs is the 'right' choice. Knowing what the optimum is, with certainty, eliminates the need for a decision. So the best anyone can do is to make a 'good' decision.

A person or a group makes a good decision when they have both sufficient authority and expertise. It must be based on good information, which is sufficient and reliable. There must be a sound process, including the analysis and interpretation of the information (see Figure 4.3.3). This needs to consider alternative interpretations and therefore compare options. And the process must allow for dissenting voices to speak out and receive proper consideration.

This gives the best chance of good judgement driving the decision. Therefore, the expectation is that a 'good' decision is more likely than an alternative to be the 'right' decision.

However, judgement can be flawed. Given the availability of reliable information and a good decision-making process, the biggest threat to good judgement is decision bias. This is a mixture of the psychological biases that affect individuals and groups when making decisions. Common biases include the anchoring, salience, framing and confirmation biases. Sadly, there are many more.

One way to improve decision-making is to involve a group. Where there is diversity of thinking and effective facilitation, groups can make better decisions than individuals. However, this requires that each person can access and assess raw data for themselves and think independently. Formal methods like the Delphi technique harness these conditions for good group decisions. But other principles also come into play, including:

- a healthy preoccupation with the possibility of failure
- reluctance to oversimplify and rush to solutions
- a real understanding and sensitivity to operations
- commitment to resilience through anticipating and foreseeing future problems
- deference to expertise and not simply hierarchy

If the project leadership or facilitation is poor, there is a real risk of groupthink. This is where the group becomes reluctant to accept divergent thinking. It can lead to the group missing important information and making poor decisions. And it can go further – groupthink can lead to risky shift, when the final decision represents a more extreme position than any one individual would have chosen.

Ultimately all project professionals need to develop good judgement in their areas of expertise. But they must also recognise the limits of their knowledge and experience. This is where artificial intelligence (AI) can assist in the decision-making process (see Chapter 6). Good project professionals know when to bring additional resources into their teams. They also know when to involve them and, crucially, how to assess their contributions.



Figure 4.3.3 Making a good decision

Recommended reading

- *Managing the Unexpected* (2015) asks: why are some organisations better able than others to maintain function and structure in the face of unanticipated change?
- *The Wisdom of Crowds* (2005) explores a simple idea: that large groups – rather than the select few – are better at solving problems and fostering innovation.
- *Active Listening Panels* (ALPs) are a good way to gather the team's and other stakeholder views. APM's People Interest Network has developed a useful ALP resource to guide users: see *Active Listening Panel: Framework* (2022).

4.3.4 Leadership of self

Self-awareness and understanding others

Leadership is not only about the interaction between the senior person and their team. There is a step before that. In order to be a good leader, an individual must be self-aware (see Figure 4.3.4). They must understand their own strengths and weaknesses, their own preferences and biases; they must understand the impression they make on others, intentionally or unintentionally. They should also be able to manage themselves to ensure they are able to work effectively and safely.

The leader is in one sense a figurehead, the person who communicates the goals, sets the standards and creates the environment – the climate in which others will be working. For this reason, the leader must be aware of the influence they are having and the type of behaviour they are driving. It is often said that ‘every leader creates a shadow’. It is important to understand what that shadow is and whether it is encouraging appropriate action and behaviour.

A leader who understands their own strengths and weaknesses, and those of individuals in their team, is better able to use the skills of others to create a successful outcome and to build effective teams by ensuring a full complement of skills and abilities. No one can do everything, and accepting the fact that others may be stronger in certain areas is an important element in achieving success.

Most individuals view others through the lens in which they, themselves, see the world. If they like to spring to immediate action, they perceive others who don't as being slow, when in fact those people may be more reflective and ultimately take a better course of action. It will depend on the circumstances. If an individual likes blunt talking, they can believe others are being ‘too sensitive’ when they react negatively to words that may have been ill-chosen. It is important to understand other people's perceptions in order to motivate and communicate effectively with them.

A good leader is prepared to listen and take advice from others. Ultimately, it may be the leader who will make the decision, but that decision will be better informed if they know where to go for advice and take time to exercise judgement on how to accept and implement what they are hearing.

Being self-aware enables a leader to understand how they like to operate and the impression they give to others. It enables them to work with what they have and to adapt their behaviour when necessary. However, it is not possible to be a different person. The leader must be authentic. That means operating in alignment with their own values and standards. Inconsistencies will be noticed and may lead to mistrust.

The importance of wellbeing and resilience is sometimes overlooked in the pursuit of achieving targets within a given timescale. For sustained success, it is essential that leaders and the individuals in their teams are able to work to the best of their ability. No one can work at full capacity all the time.

Wellbeing and resilience come from a combination of physical factors – such as getting enough sleep and exercise – and from understanding and managing responses to what is happening.

Successful leaders accept that problems and mistakes will occur. What matters is how they are handled and what can be learned from them. Once a solution is found, there is no value in ruminating on what has happened. It drains energy and takes focus away from the present.

Resilience is also built by accepting that change and uncertainty are inevitable. Very few projects run exactly according to plan and there may be a period of uncertainty before a way forward is found. Few people are truly comfortable with uncertainty – or ambiguity – and some rush to create certainty and structure at the expense of exploring better options.

A leader who is self-aware understands and should be able to manage their emotional reactions. Few people appreciate negative feedback, for example, but a person who is resilient is more likely to evaluate whether the feedback has any merit and, if it has, will learn from it. If it is unhelpful, they will be able to ignore it and move on.

Managing response comes from an element of self-discipline, but also through practice and being mindful of how and why responses are being made.

Progress comes from learning and reflection. Without it, there will be no improvement, merely a tendency to repeat old mistakes. Successful leaders understand that sitting and thinking is not 'doing nothing'. Being able to think critically in an unbiased manner and to challenge assumptions is an essential part of effective leadership.

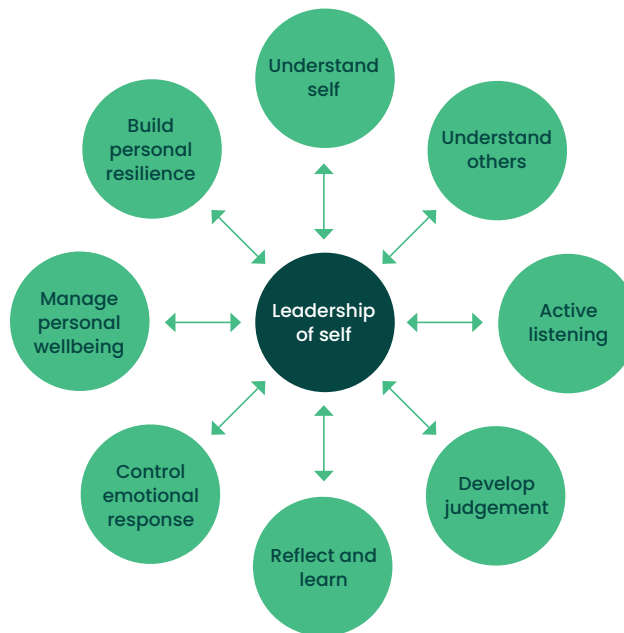


Figure 4.3.4 Leadership of self

Recommended reading

- *Evolving Project Leadership* (2021) shows how the effective project leader evaluates the self, the team and the organisational culture to cultivate fit-for-purpose project leadership.
- *How to Do the Inner Work* (2024) encourages us to better understand our thoughts, emotions and behaviours in order to reduce stress and define a sense of purpose.

4.3.5 Situational leadership

Responding to the environment

Leading individuals is a key role for project professionals. It would be easy if there were one simple formula to apply. But there is not. A good alternative is understanding the principles that determine a good leadership style for each situation. This is called situational leadership. We have described the Turner and Cochrane framework, together with the Obeng project types, in section 1.2.2. These should inform leadership styles for different types of project and different situations, but here we will focus on the Hersey–Blanchard Model.

The Hersey–Blanchard Model of Situational Leadership® is perhaps the most well-known model, which broadly tracks a leadership journey from direction (telling) to delegating via ‘selling’ and ‘coaching’.

What the models have in common is a two-step process. The leader needs to start by assessing the maturity of the team, before moving on to select a leadership style that matches the maturity of the team and the team members.

In assessing the situation, the primary consideration is how the team member relates to the work. Many models keep this simple by asking about how capable the team member is at performing the task, and their attitude to taking it on. Their capability may be anything from ignorant of what the task demands to highly competent. They might be confident and enthusiastic, hesitant and unwilling, or anything between.

At its simplest, this puts the team member into one of four states that can be matched to four leadership styles (Figure 4.3.5). These have either an instructing or coaching style of direction based on capability, and they suggest a high or low level of support based on enthusiasm. Simple labels for the styles are as follows:

- **Instructing:** A lot of instruction with little support is required for people who are enthusiastic but lack skills.
- **Persuading:** Adding in a lot more support is necessary for unskilled people who are hesitant to take on the task.
- **Encouraging:** When people are able but don't yet trust their own ability, they need a lot of support.
- **Trusting:** Capable and confident people need minimal guidance and want to be left to get on with the task.

Of course, the same person may need a different style of leadership when tackling another task where their ability and enthusiasm are at different levels.

In real life, people do not fit neatly into four boxes, so project professionals need to make judgements that take account of individual preferences, cultural norms, the specific context and many other factors. And, as they learn about the people they lead, they will adjust their approach accordingly.

A benefit of getting individual leadership right is that it develops skills and grows confidence. This not only fulfils the responsibility for professional development; it can also build resilience in the team, as people learn skills that can cover for colleagues.

To support situational leadership, project professionals need a number of other leadership skills:

- individual goal setting: agreeing what performance standards to aim for
- delegating: offering developmental work assignments
- coaching: helping learners to figure out how to do things for themselves
- feedback: giving high-quality performance information as a basis for learning

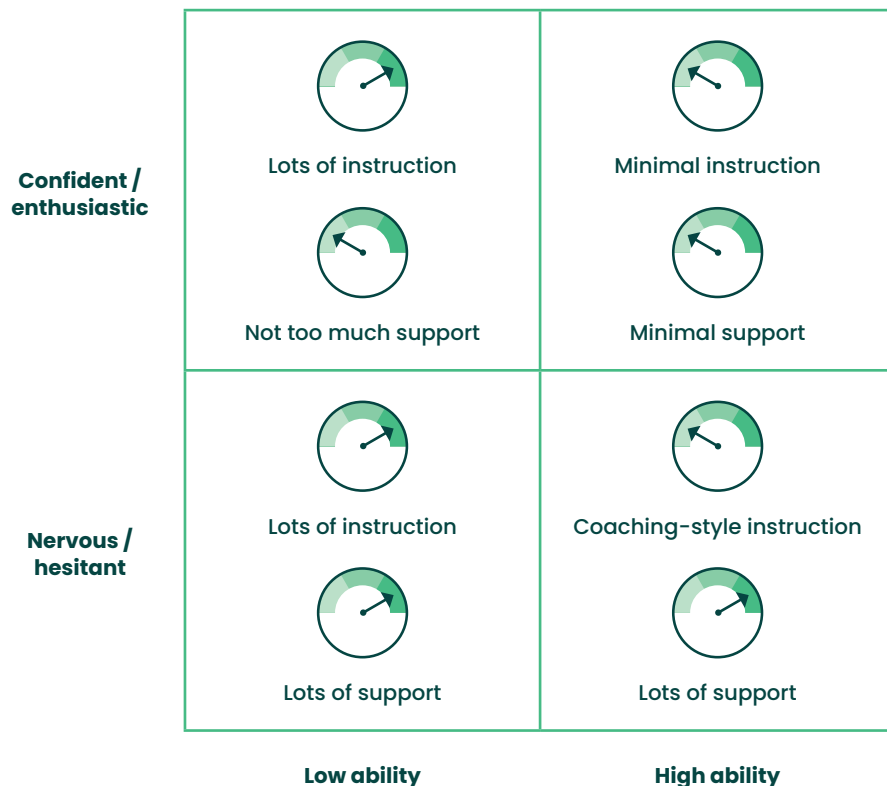


Figure 4.3.5 Matching leadership styles to team maturity

Source: Clayton (2012)

Recommended reading

- *Managing the Unexpected* (2015) deals with how organisations pivot and adapt to changing situations.
- *The New One Minute Manager* (2015), co-authored by Ken Blanchard, brings to life the environment in which many projects operate and the different situations in which leaders may find themselves.
- *Brilliant Project Leader* (2012) combines practical tools and insights to enable project leaders to adapt and motivate their team.

4.4

Team management

Project-based working involves people coming together from different functions, disciplines and organisations to deliver something of value to the investing organisation. A team is a small number of people working together towards a shared goal. In project-based working, teams are temporary, formed for a specific purpose. Their transitory nature brings different challenges from those in business-as-usual teams.

Larger projects and programmes may have team members that span multiple organisations. Examples include consultants, suppliers, partners and customer/client staff.

A key aspect of successful project-based working is bringing expertise into temporary teams. When their work is complete, those people will return to their business-as-usual roles. They will ideally leave with enhanced skills and experience, and without losing out in terms of advancement.

A particular challenge for project professionals is not having full control over who joins their team. But a project professional's reputation will strongly influence who wants to come and work for them. Often, as getting the right skills and experience is difficult, the team will be developed on the project. This development will create future capability and enhance the reputation of the team leader for future projects. This is covered in topic 4.4.1.

Some teams are co-located in the same place. Where this is possible, benefits include sharing a physical space where plans and progress are on display, and developing close working relationships. However, dispersed team working is increasingly common. The particular features and challenges of forming and working with virtual teams are covered in topic 4.4.2.

Project-based working can be exciting and fast-paced, with pressure to deliver good work to available time and budget. Sometimes pressure, which has been motivating, can become unbearable. The overload can lead to damaging stress. Project professionals have a duty to care for their team members and for themselves. This means having the skills to reduce, identify and deal with stress in the workplace. This is covered in topic 4.4.3.

Finally, sections 4.3 ('Leadership') and 4.5 ('Diversity and inclusion'), particularly complement this one.

This section is written for project professionals building and leading teams. The section includes:

4.4.1 Team development: Creating the right context for teams to perform

4.4.2 Dispersed teams: Working with people in different places and time zones

4.4.3 Wellbeing: Safeguarding self and team from the effects of excessive pressure

4.4.1 Team development

Creating the right context for teams to perform

The success of project-based working relies on teams working together effectively. They may be co-located, working at the same time, or working virtually in different places, perhaps in different time zones. But project professionals rarely start work with a ready-to-go team. They need to transform the group of people they get into a coherent team.

Team development is transforming a group into an effective team (Figure 4.4.1). In doing this, it is important to respect the team members as individuals. Each has their own capabilities, preferences, cultural norms and expectations. Team development should create a positive working culture that enables high performance and increases the chances of success.

Team leaders can lead a group through different stages of development. Their role and priorities will change as the group matures. This is how they move from stage to stage, and from group to team:

- **Stage 1 – Unstructured group:** Establish clear goals and objectives, create an inclusive environment and get everybody working.
- **Stage 2 – People assert their individuality:** Allow differences to emerge, support open discussion and resolve conflicts positively.
- **Stage 3 – Team starts to get on:** Provide structure, process, and clear roles and responsibilities, and allow the team to develop its own behavioural norms.
- **Stage 4 – Effective team:** Promote open, honest and trusting relationships. The team will collaborate, so the team leader can focus on supporting its work.

When the team has finished its work:

- **Stage 5 – Reintegration:** Help team members transition into their next roles (see 3.6.3).

Project professionals need to get the best out of their teams. Ideally, assign team members to roles that suit them, to match their strengths or offer them a chance to develop professionally. Project-based work sometimes has tasks that are neither exciting nor developmental, so ensure that roles have variety and opportunities to learn. Provide further support for people through training, coaching and mentoring.

The project professional can provide role variety, build skills and create team resilience by considering 'process' and 'technical' work. It is easy to train someone to take on a process role with little technical content. For example, a business analyst working on requirements capture could take on a risk analysis role. This helps team members appreciate the work of the whole team.

When teams are performing well, they are likely to:

- collaborate and communicate well
- pursue their own development
- innovate and challenge one another
- be efficient and productive
- become largely self-managing and self-motivating

When this happens, the need for team management reduces and the role of the team leader becomes more subtle. It is more a servant leadership role that provides the team with resources they need to continue performing well.

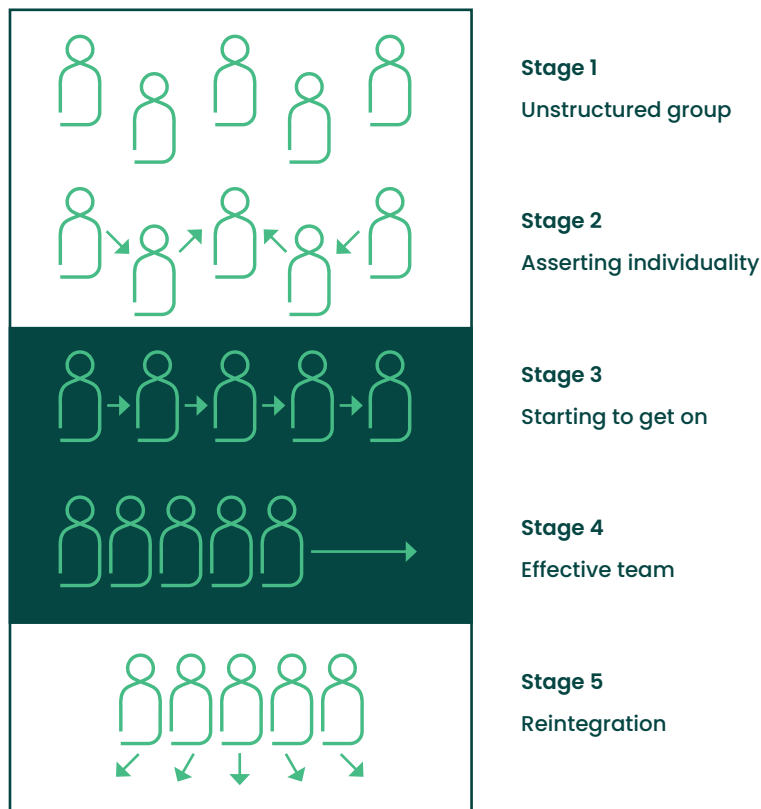


Figure 4.4.1 Development from a group of individuals into a team

Source: Inspired by Tuckman's 5 stages of team development model

Recommended reading

- *Team Development: A Practical Guide to Understanding Team Development* (2018) explores the Tuckman and Hackman models for team development. It addresses ways of ensuring team dynamics remain positive and productive, and how the leader can modify their level of involvement depending on the stage of team development.
- APM's short guide on *How to Achieve High Performance Teamwork* (2025) provides practical advice on how to create effective project teams.
- *Neuroscience for Project Success* (2022) looks at managing projects from the inside out, providing insights on team dynamics and motivations.

4.4.2 Dispersed teams

Working with people in different places and time zones

One of the strengths of project-based working is having teams of the best people, with the right skills. This is challenging when those people are in different places and time zones. Since the COVID-19 pandemic, home-based working has also increased in many organisations.

Dispersed teams are separated by geography and possibly time zones. They are also called 'virtual' or 'remote' teams. Modern communications technology is making this more common and easier. It is becoming the norm for many teams to have people who are remote from the other team members. Even when most team members are co-located, it makes sense to treat it as a dispersed team. This gives everyone the same access and opportunities, as far as possible.

While virtual teams can work well, there are challenges. It is harder to build deep relationships and trust virtually than in-person. Communication can suffer, which affects understanding and empathy. This makes it harder to spot signs of conflict developing and to resolve the conflict. The result can be less positive conflict in the form of constructive challenge.

Teams also lose the informal conversations that happen naturally over coffee or by the water-coolers. This diminishes connection and shared responsibility. Developing these is harder, but not impossible.

Despite the many software tools available, personal and professional development and productivity can suffer, and the possibility of isolation and loneliness during work hours can damage wellbeing. Project professionals can find wellbeing issues harder to spot and deal with remotely.

The final challenges arise from the diversity of many dispersed teams, as well as the practicalities of different time zones. There may be differences in culture and language, as well as generational differences. Listening to each team member and understanding their perspectives becomes even more important.

Project professionals leading dispersed teams need to find ways of overcoming all these challenges. Technology is an enabler but it is not enough by itself.

Figure 4.2.2 shows aspects of virtual leadership development. The model starts in the centre, with leadership style and a virtual mindset. Leaders need a facilitative approach to working with remote team colleagues and stakeholders. They must understand preferences and skills, and how to build and maintain trust. Teams need to agree norms for communication, collaboration and challenge which must take everyone's preferences into account.

Then technology can help implement team processes. Project professionals must evaluate different solutions for:

- virtual meetings: speaking, facilitation, information sharing, consensus building and note taking
- workload management: task identification, allocation and tracking
- collaboration: shared working, dialogue and document access

Virtual meetings will benefit from a social element, clear process and facilitative approach (see 4.1.3). If that sounds familiar, it should. A large part of leading a diverse team is finding ways to recover the benefits of people working together in the same place, even when they cannot.

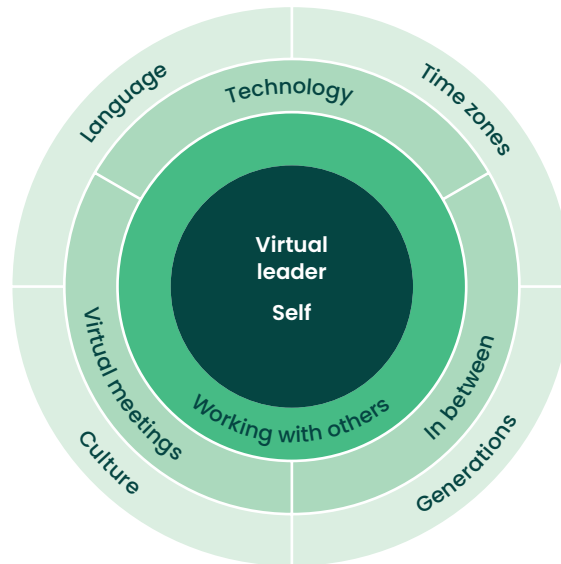


Figure 4.4.2 Steps in development of virtual leadership

Source: Pullan (2016)

Recommended reading

- *Virtual Leadership: Practical Strategies for Getting the Best out of Virtual Work and Virtual Teams* (2016) addresses how virtual working in a fast-paced world requires a new set of skills and a facilitative leadership approach from all team members to avoid isolation, disengagement, ineffective communication and counterproductive activity. The book contains numerous hints and tips that a project team can apply, and stories from those who have put it into practice successfully.
- *Leading Effective Virtual Teams: Overcoming Time and Distance to Achieve Exceptional Results* (2013) contains practical steps for each part of running a virtual team, from setting up communications for collaboration to troubleshooting tips and managing performance virtually.
- *The Long-Distance Leader: Rules for Remarkable Remote Leadership* (2018) reinforces the importance of virtual leadership starting with oneself. It is especially useful for leaders who are transitioning from purely co-located teams to working virtually for the first time.

4.4.3 Wellbeing

Safeguarding self and team from the effects of excessive pressure

Many project professionals find the challenges of a pressured environment motivating and enjoyable. They thrive on the uncertainty, variety and constant change that comes with project-based working.

Stress arises when people feel they do not have enough control over time, capability or resources. It gets worse the more the goal matters to them, or if they perceive that they don't have enough support.

People have an optimum level of pressure that matches their ability to respond at that time (see Figure 4.4.3). The region of this performance peak is eustress. It is affected by their physical, psychological and emotional state, the type of pressure the work creates and the environment.

The UK Health and Safety Executive (HSE, n.d.) defines workplace stress as 'the adverse reaction that people have to excessive pressures or other types of demand placed on them'.

Situations that can lead to excessive pressure include:

- perception of having little control of a situation
- needing to complete a volume of work in a fixed time
- relationships with colleagues or stakeholders
- concerns about career, job security or advancement
- the culture of the organisation clashing with personal values or beliefs
- family, health or commitments outside work

Project-based working inevitably leads to stressful situations. It is a crucial skill for project professionals to be able to notice signs of their own stress and of stress in their team. Everyone is distinct, so stress affects individuals differently.

Project professionals know that it is normal for people to feel stress from time to time. No one is immune to it but everyone can build up resilience, with support. Project professionals need to build their own resilience and develop coping strategies to help themselves in times of stress. They also need ways of supporting team members to cope and function well. This includes identifying and tackling the sources of stress, where possible, and allowing short and longer breaks for rest and recovery.

Some time periods will be more pressured. Project professionals can adjust schedules and allocations, manage expectations and build in contingency if team members are struggling.

Workplace stress is a growing hazard and an area of concern for many organisations. Regulatory bodies in many countries publish standards and guidance for workplace hazards. Their websites often have good advice that can be adapted to project-based working.

Beyond workplace stress, many organisations prioritise a wider focus on the wellbeing of people at work. They may promote the benefits of a healthy lifestyle and physical wellbeing to help people to be in the best shape to deal with work pressures and cope with stress.

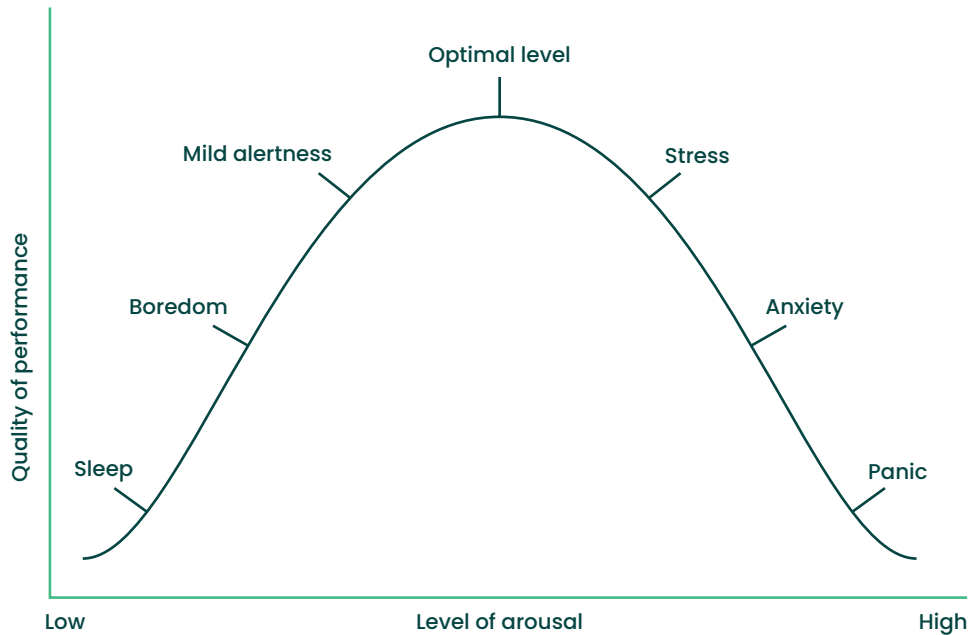


Figure 4.4.3 The balance between performance and level of arousal

Source: Adapted from the Yerkes-Dobson law (1908)

Recommended reading

- *Getting Things Done: The Art of Stress-Free Productivity* (2015) is a best-selling book and set of linked products that helps people approach personal and professional tasks.
- *Managing the Causes of Work-Related Stress: A Step-by-Step Approach Using the Management Standards* (2007) is the HSE's guide to tackling the six underlying risk factors associated with workplace stress: the demands of your job, your control over your work, the support you receive from managers and colleagues, your relationships at work, your role in the organisation, and change and how it is managed.
- *Causes and management of stress at work* (2002) is an article that looks at the causes and signs of stress at work, techniques for managing stress and organisational stress management. The article was published in a scientific journal and is available for free download.
- APM research paper *The Wellbeing of Project Professionals* (2020) identifies preventative actions that can be taken to reduce workplace stress.

4.5

Diversity and inclusion

The benefits of diversity are proven. More differences among team members can offer greater innovation, better solutions and more robust decisions. This higher level of performance delivers better results in project-based working.

But, to get the best from diversity, teams must overcome conscious and unconscious bias and prejudice. They must create a welcoming base where all people can flourish. Discrimination is illegal in many countries but project professionals should go further, taking responsibility for inclusion and belonging.

The case for diversity and inclusion is straightforward. Teams are stronger for their differences. They thrive when everyone respects each other for what they can do, rather than deprecate others for their weaknesses. Together, diversity and inclusion deliver enhanced team performance and better results.

However, the people who lead project-based working need to work to create an inclusive environment, one in which the project can harvest the benefits of diverse ideas, perspectives and ways of thinking. That environment needs to be inclusive, where everyone feels welcome, but, more than that, people should not feel the need to fit in, to be part of the team. Rather, the team culture should be welcoming of everyone for who they are. There are many ways project professionals can secure this.

What makes this difficult is the set of conscious and unconscious biases and prejudices that humans have. These are what leads to discrimination, disrespect and unfair treatment. Project professionals need to understand these biases, and they must constantly apply three strategies to protect against them.

This section introduces five terms, some less familiar than others:

- **Diversity** means having the broadest range of different people.
- **Inclusion** means creating an environment where everyone feels welcome.
- **Belonging** means making everyone feel that this is *their* environment.
- **Advocacy** means speaking up for a group that is marginalised or discriminated against.
- **Allyship** means supporting people from marginalised groups.

This section is written for project professionals who take their responsibilities for diversity, inclusion and fairness seriously. The section includes:

- 4.5.1 Benefits of diversity and inclusion:** Diversity with inclusion delivers higher performance and better results
- 4.5.2 Creating inclusive working environments:** Making people feel welcome in the workplace
- 4.5.3 Conscious and unconscious bias:** Addressing prejudice and bias in project-based working

4.5.1 Benefits of diversity and inclusion

Diversity with inclusion delivers higher performance and better results

Our workplaces consist of people with different backgrounds, abilities and ways of working. This is a good thing that should be encouraged. Diversity refers to the range of different people. It includes race and ethnicity, class, age, gender and gender identity, sexual orientation, language, mental health, religion, physical and mental ability, and neurodivergence.

There is plenty of evidence that diverse teams can deliver higher performance. This is because individuals from different backgrounds and cultures bring different experiences and perspectives to the team. However, diversity alone is not enough.

Inclusion is also necessary. Without it, a team cannot harness the benefits of the different perspectives. In an inclusive environment, everyone feels welcome. People respect and listen to each other, allowing the diverse perspectives to thrive.

There are many benefits that flow from a diverse and inclusive team. They arise from the nature of diversity. Different people see the world in disparate ways and they interpret situations through different filters. This gives different perspectives on every aspect of a project, programme or portfolio.

It also offers a broader set of skills. Irrelevant-seeming capabilities can find useful applications in project-based working, which, in turn, makes for a more adaptable team. And, when project leaders create an inclusive environment (see 4.5.2), there is a stronger sense of community within the team.

These strengths deliver enhanced performance and, thus, better results. A more diverse team can better represent the community of end users or customers, so they will be more able to design and deliver products that meet users' needs.

When team leaders get the best from a diverse team, the variety of skills and perspectives will also deliver:

- enhanced problem-solving capabilities
- greater levels of creative thinking and innovation
- more robust decision-making

And finally, a broader range of personalities, experiences and skills makes the team more adaptable. This will make for a more resilient team. It can manifest as an ability to both:

- tolerate and support one another, and to thrive under stress
- handle shocks with novel thinking, rapid execution and mutual support

These benefits are represented in Figure 4.5.1.

In project-based working, leaders often have an opportunity to bring together diverse teams. They have the level of control to create an inclusive environment. Diversity without inclusion, however, will not allow leaders to get the best from people. Inclusion without diversity creates a comfortable environment with few new ideas, little challenge and a poor understanding of the society that the project serves.

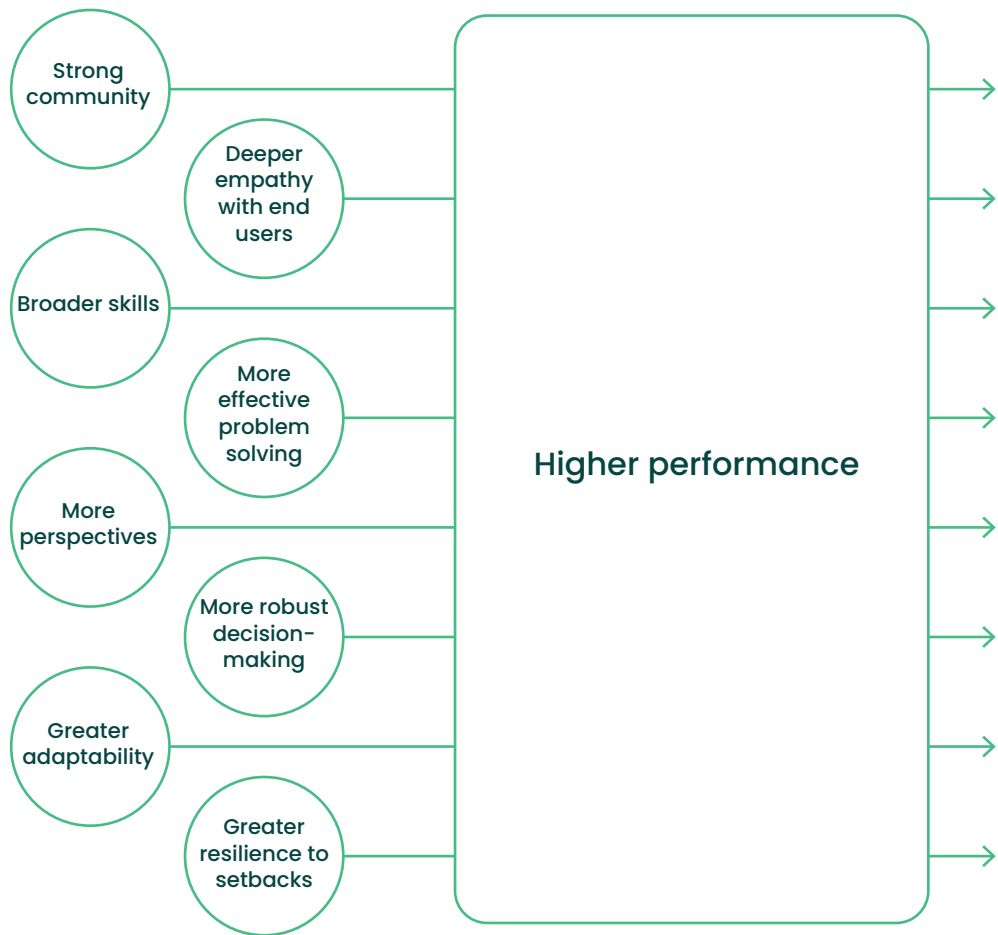


Figure 4.5.1 Diversity leads to higher performance

Recommended reading

- *Developing effective interventions for gender equality in UK construction project organisations* (2024) is APM research that looks at the root causes of gender disparities in the project workplace.
- *Riding the Waves of Culture: Understanding Diversity in Global Business* (2020) is a guide to cross-cultural management that takes readers beyond cross-cultural awareness and on to issues to help professionals take strategic advantage of cultural differences in the business environment.

4.5.2 Creating inclusive working environments

Making people feel welcome in the workplace

A diverse team can deliver a higher performance than a homogeneous one. But this is only true when people feel comfortable, respected and valued. This needs an inclusive workplace. Inclusion is making an environment where everyone feels welcome.

Some people feel there is little project professionals can do to affect organisational culture. However, they have a special opportunity. Project-based working allows the creation of a team with its own culture, so project professionals are responsible for the wellbeing of their teams.

Inclusion starts with ensuring that there are no barriers to entry into the organisation. This will probably need to include measurement of job applications, and the appointment and retention of underrepresented groups. Inclusion should allow everyone to feel they 'fit in' and are welcome in the team. Ideally, this should lead to belonging, making everyone feel this is *their* environment. People don't feel the need to 'fit in' – the team accepts and values their differences.

Project professionals must craft a team culture that welcomes everyone, so they feel at home (Figure 4.5.2). This should start with hiring, with capability-based requirements set for each role and open recruitment run whenever possible. If possible, blinded CVs should be adopted: that is, names, genders, ages and other personal identification is removed from them before review. It is important to consider whether it is better for the team to hire the best individuals or to aim for the best team, balancing different perspectives.

In some jurisdictions, training to better understand diversity, inclusion and cultural differences is mandatory, but it is also good practice to train the project team in this subject. Help them understand that fairness is not about treating everybody in the same way – it is adapting to each person's needs and preferences. Periodic surveys should be considered to get feedback from the team on how they feel.

There are many small things teams can do that add up to a big difference. Project professionals should think about the structure of meetings and how facilitation encourages everyone to contribute, ensuring that people listen. Where a contribution is not acknowledged, echo the point and attribute it to the speaker. Give recognition for collaboration and celebrate wins together.

A valuable initiative is mentoring and sponsorship. This can help people navigate the culture and talk about challenges. Self-promotion is less comfortable for some people, so sponsorship can help them to celebrate their own achievements.

Sometimes people struggle to be heard or feel comfortable. Advocacy is speaking up for a group or individual who is not getting the respect they deserve. Be aware that this can be disempowering, even when well meant.

Allyship is supporting people with differences, but not by speaking for them. Instead, it is helping them to secure a platform to speak for themselves. This can be hard, so ask people whether they want your help and, if so, how.

Finally, if you see prejudice, discrimination or disrespect, speak out. Leading is about setting standards and challenging breaches. We must all find a line to hold and challenge anyone who crosses it.



Figure 4.5.2 Inclusion and belonging in a project team

Recommended reading

- APM has a number of resources available including *Fairness and Unfairness in Projects* (2022) and *Joining the Dance?* (2018), a thought leadership paper on an inclusive profession.
- *BS 76005 Valuing people through diversity and inclusion* (2017) has been developed to support organisations in embedding diversity and inclusion into everyday practices.

4.5.3 Conscious and unconscious bias

Addressing prejudice and bias in project-based working

People come into the workplace with different backgrounds, cultures, abilities and perspectives. It is natural for each of us to feel most comfortable with people who are most like ourselves.

This can lead to discrimination. Discrimination is when one person is treated more or less favourably than another. The sources of this are prejudice and bias. Prejudice is forming an opinion before having all the relevant facts, often based on stereotypes or outdated perceptions. Bias is a preference for or against a person or their ideas without reference to their specific merits. Both can have an adverse impact on fairness, respect and wellbeing.

Bias can be either conscious or unconscious. Understanding biases, and the strategies to address them, is crucial for creating a fair and inclusive project-based environment.

Conscious bias, or explicit bias, involves stereotypes and attitudes that people are aware of and express deliberately. These biases are often rooted in upbringing, social conditioning, experiences and cultural norms.

Conscious bias manifests in discrimination, exclusion or harassment of individuals or groups, based on their identity. In many jurisdictions, some expressions of conscious bias are illegal. It can expose individuals and organisations to the risk of prosecution. If unsure, project professionals should always seek advice. They must:

- enforce strict anti-discrimination policies to combat conscious bias
- set up ways people can report misconduct
- deal rapidly with any causes for concern

Unconscious bias, or implicit bias, is more subtle and harder to identify. It arises from beliefs, stereotypes and assumptions below the level of conscious awareness. These can affect selection, recognition, working relationships and decision-making. And they can do so without the people knowing it is happening.

Common causes of unconscious bias include:

- fear: of difference, competition or failure
- shortcuts: applying hasty judgements without scrutiny
- reluctance: to learn about people, to relearn behaviours or to change
- perceptions: mistaken understanding of what is right or expected

Bias can lead to unfair hiring, opportunities and advancement. This not only impacts individuals: it also misses out on access to talent. And it can create a hostile work environment. Even seemingly minor instances of bias can make people feel unwelcome and undervalued. This can damage morale, productivity and welfare.

Biases can affect choices that can impact delivery, alienate stakeholders or lead to misrepresentation of a situation. They put our projects, programmes and portfolios at risk.

Dealing with bias draws on approaches within three strategies:

- Awareness and education: like discussing diversity and inclusion, and securing expert training
- Inclusive processes: like structured decision-making, hiring practices and communications planning
- Constant vigilance: by maintaining a culture of belonging, with feedback, accountability and personal responsibility

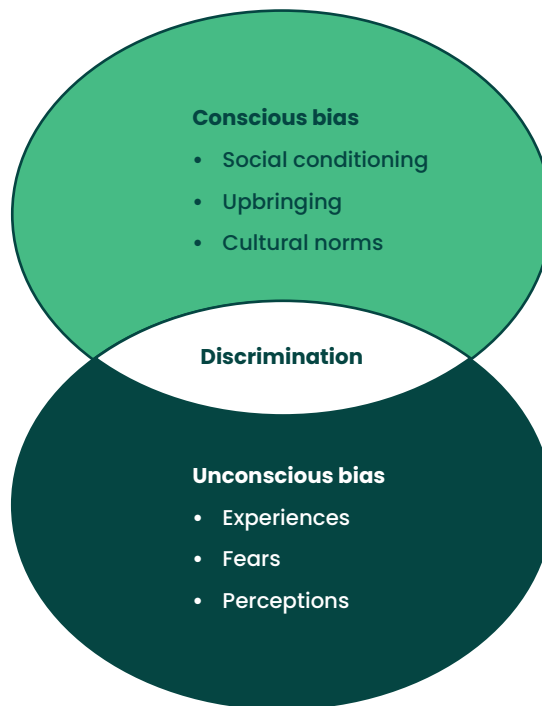


Figure 4.5.3 Impact of conscious and unconscious bias

Recommended reading

- *Riding the Waves of Culture: Understanding Diversity in Global Business* (2020) is a guide to management that takes readers beyond cross-cultural awareness and on to issues to help professionals take strategic advantage of cultural differences in the business environment.
- *Improving the Early-Career Experiences of Racially Diverse Project Professionals* (2024) looks at how project-based organisations can better facilitate diverse project practitioners' transition into the project management profession and meet their early-career needs.

4.6

Ethics, compliance and professionalism

The Association for Project Management (APM) works hard to champion the professionalism of project-based working. It expects a lot of its members. But the reward is an increasing respect for the profession from wider society. Professionalism is a commitment to developing and maintaining the highest standards of competence and integrity.

APM sees professionalism in terms of five things: knowledge, competence, career development, continuing professional development (CPD) and professional conduct.

CPD is developing and improving professional practice. It involves the employee, along with their manager and employer. However, the responsibility lies firmly with each project professional. There are many ways to pursue CPD. They can include informal and formal learning, along with accredited qualifications.

Project-based work sits within a range of legal and regulatory frameworks, so project professionals must be able to understand and comply with them. Many of these regulatory frameworks include codes of practice, which can be mandatory. On top of these, there are also discretionary standards that organisations can elect to follow. All of this is made more complex by the nature of project-based working. It often spans industries and jurisdictions, which can each have their own standards, regulations and laws. Topic 4.6.2 addresses knowledge as well as focusing on professional conduct.

Like all professions, project-based work is embedded in society. Project professionals must act ethically at all times. APM, like other professional bodies, publishes a code of professional conduct. It also advocates for responsible project management (see 2.2). This code sets high standards for its members. It is a responsibility on all professionals to speak out if they become aware of wrongdoing. It is also a responsibility for those leading project-based work to make it easy and safe for others to speak out.

This section is written for all project professionals and includes the following topics:

4.6.1 Continuing professional development: Continual pursuit of professional excellence

4.6.2 Regulatory environment: Navigating the legal and regulatory environment

4.6.3 Ethics and standards: Maintaining trust in the profession

4.6.1 Continuing professional development

Continual pursuit of professional excellence

Continuing professional development (CPD) is a central part of professional life. The Chartered Institute of Personnel and Development defines CPD as 'learning experiences which help you develop and improve your professional practice' (CIPD, n.d.).

Ongoing CPD involves:

- identifying current and future needs
- setting learning objectives and targets
- planning activities to support personal and professional development
- recording activities and achievements

Figure 4.6.1 summarises a typical CPD cycle.

There are three participants in CPD: the learner, their manager and their organisation. Each professional must take responsibility for identifying their own competence gaps. They must pursue learning or opportunities to widen their skills and fill those gaps. Their manager has a responsibility to offer good performance feedback. This helps professionals set the direction for their learning and development. For employees, the employer also has a responsibility to support CPD activities. This may include funded training or paid time off.

Qualifications and accreditations are an important part of a project professional's CPD. APM offers its own qualifications and accreditations, while other awarding bodies offer qualifications in project, programme and portfolio management. These include further education and higher education institutions and professional bodies. Project professionals will often hold accreditations from more than one professional body or academic institution.

Professionals who hold formal certifications or qualifications may need to pursue CPD to maintain their status. Some professional bodies set CPD expectations, so each one publishes its own CPD requirements. Often, these are expressed in CPD units or professional development units (PDUs).

APM offers a structured sequence of professional project management qualifications. These range from the Project Fundamentals Qualification (PFQ) through the Project Management Qualification (PMQ) to the Project Professional Qualification (PPQ). They culminate in the Chartered Project Professional (ChPP). This is the highest attainment of project management knowledge, professional practice and ethical behaviour. APM places members with ChPP status on its Register of Chartered Project Professionals.

Learning is more effective if it ranges beyond formal courses and qualifications. There are many formal and informal information sources. Papers in academic journals like the *International Journal of Project Management* highlight research findings. Articles in practitioner magazines, such as *Project*, explore the profession in practice. Podcasts and YouTube channels offer a rich mix of knowledge and opinion.

Informal resources like blogs and social media posts are a good way to keep up with trends and challenges. Conferences are platforms for sharing and learning. Social media and events are good ways to build contacts with others in the profession.

The term 'reflective practitioner' describes people who actively reflect on their experiences. This is an excellent approach to continuous learning and development. Reflective practitioners do not rely on digested knowledge, such as training. Instead, they find learning opportunities in their daily experiences, successes, setbacks and unresolved issues.

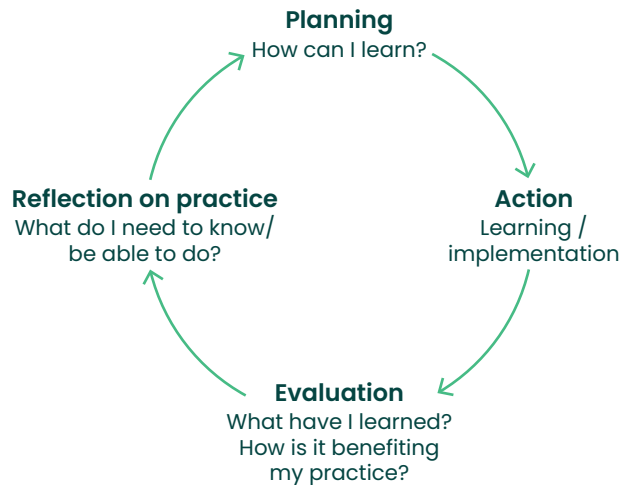


Figure 4.6.1 A typical continuing professional development (CPD) cycle

Source: Adapted from Friedman (2012)

Recommended reading

- *Continuing Professional Development* (2012) analyses the barriers to CPD and how, with greater coordination between professional bodies, this can change.
- *How to Be an Effective Mentor* (2025) is a short guide by APM giving practical tips and advice to mentors and mentees.
- *APM Competence Framework* (2022) lists the core competences for professionals working in project management.

4.6.2 Regulatory environment

Navigating the legal and regulatory environment

All project-based work operates within a legal and regulatory framework. This arises from the industry and countries where the work happens and where the products end up (Figure 4.6.2). Project professionals must understand the regulatory environment. In law, ignorance is not a defence.

Laws and regulations change over time, so project professionals need to establish a mechanism for monitoring these changes. They need qualified and experienced experts, some of whom will come from organisational functions to advise the project. Examples include procurement and project finance law specialists. Others may be part of the core project team, such as health and safety managers in a construction project. In an unfamiliar regulatory environment, projects may bring in legal or consultancy advice.

Regulations generally fall into three categories:

- regulations that set out specific actions the project must take, like those relating to the control of substances hazardous to health
- regulations requiring competence of some key members of the project team to be pre-established
- regulations that require employers to set goals, but let them decide how to meet them, for example, for manual handling operations

Many regulations are specific to the type of work undertaken by the project. Examples include construction design and management regulations. Regulations that are relevant to all work include those relating to data privacy or safeguarding people at work. But some situations require that project team members or specialist advisors for specific pieces of work are appropriately qualified and/or certified too.

Some organisations aim to meet discretionary international standards. Reasons include risk aversion, continuous improvement or as a marketing differentiator. An example is ISO 14001:2015 to improve environmental performance.

The regulatory environment influences many project, programme or portfolio processes, including:

- requirements capture, quality planning and integrated planning
- scheduling
- risk and issue management
- change control
- project audit and assurance

Failure to comply with legally binding regulations can result in corrective actions or fines for organisations. Some even carry custodial sentences for individuals. The severity of a sanction will depend on the extent and impact of the non-compliance. It will also depend on whether there was an individual or systemic failure. Failings may be due to accident, oversight, carelessness or deliberate commission. This, too, will affect the sanction.

Sponsors and the wider governance board (see 2.6.2 and 2.6.4) have important roles. They oversee and challenge project practice, and provide support for project professionals. They may also commission expert guidance or legal support if needed.

Most organisations aim to achieve minimum compliance with relevant legislation and regulations. But some go further by aiming to achieve or exceed the spirit of the law as part of their corporate social responsibility (CSR). Many organisations have specific objectives and a defined appetite for compliance risk. A common example is a desire to 'do no harm'.

Where projects cross jurisdictional borders and geographical locations, regulations and standards may conflict. In these cases, governance structures decide what standards to align to, taking specialist advice if needed.



Figure 4.6.2 A hierarchy of legal and regulatory influences

Recommended reading

- *Understanding Regulation: Theory, Strategy and Practice* (2013) is an in-depth and multidisciplinary discussion of an area of public policy crucial to modern government. It is a clear and concise introduction to key issues in regulation and deals with both the theories and practice of regulation.
- *Essentials of Business Law* (2018) delivers a succinct exposition of the core aspects of business law for those seeking an understanding of the legal principles and regulations that apply to business. Topics covered include contracts, misrepresentation, sales of goods, agency, negligence, nuisance, criminal law, employment, partnerships and company-related matters.
- *The Paradox of Regulation: What Regulation Can Achieve and What it Cannot* (2012) takes a deeper and questioning look at the role of regulation and its capacity to reduce risk and enable new activities. Industrial, financial and social risks are handled through regulations, but their use opens up new questions about the complex interplay between risk and regulation.

4.6.3 Ethics and standards

Maintaining trust in the profession

Any profession should expect that its members act ethically. Trust and respect are vital to the success of anyone who wants to be regarded as a professional. Trust comes from showing competence and reliability, and building good working relationships. But it also needs a consistent moral, legal and socially responsible attitude. A commitment to organisational and professional codes of conduct reinforces trust.

Professionalism is a commitment to developing and maintaining the highest standards of competence and integrity. The APM FIVE Dimensions of Professionalism provide a framework of standards that guide the development of project professionals. The dimensions are:

- **Breadth:** The *APM Body of Knowledge* defines the knowledge needed to manage any kind of project.
- **Depth:** The *APM Competence Framework* provides a guide to competences in project-based working. It maps levels of knowledge and experience to help progress skills and abilities.
- **Achievement:** APM qualifications are recognised across the profession.
- **Accountability:** The APM Code of Professional Conduct outlines the ethical practice expected of a project professional.
- **Commitment:** CPD helps develop project management practice.

APM, like all leading professional bodies, has a code of conduct. This sets standards, guides professionals and raises trust and confidence in the profession.

The APM Code of Professional Conduct sets standards for project professionals in four areas:

- **Professional conduct:** Competence, integrity and regard for legislation, regulation, health, safety and the environment.
- **Personal responsibility:** Honesty and probity, confidentiality, sound judgement and acting in the best interests of employers and clients. Also maintaining CPD, declaring interests, rejecting inducements, acting with due diligence and taking responsibility for actions.
- **Responsibility to the profession and to the Association:** Upholding and building the reputation of APM and the wider profession.
- **Ethical conduct:** Doing things 'right' and reporting ethical concerns.

The last of these underlines a professional duty to report any wrongdoing. Everyone has a wider responsibility to 'speak up' and report any form of wrongdoing. This is sometimes called 'whistleblowing'. The responsibility is not only to a professional body: it may have wider legal implications.

Organisations need to have a policy and process to enable individuals to speak up. Many countries have legislation that protects people who report cases of perceived wrongdoing. Examples include fraudulent or corrupt practices, and failure to safeguard health and safety. Protection for people who speak up can extend to past, present or potential future events.

Policies can also protect employees speaking up about something in their own company or another company. It could be part of the project, programme or supply chain. Project professionals therefore have several responsibilities. They must create an ethical culture where it feels safe for people to speak up. And they must ensure all team members know how to speak up, if they need to. It is good practice for an organisation to create, communicate and live by a clear set of values that guides decision-making at all levels.

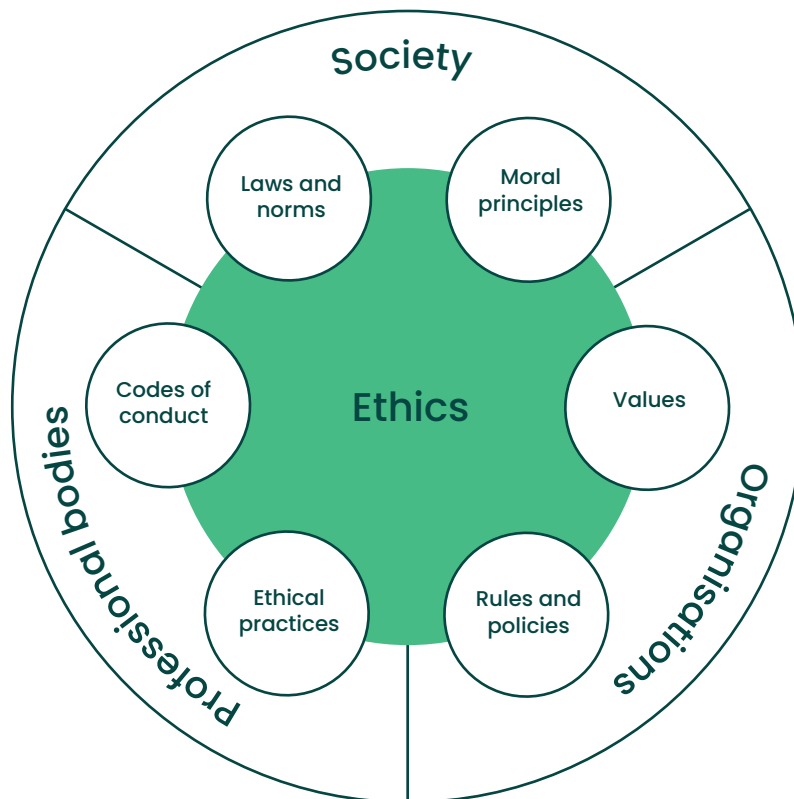


Figure 4.6.3 The scope of ethics

Recommended reading

- *The Blind Spot* (2023) is an APM study on the ethical dilemmas that senior leaders and experts face.
- *Eliminating Modern Slavery from Projects* (2020), published by APM, aims to give individuals working on projects the competence and confidence to spot modern slavery practices.

5 Planning and managing deployment

Projects have to be planned and managed. This starts with creating a clear understanding of their requirements, including what success looks like. Once the requirements have been established, the scope needs to be defined, and options created and then evaluated.

As the project moves forward, quality must be defined and then controlled. The planning process must integrate risks, contingencies and dependencies before scheduling the project and capacity planning. This requires resource management that reflects the work breakdown structures, and an understanding of resource constraints and bottlenecks.

Managing the project requires controls – often referred to as project controls. These controls can be formal or informal; they can be loosely applied or tightly monitored and audited. The use of these controls should be governed by their purpose in managing the project. There are controls within the project, around quality, planning, schedule adherence, resource usage, budgets and spend, contract adherence, risk and issue management, and change. But there also need to be controls outside the delivery of the project, such as controls on the development of the requirement, the solutions and stakeholder interactions. This chapter covers the elements that are at the core of planning and managing deployment, the topics where controls are often applied.

This chapter is composed of 11 sections:

- 5.1 Requirements management**
- 5.2 Solutions development**
- 5.3 Quality management**
- 5.4 Integrated planning**
- 5.5 Schedule management**
- 5.6 Resource capacity planning**
- 5.7 Resource management**
- 5.8 Budget and cost control**
- 5.9 Contract management**
- 5.10 Risk and issue management**
- 5.11 Change control**

5.1

Requirements management

Requirements management is the process of capturing, analysing and prioritising the wants and needs of the project's stakeholders. Comprehensive and attainable requirements are a critical component of all changes, as they help set the benchmark for the project's success, including defining the benefits, objectives and acceptance criteria.

In the early stages of the life cycle, high-level requirements demonstrate how the change aligns with the organisational strategy and they help to define what success would look like at the end of the project. They also help sponsoring stakeholders to draw out the benefits on offer, confirming they're realistic, attainable and measurable. All this information comes together to enable early decision-making and the creation of key artefacts, such as the business case.

As the project progresses, the requirements management process delves into the next level of detail. More detailed requirements are produced through collaboration between project professionals and subject-matter experts. This allows project teams to define their objectives and acceptance criteria while ensuring that everything links to the previously identified benefits and success criteria.

While it is not covered in this chapter, detailed requirements also inform other parts of the project management process, including defining the project's scope and driving the solutions development process.

Given the critical role of requirements in a project's success, it's best practice to follow a consistent requirements management process where project teams continually gather, analyse, prioritise and review the project's requirements, updating them to reflect new information as it emerges. The application of the requirements management process can differ from project to project, with more or less detail captured at different times, depending on the chosen project life cycle.

This section is for project professionals tasked with supporting, leading or overseeing requirements management. It includes:

- 5.1.1 Success and benefits:** Understanding what success means for different stakeholders
- 5.1.2 Objectives and requirements:** Comprehensive and measurable requirements are critical to project success
- 5.1.3 Requirements management process:** The activities and considerations within the requirements management process

5.1.1 Success and benefits

Understanding what success means for different stakeholders

Project success is the satisfaction of stakeholder needs and is measured by the success criteria agreed at the start of a project. Benefits are the positive and measurable improvements delivered by change and are focused on strategic intent. For example, success criteria for a project could be the refurbishment of an office building and the associated benefits could be a reduction in maintenance costs (a tangible benefit) and improved staff morale (an intangible benefit).

Benefits are therefore different to success criteria. History tells us that it is possible to have a project that delivers the required success criteria but fails to deliver the associated benefits. Conversely, a project may deliver benefits but be perceived to be a failure. Stakeholders may have different views of success, and the role of the project professional is to:

- explore perceptions of success and benefits
- facilitate an agreed position, as far as possible

Success criteria are agreed with stakeholders as early as possible but can be changed subject to formal approval (see 5.11.1). In an iterative working environment, the team may find it easier to respond to changing success criteria, as regular engagement with end users allows for gradual learning and adaptation.

Achievement of project management success criteria is known at project handover (see 3.6.1) and accountability lies with the project manager. In a linear life cycle, benefits are often only realised once the product has been adopted and embedded into business as usual (BAU) (see 3.1.3). Accountability rests with the project sponsor.

Benefits must be structured using a clear plan. They must be identified, agreed, actively managed and monitored throughout the project until they are realised. This requires assigning ownership and accountability. If the project excludes benefits realisation, the project professional needs to engage with the entity responsible for delivering the benefits. This may be a programme, a portfolio or BAU.

The investment appraisal and business case for the change depend on the attribution of benefits at the right level. This avoids things being missed or double-counted. To facilitate this, stakeholder consultation and benefits mapping can be an iterative process, requiring good business analysis skills and an ability to engage with stakeholders (Figure 5.1.1).

A consistent approach to benefits identification across an organisation helps to assess the impact on business performance and ensures that investment decisions contribute to the business strategy.

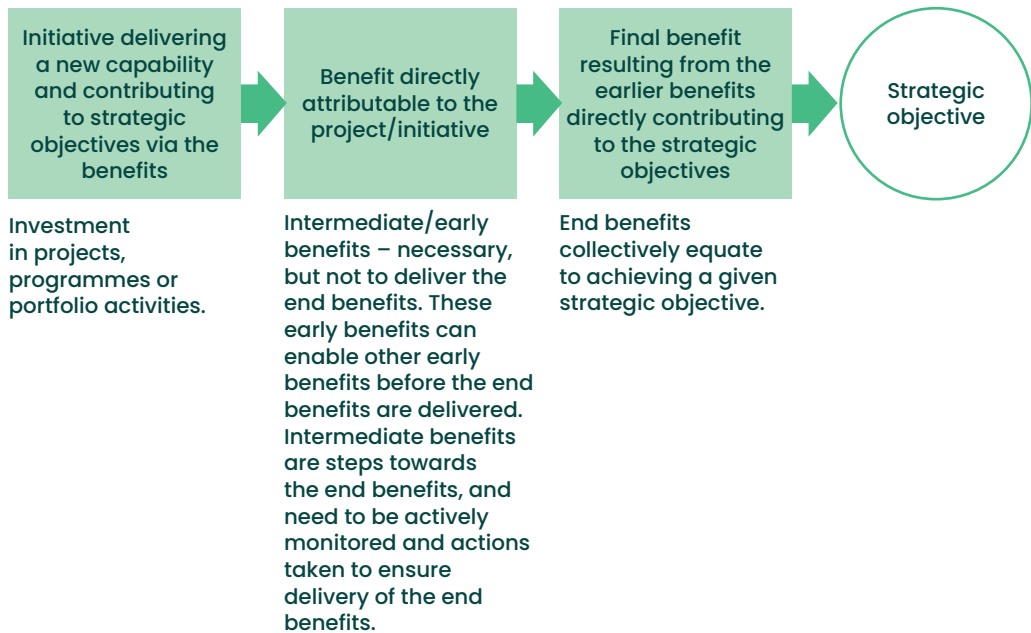


Figure 5.1.1 Benefit mapping process

Recommended reading

- *A Guide to Using a Benefits Management Framework* (2019) explains what is involved in introducing benefits management as a discipline into an organisation or a large change portfolio. The guide can also help those who are looking to improve their overall organisational benefits management capability.
- *Sponsoring Change: A Guide to the Governance Aspects of Project Sponsorship* (2018) guides board members and senior managers to adopt good practices regarding sponsorship of change projects. The guide explains why a sponsor needs to be accountable for project success, and how they can reduce risks to the organisation and maximise the benefits realised from projects and programmes.
- *Benefit Realisation Management: A Practical Guide to Achieving Benefits Through Change* (2010) includes a step-by-step benefits realisation process, explaining along the way how to define projects and programmes by mapping the benefits, produce a convincing business case, communicate and get stakeholder support, agree the measures to monitor progress, and assess the ultimate success of the project or programme.

5.1.2 Objectives and requirements

Comprehensive and measurable requirements are critical to project success

Project objectives can be derived from the strategic aims of a programme or portfolio. For a standalone project, objectives may directly relate to an organisational strategy.

Specific project objectives and requirements are informed by the success criteria and benefits desired by stakeholders (see 4.1.1).

Project professionals ensure that:

- there is a clear link between benefits, project success criteria, project objectives and project requirements
- objectives are articulated and agreed between the stakeholders
- requirements are clear, unambiguous and expressed as simply as possible

A good test is to check that requirements are SMART – Specific, Measurable, Achievable, Relevant and Time-based.

Project professionals facilitate the process to gather, analyse, justify and baseline requirements. The agreed functional and non-functional requirements are the starting point for selecting an optimal solution. In a linear life cycle, the detailed scope of work can then be defined, together with acceptance criteria.

Detailing requirements (e.g. speed of ship), together with measures (e.g. greater than 12 knots), enables acceptance of deliverables during the transition to users. Sometimes, more requirements are requested than it is feasible to deliver.

Some requirements are easier to measure than others. The benefits desired by the sponsor may lead to requirements that cannot easily be measured, for example, work to be delivered in an environmentally sustainable way.

The definition of requirements leads logically to the design of test and evaluation criteria to determine acceptability. This can be shown as a 'V' depiction of a project life cycle (Figure 5.1.2).

The level of rigour in the requirements process mitigates the risk that later in the life cycle there is a dispute between the project, programme, portfolio or organisation about the acceptance and transition of the deliverables into use. For projects delivered as part of a commercial contract, this is particularly important to safeguard both the investing and supplier organisations. This is easier to achieve for projects where there is a high degree of certainty and a stable commercial environment.

In iterative life cycles, requirements can be identified and further refined through analysis to reach agreement and clarify the understanding of what needs to be achieved. Some methods express each requirement as a user story. This is a description of the specific features requested by stakeholders, captured in natural language. They will contain just enough information to enable a simple estimate of the development effort.

Some projects benefit from higher levels of initial rigour, others from evolutionary requirements development.

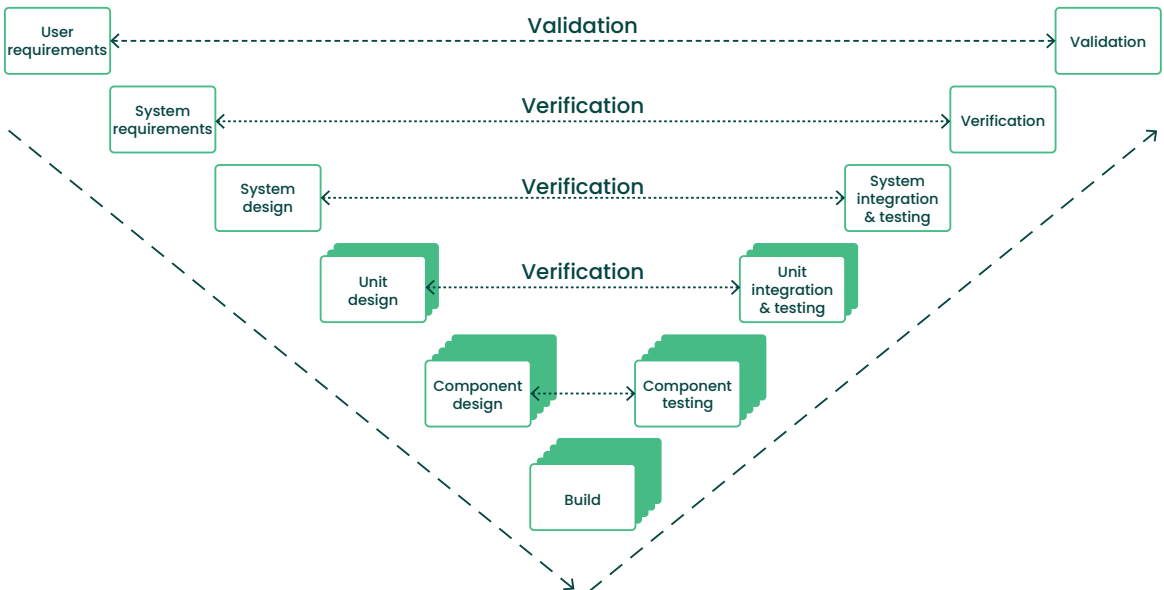


Figure 5.1.2 An example 'V' depiction of a project life cycle

Recommended reading

- *Business Analysis and Leadership: Influencing Change* (2013) provides practical guidance to people involved in engaging stakeholders to agree objectives and requirements. The book includes case studies, practical advice and contributions from leading practitioners and thinkers.
- *Requirements Engineering* (2017) is a commonly used reference and general text for requirements writing. The book gives useful hints to practitioners on how to write and structure effective requirements, and offers a good understanding of the requirements process.
- *Scrum: How to Leverage User Stories for Better Requirements Definition* (2015) has some good hits and tips, together with a selection of samples and examples for applying an iterative life cycle. Scrum is one of the few approaches that make use of the concept of user stories. User stories utilise text narratives to describe how a user interacts with the system, and can potentially be applied to other types of projects.

5.1.3 Requirements management process

The activities and considerations within the requirements management process

As the world around a project or programme changes, so do the wants and needs of stakeholders. To ensure teams can manage those changes, it's best practice to follow a requirements management process. This process helps identify, analyse and validate stakeholder requirements against the aims of the project, programme and broader organisation.

Stakeholder requirements and project scope go hand in hand. After all, to define the work and tasks that will be performed (the scope), you need to know what stakeholders want from the project (the requirements) – you can't have one without the other.

The requirements management process first establishes a requirements baseline and then regularly reviews it as the project or programme progresses:

- **Identify stakeholders and gather their requirements:** To begin, teams identify their stakeholders by analysing the business areas that will benefit from or be affected by the change (3.1.1). Then, once an approach to capturing the requirements is agreed (e.g. via workshops or surveys), teams engage the identified stakeholders to gather their initial view of the requirements.
- **Analyse, validate and align the requirements:** Just because a stakeholder proposes a requirement, it doesn't automatically mean it should be accepted. Teams should analyse the requirements thoroughly, considering any dependencies, risks and constraints. Then they should validate each requirement and check that it aligns with the project or programme objectives, and contributes to the broader organisational aims.
- **Prioritise the requirements:** Often, there are more requirements than it is feasible to deliver. To help with this, teams use techniques such as MoSCoW (Figure 5.1.3) to prioritise each requirement based on criteria such as value, benefit, cost, risk and size. The relevant stakeholders agree upon the prioritisation, and the initial requirements baseline is set.
- **Review and reprioritise the requirements regularly.** As the project or programme progresses, the existing requirements should be regularly reviewed and reprioritised as more information becomes available. Newly identified requirements should follow the same analysis, validation and alignment process before being consolidated with the existing requirement to form an updated baseline.

The frequency of requirements reviews should be project- or programme-specific, but they most commonly occur at critical events such as stage gates or iteration cycles.

The requirements management process continues throughout the project life cycle, supporting the emergence of new information and the evolution of stakeholder needs. When requirements change or new requirements are introduced, they should be aligned with the relevant configuration management and change management processes to ensure the correct governance, control and decision-making are applied.

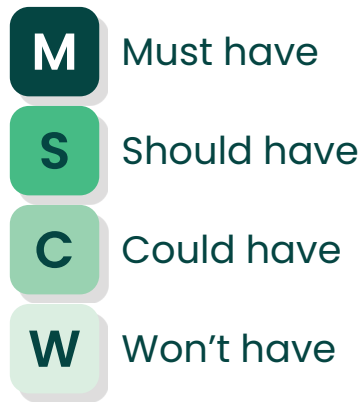


Figure 5.1.3 MoSCoW approach used to highlight the most essential requirements

Recommended reading

- *Mastering the Requirements Process* (2024) is a guide suitable for project leaders, and those operating in a project environment, tasked with coming up with unambiguous and testable requirements to meet customer wants and needs.

5.2

Solutions development

Solutions development is about looking for the optimal solution to meet agreed requirements. The process starts with creating clarity about the problem the project will solve. This implies a need to prioritise the requirements from different stakeholders. Next, the project team generates and explores multiple options to find a preferred solution. That solution will then be subject to refinement and change control.

Finding and selecting a solution mixes creative thinking with rational assessment. Finding options for potential solutions requires good facilitation skills and technical expertise. Project managers need their teams to think creatively and utilise their experience. Once options are available, rigorous analysis finds the option that offers greatest value. This evaluation takes account of multiple criteria.

With a solution in place, the next step is to develop a scope that articulates the parts of the chosen solution. This needs to maximise clarity and minimise ambiguity. The primary tools for articulating this are the:

- product breakdown structure (PBS): this sets out the scope in terms of the products specified
- work breakdown structure (WBS): this sets out the scope in terms of the activities needed to deliver the products

Iterative approaches tackle solutions development and scoping differently. They identify options in much the same way but select solutions and fix scope later, and in stages. Potential scope items form a product backlog of possibilities. This approach is best suited to circumstances where it is not desirable to commit early, for example, factors like:

- changing requirements
- developing technology
- learning from the development process

This section is for people developing a solution and scope definition. It covers linear and iterative projects, which can be standalone or part of a wider programme or portfolio. Indeed, much of this information can apply at programme level. The section includes:

5.2.1 Scope definition: Turning requirements into the parts of the chosen solution

5.2.2 Exploring options: Building in time to look at options before rushing to solutions

5.2.3 Options and solutions: Exploring multiple options to identify a preferred solution

5.2.4 Solutions development process: Determining the optimal solution to satisfy agreed requirements

5.2.1 Scope definition

Turning requirements into the parts of the chosen solution

In project-based working, 'scope' refers to the entirety of the outputs, outcomes and benefits, and the work required to produce them. Scope management is the process for defining and controlling scope.

The business case (see 2.4.1) will contain a high-level scope statement for each option. This forms the basis for costing in the investment appraisal.

In defining scope, it is important to be clear about what is 'in scope' and what is 'out of scope' – or 'exclusions'. This is crucial for mitigating the risk of 'scope creep' later in the project.

An important aspect of scope definition is understanding interfaces with other projects. This is critical to avoid duplication, conflicts or omission of work, either:

- within a programme or portfolio
- with other standalone projects
- with BAU work

Scoping relies on understanding the organisation's requirements. To do this, survey the client or sponsor, the users or customers, and other stakeholders. The challenge is reconciling conflicting priorities. This is harder because of a need to work with limited resources, budget and time.

Scoping can start when the project team has a preferred option to meet requirements. This involves breaking down the main parts of the solution into their components. Project managers often allocate these parts to experts who create detailed scope definitions.

In doing this, the team records assumptions and exclusions from scope. Exclusions might include, for example:

- operational transition resources that may come from BAU, not the project
- customisation of the reporting functionality of a software implementation
- provision of ground-source heating in a building refurbishment

A clear statement of what is in and out of scope reduces the risk of misunderstanding and scope creep. It also creates a baseline for assessing change requests. Poor scope definition is a common cause of delays, cost overruns and lost benefits.

In projects with a linear life cycle, breakdown structures formally define scope. These can define:

- things to create: product breakdown structure (PBS)
- activities to complete: work breakdown structure (WBS)
- allocation of people: organisation breakdown structure (OBS)
- project budget: cost breakdown structure (CBS)

A 'work package' is a part of a WBS that a project or programme manager allocates to a team member (see Figure 5.2.1).

In linear life cycle projects, approval of a scope definition fixes it. Project planning then optimises the time, cost and quality, and the scope will only change through formal change control (see 5.11).

In projects using an iterative life cycle, a product backlog contains potential scope elements, often in the form of user stories. In each iteration, the team draws down the highest-priority scope items for development. Later iterations extend the scope until the marginal benefit no longer exceeds the cost of work.

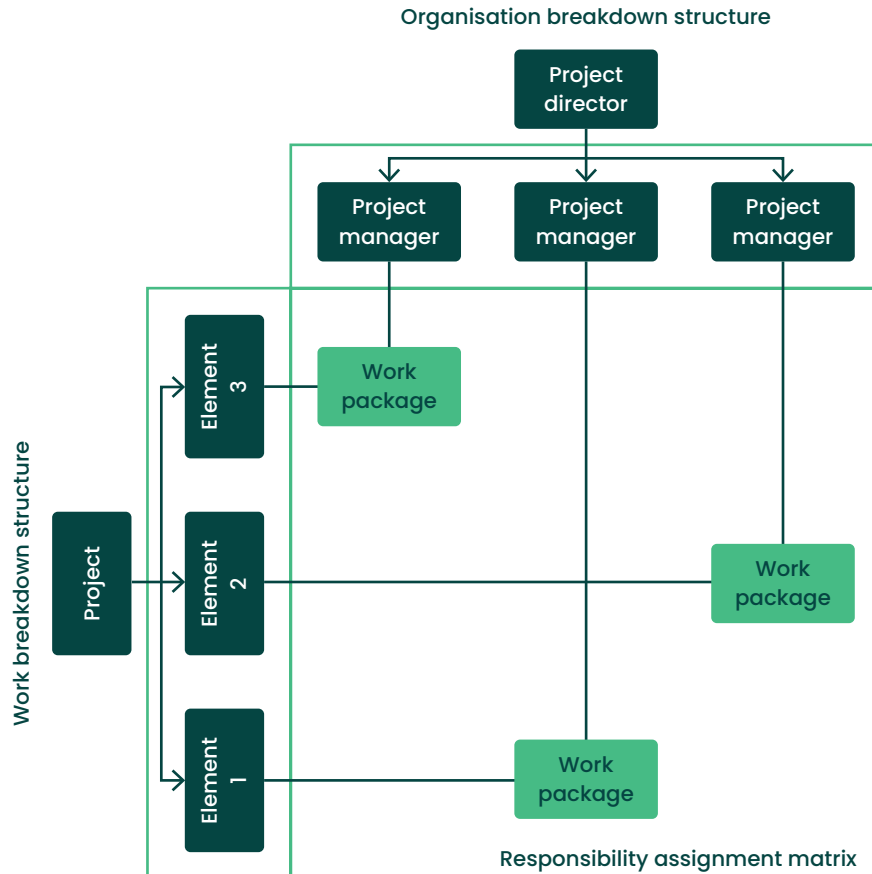


Figure 5.2.1 Relationship between breakdown structures

Recommended reading

- APM's Planning, Monitoring and Control Specific Interest Group guide *Planning, Scheduling, Monitoring and Control* (2015) provides a detailed explanation of scope definition and how various breakdown structures relate to each other. This is important to ensure that the lowest level of scope definition enables effective monitoring of time, cost and quality.
- APM's Earned Value Management Specific Interest Group guide *Earned Value Management Handbook* (2013) provides additional information regarding the breakdown structures and the establishment of the control accounts if earned value approaches are being used.

5.2.2 Exploring options

Building in time to look at options before rushing to solutions

At the beginning of a project, it is a good discipline to set aside time to generate options and alternatives. Too often, there is a rush to solutions, so, in a larger and more complex project, this set-aside time shows good governance. Dedicating a phase to option generation improves the transparency of the governance, clarifying to others why the eventual final approach was taken.

Framing is important. If the objective of a project is described as building a bridge between A and B, then the focus of the option development will be on the location and type of bridge to be built. If the objective is stated as facilitating the movement of people between A and B, then the options to be considered should include ideas beyond simply building a bridge. Ideally, the framing should clearly set out the problem to be solved, which will eventually lead to the outcomes to be delivered and the benefits to be realised.

Options generated should cover multiple dimensions, and these should include:

- doing nothing
- a minimum viable solution to a complete solution
- a single-phased project or a multi-phased project
- the level of technology maturity to be used

A minimum viable solution is always one option that should be considered. It will be the cheapest and quickest to deliver, and will require the project designers to really focus on identifying the absolute minimum requirements for a successfully delivered project. However, it is also possible that stakeholders will want the project to go further than this and add more features to give greater benefits in the longer term.

Options for phasing are critical too. A simple, short project can be designed and delivered in a single phase, but with more complicated projects, phasing is important. Should a pilot to provide proof of concept be undertaken? Should the project concept be tested in one area and then rolled out more widely after initial evaluation and lessons learned? Or do time constraints or demonstrating wider commitment demand a single phase? Further, are all the details of the project known at the outset? If they are not, a discovery phase may be needed to reach a point of full clarity. Then the project can be either terminated or replanned in light of the information acquired.

Different options also take on different risks. One crucial question can be 'What level of technology maturity is required?' Most project stakeholders don't want to rely on old technology, but there should be a debate when the technology being considered has not been fully tested or when the standards for the new technology have not been finalised.

Finally, starting a project requires making a decision. The decision will favour one of the options and, as a result, cut off other options and opportunities for the future. Some options, because of their nature, will allow a wider variety of future opportunities, while others will be very specific and take an organisation down a tightly defined path. Projects therefore need to be considered in line with the broader strategic direction of the organisation and not simply evaluated in isolation.

Not all options can be fully worked up into solutions, but they should be identified and broadly evaluated against the purpose of the project (the problem to be solved), their contribution to the organisation's future, the risks involved and chances of success and failure, the capability to deliver the project, and the usual criteria of cost, time and quality. This should provide a viable shortlist.

The next phase, where options are considered with their solutions, is considered in 5.2.3.

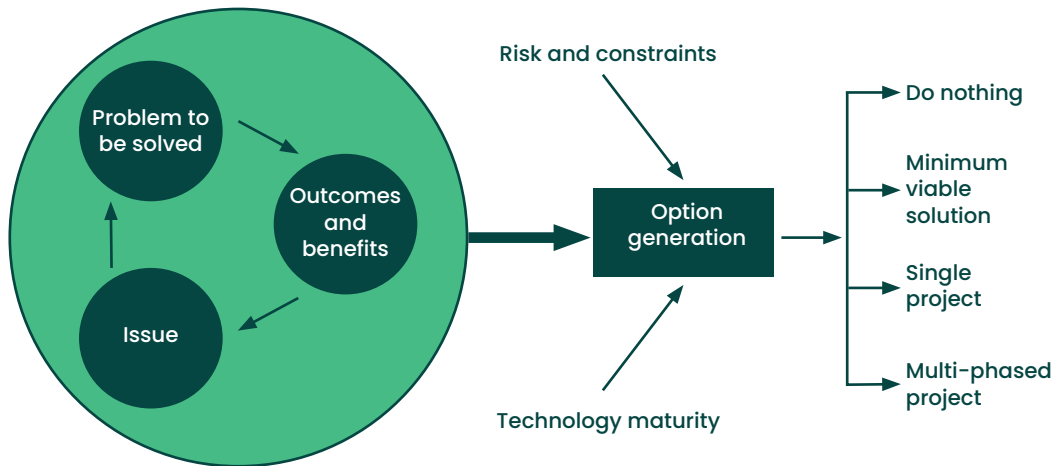


Figure 5.2.2 Option generation

Recommended reading

- The APM Learning module *Developing effective approaches and solutions* (2022) enables users to understand the problem that needs solving, capture requirements and explore multiple options, so you can identify and develop an effective approach and solution.

5.2.3 Options and solutions

Exploring multiple options to identify a preferred solution

At the start of a project, a key objective is finding the optimum solution to satisfy the project requirements. Often, clients will fund a feasibility study to explore multiple options that could satisfy the sponsor's requirements (Figure 5.2.3).

These options feed into a business case. This documents the primary options for consideration. It is good practice to include the 'do nothing' or a 'do minimum' option. The business case is a record of the options that decision makers consider, with reasoning and evidence to support their decision.

A detailed evaluation of alternatives, to find the preferred option, is sometimes called 'optioneering'. This follows a two-step process of:

1. identifying options for evaluation
2. conducting an objective assessment

Identifying options requires creativity. The project manager needs to act as an effective facilitator to secure a broad range of ideas from their team. There can be many elements in defining options, for example:

- technical: process, material, quality, sustainability
- stakeholder: needs, priorities, preferences, desires
- procurement: build, commission, buy off-the-shelf, lease, service
- transition: phased, 'big bang', dual running
- operational: management overhead, complexity, staffing, maintenance

Once the project team has a full set of workable options, the next step is evaluation. Value management principles are critical for option evaluation. They put a focus on maximising the surplus of benefits over costs.

Evaluation needs to follow a rigorous process, to ensure it is free of subjective bias and, worse, undue influence by stakeholders with a strong agenda. This means:

- assessing options against criteria that key stakeholders determine in advance
- using an evaluation that is consistent across all options

A robust approach for comparing options is an investment appraisal. This considers financial costs and benefits. Many organisations have a preferred methodology, which can be as simple as a comparison of total costs and benefits, or it may use a simple return on investment calculation. The most rigorous approach is a discounted cash flow (DCF). This considers development, operational, and decommissioning costs and revenues. It then applies discount factors to account for the change in the value of money over time.

There are other criteria, such as:

- strategic requirements
- non-financial benefits
- implementation risk
- operational considerations

From their evaluation, the project team will make their final recommendations.

For linear projects, analysis and selection of options takes place early in the life cycle. In projects with an iterative life cycle, the project can keep options open for longer to allow the team to assess options later in the process, following completion of early work. Because the project releases products in increments, user feedback and unfolding requirements can support evaluation of options.

The investment decision confirms the solution that will go into development. Any changes after this will need to go through formal change control (see 5.11).

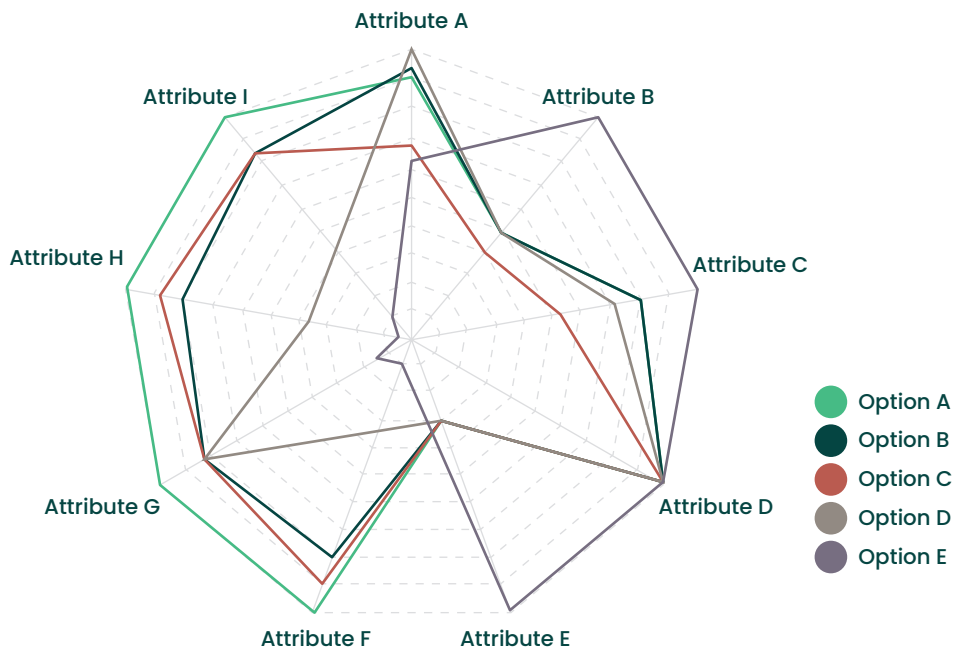


Figure 5.2.3 Comparing multiple options using a spider plot

Recommended reading

- *Project Controls in the 21st Century* (2025) includes an overview of optioneering, plus examples of how an optioneering review, and the option selected, link to decisions about resource utilisation and optimisation.

5.2.4 Solutions development process

Determining the optimal solution to satisfy agreed requirements

Requirements management is about what stakeholders either need or want to be able to do. Once the project team has identified requirements, the solutions development process explores how to meet them. Solutions development is the ability to determine the optimal solution to satisfy agreed requirements.

You need to find solutions that meet both:

- functional requirements: these are what the product or service needs to be able to do
- non-functional requirements: the qualities of the solution – this includes aesthetics, durability, sustainability, reliability, security, safety, performance and maintenance regime

The solutions development process has six principal components, as shown in Figure 5.2.4.

The first step is to identify options that can deliver the requirements. With more options, you increase the likelihood of project success. Engage your team to research the topic and think creatively. However, each option must address your stakeholders' priorities, and must also be capable of delivering value to the sponsoring organisation.

Next, evaluate how each option meets your stakeholders' priorities and delivers value. Evaluation is most robust if you agree evaluation criteria with stakeholders before identifying options. They may include:

- a cost–benefit analysis linked to the business case
- commercial, operational, technical, environmental and strategic impacts

From the evaluation, select the best option to meet stakeholders' requirements and deliver value.

The whole process must be subject to good governance. Document:

- the option you selected
- the decision process you followed
- the specification for the solution

Also be sure to communicate the choice to stakeholders.

During delivery, put in place a monitoring and control cycle. Part of this will monitor solutions development. It needs to include a quality assurance process and quality control of products. Use the Shewhart cycle (plan–do–check–act) to adapt your delivery process and optimise efficiency. Assess what data to collect to help you understand progress and anticipate issues.

The last step in the solutions development process is change control. This ensures all requests for change regarding features or functionality get a robust assessment. Following approval, create a full record of the new solution configuration. Later in this chapter are topics on the change control process (5.11.1) and configuration management (5.11.3). You also need to communicate the changes to stakeholders.

At the end of delivery, you need to secure sign-off before handing over the deliverables. This is the last element of the governance part of the solutions development process.

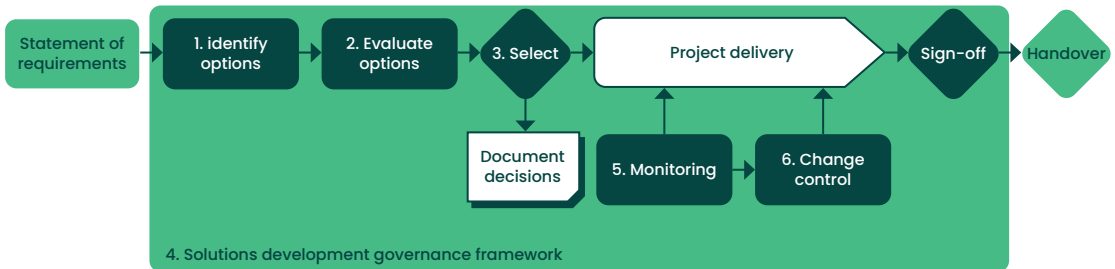


Figure 5.2.4 Solutions development process

Recommended reading

- *Lateral Thinking: A Textbook of Creativity* (2016) encourages readers to look at problems from a variety of angles and offer up ingenious and effective solutions.

5.3

Quality management

Quality is the responsibility of everyone who leads project-based work. In turn, they need to create a culture of quality, where their teams share that sense of responsibility. Quality management is the set of processes by which this is done.

APM defines 'quality management' as the ability to ensure that outputs are delivered in accordance with requirements. This means that project teams need to understand what their customers, users and other stakeholders need. They must represent this understanding in terms of quality standards and acceptance criteria by engaging closely with those stakeholders. They also need to understand any external constraints, such as:

- industry standards
- regulatory or legal requirements
- organisational policies

Next, the project team must determine how it will deliver its products to meet the acceptance criteria. A big part of this is building a culture of quality within the team. The rest is process: how and when oversight will occur, and who will be responsible. Along with the standards, this forms the quality plan.

The two core processes are:

- quality assurance (QA): to ensure delivery follows the quality plan, to reduce the risk of defects, errors or waste
- quality control (QC): to ensure the outputs meet the specifications, by detecting and rectifying any defects that get through QA

Closely allied with quality management is continuous improvement. This is about constantly working to make products, services and processes better. There are many continuous improvement approaches used in industry, commerce and public service, so project professionals need to understand the basics of these and apply them to their project-based working to improve the efficiency of their teams and make them more effective professionals.

The fundamental model that underlies continuous improvement is the Shewhart cycle. This is most often represented as a cycle of plan–do–check–act. It is one of the most important models from outside project-based working for project professionals to know and is a good model for quality management in general.

This section is for project professionals concerned with the quality of the work their teams do. It covers linear and iterative project work at all levels. The section includes:

5.3.1 Quality planning: Ensuring the project delivers outputs to meet requirements

5.3.2 Quality assurance: Ensuring delivery follows the quality plan

5.3.3 Quality control: Ensuring the outputs meet the specifications

5.3.4 Continuous improvement: Making things better, continuously

5.3.1 Quality planning

Ensuring the project delivers outputs to meet requirements

APM defines 'quality' as the fitness for purpose or the degree of conformance of the outputs of a process, or the process itself, to requirements.

Assurance and audit focus on the overall governance and management of a project, programme or portfolio. Quality planning focuses on how the outputs of a project meet the customer's standards (Figure 5.3.1).

Quality planning determines:

- the organisation's attitude to quality
- the acceptance criteria against which the project will assess quality (quality indicators)
- how the project will ensure that the output quality meets the criteria (QA)
- how the project will verify that the output quality meets the criteria (QC)
- how the project will communicate with its stakeholders about quality

At the heart of the quality plan are the acceptance criteria and how the project will meet them. To do this, it references applicable:

- customer requirements
- legislation, regulations and standards applicable to the products
- values and policies of the investing organisation
- quality management standards and methodologies
- testing and assurance methodologies and toolsets

Acceptance criteria are formal statements that set out the essential requirements for each deliverable. Reconcile these to any quality statements that the project's business case contains. They also guide the planning of the measurements and tests that check that outputs meet specification. It is important to do this after scope definition and before further planning. QA and QC activities take time and consume resources that teams must schedule and cost.

The quality plan documents might include:

- acceptance (pass/fail) criteria for each output
- the QA process
- methods to verify that the outputs meet specification
- test scripts
- the frequency of the tests, checks or audits
- sample sizes or sampling percentages
- levels of failure that the project can tolerate
- resource requirements, such as external facilities, test equipment or staff
- governance procedures, such as formal sign-off or stakeholder approvals

Getting stakeholder agreement makes handover of project deliverables far easier. Planning how the project will do this is a key success factor.

In a linear life cycle, scope and acceptance criteria are documented and signed off before project planning. Sign-off is usually a responsibility of the project sponsor or client. In an iterative life cycle, QA happens throughout each iteration and QC happens at the end of each iteration. Sign-off will take place on behalf of the user, who may be a sponsor, client or product owner.

A sponsor or client – along with any wider governance board – will also agree the quality plan. This will then form part of the integrated project management plan. Where a project is part of a programme, some of the overarching quality requirements may be planned and controlled at programme level. Then, projects will follow programme standards.



Figure 5.3.1 Quality planning in the context of wider quality management

Recommended reading

- The APM Learning module *Quality management* (2024) explores what is meant by quality and how it affects the work of a project manager. It includes key definitions, principles and processes.
- APM, *What is quality management and control?* (2020) signposts readers to a range of further reading resources, including blogs and books on all aspects of quality planning and control.

5.3.2 Quality assurance

Ensuring delivery follows the quality plan

Quality assurance (QA) is the process that ensures consistent application of the procedures and standards defined in the quality management plan (see Figure 5.3.2). The purpose of QA is to give confidence that the project or programme will meet relevant quality standards.

QA differs from quality control (QC). QA focuses on preventing defects, errors and waste, and QC focuses on detecting and rectifying defects. Effective QA reduces risk and enhances team morale, project success and stakeholder satisfaction.

The quality plan sets out quality standards that align with customer requirements. It also describes how the team will ensure it meets these objectives. There must be a regular cycle of reviewing project work to assess conformance to:

- product specification and quality standards
- project procedures and processes
- organisational policy, procedures and values
- industry standards and legal or regulatory requirements

QA applies to both the outputs of the work and the processes used to achieve them. Following evaluations, a lessons-learned review will identify the actions that are needed to remedy issues and strengthen project processes.

A good quality assurance team is indispensable for implementing QA effectively. Clear roles and responsibilities are necessary to ensure accountability and foster collaboration. But attitudes are more important. Cultivating a team culture that values quality encourages awareness, problem solving, knowledge sharing and continuous improvement.

Effective project working needs team members with the training and support to fulfil QA roles, but not everyone will do this. The culture needs to prepare all team members to expect their work to be subject to proper oversight.

Another aspect of QA is communicating findings and action plans to stakeholders. QA should give them confidence that the project will meet its quality requirements.

QA is an important part of project governance. It oversees the consistent application of processes, procedures, policy and standards. QA is a project or programme activity, so it differs from project or programme assurance and audit, which the organisation imposes on the project or programme. In this way, it reduces the risk of external scrutiny uncovering issues.

There is an intersection between QA and continuous improvement. What the team learns through QA oversight should feed into continuous improvement of project processes. The Shewhart plan-do-check-act cycle (see 5.3.4) is equally applicable here.

Quality assurance applies to all tiers of project-based working and any life cycle approach. In each of these, it is necessary to tailor the QA process to meet the specific situation.

Embracing a culture of quality empowers project-based teams to adapt to changing circumstances, to innovate and excel. But, at the same time, it ensures that their working practices, and the products they create, conform to essential standards.

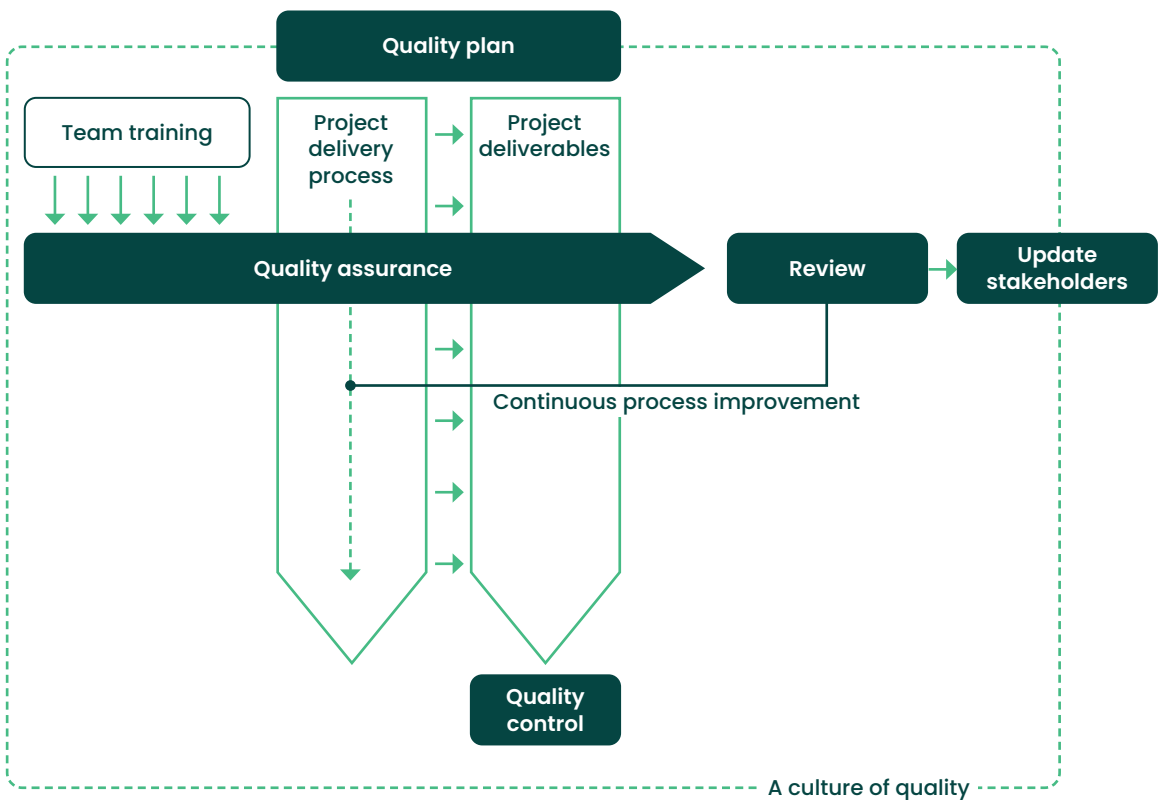


Figure 5.3.2 Quality assurance in context

Recommended reading

- APM's short guide *How to Learn Lessons Effectively* (2024) talks about the principles for sharing knowledge, including a policy for continually reviewing older lessons.
- *A Guide to Integrated Assurance* (2014) is designed to support those who sponsor or manage projects by describing principles and practices for providing efficient and effective assurance of projects and programmes.

5.3.3 Quality control

Ensuring the outputs meet the specifications

The purpose of quality control is to verify that project outputs meet the acceptance criteria defined in quality planning (see 5.3.1). This happens through inspection, measurement and testing.

QA builds quality into delivery using standard processes and procedures, supported by training and feedback. QC aims to find problems that get through production, and stop them from reaching the customer.

Effective QC needs:

- configuration management (see 5.11.3) of specifications
- test plans and an inspection regime
- documentation of defects
- an implementation process to remedy defects

Test plans are critical. They should include:

- the sample size of tests (either the whole product or a random sample)
- test protocols, what to test and how to test it
- resources requirements: facilities, people, expertise or equipment
- independent performance or witnessing of tests by a regulator or operational owner

In many project scenarios, products are complex and technical, and verifying that outputs conform to specifications can be a large task. In these scenarios, testing is well understood. However, all projects need to deliver outputs that are fit for purpose, so QC is always important. It applies equally to final and interim outputs (interim products include reports, communication materials and financial models).

A key concern will always be the degree of conformity that the customer requires. In quality planning, decisions are made about acceptance criteria and what action to take in the event of non-conformance.

In some projects, such as those building safety-critical products, using a non-compliant item is unacceptable. Rework will be necessary, triggering a change to the plan. In other cases, for example, in user acceptance testing of a system, some deviations from the specification may be tolerable at initial use. Decision makers may choose to press ahead to 'go-live' with known issues and decide that remedial work will happen after handover, as part of BAU maintenance. These issues will appear on a snagging list.

Projects vary hugely, so there are many forms of QC, depending on:

- the technical nature of the work
- regulatory provisions
- the particular requirements of each industry

Project professionals should avoid following a generic process. Instead, the QC regime should be set with guidance from technical experts. Ensuring that outputs conform to acceptance criteria should never rely on QC; rather, it is the final verification (Figure 5.3.3). Project managers deliver quality by:

- planning for quality
- investing in stakeholder relationships
- applying strong QA processes
- in the last instance, verifying quality through inspection, measurement and testing

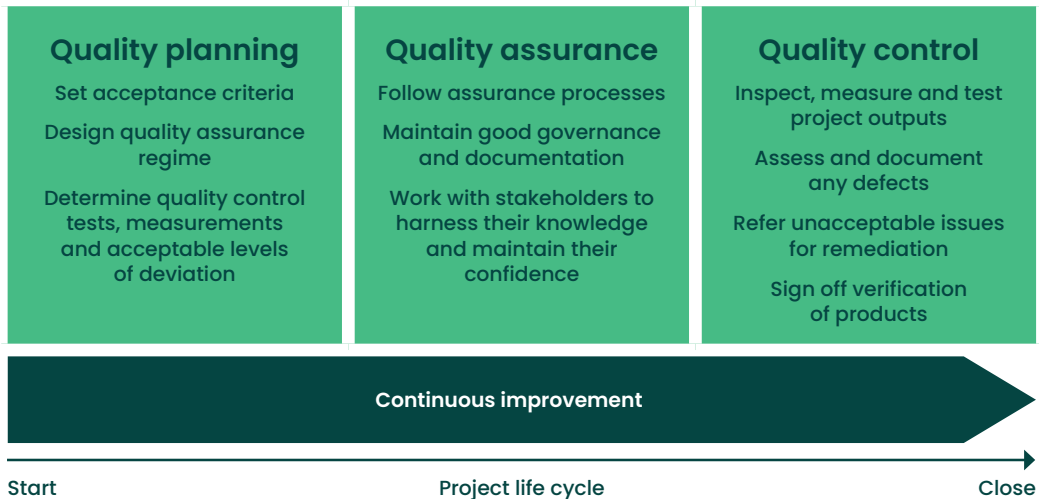


Figure 5.3.3 Quality control is the final verification

Recommended reading

- *The Essentials of Managing Quality for Projects and Programmes* (2017) dedicates Chapter 6 to the topic of controlling quality. It offers advice on using the quality plan to maintain control, measuring project quality and issues of governance.
- *Project Quality Management: Why, What and How* (2014) includes a significant discussion of project quality control and continuous improvement issues in Chapter 6. The discussion includes a detailed case with lessons learned.
- *Agile Testing: A Practical Guide for Testers and Agile Teams* (2008) is an important source which attempts to reposition the role of testing in the context of agile projects. The book defines agile testing and illustrates the tester's role, using practical examples, showing how to complete testing activities within short iterations and how to use tests to guide development.

5.3.4 Continuous improvement

Making things better, continuously

Continuous improvement is an ongoing process to improve products, services or processes. Improvements can be incremental (a bit at a time) or breakthrough (a big change in one go). The immediate goal is to improve efficiency and reliability, and to reduce waste. The ultimate aim is delivering greater value by increasing benefits and reducing cost. It is also known as 'kaizen' in Japan.

The process is a loop. The stages are:

- gathering performance data – quantitative and qualitative
- analysing performance data
- identifying opportunities for improvement
- implementing changes
- evaluating the outcomes

Continuous improvement takes place at two levels, working on:

- organisational project, programme or portfolio management processes
- how project professionals manage, and how their teams collaborate on, the current project or programme

The Shewhart cycle (plan–do–check–act) is the basis for iterative refinement of the delivery process (see Figure 5.3.4). It will help to optimise efficiency and reduce waste.

- **Plan:** defining standards, establishing procedures and allocating resources.
- **Do:** implementing processes and monitoring outcomes.
- **Check:** evaluating project performance against defined standards.
- **Act:** incorporating lessons learned and implementing corrective actions.

The tools for basic continuous improvement include:

- setting performance indicators (quality standards, key performance indicators (KPIs), objectives and key results (OKRs))
- prototypes and piloting innovative ideas
- quantitative evaluation (data gathering and analysis)
- qualitative evaluation (surveys, interviews and observation)
- team reflection (through lessons-learned reviews or retrospectives)

At the core of continuous improvement is gathering and acting upon feedback from our processes. Many organisations employ continuous improvement toolsets to support this, drawn from manufacturing industries. These frameworks include total quality management (TQM), Lean and Six Sigma.

Both iterative and linear project approaches can adopt continuous improvement equally. However, continuous improvement is often part of the iterative life cycle. Some project professionals assert that iterative working is a continuous improvement method. Indeed, agile practitioners have contributed a wide range of innovative retrospective frameworks to the profession.

Effective continuous improvement has systematic processes and is a high priority at leadership levels. It is a strong indicator of an organisation's project-based working having a high level of maturity. Continuous improvement is a cultural choice within the organisation as a whole and within an individual project, programme or portfolio. It works best in cultures that value curiosity, continual learning and developing people. It also thrives in organisations and teams that put blame aside in favour of solving problems and learning from them.

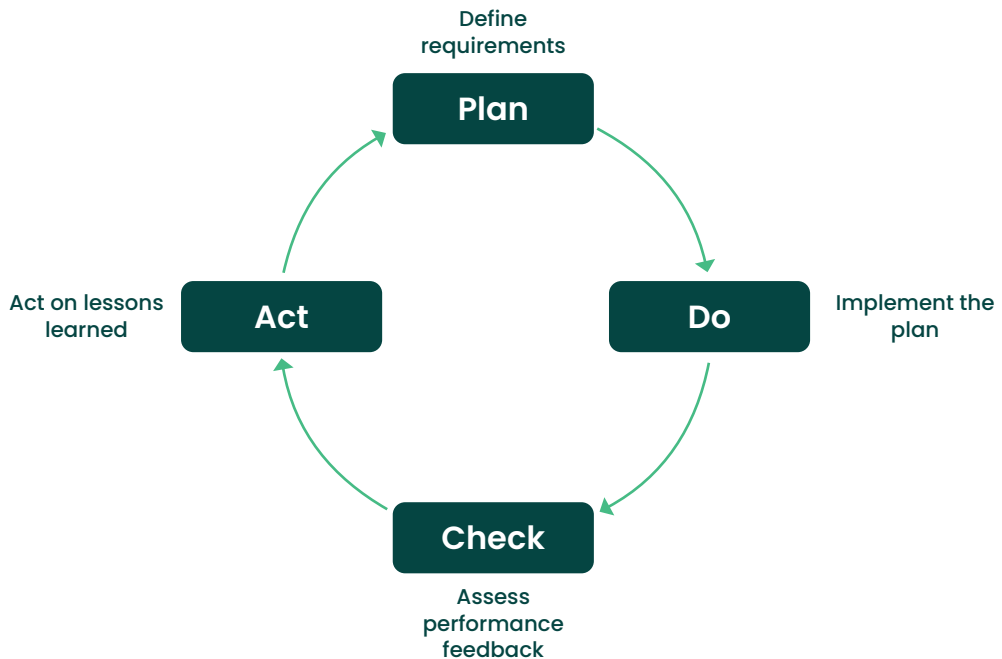


Figure 5.3.4 Shewhart cycle for continuous improvement

Recommended reading

- *The Toyota Way: 14 Management Principles from the World's Greatest Manufacturer* (2014) explores the management principles of building quality and reliability into workplace systems.
- *The Goal: A Process of Ongoing Improvement* (2022) talks through the process of finding constraints (the bottlenecks) and how to overcome them.

5.4

Integrated planning

Integrated planning is the process of collecting, aligning and managing a suite of constituent plans that come together to create a holistic project management plan. The aim of integrated planning is to align all the components of a successful project, to provide stakeholders with confidence that the business case can be achieved.

These component parts include the project's scope, time and schedule, cost, resources, risks and issues, success and completion criteria, and benefits. The format of an integrated plan will vary based on the project's complexity, the wider organisational processes and the preferences of key project stakeholders. But, in most circumstances, an integrated project management plan will succinctly summarise all the project's key information for use in forums such as a steering committee or governance board.

The project's approach to integrated planning often differs depending on the chosen life cycle, with more or less detail available at different times.

- **Linear:** The working assumption is that the work can be defined early, forming a detailed baseline from which the project can progress. While issues will inevitably arise during delivery, this does not negate the need for the best and most accurate plan possible before work starts.
- **Iterative:** While a baseline plan is still required, it's much less defined, as the working assumptions are based on flexibility and agility. Instead, the component parts of an integrated plan, and so the integrated plan itself, will gain further detail as the project progresses.

Creating an initial integrated plan is a complex undertaking, requiring input from many different stakeholders. To help overcome that complexity, project professionals should use robust estimation, dependency management processes and risk identification to provide stakeholders with the initial confidence to sign off and proceed with the project.

As the project progresses and new information emerges, the integrated plan should be monitored closely using data gathered about each component, to demonstrate control. Agreed governance forums should be used to report on progress and release contingencies as required if things don't go to plan.

While the processes for managing the constituent parts of an integrated plan are covered in other sections (for example, schedule management in section 5.5), this section is for project professionals tasked with supporting, leading or overseeing integrated planning. Specifically, it will cover:

5.4.1 Estimation: Forecasting the project's time, resource and costs

5.4.2 Dependency management: Identifying, planning and managing interrelated activity

5.4.3 Risk identification: Identifying and documenting project risks

5.4.4 Contingency planning: Setting aside resource to respond to uncertainty

5.4.1 Estimation

Forecasting the project's time, resource and costs

Estimation is the process of forecasting the time, resource and costs required to deliver the project's scope to the agreed level of quality. All estimates are predictions of the future, but project professionals should use a range of inputs, tools and techniques to make them as accurate and realistic as possible.

In the context of project management, estimates have multiple purposes:

- to inform investment appraisals and option selection as part of a business case (see 2.3.1 and 2.4)
- as an input for creating a resource plan and project schedule (see 5.5.1, 5.5.2 and 5.6.1)
- to enable budget setting and an assessment of affordability (see 5.8.1)
- as the starting point for risk analysis and contingency planning (see 5.10.3 and 5.4.4)

When they are conducted in the early phases of the project life cycle, estimates may be used to forecast the project to the point of its transition into BAU or the entire extended life of the product or service. As time progresses and new information emerges, estimates become increasingly accurate, helping stakeholders make better-informed decisions.

Estimates are typically made using either a top-down or bottom-up approach. In some instances, a combination of techniques may be used to build user confidence (see Figure 5.4.1).

Top-down estimates are useful in the early stages of the project life cycle, as they take a high-level view when detailed information isn't available. Methods include the following:

- **Parametric:** Uses historical data and compares it with known project variables to calculate an estimate. For example, historical construction data shows that 1 litre of paint will cover a 5 square metre wall. If a new project needs to cover a 550 square metre wall, this establishes an estimate of 110 litres of paint.
- **Analogous:** Uses a like-for-like comparison against historical projects. For example, historical construction data shows that 2,000 bricks are needed to build a house. If a new house was built to the same specification, the estimated number of bricks required would still be 2,000. Analogous is also known as comparative estimating.

Bottom-up estimates are often used to validate top-down estimates, using more detailed information as it becomes available. Methods include the following:

- **Analytical:** Breaks the project down into parts and uses internal and external data sources to estimate them in detail. For example, the component parts of a new computer are defined, and the cost of each one is estimated using an external benchmark.
- **Delphi:** Breaks the project down into parts and uses the knowledge of subject-matter experts to estimate them through consensus. For example, the tasks required for the computer build are defined, with a group of engineers agreeing how long each task will take.

To improve accuracy, project professionals often use three-point estimates to determine a range. They can then provide an informed opinion on the most likely estimate within that range, while noting any estimating assumptions that have been made. It's important to highlight these assumptions, as they go on to support risk analysis (see 5.10.3) and contingency planning (see 5.4.4).

Iterative and linear life cycles require the same level of estimation data, documentation and rigour, but may approach the process differently. It's common for linear projects to estimate in greater detail during the earlier stages of the life cycle, whereas iterative projects build detail as they progress.

Estimating is not a one-off activity reserved solely for the project's business case. Instead, it should be seen as a continuous process that evolves as more information comes to light. Updated estimates should be incorporated into the project's change control process, driving adjustments to the integrated project plan as required.

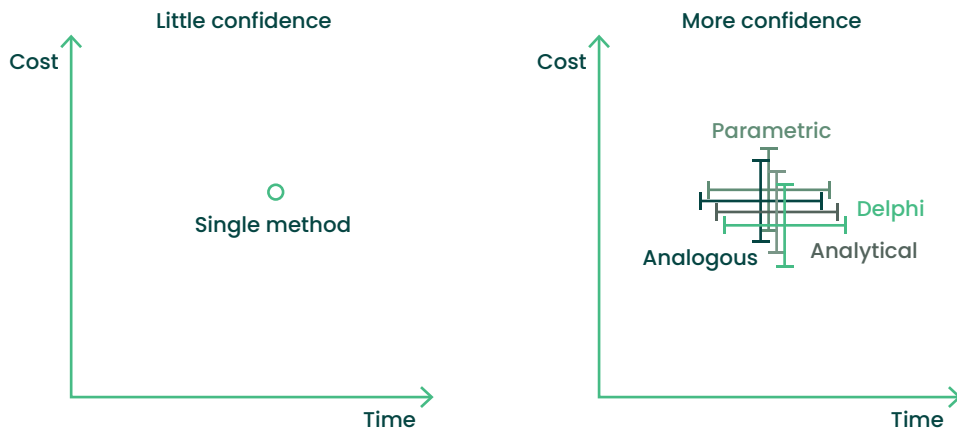


Figure 5.4.1 Combining estimation techniques to build greater confidence

Recommended reading

- The APM/Association of Cost Engineers' (ACostE) *Estimating Guide* (2019) is a practical document for project managers on approaching estimating, types of estimates and the process.
- *Working Guides to Estimating and Forecasting*, Vols 1–5 (2018) is a more detailed academic, yet humorous, guide to estimating, with a large body of work dealing with the theories essential to estimating.
- *Systems Cost Engineering: Program Affordability Management and Cost Control* (2009) provides a number of practical applications for the use of parametric cost models within projects, including supplier assessment, technology insertion and software estimating.

5.4.2 Dependency management

Identifying, planning and managing interrelated activity

An integrated plan must ensure that each part of the project comes together at the right time to maximise the chances of success. This alignment is especially important because those different parts often have a relationship with each other, so that a change to one will inevitably have an impact on another.

A dependency is the relationship between activities and resources within a project. While dependencies are present in many areas of project management, including across different projects, programmes, and portfolios, in this section we'll focus on dependencies associated with a particular project's schedule (see 5.5), where the ability for a task to start or finish is dependent on the completion of another.

Dependency management is the process of identifying, planning and managing a project's interrelated tasks and resources. Failure to manage dependencies often causes projects to pause and replan, leading to an increase in costs and a delay to business benefits.

The process of dependency management begins with identifying the dependencies between activities (Figure 5.4.2). Dependencies typically fall into one of three categories:

- **Logical/hard dependencies:** Activities that cannot start until other activities have started or completed. For example, a wall cannot be painted until it is built.
- **Preferential/soft dependencies:** Activities that should not start without other activities starting or completing. This is often driven by governance, process or quality control. For example, a project communication should not be sent until the peer review is complete.
- **Resource dependencies:** Activities that cannot start until another activity is complete, as they use the same resource. For example, a project has only one developer, so the product page cannot start to be developed until the homepage has been completed.

While the identification of dependencies often happens naturally as part of other scoping activities (e.g. when defining a work breakdown structure (see 5.7.2)), project professionals should utilise the knowledge of subject-matter experts to define and agree how tasks relate to one another.

Once identified, dependencies should be fed into the planning process, to ensure that the sequence of events is agreed and their impacts are planned for. Scheduling techniques such as critical path (see 5.5.1) and critical chain (see 5.5.2) are useful here, as they consider dependencies through the lens of, respectively, the project's activities and resources.

As the project progresses and new information emerges, dependencies should be managed through regular review and replanning. However, a project with many dependencies is inherently risky. To reduce that risk, project professionals should take action to reduce preferential or resource dependencies where possible, considering the impacts on other factors such as cost, benefit and quality in the process.

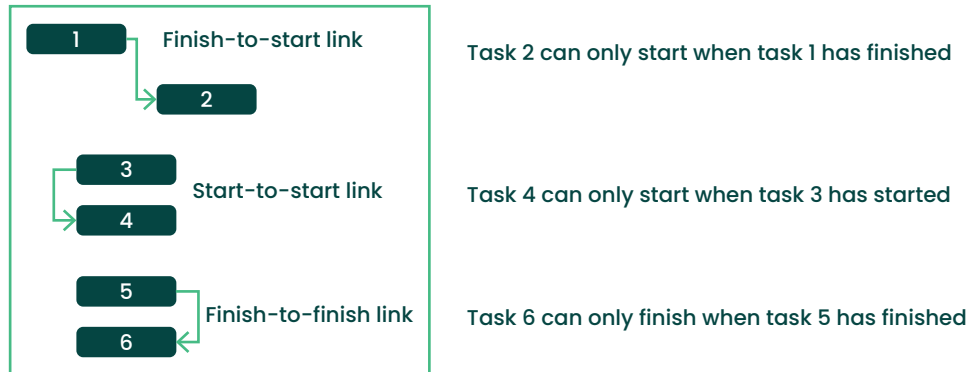


Figure 5.4.2 Types of dependencies

Recommended reading

- APM's *Project Controls in the 21st Century* (2025) considers the impacts and management of dependencies between activities and resources throughout the project.

5.4.3 Risk identification

Identifying and documenting project risks

Uncertainty and risk are ever-present in the project profession. Organisations are most likely to succeed when executives and project teams understand the uncertainties that surround them. This is especially important when these uncertainties have the potential to impact a project's objectives, either positively (as opportunities) or negatively (as threats).

Risk identification is the process of determining the potential risks to a project's objectives. This is a collaborative process that requires the input and expertise of a wide range of stakeholders. It sets the foundation for the project's risk management approach, supporting follow-on processes such as risk management (5.10.2) and risk analysis (5.10.3), and helps develop the risk development plan.

To begin the process, teams must identify the project objectives that could be at risk and collectively agree their appetite for risk moving forwards. Different people have different perceptions of what is and isn't risky, so a set of definitions and scales can help everyone agree on what is acceptable and form the basis of strong governance. For example, would an event with the potential to impact the project's cost by 5% be considered low, medium or high risk?

Once definitions and scales have been agreed, project professionals can begin facilitating the identification of risks. The objective is to leverage stakeholders' expertise and experiences to draw out all knowable risks to the project's objectives.

Group workshops are commonly used for this purpose, often taking place in the early stages of the project life cycle. To align differing stakeholder opinions and address any group bias (see 4.3.3), it is important to describe identified risks clearly, separating the:

- **cause:** environmental factors that may create uncertainty
- **event:** situations that may occur within or around the project
- **impact:** the effect an event could have on one or more project objectives

For example, an upcoming general election (cause) could create a rise in economic inflation (event) that could lead to the cost of project resources increasing (impact).

As each risk is identified, it must also be assigned a risk owner. Risk owners are the individuals or groups who are best placed to assess and manage the risk. Working with the risk owner, project professionals ensure each risk is clearly documented in the risk register. This helps make it ready for future analysis and management as part of the broader risk management plan.

While risk identification is commonly completed in the early stages of the project life cycle, project professionals should adopt a risk-based mindset throughout the project. This facilitates the identification of further risks as the project evolves, additional stakeholders come on board and new information becomes available.

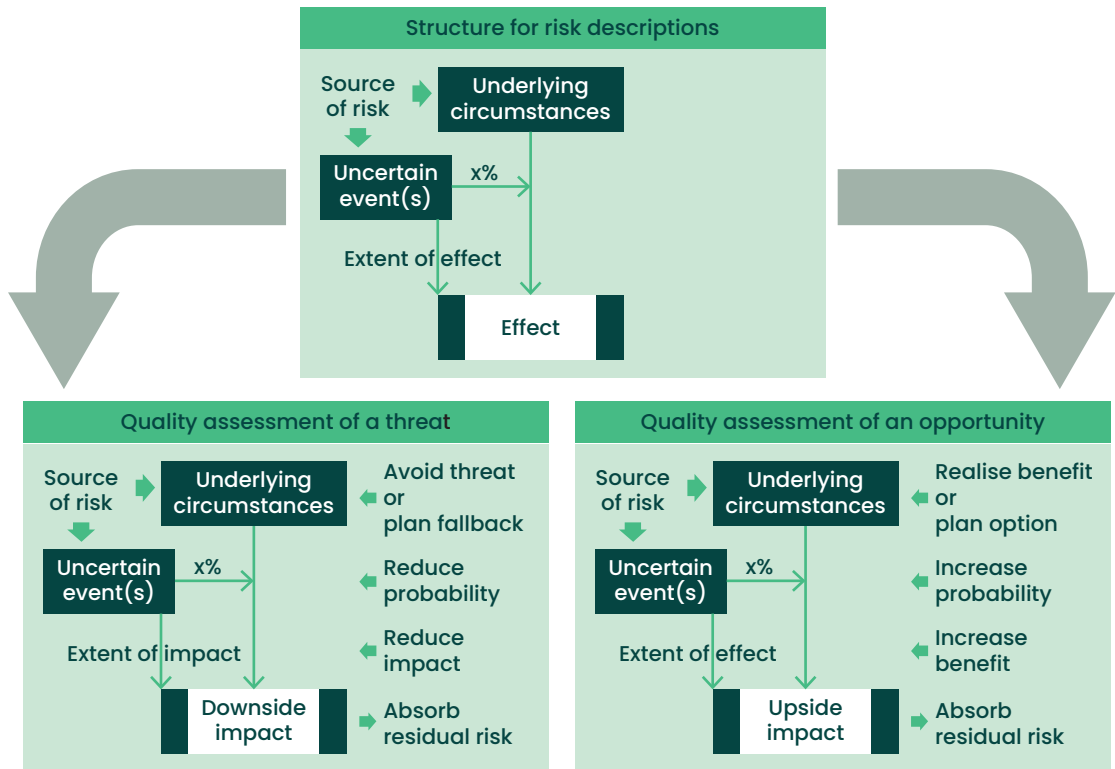


Figure 5.4.3 Describing risks using a defined structure

Recommended reading

- APM's Risk Interest Network's *Project Risk Analysis and Management Guide* (2025) contains a detailed explanation of why it is important to identify risk events and how to do this in a way that is as unbiased as possible.
- *A Short Guide to Facilitating Risk Management: Engaging People to Identify, Own and Manage Risk* (2011) is a practical and easy-to-read book that addresses the many pitfalls associated with making risk management work in practice, including tips regarding the risk identification step in the overall risk management process.
- *Managing Risk in Projects* (2009) places risk management in its proper context in the world of project management, and emphasises the central concepts that are essential in order to understand why and how risk management should be implemented. The book compares different standards and perspectives, showing the role and positioning of risk identification.

5.4.4 Contingency planning

Setting aside resource to respond to uncertainty

While it's fine to have an optimistic project plan, it would be remiss not to consider the impact of risks. Uncertainty is ever-present in the project profession, so contingency is there to provide a safety net, giving stakeholders confidence that project objectives can still be achieved even when things don't go to plan.

Contingency planning is the process of setting aside resources that are solely reserved for responding to risks and uncertainty. Contingency is used to bridge the gap that risk creates, planning for the impacts of known and unknown risks, and the uncertainty of early-project estimates (Figure 5.4.4).

For example, if a construction project's schedule is estimated at 40 weeks, a contingency of two weeks (5%) may be added to the plan to offset the risk of poor weather delaying the build phase.

Contingency is typically expressed in three ways:

- **Monetary value:** an allowance for impacts to the project's cost or benefit
- **Time:** an allowance for impacts to the project's schedule
- **Scope:** an allowance to change or remove non-essential work

Historically, linear projects have held additional budget or factored extra time into schedules to provide the desired contingency. More recently, especially within iterative projects, low-value scope items are often changed or sacrificed to ensure higher-priority deliverables remain on plan. But, regardless of the type, contingency should be clearly defined, as a dedicated budget line, scheduled activity or scope item.

To calculate contingency, organisations typically use either a deterministic or probabilistic approach:

- **Deterministic:** A simple approach that uses subject-matter expertise and industry benchmarks to provide a single contingency percentage for each area of risk (e.g. 10%).
- **Probabilistic:** A more complex, statistical approach, such as Monte Carlo, that provides a range of contingency options. Stakeholders can choose the best option based on their desired confidence level and risk appetite (e.g. multiple options between 5% and 15%).

Contingency may be held at different levels of a project, programme or portfolio, and also across the organisation as a whole, and is sometimes referred to as a management reserve. No matter the level at which contingency is held, a defined process should be in place to control the requesting and allocation of contingency throughout the project life cycle.

The management and release of contingency, including the links to risk management, issue management and change control, are covered in topic 5.8.3.

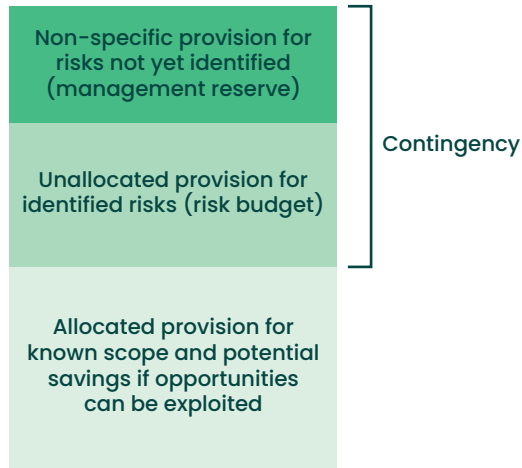


Figure 5.4.4 Provision for known and unknown risk

Recommended reading

- APM's Risk Management Specific Interest Group's *Prioritising Project Risks* (2008) covers a wide range of techniques for analysing overall project risk as an input to contingency determination. It also refers to other specific texts.
- *Practical Schedule Risk Analysis* (2009) provides detailed guidance on how to build a competent risk model and perform a project schedule risk analysis using Monte Carlo simulation so that schedule contingency can be determined.
- *Why Can't You Just Give Me the Number? An Executive's Guide to Using Probabilistic Thinking to Manage Risk and to Make Better Decisions* (2014) covers the science behind quantitative risk analysis in an accessible way, addressing the challenges of decision bias and communicating outputs to stakeholders in ways such that they can be understood sufficiently to make contingency plans.

5.5

Schedule management

Schedule management is the process of developing and maintaining schedules. It identifies when activities are planned to be performed. It considers any dependencies and can be used for internal and/or external resources and activities.

It is a key process to show that the timeline for the change is achievable. It confirms the expectations of project professionals and customers and forms the basis for monitoring and control. There are different levels of schedules, ranging from a plan on a page to a more detailed control schedule.

The project professional will need to determine which scheduling technique is most appropriate.

The critical path approach places an emphasis on activities. It identifies the series of tasks that must be completed promptly to ensure the project's timely and successful completion. In other words, it doesn't consider resource constraints.

The critical chain approach places an emphasis on resources. It considers both task dependencies and external factors such as resource constraints. This approach ensures limited resource is considered in constructing a realistic schedule.

Both approaches will require input from technical and BAU stakeholders. Their perspectives will inform of any potential impact in terms of time and resource on the schedule.

The resulting timeline will form an important part of the deployment baseline, which needs to be documented in sufficient detail to inform the direction of work and monitoring of progress.

The agreement of the deployment baseline is a key milestone for any change initiative. It clarifies and confirms the understanding of what will be achieved, when and at what cost.

Throughout the deployment phase, data on actual performance will be gathered and be compared with the baseline. Any deviations will be reported and appropriate action taken. The critical chain technique doesn't focus on all deviations but specifically reports excessive buffer consumption so that a recovery of the critical path or chain can be prioritised.

The section includes:

5.5.1 Scheduling – critical path: Time-based planning with an emphasis on activities

5.5.2 Scheduling – critical chain: Time-based planning with an emphasis on resources

5.5.3 Deployment baseline: Agreeing the integrated plan to enable managed deployment

5.5.4 Progress monitoring and reporting: Tracking performance against the deployment baseline

5.5.1 Scheduling – critical path

Time-based planning with an emphasis on activities

Time-based planning is used to produce schedules that show when activities within a project are due to take place. The schedule can reflect work at project, programme or portfolio level.

The two main types of scheduling are critical path and critical chain (5.5.2). Critical path places an emphasis on the activities within a project. It helps to highlight key tasks that define how long the project takes. It is not set in stone, and so will be constantly reviewed to ensure that the project is delivered within the agreed time constraints.

To use this technique, interdependencies between activities need to be agreed. Establishing the logic between the activities enables a precedence network, or network diagram, to be created so that estimates of duration (based on effort) can be made (see Figure 5.5.1). Ideally, three-point estimates of duration are produced for each activity: best case, worst case and the most likely point in the range. This information is then used in a critical path analysis (CPA).

Project professionals within the team are key players within this process. Their input enables a more realistic schedule to be produced and agreed upon.

When activities are being planned using CPA, float will be identified. This will indicate whether the activity is time critical or there is some flexibility in the start date.

Although CPA is almost exclusively performed using scheduling software, it remains a key skill for project professionals. It is important to understand the technique so that the output of the software can be verified.

For linear projects, a rolling wave approach to scheduling is often used. Near-term activities are considered in detail, with later stages of the project being considered at a higher level.

For large or complex projects, several schedules can be produced. Each deals with different aspects of the project. A master schedule will combine, coordinate and track the subordinate schedules.

For an iterative life cycle, time and cost are fixed but scope and quality are flexible. Therefore, rather than estimating time and cost, a timebox approach is used. Here, a fixed period with determined resources is agreed. Within the timebox, work to fulfil the scope is completed as efficiently as possible. Several iterations may take place before the output is considered to be fit for purpose.

The result of all these techniques is usually presented as a Gantt chart, which shows activities as bars on a timeline.

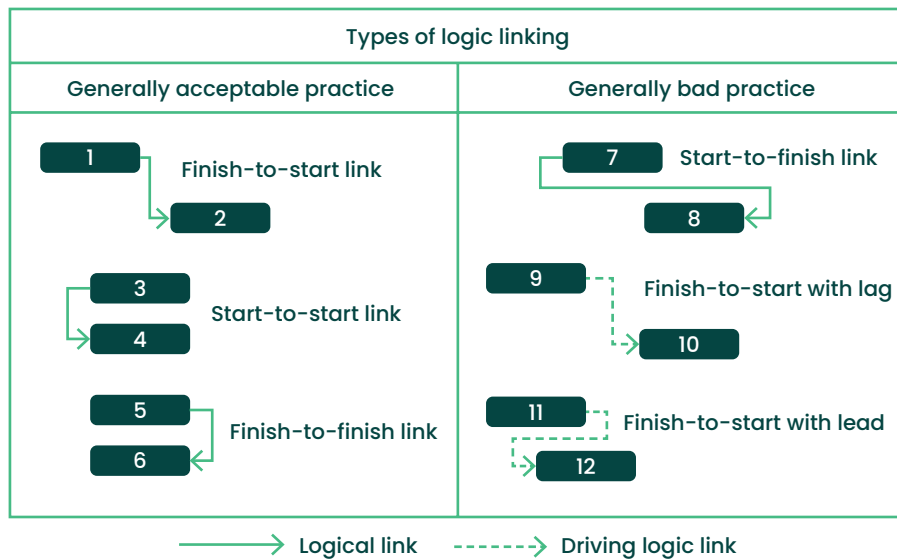


Figure 5.5.1 Precedent relationships in critical path analysis

Recommended reading

- APM's Planning, Monitoring and Control Specific Interest Group guide *Planning, Scheduling, Monitoring and Control* (2015) explains in section 6.1 how to schedule projects in detail, with definitions of key terms related to the critical path method.
- *The Scheduling Maturity Model* (2012) provides insight into the practices needed to build a fit-for-purpose project schedule.
- *APM Introduction to Programme Management* (2016) provides the relationship between the project schedule and the master schedule and programme.

5.5.2 Scheduling – critical chain

Time-based planning with an emphasis on resources

Topic 5.5.1 describes scheduling using the critical path method. This topic focuses on the critical chain method.

The critical chain approach aims to avoid common working practices such as:

- multitasking between activities
- starting planned work at the earliest start date and committing time until it is finished
- starting tasks which aren't fully enabled, to avoid stopping and starting while waiting for missing information, materials or approvals

These practices often result in task durations exceeding their estimates, which didn't account for them. Critical chain drives a set of behaviours to make these poor practices far less common.

The critical chain method uses the most optimistic estimates (the best case in a three-point estimate). Float (also known as safety time) is not held within an estimate for an activity but is stripped out and included as a buffer at the end of a critical chain of activities (see Figure 5.5.2).

For this method to work, resources need to be available when an activity is due to start. To achieve this, the team and other stakeholders who may need the same resource need to be consulted and reach an agreement.

Critical chain depends on a culture being created where it is accepted that best-case estimates may not be achieved. There needs to be an understanding that work should be completed as soon as possible. However, the buffer for the chain is available as a safety net for the whole project.

Once resources are working on a project task, they focus solely on completing the activity to the required standard. The aim is to overcome the temptation to delay activities, so, rather than monitoring start and end dates, the focus is on encouraging resource to act as quickly and efficiently as possible.

To control the project schedule, the project professional needs to monitor the use of the buffer.

In projects where time is critical, there is empirical evidence that out-of-the-ordinary results can be achieved in terms of improved on-time delivery, quality and cost.

The purpose of critical chain is the same as that of critical path analysis, i.e. to:

- enable analysis of the work to be conducted
- determine the start and end dates for the work
- use scarce resource efficiently

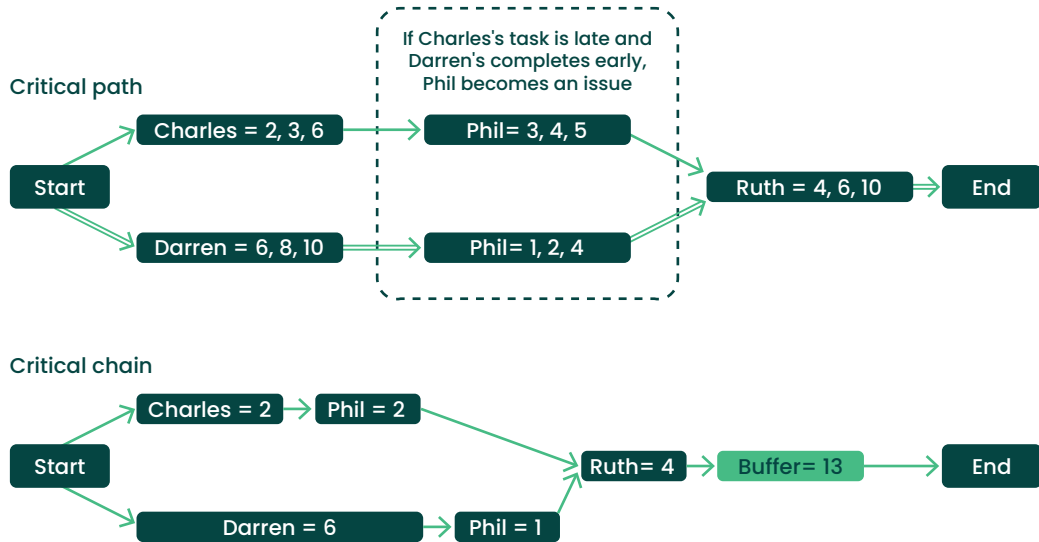


Figure 5.5.2 Comparing scheduling approaches

Recommended reading

- The APM *Senior Managers' and Project Managers' Guide to Critical Chain* (2024) provides an accessible introduction to the technique, the rationale for using it and practical guidance on how to implement it.
- *Theory of Constraints Handbook* (2010) covers critical chain in Section II Chapter 3 ('Critical chain project management primer') and Chapter 4 ('A field report'). It provides practical examples and lessons learned.
- *Advanced Multi-Project Management: Achieving Outstanding Speed and Results with Predictability* (2012) is a well-argued manifesto for critical chain in multi-project environments, providing practical guidance on implementation (including software requirements), case studies and a clear description of agile versus critical chain (in Appendix D).
- *The Critical Chain Implementation Handbook* (2014) is a comprehensive step-by-step 'how to' guide which also clearly explains the 'why'.

5.5.3 Deployment baseline

Agreeing the integrated plan to enable managed deployment

All the planning work for a project results in the production of a deployment baseline (Figure 5.5.3). This is the plan for the execution of the project and supports the integrated project management plan (PMP). It typically sets out the baseline plans for the project, i.e.:

- scope
- timeline
- quality
- budget

The deployment baseline and PMP are approved at the decision gate prior to entering the deployment phase.

For a linear life cycle, the scope is fixed at this point, so any changes identified in deployment must go through the agreed change control process. It is the starting point for progress monitoring. This approach is useful for projects with well-understood and clearly defined outputs. Scope and quality are the drivers while cost and time are calculated on the basis of the scope.

With iterative projects, there is more uncertainty and therefore the approach is more evolutionary. Iterative projects commit resources over limited periods (timeboxes). The deployment phase is split into timeboxes, each with a fixed end date. They deliver products that are developed over successive cycles (iterations).

The deployment baseline may include:

- the number of iterations
- the necessary functionality
- baseline resources and schedule

The achievement of scope and quality may vary from the plan as new knowledge becomes available. The team may have the authority to prioritise and act on new knowledge. Any work not achieved in the timebox is returned to the backlog.

The approval of the deployment baseline is a good time to reconfirm the boundaries of the project. For example:

- what is in and out of scope
- interfaces with other projects in a programme or portfolio
- interfaces with BAU

Any lack of fit would require the PMP to be reworked prior to approval. This could involve adjusting the scope or making contingency available to fund risk responses which weren't included in the original scope.

If a critical chain approach to scheduling has been used, the buffer sizes are approved and frozen in the deployment baseline. This enables performance to be monitored.

It is good practice for an integrated baseline review to be completed by an independent reviewer. This is a risk-based review which will give assurance that the deployment baseline is realistic and can be completed within budget and schedule.

The approved deployment baseline and PMP provide the starting point for successful deployment.

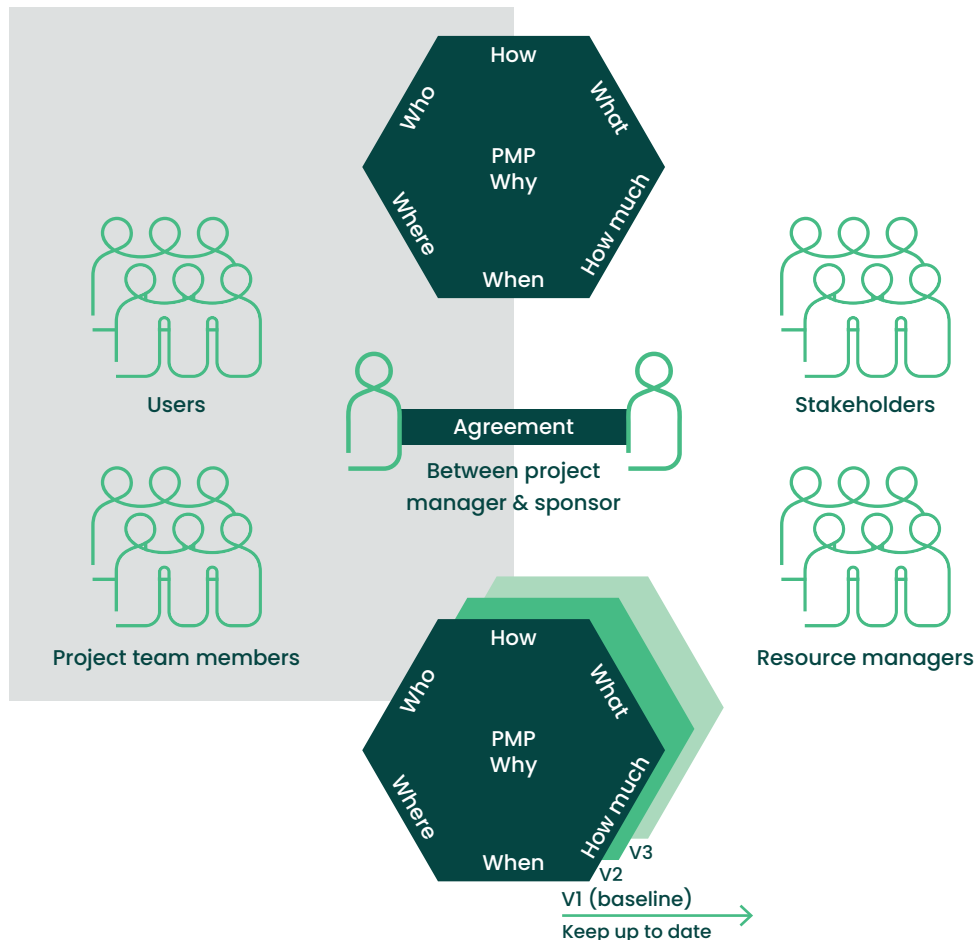


Figure 5.5.3 The project management plan as the baseline for managed deployment

Recommended reading

- APM's Planning, Monitoring and Control Specific Interest Group guide *Introduction to Project Control* (2010) provides the 'what, why, when, who and how' of project controls, covering scope, quality, time, resource and cost planning, and the creation of the project baseline.
- APM's Planning, Monitoring and Control Specific Interest Group guide *A Guide to Conducting Integrated Baseline Reviews* (2016) has a detailed step-by-step approach to integrated baseline reviews and provides insight into the timing, roles and responsibilities.
- APM's Earned Value Management Specific Interest Group guide *Earned Value Management Handbook* (2013) explains the relevance of the deployment baseline to earned value management.

5.5.4 Progress monitoring and reporting

Tracking performance against the deployment baseline

The three elements required for performance measurement are:

- a baseline to measure against (see 5.5.3)
- data on the actual performance
- an assessment of the implications of performance to date

Progress monitoring enables the production of meaningful reports to facilitate decision-making; these may be required for wider reporting to sponsors or boards.

Project professionals agree and establish methods to monitor all aspects of the work, including:

- progress:
 - achievement of planned scope to the required quality
 - committed costs and cash flow
 - performance of contractors
- opportunities and risks:
 - changes in the project environment
 - changes to the risk profile and their impact on time buffers or cost contingency
 - opportunities identified and their impact on time buffers or cost contingency
- health of the project:
 - stakeholder sentiment and effectiveness of stakeholder engagement
 - motivation and satisfaction of team members
 - health of the relationships in the supply chain

The methods used are often determined at programme, portfolio or organisational level. Earned value analysis (EVA) is one way of tracking actual spend and actual work (see Figure 5.5.4). It provides opportunities to look at the efficiency of spend through the cost performance index (CPI) and productivity through the schedule performance index (SPI). Critical chain does not use EVA, CPI and SPI. Instead, the rate of buffer consumption (the percentage of buffer consumed versus the percentage of the critical path/chain completed) is monitored. If required by the customer, EVA, CPI and SPI can be produced using an unbuffered schedule.

Primary progress and performance tracking is best done by those responsible for the work. However, in many organisations it may be a function of the project management office (PMO). They will use established project controls to monitor progress in terms of time, cost and risk.

The project manager and team will be responsible for monitoring progress in other non-quantifiable areas of the project, for example, stakeholder engagement and team morale.

The frequency of monitoring and reporting depends on the circumstances and may change throughout the life cycle. When CPI or EVA is being used, monitoring monthly may be appropriate for many projects but some critical activities may warrant more frequent tracking. When critical chain is being used, progress data is captured and reviewed at least weekly and often daily.

The outputs of progress monitoring are typically presented quantitatively. Traffic-light approaches can be used to flag areas that are in or out of control. Elements of the plan may be reported individually or in a dashboard format that covers multiple areas of performance. However, on projects of any length, it is important to report performance graphically so that trends can be identified and issues detected early. It is important that progress is clearly identified in enough detail to pinpoint any issues and address them.

Iterative approaches use burndown charts to track completion and measure progress. These are used to calculate the efficiency of the team and improve future estimates.

Where progress monitoring and reporting highlights issues that can't be recovered, replanning is required to establish an amended baseline.

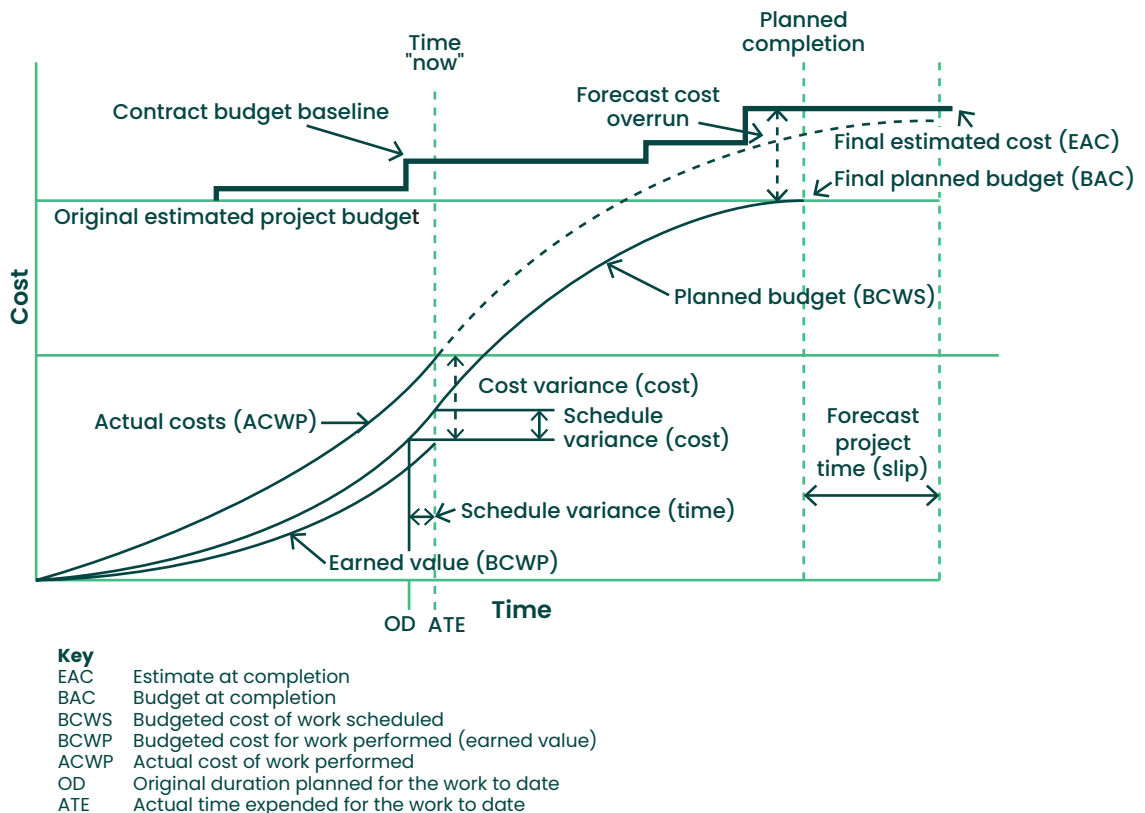


Figure 5.5.4 Insights available through earned value analysis

Recommended reading

- *Project Controls in the 21st Century* (2025), published by APM, explains how a baseline to measure against is critical for any form of project control.
- APM's Earned Value Management Specific Interest Group guide *Earned Value Management Handbook* (2013) provides a detailed account of how to establish a commitment to use earned value management and how to establish data monitoring and reporting.

5.6

Resource capacity planning

Resources are a critical part of project management, requiring careful planning to ensure the right people are in the right place at the right time. As no organisation is blessed with an abundance of resources, project professionals must work to optimise their resources to enable the achievement of organisational objectives.

Resource capacity planning establishes the resource needs in line with the strategic direction of the organisation to ensure that resource utilisation is maintained at an appropriate level for optimal efficiency. In its simplest form, resource capacity planning continually compares the current resource capacity with upcoming demand, with project professionals taking action to match capacity and demand as best they can.

This begins with establishing a clear picture of the future resource requirements, at either programme, portfolio or organisational level. Project professionals should assess each project, working to establish the specific resource requirements, including the:

- skills, competences and experience required
- level of capacity required, e.g. full-time or part-time
- impacts of cost, time and quality constraints
- influences of the chosen project life cycle

Once the requirements for each project are defined, they are combined to create a view of the total resource demand. A similar assessment is also completed for the existing resources, creating a clear picture of the similarities and differences between the current resource capacity and the future demand.

To manage the constant balance between resource capacity and demand, project professionals should look for creative ways of optimising the resources they have in a way that delivers the most value. At a project level, techniques such as resource levelling and smoothing are used to maximise resource efficiency. More strategically, project and programme phasing, and the identification of cross-initiative synergies, help ensure projects are aligned and resource demand is optimised.

However, despite everyone's best efforts, sometimes capacity cannot meet demand and trade-offs must be made. Project professionals should establish a clear view of each initiative's value, priority and dependencies to enable stakeholders to make informed priority calls when required.

This chapter is for project professionals tasked with supporting, leading or overseeing resource capacity planning. Specifically, it will cover:

5.6.1 Identifying resource requirements: Establishing each project's resource requirements

5.6.2 Assessing total resource capacity: Comparing demand and capacity across programmes and portfolios

5.6.3 Resource capacity optimisation: Optimising resource capacity to create maximum benefit

5.6.1 Identifying resource requirements

Establishing each project's resource requirements

While typically managed at programme, portfolio or organisational level, resource capacity planning starts with establishing the resource requirements of each individual project (Figure 5.6.1). After all, differences in scope, requirements and stakeholder needs mean that no two projects are truly the same, so each one often needs specific skills, experiences and competences to ensure it succeeds.

The identification of resource requirements usually begins after the high-level project scope has been established, as this provides clarity on the work to be completed. From there, project professionals can begin identifying the resource requirements, specifically focusing on:

- the skills required to complete the work, including particular hard skills (e.g. coding languages) and soft skills (e.g. communication)
- the seniority level required to manage the project's demands and complexity, e.g. junior, mid-level, senior
- whether any previous domain expertise is required, e.g. financial services, construction
- whether any industry-specific qualifications are required, e.g. CEng, ChPP

To determine these requirements, project professionals should consult subject-matter experts, review the resource profiles and lessons learned of similar projects, or complete a skills analysis exercise, such as a job role mapping.

Once the skills and competences are established, project professionals should estimate the capacity required from each resource by determining whether there is a full-time or part-time requirement, estimating the effort in hours or days per week. Historical data and expert insights can help with these estimations, alongside using a mix of top-down and bottom-up estimating techniques.

As more clarity about the resource requirements emerges, it's best practice to consider the impacts and constraints of cost, time and quality. Factors such as resource availability, budget, seniority and hiring lead times may go on to influence the decision on whether to use internal or external resources, with location requirements also a key consideration in a hybrid working world.

To finish the resource identification process, project professionals should also consider how the chosen project life cycle influences the resource requirement over time. For example, linear projects may require a front-loading of design and planning skills, whereas a similar iterative project may require those skills throughout the entire project life cycle.

Project professionals should work with human resources experts to understand how their resource requirements align with established job roles within their organisation. This helps to standardise resource profiles across the board and provides a strong foundation for the next stages of the resource capacity planning process.

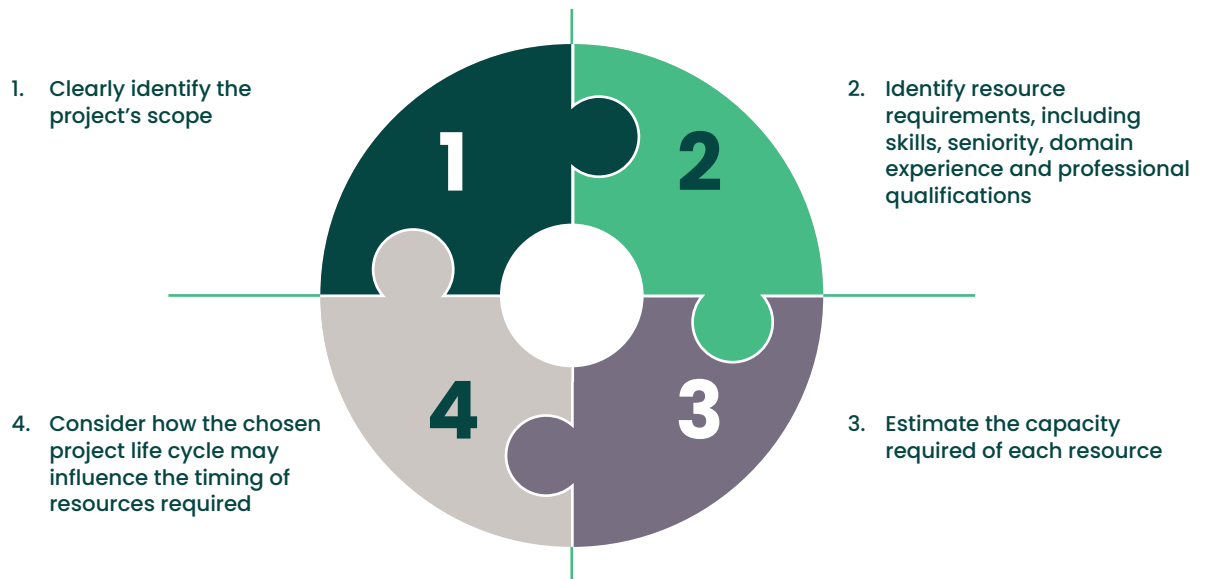


Figure 5.6.1 Assessing resource requirements

Recommended reading

- *Project Controls in the 21st Century* (2025) includes a section on aligning competencies to a resource plan. It describes an initial plan of resources required for the project, which will be further defined via the scheduling and scope of the project.

5.6.2 Assessing total resource capacity

Comparing demand and capacity across programmes and portfolios

Once the resource requirements of individual projects have been identified (5.6.1), project professionals must work to establish their current resource capacity and assess how it meets resource demand across the programme, portfolio or wider organisation.

This may seem like a simple exercise, where the supply and demand are matched to meet the strategic priorities, but in reality this is a complex, ongoing process, as project professionals navigate organisational changes and competing priorities to ensure resources are best assigned to generate maximum value.

To achieve this, project professionals must start by defining their current resource capacity. This means establishing a clear view of each resource's skills, experience, competences and capacity, often creating a high-level profile for each team member. As with resource demand (see 5.6.1), this information should be aggregated to produce a collective view of the total resource capacity across a particular programme or portfolio, or the wider organisation (see Figure 5.6.2).

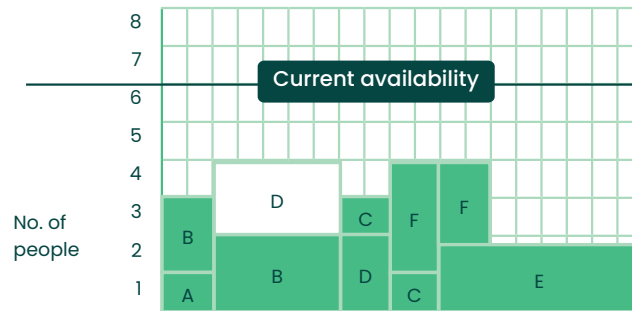
Once the resource demand and capacity are both defined, project professionals must consider several factors that will influence how resources could be assigned to meet the demand. Organisations have finite resource capacity, so trade-offs will often need to be made to maximise value and best enable the organisation's strategic objectives to be met.

To enable these priority calls to be made, project professionals must ensure they know:

- **The value of each change initiative.** This should consider short-term, long-term, financial and non-financial benefits, which are typically derived from each project or programme's business case.
- **The priority of each change initiative.** This is especially important for those projects that sit within the same programme or portfolio, where a common project sponsor can take responsibility for prioritisation decisions.
- **The dependencies between change initiatives.** Dependency mapping (see 5.4.2) helps project professionals to clearly understand the impact that a resource capacity decision may have on the time, cost, quality and benefits delivery of other change initiatives.

These three factors will constantly change as each project progresses and more information becomes available. To manage this moving picture, it is best practice for project professionals to use a resource management tool to assist with the collection and aggregation of resource capacity planning data. This creates a single source of truth that can be used to increase transparency and enable complex capacity planning decisions to be made.

The resource requirements can be graphed in a resource histogram to show the demand profile



The resource histogram demand profile will be reviewed against current resource availability (6 in this example)

Figure 5.6.2 Resource histogram showing total resource capacity

Recommended reading

- APM's *What is resource optimisation?* (2020) is a web resource that lists a range of relevant materials, including case studies and blogs, that explore resource planning and assessment in more detail.

5.6.3 Resource capacity optimisation

Optimising resource capacity to create maximum benefit

Having established the resource demand and programme or portfolio constraints (5.6.1), and resource capacity (5.6.2), project professionals must work to optimise their resource capacity to generate maximum value.

In most situations, resources are scarce in organisations and so they must find creative ways to make their capacity go further. Optimising resource capacity seeks to avoid individual projects needing to:

- change their scope
- change their acceptance criteria, often leading to reduced quality
- delay timelines, so that outputs, outcomes and benefits are delivered later than promised

To help with this, several techniques can be used to optimise project-level resource capacity, with the two most common examples being resource levelling and resource smoothing (Figure 5.6.3).

Resource levelling works to optimise the timeline when resource constraints are fixed, answering the question: 'With the resources available, when can the work be finished?' Project professionals use other techniques such as critical path (5.5.1) or critical chain (5.5.2) to optimise project tasks and schedules and to get the most from the resources that are currently available.

Resource smoothing works to ensure scope, quality and time are achieved while accepting an impact on cost, answering the question: 'What resources are required to deliver the expectations?' This doesn't necessarily mean just adding more resources, but it also identifies ways to increase current resource capacity or better manage work to avoid peaks and troughs.

The result of project-level resource optimisation is a curve that shows the planned deployment of resources (and therefore cost) to complete the scope and achieve the required quality over time.

At a broader programme, portfolio or organisational level, optimisation can be achieved by reordering or phasing change initiatives to avoid clashes of resource requirements. In addition, project professionals should look to identify synergies between initiatives, such as using a common infrastructure, leveraging joint supplier contracts or standardising processes, in an attempt to reduce the collective resource demand.

Despite everyone's best efforts, sometimes capacity cannot meet demand and trade-offs must be made. Resource capacity decisions should be based on the impact on business benefit, with project professionals needing to leverage conflict management, relationship management and negotiation skills to reach an agreement with stakeholders.

Resource capacity is one of the more complex disciplines within the project profession, with industry surveys often showing it to be the biggest challenge organisations face. For this reason, project professionals should apply strong governance and control around resource capacity planning, working collaboratively with stakeholders to support the organisation's current and future needs.

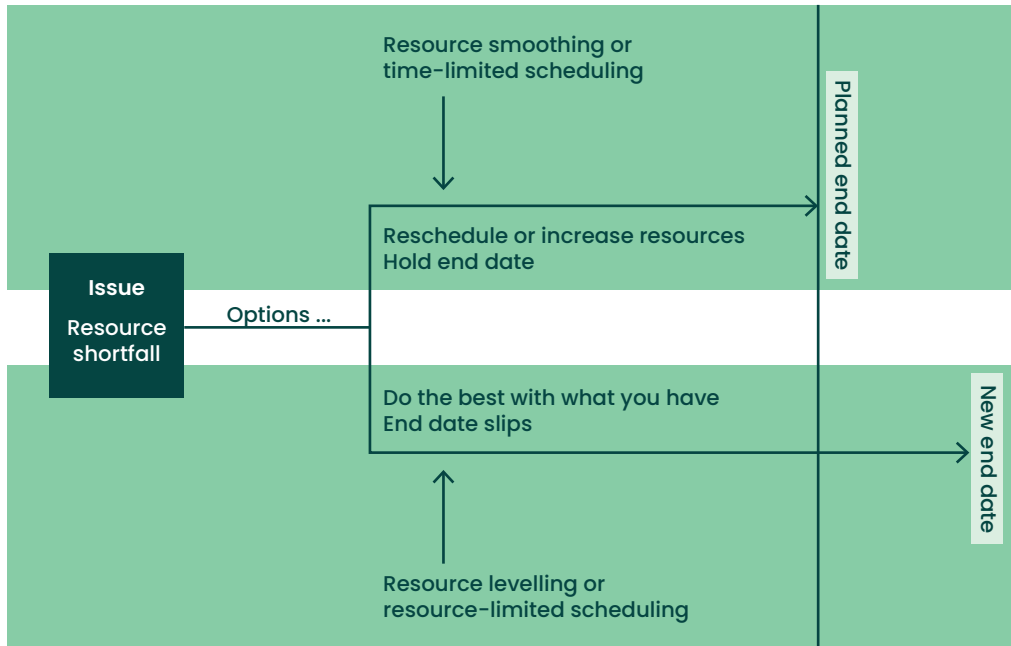


Figure 5.6.3 Resource levelling and smoothing options

Recommended reading

- APM's *Project Controls in the 21st Century* (2025) includes specific sections on resource levelling and resource smoothing.
- *Project Management* (2013) covers most aspects of project management, with two chapters dedicated to scheduling resources. Chapter 15 covers the basic principles, explaining resource-limited and time-limited scheduling, while Chapter 16 offers practical advice on which resources to optimise.
- *The Handbook of Project-Based Management: Leading Strategic Change in Organizations* (2008) looks at delivering beneficial projects. The chapter on performance also looks at resources, offering alternative views of resource smoothing for a project scheduled by 'early start' and 'late start', and for smoothing that focuses on prioritising different types of resources.
- *The Resource Management and Capacity Planning Handbook: A Guide to Maximizing the Value of Your Limited People Resources* (2014) is a dedicated guide for practitioners. The book begins by exploring the current state of resource planning, while Chapter 3 addresses things that cause havoc with resource efficiency and suggests approaches for dealing with the issues.

5.7

Resource management

Managing resources to achieve the project's objectives

The outcome of many projects depends on the project professional's ability to effectively manage resources. This is especially true for 'people resources', where leveraging the skills and experiences of subject-matter experts is essential for project success.

Resource management is the acquisition and deployment of the internal and external resources required to deliver the project, programme or portfolio. Failing to establish sound resource management processes and controls can lead to schedule delays, poor quality, increased costs or a reduction in business benefits.

Effective resource management starts with ensuring the project's structural foundation is in place. Project professionals must decide on the project's organisational structure, defining how roles, responsibilities and authority will be assigned, and how resources will work together to achieve the project's objective.

In most instances, project professionals choose one of three structures: functional, matrix or project (see 5.7.1).

This decision creates the project's organisational breakdown structure (OBS), enabling the identification of resource requirements in line with the resource capacity planning process.

To allocate the right resources to the right tasks, project professionals often create a work breakdown structure (WBS), detailing the work of each discrete element of the project's scope. Then, a responsibility assignment matrix (RAM) is formed by combining the WBS and the OBS. This clearly shows each individual work item, the resources assigned and the role each resource holds.

Project professionals must ensure allocated resources generate the expected value, so they should put controls and governance in place to monitor resource performance, availability and utilisation. This is done by collecting resource management data, using the results to report progress, escalate issues and align expectations with those of resource managers.

This chapter is for project professionals tasked with supporting, leading or overseeing resource management. Specifically, it will cover:

5.7.1 Defining an organisational structure: Choosing a structure that enables resource success

5.7.2 Planning and allocating resources: Planning the work and allocating the right resources

5.7.3 Managing resources effectively: Managing resources for optimal utilisation

5.7.1 Defining an organisational structure

Choosing a structure that enables resource success

While ‘resources’ can be defined as ‘anything that’s needed to complete a project’, it’s important to focus on ‘people resources’ and how to best leverage their skills and experience to drive project success. To achieve this, projects must first be structured in a way that enables those people resources to operate most effectively.

Organisational structures define how roles, responsibilities and authority are assigned within a team and how resources work together to achieve a particular objective (Figure 5.7.1). While all organisations have their own permanent structures that are designed to manage BAU activity, projects are temporary endeavours requiring unique temporary structures.

Functional, matrix and project are the most popular types of temporary structures for projects, each taking a different approach to resource distribution:

- **Functional:** Resources are embedded in their functional area (e.g. finance), reporting to the functional lead (e.g. the finance director), but are focused solely on completing project-based work. This approach fosters the development of expert domain knowledge but may cause people to feel isolated as lone project workers in an otherwise BAU team.
- **Matrix:** Resources are embedded in their functional area (e.g. finance) but split their time reporting to a functional lead (e.g. the finance director) and a dedicated project lead (e.g. the project manager). This approach helps the project remain visible within the BAU operation, but often causes people to struggle with conflicting priorities.
- **Project:** Resources are dedicated to the project, reporting exclusively to a project lead (e.g. the project manager). This approach ensures a complete focus on the project objectives, but can cause the project to become disconnected from the BAU operation.

Project professionals should choose the structure that they believe best enables them to deliver the project, but they will inevitably have to operate within the project’s time, cost and quality constraints. Deciding on a structure leads to the creation of an organisational breakdown structure (OBS), which helps visualise the project hierarchy and is useful for later stages of the resource management process.

Once the structure is set, project professionals should then identify the resources required to deliver the project’s scope. In most instances, this will have already been completed as part of the resource capacity planning process, in which the skills, competences, type (i.e. internal versus external) and capacity of each resource are determined. If not, the process for identifying resource requirements is covered in topic 5.6.1.

Once resource requirements have been identified, project professionals should engage with the relevant resource managers to begin requesting the allocation of project resources. While the detailed planning and allocation of resources is covered in the next topic (5.7.2), it’s best practice to raise resource requests as soon as possible to avoid delays.

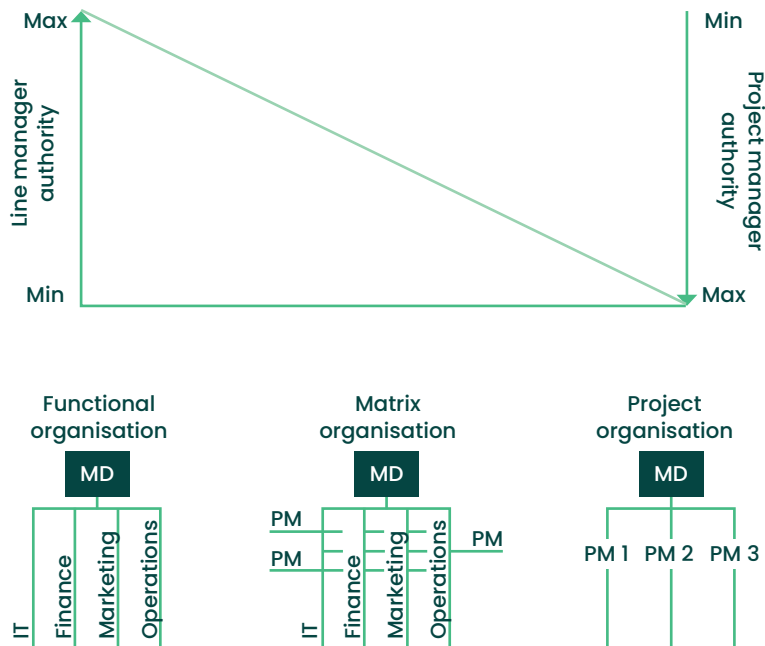


Figure 5.7.1 Organisational structures

Recommended reading

- *Reframing Organizations* (2017) includes the latest update of Bolman and Deal's four-frame model, with coverage of cross-sector collaboration, generational differences, virtual environments, globalisation, sustainability, and communication across cultures.

5.7.2 Planning and allocating resources

Planning the work and allocating the right resources

Once the organisational structure is in place and the resources have been requested, project professionals must clearly define the project's work and schedule to determine how, when and where resources will be allocated.

This definition of the project's work is most commonly completed using a work breakdown structure (WBS). Here, each element of the project's scope is broken down into work packages that detail the lowest level of work to be completed. By the end of the process, each item of work should be defined, providing enough granularity for each one to be estimated and scheduled.

While estimation and schedule management are covered separately, in topic 5.4.1 and section 5.5 respectively, project professionals should aim to establish with as much accuracy as possible when each work package will start and end. This will not only highlight when the associated resources are required but also further validate the project's overall resource requirements.

Once the work has been established, project professionals must allocate the right resources to the work. Each work package should be reviewed and a decision made about who in the project will take responsibility for carrying out the task, supervising the activity and reporting on its progress.

To help with this, it's best practice to combine the work breakdown structure and organisational breakdown structure to create a responsibility assignment matrix (RAM). When used in this way, a RAM clearly shows the individual work items, the resources assigned to each one and the role each resource holds (Figure 5.7.2). These roles are shown using RACI coding, to help define who is:

- **R**esponsible for conducting each piece of work (R)
- **A**ccountable for owning the work and approving it when it is complete (A)
- **C**onsulted during decision-making and supporting the work being completed (C)
- **I**nformed about the work's progress (I)

In line with the project's schedule, the RAM can be used to allocate resources to the project at the correct times. The defined RACI for each task is useful for setting resources' expectations of what is required, and for connecting them and their work to the project's broader objectives.

As the project progresses and new information emerges, the structure and allocation of resources may need to change. This is why, after starting during the resource identification process, project professionals should continue to regularly engage with the resource managers to ensure any new or updated allocations are agreed and actioned.

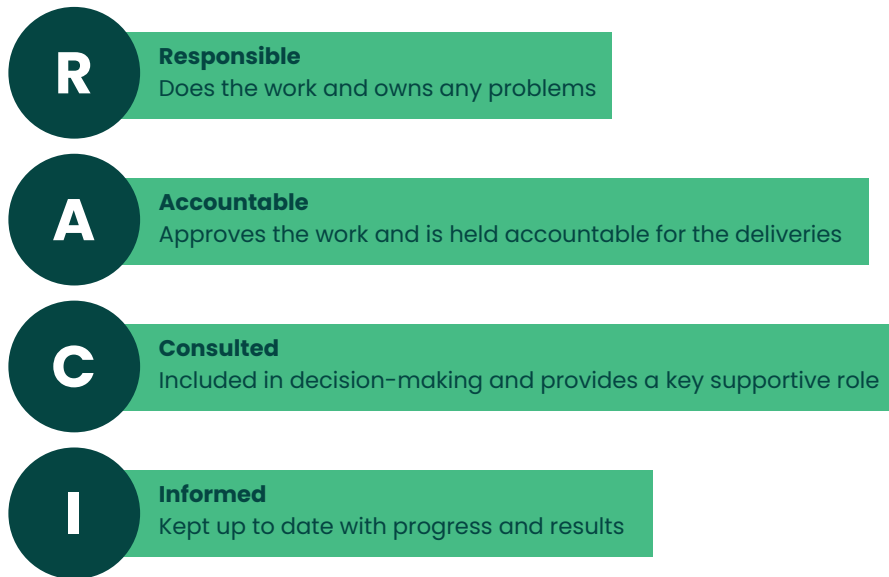


Figure 5.7.2 RACI roles and responsibilities

Recommended reading

- *Senior Managers' and Project Managers' Guide to Critical Chain* (2024) explains the benefits of the critical chain method, including increased throughput with the same resources, reducing pinch points and demand for extra resources.

5.7.3 Managing resources effectively

Managing resources for optimal utilisation

Resources directly contribute to the project's deliverables, so project professionals need to manage them closely for the best chance of success. Failure to do so can lead to schedule delays, and poor quality or a reduction in benefits, so it is best practice to establish sound processes, including reporting and governance, to manage allocated resources effectively.

Following a resource's allocation to the project (5.7.2), project professionals should establish ways to monitor their performance, availability and utilisation. There are many ways to do this, but four of the most common include:

- using timesheets to track the time each resource spends on project work
- monitoring tasks to assess whether the work is completed on time and to quality
- gaining feedback from stakeholders on resource performance
- using a resource histogram (Figure 5.6.2) or demand profile (Figure 5.7.3) to visualise utilisation

As resource management data is collated, it can be visualised within a resource management report, such as a resource histogram, to clearly show performance, availability and utilisation against the project's expectations.

Resource management governance should be established as early as possible to provide project professionals with a suitable mechanism to report progress, escalate issues and align expectations with those of resource managers. Specifically, resource management governance should:

- report on resource performance, establishing clear escalation routes and contingency plans to resolve underperformance
- monitor constraints which are affecting resource availability, feeding into the organisation's resource capacity planning (see 5.6) and risk management (see 5.10.2) processes as a result
- track resource utilisation, providing transparency on actual utilisation against the project's forecast

Even though good governance enables effective resource management, project professionals should know that they are always competing against other projects and programmes for resources. This often forces project professionals to seek creative ways to make their resource capacity go further. While covered in more detail in topic 5.6.3, techniques such as resource levelling or resource smoothing can be used to optimise project-level resource capacity when constraints cannot be overcome.

If a project needs to adjust its resource requirements, project professionals should assess the impacts against other factors (such as quality or cost) before taking action to formally request a change to their baseline. It is best practice for this to change to be raised through the organisation's change control process (see 5.11) to ensure change requests and impact assessments are reviewed with the proper transparency and control.

Managing resources is not a one-off process. Instead, project professionals must continually monitor, optimise and replan their resources as new information emerges during the life of the project.

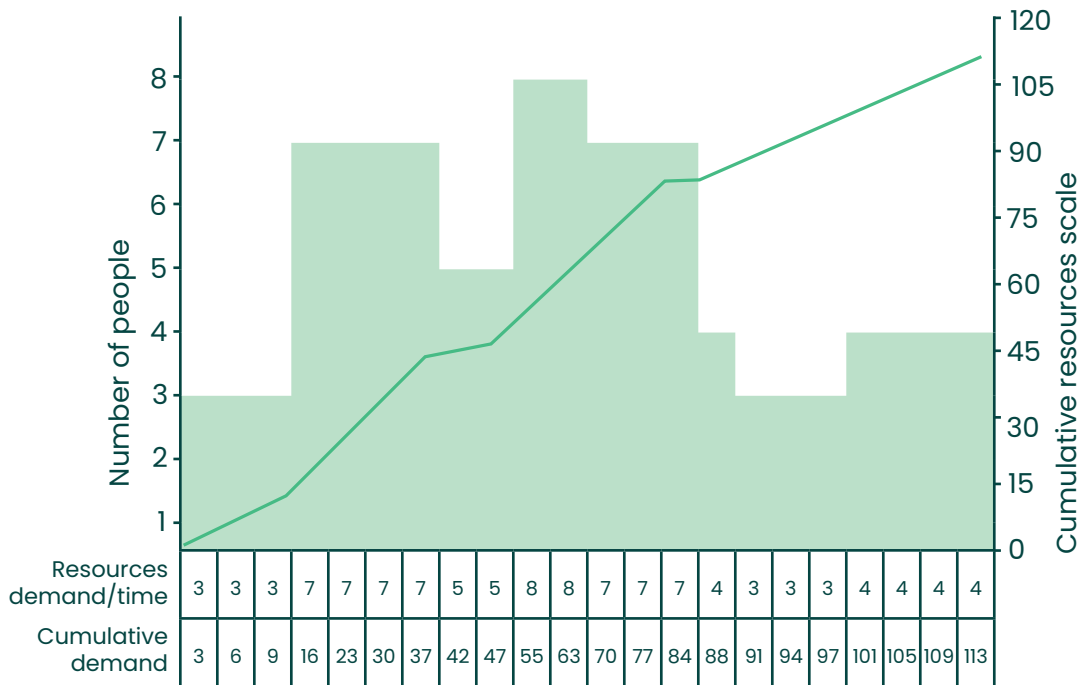


Figure 5.7.3 Resource demand profile and cumulative curve

Recommended reading

- The APM Information Sheet *Resource smoothing and levelling* (2024) provides a short summary of the main differences between the techniques and when to use them on a project.

5.8

Budgeting and cost control

Budgeting and cost control is as important as managing the schedule, quality, scope, risk or value of a project. It has several elements:

- cost estimation
- setting an agreed budget
- managing actual and forecast costs against budget
- closing down finances

These processes are described in the four topics in this section.

The first process is cost planning. This covers estimating costs, building a budget and scheduling the funding through the initiative. It is closely related to resource planning and project scheduling. To do this, project professionals need to identify the types of resources they will use. They must also understand the various ways different costs behave. A core skill in cost planning is estimating.

During the delivery of a project, programme or portfolio, project professionals control expenditure. This requires them to forecast costs, monitor spending, control variances and report on status. This is a fundamental governance activity that ensures the project or programme team is spending the sponsoring organisation's money wisely.

All project professionals experience events that are not part of the core plan. Costs can diverge from the budget when risks, issues and changes to scope or specification need to be dealt with. To fund this, projects draw on contingency funds. There is no magic purse with unlimited funds, but experienced professionals ensure their budgets include contingency and a management reserve.

When a project or programme is complete, one of the closure tasks is an orderly financial closedown. This means taking care to ensure all transactions are complete; the team will then analyse and report on the overall financial performance of the project. The processes for managing schedule contingency are similar.

This section is written for project professionals who need to manage project, programme or portfolio costs. The section includes:

5.8.1 Cost planning: Anticipating costs and timing

5.8.2 Cost control: Ensuring projects spend money wisely

5.8.3 Contingency management: Controlled release of management reserves

5.8.4 Financial closedown: An orderly shutdown of project financial management

5.8.1 Cost planning

Anticipating costs and timing

Budgeting and cost control starts with cost planning. Project professionals estimate costs, establish a budget, and consider how and when they will draw down and commit funding. This helps manage resource demand, supplier payments and funding requests. The resource-optimised schedule is an input to cost planning.

Project professionals consider all types of costs:

- **labour costs** (people), both internal or contracted
- **services** such as software development or legal advice
- **materials** and consumable items, including both finished and raw materials
- **assets**, including equipment, vehicles and infrastructure that can be bought, leased or rented

Some organisations reflect all resources in the schedule, including the use of materials and procurement phasing. Others only schedule labour: non-labour costs only appear in the budget.

There are many approaches to estimating, with some being more appropriate to different types of cost. For example, the materials cost comes from published prices and volume estimates. Subcontracted staff may be commissioned on a 'time-and-materials' basis that requires estimators to consider how many hours they will work, or they may be on a fixed-price contract. Here, suppliers estimate the time needed and include contingency or a risk premium.

Project professionals need to be aware of the risks of bias in their estimating. The most prominent is optimism bias, which leads to overconfidence that events will follow an ideal course. Review and challenge processes counter this bias, and include contingency for each cost area (see 5.8.3).

As well as budgeting, project professionals profile costs over the life of the initiative. This is a cash flow forecast. Earned value management (see 5.5.4) calls this cost profile the budgeted cost of work scheduled (BCWS).

Sustainability is important in project-based working. One way to ensure that the long-term impacts of an initiative are considered is whole-life costing. Increasingly, project professionals plan budgets to the end-of-life and decommissioning stages of initiatives.

Additional cost planning aspects include the following:

- **Fixed and variable costs:** Fixed costs do not depend on scale or quantities. Variable costs change with use or quantity.
- **One-off and recurring costs:** One-off costs are incurred once, like buying a computer. Recurring costs are incurred repeatedly, like renewing a software licence.
- **Capital and revenue costs:** Treatment of capital and revenue costs depends on the accounting rules that the organisation follows. Often (particularly in the public sector), there are controls on capital expenditure.
- **Timing of funding release:** Release of funds may depend on the release of products, test certification or stage-gate processes. In iterative life cycles, release of funds may be linked to iterations (Figure 5.8.1).

All expenditure must be subject to governance oversight and decision-making. This places cost planning at the heart of governance. The level of detail and rigour that needs to be applied will depend on the organisation, the risk and scale of the initiative, and the workstream or phase. Iterative life cycles can control expenditure more easily, and so can tolerate reduced rigour at the outset.

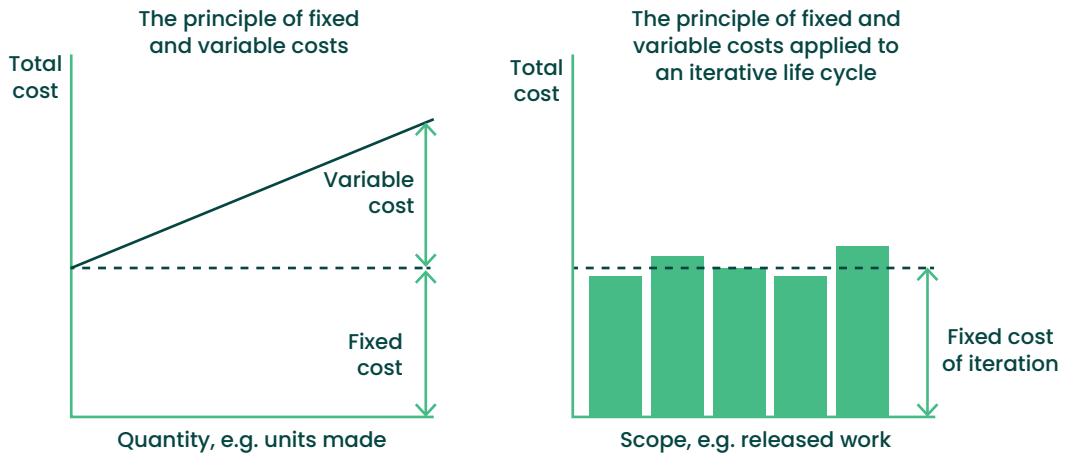


Figure 5.8.1 Fixed and variable costs for linear and iterative life cycle approaches

Recommended reading

- APM's Planning, Monitoring and Control Specific Interest Group guide *Planning, Scheduling, Monitoring and Control* (2015) explains how to construct the planned value (budgeted cost of work scheduled) forecast.
- APM's Planning, Monitoring and Control Specific Interest Group guide *A Guide to Conducting Integrated Baseline Reviews* (2016) has a detailed step-by-step approach to integrated baseline reviews and provides insight into the timing, roles and responsibilities.
- The APM/Association of Cost Engineers' (ACostE) *Estimating Guide* (2019) is a practical document for project management on approaching estimating, types of estimates and the process involved.

5.8.2 Cost control

Ensuring projects spend money wisely

Cost management consists of cost planning and cost control (Figure 5.8.2). Having established a budget and scheduled expenditure (see 5.8.1), it is time to deliver the project and spend the money. The project professional will manage the actual costs against that budget – this is cost control.

The cash flow forecast that was the basis of project approval is the initial baseline for project expenditure. Monitoring takes place against this baseline and the approved budget. Tracking actual expenditure should be as near to real-time as possible, so the faster the project spends money, the more frequently monitoring should happen. Modern project management information systems can connect to enterprise accounting software, which can provide real-time expenditure reports.

However, these approaches can still lag behind commitments made by project professionals. Keeping an assessment of budget status accurate and up to date can be a big challenge, so, on large projects and programmes, project professionals will seek the support of dedicated finance and cost management experts.

The primary reason for monitoring is to be able to take quick action to correct deviations from the plan. When expenditure looks likely to exceed budget, or has already done so, the project team investigates and intervenes. If it is not possible to recover the project back to the original budget or cost schedule, the team will create a revised expenditure forecast. This may need approval from a client, sponsor or governance structure.

Monitoring also allows project professionals to know when contractors have met their obligations so that they can authorise contractual payments, subject to any outstanding quality control.

Equally, monitoring is essential to support project oversight by, for example, a stage-gate process. In cost control terms, this may be an essential step in securing drawdown of further tranches of budget. If there are overspends that cannot be recovered, the project leader may seek authorisation to draw down budgeted contingency (see 5.8.3). In extreme cases, they may require additional, unbudgeted funding to be authorised.

Good governance needs good information. Project teams need to provide timely and accurate reports of financial status, which will be a combination of:

- current status reports and trends
- backward-looking accounts of reasons for deviations from the plan
- forward-looking forecasts and action plans

Cost control is an aspect of wider progress monitoring and reporting (see 5.5.4). Financial status and forecasts contribute to the data set that earned value analysis (EVA) uses. This allows project professionals to assess the performance of the schedule and budget together.

Projects following an iterative life cycle are usually easier to monitor and control. The largest part of the expenses of each iteration is fixed costs (Figure 5.8.1), and clients and sponsors can manage future financial commitment by reducing the number of iterations.

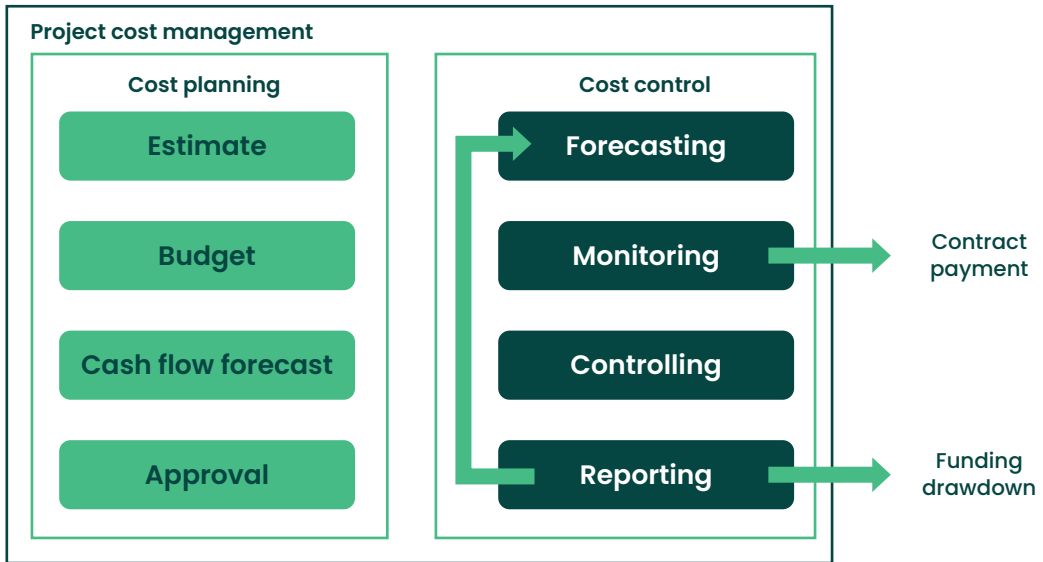


Figure 5.8.2 Cost control in the context of cost planning and project cost management

Recommended reading

- *Project Controls in the 21st Century* (2025), the follow-on to the *Planning* guide below, provides the latest thinking on cost planning and control.
- APM's Planning, Monitoring and Control Specific Interest Group guide *Planning, Scheduling, Monitoring and Control* (2015) includes specific chapters on planning and managing costs.
- The APM Earned Value Management Specific Interest Group's *Earned Value Management Handbook* (2013) is APM's definitive guide to earned value management.

5.8.3 Contingency management

Controlled release of management reserves

Contingency is allocating additional time or money to deal with risks, in case they occur. In creating a budget, project professionals include contingency either for the whole initiative or for each major part of it. Topic 5.4.4 covers including contingency within integrated planning. References here to budget contingency are equally valid when applied to schedule contingency.

In budgeting and cost control, contingency can appear as line items in a budget, as additional timeboxed iterations or as a management reserve across the whole project. Contingency is not hidden extra money that has been put aside to magically solve problems. It is a management tool, subject to governance processes.

Projects, programmes and portfolios can hold contingency at different levels to deal with different sorts of risk. Allocating all the contingency to a project professional requires high levels of trust from the sponsor and governance board. They must be confident that the extra resource will only be used if necessary. Some risks, like currency and interest rate risks, are better managed at organisational level.

During deployment, the project consumes resources and so incurs costs. Monitoring highlights when the project is likely to deviate, or has deviated, from plan. Deviations result from risks, issues and changes to scope or specification.

There are two main types of contingencies:

- **Planned contingency** is set aside to manage known risks and anticipated events.
- **Management or project reserve** is extra funds that are available to deal with unanticipated issues.

Where a risk was identified and analysed effectively, there will be enough planned contingency to deal with the deviation. Then the project professional requests authority to use (draw down) the contingency – this is contingency drawdown (Figure 5.8.3).

Where the deviation arises from an unanticipated issue, there may not be sufficient contingency to meet the cost. The project professional would then ask to draw upon unallocated management reserves.

However, it is possible that the management reserves are insufficient. The project professional would then raise a change request (see 5.11.1) to seek approval for additional funds to manage the situation.

Project professionals expect to use some of the contingency, but they aim to minimise drawdown. Unused contingency results from overestimating, luck or good risk management. Insufficient contingency results from optimistic estimates, bad luck or poor risk management.

In an iterative life cycle, timeboxed cycles have fixed time, resources and cost. Contingency relates to scope, quality and number of iterations. The team works through a series of cycles, drawing down backlog work for each. After the planned iterations, the solution may meet a minimum viable standard, but it may not meet all the 'should-have' requirements. In this case, the team may raise a change request either to vary the scope and quality, or to secure additional iterations. The sponsor and governance board review the change request and either authorise or reject it.

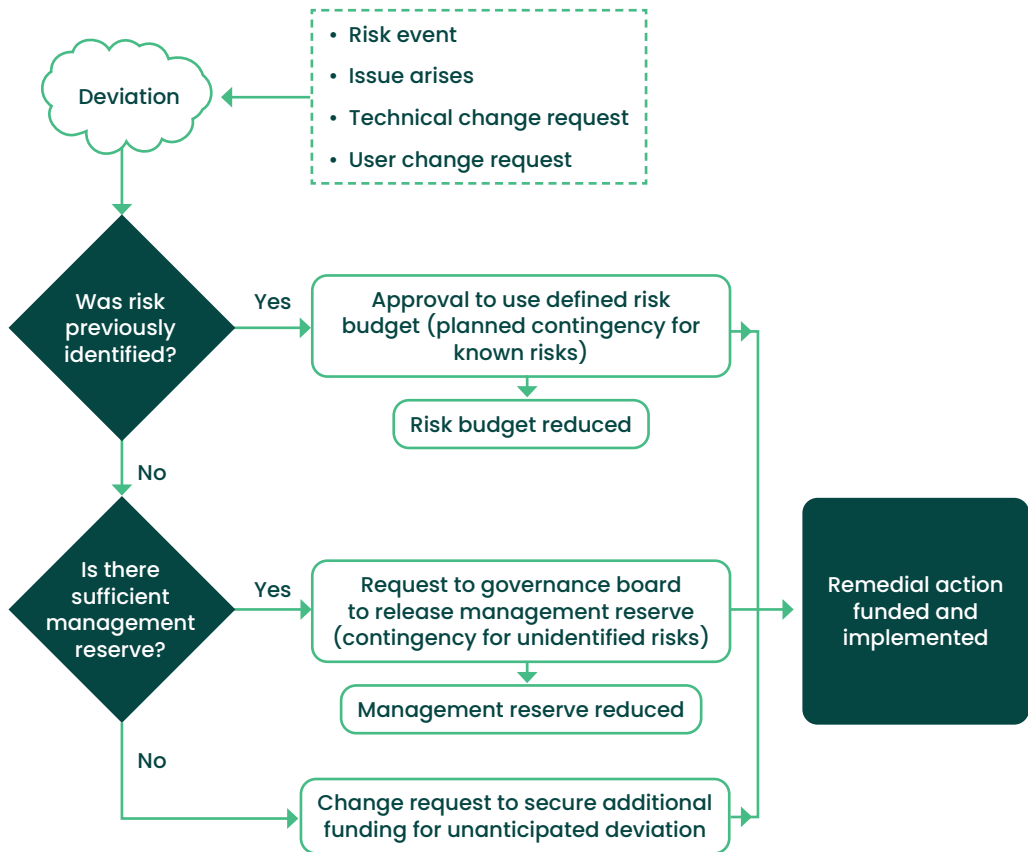


Figure 5.8.3 Process flow for contingency drawdown

Recommended reading

- The APM Risk Interest Network's *Project Risk Analysis and Management Guide* (2025) explains how governance can monitor and control the use of contingency.
- APM's Programme Management Specific Interest Group guide *APM Introduction to Programme Management* (2016) has a good description of the programme-level risk management reserves approach and its relationship to project reserves.
- *Practical Cost Control Handbook for Project Managers: A Practical Guide to Enable Consistent and Predictable Forecasting for Large, Complex Projects* (2020) is a practical handbook for managing large, complex projects. It addresses the traps of cost control and forecasting to support decision-making and addresses contingency through the lens of forecasting.

5.8.4 Financial closedown

An orderly shutdown of project financial management

As a change initiative comes to an end, project professionals must ensure an orderly shutdown of all aspects of the work. There are many details to complete before leaving for the next role, including financial management. Structured financial closedown is a responsibility for leaders of projects and programmes.

Perhaps the most important financial closedown task is ensuring that the project meets all outstanding financial commitments (see Figure 5.8.4). This means processing and paying invoices from contractors, consultants and suppliers. Only after all bills have been paid can the team consider contracts to be complete.

Beyond the project, there will sometimes be transition activities as the organisation implements the changes. Where budgets include these activities, it is important to secure drawdown of these funds.

Project professionals aim not to draw down all contingency funds or management reserves. They may need to ensure these contingency funds are returned to the organisation, but, within a wider portfolio or programme, it may be possible to hold them as reserves for other initiatives.

Final costs often lie in outstanding timesheets and expense claims, so it's wise to set a deadline for team members to submit them because they must be processed before finalising the project financial records.

When these are in place, the project can make a final assessment of financial performance, comparing actual expenditure against the business case budget. It may include a record of forecasts and there is likely to be a need for a final financial report on project performance.

Once all financial transactions and evaluations are complete, any project-specific cost centres need to be closed. At the same time, project team members' access to finance systems should be reviewed. Then, all financial records need to be secured and archived according to the organisation's archiving policies and procedures. This will ensure that they are available for any potential future audit.

The last thing to think about is stakeholders. What do they need to know and want to know about the financial performance of the initiative? This will range from highly formal notifications and preparing statements for the organisation's statutory reporting to informal public relations, perhaps including high-level financial information in newsletters, articles and management briefings.

For projects within larger programmes, or programmes within portfolios, a programme or portfolio management office (PMO) may do some or all of this. PMOs often have a role in the financial oversight of the projects and programmes within their remit.

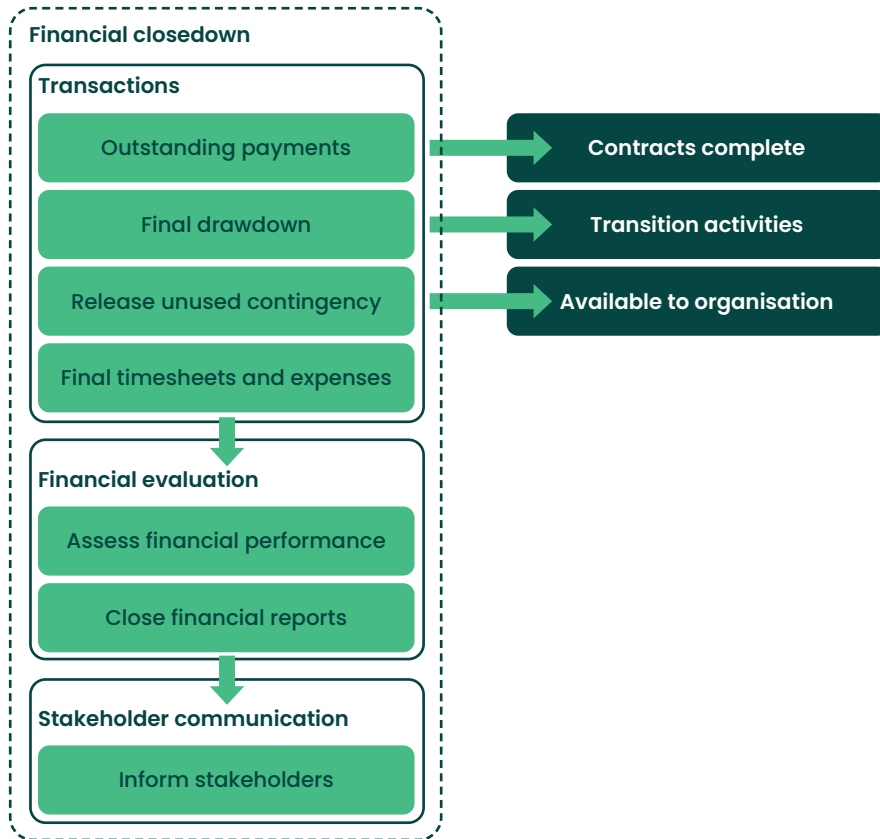


Figure 5.8.4 An orderly financial closedown

Recommended reading

- APM's short guide *How to Close Projects Successfully* (2024) considers the challenges of closing projects and provides tools and techniques to manage the process.

5.9

Contract management

Working with suppliers to enable project success

Contract management is the process of creating and awarding a contract, and managing the relationship between a project and a third-party supplier. A robust contract management process unlocks access to goods and services which contribute to the successful delivery of project outcomes.

Once the project's procurement strategy is agreed (see 3.3.2 and 3.3.3), the contract management process begins with identifying the type of contract that will best serve the project's requirements. To achieve this, project professionals first determine how many suppliers they need to engage with, before moving on to agree how a supplier's work should be delivered and charged for.

With the foundations set, the project team moves into the contract award process. In most circumstances, this takes the form of a competitive tender, where suppliers are assessed on their ability to support the project's objectives. Alongside procurement and legal experts, project teams shortlist suppliers based on their submissions, and then enter into detailed negotiations before signing a contract.

Once the contract is in place, project professionals must manage the supplier throughout the life of the contract, working collaboratively to ensure the agreed work is delivered. Clear roles and responsibilities, robust governance and regular performance reviews should be put in place to enable effective supplier management, with processes to manage contract changes and disputes if they arise.

When the project team is satisfied that all agreed work has been delivered, both parties can agree to close the contract, ensuring that closure activities such as handovers and financial settlements are completed.

Suppliers play a crucial role in many projects, no matter the industry, country or life cycle they operate in. Following a defined contract management process helps teams unlock the goods and services they need to succeed, while ensuring compliance with local legislation, policies and procedures.

This chapter is for project professionals tasked with supporting, leading or overseeing contract management. Specifically, it will cover:

5.9.1 Types of contracts: Selecting the right contract for the project's requirements

5.9.2 Contract award: Selecting suppliers and setting up contracts for success

5.9.3 Managing contract performance: Monitoring and evaluating contract performance

5.9.1 Types of contracts

Selecting the right contract for the project's requirements

Using suppliers' goods and services is a critical component of many projects. Contracts create legally binding agreements between a project and its suppliers, helping to define the scope, set expectations, build trust and, ultimately, increase the chance of project success.

But no two contracts are the same. The size, scale and complexity of a contract will depend on the requirements of the project, as well as country and industry norms. For example, a contract for a one-off purchase of materials is likely to be simple, whereas the year-long outsourcing of a software development team will be more complex.

The type of contract a project chooses often depends on the answers to two key questions:

- How many suppliers are needed to meet the requirements?
- Which cost and delivery approach would be best for both parties?

Depending on the number of suppliers, the following types of contract can be selected:

- **Comprehensive contract:** One supplier is responsible for all contract requirements.
- **Sequential contract:** Two or more suppliers share sequential parts of the contract, e.g. one completes the design work, then another builds.
- **Parallel contract:** Two or more suppliers share the contract and deliver in parallel, e.g. one completes the ground floor, the other the first floor.
- **Subcontract:** The project contracts with one supplier, but that supplier contracts the delivery to another supplier.
- **Partnership:** Two or more suppliers come together as one joint venture to share all contract requirements.

Once the supplier approach is agreed, attention turns to the cost and delivery approach. There are many options here, each specifying how work is completed, when monies are paid and how changes will be managed. These are the three most common approaches:

- **Fixed price:** Parties agree a set price for a defined scope of work, with changes managed via change requests.
- **Time and materials:** Parties agree a unit price for materials and labour. Costs increase as work is completed against the scope, often with a 'not-to-exceed' limit in place.
- **Cost plus:** This is similar to 'time and materials', but supplier profit is charged separately, typically as an agreed percentage of the project's total cost.

The choice of cost and delivery approach will be determined to a great extent by how well the project can be specified at the outset. Simpler, well-defined projects favour a fixed-price approach, while projects that are ill-defined or are expected to develop require greater flexibility and so lend themselves to time and materials or cost plus. The choice of project approach will also partly depend on the risk appetite of the contracting parties. A fixed-price contract gives certainty to the buyer but transfers some risk to the supplier, which the supplier will cost into their quote. Time-and-materials and cost-plus contracts put considerable risk on the buyer, as they will need to tightly control what is happening. The buyer never completely transfers the risk to the supplier, as suppliers may walk away, or in some circumstances, become insolvent so they can't continue.

Selecting the right type of contract is critical for success, so it's best practice for project professionals to leverage the expertise of procurement and supply chain experts when making contract decisions.

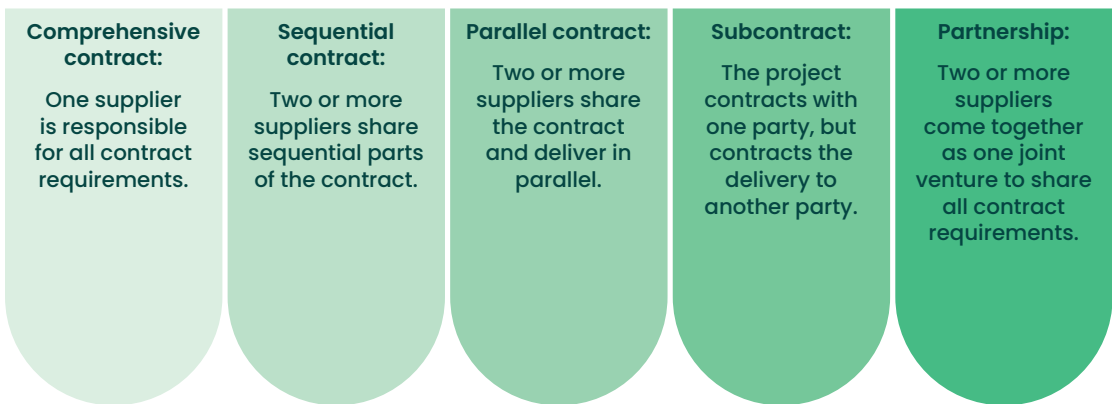


Figure 5.9.1 Five procurement contract types

Recommended reading

- *APM Guide to Contracts and Procurement* (2017) offers a basic understanding of 'how to' procure sub-project works and to manage delivery through the phases of the procurement life cycle.

5.9.2 Contract award

Selecting suppliers and setting up contracts for success

Contracts are awarded to successful suppliers following a defined procurement process (see 3.3.2 and 3.3.3). In most circumstances, this is a competitive tender exercise where suppliers are assessed on their ability to support the project's objectives.

To select the participants for a competitive tender, a prequalification process is often followed. Here, a broad selection of suppliers are invited to apply, before they are narrowed down to a shorter list. In rare circumstances, this process might result in only one suitable supplier, meaning the project can progress straight to a negotiation. But, in most instances, multiple suppliers will be chosen to progress to the next stage and issued with a formal invitation to tender. These suppliers submit their best offers against a common scope of requirements.

Depending on the size of the tender, this process may take many weeks or months, with an agreed deadline for submissions. To ensure fairness, good practice dictates that late bids are disqualified and responses to clarification questions are shared with all suppliers.

Procurement professionals and project stakeholders agree criteria for objectively judging the tender submissions, including these key considerations:

- **How will time, cost and quality be prioritised?** Best value is not always represented by the lowest cost. A scoring system should assign relative weightings to different factors and be shared with suppliers.
- **Who will assess tenders?** An assessment panel should be appointed which includes technical and commercial experts, the sponsor and other stakeholders representing the investing organisation.
- **How will confidentiality be assured?** To ensure the competitive nature of the tender process, as well as for ethical reasons, confidentiality between suppliers should be maintained.
- **Is there specific legislation governing the contract award?** Project professionals need to be aware of any specific regulations surrounding the awarding of the contract. They must ensure compliance to minimise the risk of penalties or legal challenge later.

Having assessed the submissions, it is usual to shortlist suppliers and invite them for negotiation. When a preferred supplier or suppliers is or are selected, final negotiations are completed and the contract is signed. It is recommended that a back-up supplier is retained in case negotiations break down.

Contract award processes can be complex and impeded by legal complexity, so it's best practice for project professionals to use the expertise of procurement and legal experts when awarding contracts.



Figure 5.9.2 Essential steps in contract award

Recommended reading

- *The Project Manager's Guide to Purchasing: Contracting for Goods and Services* (2010) is a practical guide to the process of contracting. The book focuses on the steps from selecting a tender to placing a contract. Chapter 8 is dedicated to selecting the tenderers, Chapter 9 covers the enquiry process and Chapter 11 is concerned with evaluating the tenders.
- Section 5 of the *APM Guide to Contracts and Procurement: For Project, Programme and Portfolio Managers* (2017) looks at preparing the contract terms and requirements, while section 6 is concerned with selecting the provider and awarding the contract. Developed by APM's Contracts and Procurement Specific Interest Group, the guide covers each step in detail, utilising an input, activity and output structure.
- *Bids, Tenders and Proposals: Winning Business through Best Practice* (2015) adopts the supplier perspective, offering guidance on structuring bids and tenders. After an explanation of how to write and structure the documents and the process, Chapter 23 explains how clients evaluate tenders.

5.9.3 Managing contract performance

Monitoring and evaluating contract performance

Once an agreement is in place, suppliers must be carefully managed to ensure they deliver the work detailed in the contract. To achieve this, project professionals and suppliers should work together to define clear roles and responsibilities, set up robust governance, and conduct regular progress and performance reviews.

Contract management is a people-focused activity, so it's common to begin by bringing both parties together to agree roles and responsibilities. On the project side, a contract manager should be appointed to lead the supplier relationship and report on progress. In many instances, this is the project manager themselves. Supplier roles and responsibilities vary by contract, but project professionals should always ensure they have an agreed point of contact to work with during the project.

Once roles and responsibilities are set, both parties should agree an approach to performance management. This comprises two key components:

- the key performance indicators (KPIs) the supplier will work to
- the governance structure used to report on the supplier's progress and performance

Most contract performance indicators will relate to time, cost or quality, ensuring the deliverables align with the broader timeline and parameters of the project. The structure, format and regularity of contract governance often depend on the size, scale and risk appetite of the project. But, in most circumstances, regular performance review meetings are set up to report on progress, budget, risks and issues, and to plan activities for the next period.

Alongside the governance approach, both parties should agree clear processes for:

- requesting changes to the contract (Figure 5.9.3)
- managing disputes

As the project progresses and new information develops, changes to the original contract may be required. It's important that both parties agree on a process for requesting, evaluating and agreeing changes to the contract, including how any impacts on the contract's scope, time, quality and cost are managed.

While the win-win nature of most contracts encourages parties to resolve issues collaboratively, disputes may occur when contract KPIs aren't met. Project professionals should ensure a clear mechanism to raise and resolve disputes is in place, alongside an understanding of the legal ramifications of doing so.

When all these components are in place, the project is well equipped to manage the supplier through the life of the contract. Once the project team is content that all contract work has been delivered, both parties can agree to close the contract, ensuring closure activities such as handovers and financial settlements are completed.

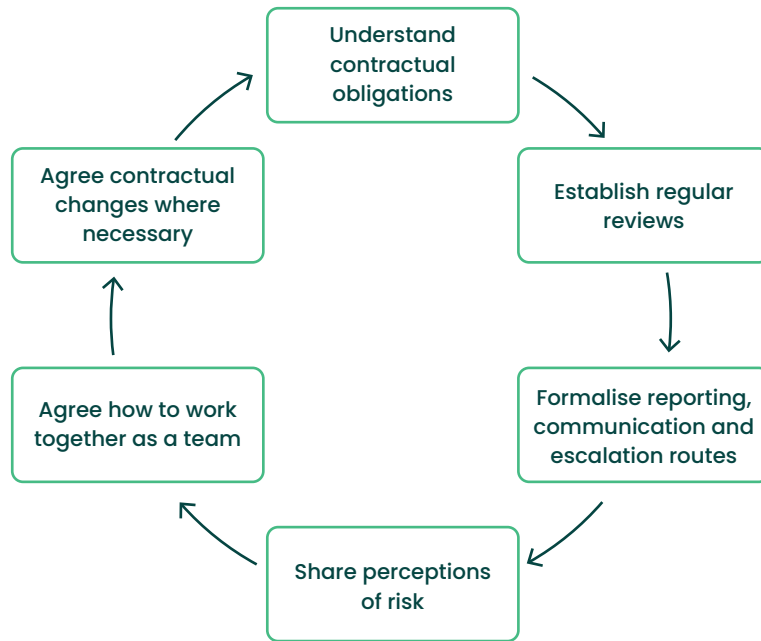


Figure 5.9.3 Controls that support contract management

Recommended reading

- *APM Guide to Contracts and Procurement: For Project, Programme and Portfolio Managers* (2017): Chapter 7 looks at how to manage and deliver the contract once an agreement is in place.

5.10

Risk and issue management

Risk management is the process that allows project professionals to identify, assess and respond appropriately to risks. It involves deciding which threats to actively minimise and which opportunities to maximise or pursue.

Issue management is about having the flexibility to react to issues in appropriate ways. This includes escalation to the relevant authority.

All projects bring change and therefore uncertainty, but risk and issue management minimises the effects of that uncertainty. As a result, the deployment baseline and project management plan will be more robust.

Issue management relies on appropriate tolerance levels being agreed. While the project manager and their team will be able to deal with day-to-day problems, issues will need to be escalated so they can be dealt with by those who have a more strategic view of the organisation, who can commit more resource to the issue if required and authorise changes to the scope if needed. The project can be cancelled if the issue makes the project unviable.

Risk management considers the risk appetite of the organisation, the governance in place, and the size and complexity of the change. It gives confidence to stakeholders that risks are being proactively managed. The process protects the reputation of the organisation and leads to realistic plans and appropriate contingency.

Risk analysis assesses the probability and impacts of risks using qualitative and/or quantitative techniques. It is unrealistic to believe that every risk can be managed, so it is necessary to prioritise risk events so that the top risks can be addressed. This shows which risk events may need to be escalated to the project sponsor or governance board.

Opportunity management recognises that uncertainty can have a beneficial effect on the achievement of objectives. This is the positive side of risk. Project professionals are encouraged to consider opportunities throughout the project life cycle, using the same process and tools that are used for threats. Systems thinking (1.3.5) also gives the project professional a range of tools to help think through issues and risks.

This section is written for project professionals who need to undertake risk and issue management at project, programme or portfolio level. It includes:

5.10.1 Issue management: Adapting the plan to resolve issues

5.10.2 Risk management: Being ready to respond to minimise threats and maximise opportunities

5.10.3 Risk analysis: Ensuring project plans take account of variability and risk events

5.10.4 Opportunity management: Actively identifying, planning and implementing responses to opportunities

5.10.1 Issue management

Adapting the plan to resolve issues

In project management, an issue occurs when the tolerances of delegated work have been exceeded, or when the tolerances will definitely be exceeded.

Issues are different to problems. Issues require support from the sponsor to agree a resolution (Figure 5.10.1). Problems are dealt with on a day-to-day basis by the project manager and the team.

There is often a tendency to mix up the identification, analysis and management of risks with the management of issues. They are related but are not the same. Issues may develop when particular risks or groups of risks occur. Issues happening now may also be the causes of new risks and may result in a reassessment of the likelihood and impact of previously identified risks.

It is understandable that project professionals prioritise the management of issues over the management of risks. Issues are happening now, whereas risks are potential threats or opportunities. In a project where issues continually happen, there may be an underlying problem with project plans and controls.

Irrespective of the source of the issue, the process to manage it is the same. The project professional ensures that the following activities happen:

- When an issue is detected, it is logged in an issue register. Analysis takes place quickly to understand the nature of the issue. This includes its cause and the impact if it is not resolved.
- Issues are escalated to the sponsor, who in turn may escalate them to the governance board for resolution.
- Actions are assigned to the person or group best placed to take ownership of the issue.
- Issues may result in changes to scope, so the agreed change control process must be followed.
- The management of issues is tracked from identification through to resolution.

The issue management process is a simple concept, but there are barriers to effective adoption. For example, there may be a reluctance for project professionals to escalate issues early, or the governance board may treat the symptom (outcome) as opposed to addressing the root cause.

Issue management is an important project control. Used correctly, it ensures that project professionals work within their delegated authority and decisions on resolution will be made at the appropriate level.

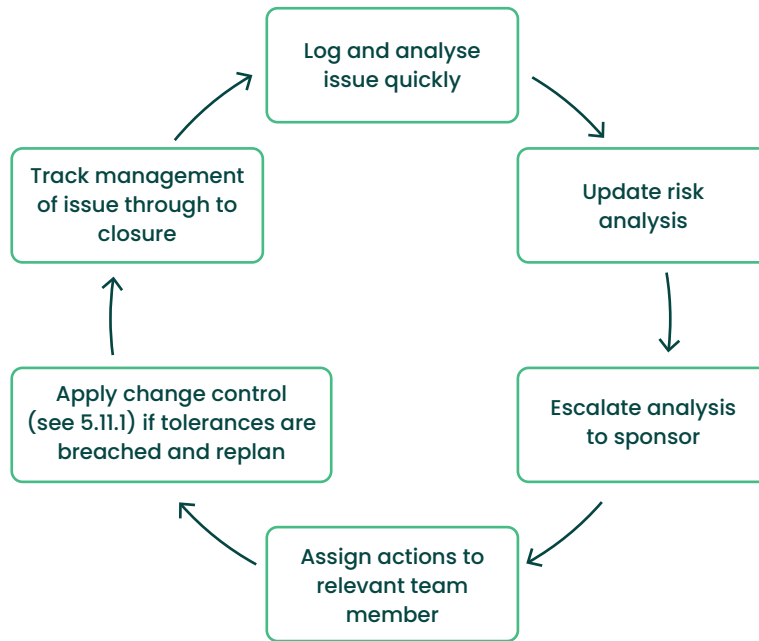


Figure 5.10.1 Key aspects of issue resolution

Recommended reading

- APM's *Project Risk Analysis and Management Guide* (2025) demonstrates how issue management fits into the wider discipline of risk management.
- *The Project Workout: The Ultimate Guide to Directing and Managing Business-Led Projects* (2019) dedicates a chapter to the discussion of what went wrong, including procedures for dealing with issues and advice for managing the issue log (register).
- *Decision Making & Problem Solving: Break through Barriers and Banish Uncertainty at Work* (2019) offers a set of techniques and insights for resolving problems in the workplace. The book includes practical exercises, templates and advice on how to generate ideas, solve problems and inspire confidence within a team.

5.10.2 Risk management

Being ready to respond to minimise threats and maximise opportunities

Information collected during risk identification and risk analysis (see 5.10.3) informs the next steps in risk management.

A decision needs to be made to proactively spend time and money to reduce risk exposure. The tolerance level for risk exposure will be determined by the risk appetite of the investors and wider stakeholders.

People perceive risk in different ways. There are many conscious, subconscious or emotional factors that influence the perception of risk and risk attitude, so the project professional needs to work with a range of stakeholders to understand this. Their engagement is necessary to achieve pragmatic risk responses.

If there is a justification for investing time and money to increase certainty:

- the required resources will need to be planned and agreed
- the deployment baseline and PMP will need to be updated

There are two main responses to threats and opportunities: proactive and reactive.

A proactive response is a planned and implemented response: 'We do something now.' The available responses for threats are:

- **avoid** by changing the scope to eliminate the cause
- **reduce** the probability and/or impact of the threat
- **transfer** the cost by taking out insurance or placing the risk with a better-equipped owner

Proactive responses for opportunities are:

- **exploit** by changing the scope to make sure that the opportunity happens
- **enhance**: increase the probability and/or impact of the opportunity
- **share** the benefit, possibly through a joint venture

In the case of reactive responses, a response is identified but will be implemented only if the risk materialises. A threat could be accepted with a contingent response, and an opportunity could be rejected as it may not be worthwhile.

Keeping the risk conversation alive is crucial to the ongoing delivery of any project. The risk management process is iterative – it reflects the dynamic nature of project work. It captures and manages emerging risks. New knowledge will be acted upon and contingency estimates refined.

A risk register is used to document risks, analysis and responses. It will assign a clear ownership of actions.

Information on priority risks is escalated to the sponsor or governance board. This enables stakeholder expectations to be managed and evidence-based decisions to be made.

The final part of the process is to ensure that all risks are closed when they have occurred or when there is no possibility of them occurring.

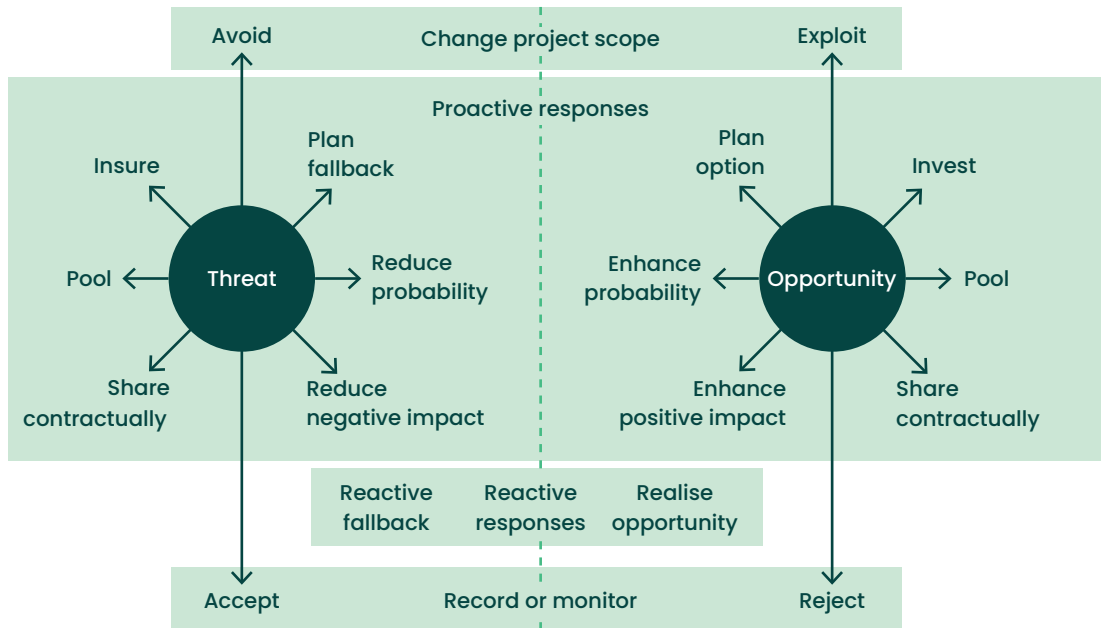


Figure 5.10.2 Generic response strategies for threats and opportunities

Recommended reading

- APM's *Project Risk Analysis and Management Guide* (2025) contains a detailed explanation of how to respond to risks efficiently and effectively, and describes the importance of establishing a culture in which the risk conversation continues through the life of the project or programme.
- *Practical Project Risk Management: The ATOM Methodology* (2012) provides a detailed guide to the application of a risk management process to a project.
- *Understanding and Managing Risk Attitude* (2007) brings together leading-edge thinking on attitudes to risk and emotional literacy, to guide those wishing to move from a process-only conception of risk management to one that addresses the influences of people in the process.

5.10.3 Risk analysis

Ensuring project plans take account of variability and risk events

Risk analysis assesses uncertainty. To do this, risk events need to be analysed and then their individual significance and/or combined impact on objectives needs to be determined.

There are two analysis techniques. Qualitative analysis allows risk events to be prioritised, while quantitative analysis focuses on specific risk-based decisions. Techniques such as decision trees or sensitivity analysis can be used, or a more holistic approach can be taken to determine how overall project risk may impact the achievement of objectives.

As a minimum, basic qualitative analysis needs to be carried out to identify and prioritise risk events based on the:

- probability/likelihood of occurrence (Figure 5.10.3), and
- size of impact on schedule, cost, benefits and potentially other objectives

For qualitative risk analysis to be as useful as possible, it is important to focus on the objectives that are at risk. Project-specific impact scales can then be fine-tuned. This enables meaningful risk prioritisation and appropriate risk responses (see 5.10.2).

Different people have different perceptions of what is risky and why. It is therefore important to engage a range of different project professionals in risk analysis. Their different perspectives will add value and integrity to the process.

For many projects, a qualitative approach is sufficient. For large or complex projects, a more sophisticated approach may also be required to support the investment decision.

Where an assessment of overall project risk is required, a probabilistic approach can be used. A Monte Carlo simulation is an example of this. It is a model used to predict the probability of a variety of outcomes to help explain the impact of risk and uncertainty in prediction and forecasting models.

Outputs from a probabilistic risk analysis help the project professional to:

- understand the probability of achieving certain out-turn dates, costs or benefits
- inform and influence decision-making about the chances of achieving the business case and plan
- agree the level of contingency to provide the required level of confidence

All risk analysis relies on a good understanding of stakeholder perception of risk. Irrespective of whether a qualitative or quantitative approach is used, risk analysis is a key process for the project professional to undertake. It is a necessary input to all business cases and plans, at project or programme level, regardless of the life cycle approach chosen.

Probability	0.9	VHI	0.045	0.09	0.18	0.36	0.72
	0.7	HI	0.035	0.07	0.14	0.28	0.56
	0.5	MED	0.025	0.05	0.10	0.20	0.40
	0.3	LO	0.015	0.03	0.06	0.12	0.24
	0.1	VLO	0.005	0.01	0.02	0.04	0.08
			VLO	LO	MED	HI	VHI
			0.05	0.1	0.2	0.4	0.8
			Impact				

Figure 5.10.3 Example probability/impact grid to qualitatively prioritise risk events

Recommended reading

- APM's Risk Interest Network Group has three relevant guides on the analysis of risk: *Project Risk Analysis and Management Guide* (2025), *Project Risk Analysis and Management Mini Guide* (2018) and *Prioritising Project Risks* (2008). These guides summarise the tools and techniques for qualitative and quantitative risk analysis.
- *Practical Project Risk Management: The ATOM Methodology* (2012) provides a step-by-step guide to how to implement a risk management process for any project. This includes detailed advice on how to make a qualitative risk analysis process as objective as possible, and practical advice on building risk models for Monte Carlo simulation.

5.10.4 Opportunity management

Actively identifying, planning and implementing responses to opportunities

The definition of 'risk' in the project management world has changed to include opportunities – the positive side of risk. Even though the definition has changed, the idea of positive risk has not been accepted by all project professionals. To some, opportunities are perceived as being quite different to threats. They are often not seen as two sides of the same coin. However, opportunity management can often lead to better ideas from the outset and during the project.

Opportunities can fall into several different categories including:

- technical
- business, economic or commercial
- environmental, legal and social
- logistics and supply chain

These categories should be considered to gain a full understanding of the opportunities that have the potential to arise.

To maximise the success of a project, opportunities need to be actively managed. As with threats, opportunities need to be identified early in the project life cycle so there is ample time to address them effectively.

The project risk management process is the same for threats and opportunities:

- Opportunities are identified by looking for positive impacts on the project objectives. This can be a creative process, drawing on the experiences of a wide range of stakeholders.
- It is important to consider both internal and external factors. Opportunities need to be defined clearly, separating cause from effect. This facilitates their subsequent analysis and management (see Figure 5.10.4).
- Assessment can be either qualitative or quantitative (see 5.10.3).
- The subsequent responses will be to either exploit, enhance, share or reject (see 5.10.2).

However, care needs to be taken, as opportunities are uncertain. Therefore, although they could have a positive effect on the project objectives, they should not be made part of the project cost and schedule, as they may not occur. The potential savings are typically held as a below-the-line figure.

A culture needs to be adopted that places adequate focus on the pursuit of opportunities. Encouraging identification of both positive and negative risks throughout the life cycle will assist with this.

Opportunities should be made visible to the sponsor. Opportunities are a key factor in determining the health of the project and informing decisions. The importance of opportunity management needs to be emphasised and embraced at all levels within the organisation.

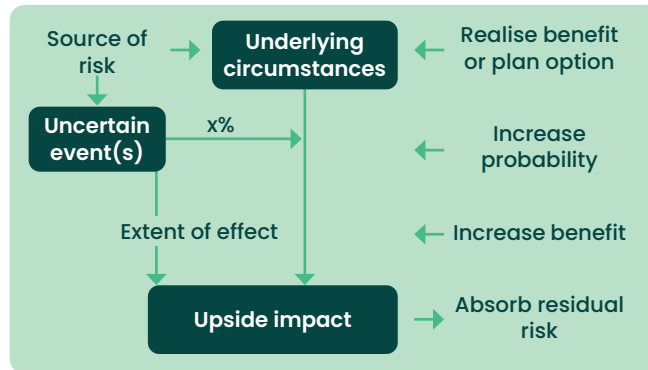


Figure 5.10.4 Quality assessment of an opportunity

Recommended reading

- APM's Risk Interest Network Group has three relevant guides on the analysis of risk: *Project Risk Analysis and Management Guide* (2025), *Project Risk Analysis and Management Mini Guide* (2018) and *Prioritising Project Risks* (2008). These guides consider tools and techniques for identifying opportunities, as well as threats.
- *The Asymmetry between Threats and Opportunities in Risk Management* (2023) reveals attitudes to opportunity management among APM Corporate Members.

5.11

Change control

Managing changes to the project's baseline

Change is an inevitable part of any project, but it needs to be managed effectively to avoid misalignment, confusion and unwanted consequences. Change control refers to the governance, process and controls that identify, evaluate and action changes to a project's baseline.

Effective change control is underpinned by a project's change control process. A defined change control process ensures project sponsors understand the impacts of a requested change, providing them with the information they need to make an effective decision.

While all change control processes differ slightly, in most circumstances they follow a common set of steps:

- Change requests are raised and documented in a change log or register.
- Changes are evaluated, first at a high level and then in more detail if required.
- A recommendation is made to the project sponsor, who decides whether to approve, reject or defer the change.
- If the change is approved, the project's plans are updated, and action is taken to implement the change.

To make the change control process effective, project professionals need to quickly and effectively assess the impact of a requested change. This impact assessment is a complex analysis that requires detailed information gathering and input from subject-matter experts. The impacts should be quantified, specifically focusing on other areas of the integrated PMP, such as the scope, benefits, cost and quality.

Project professionals should work hard to present the evaluation in a way the project sponsor can understand, while ensuring they provide enough information to enable effective decision-making. If the change is approved, plans should be updated, actions taken to implement the change and controls put in place to monitor the impacts.

For highly technical projects, approved changes directly impact the configuration of the project's deliverables. A configuration management process should be implemented to maintain the accuracy, visibility and traceability of the project's configuration items, ensuring constant compliance with organisational and regulatory standards.

This chapter is for project professionals tasked with supporting, leading or overseeing change control. Specifically, it will cover:

5.11.1 Change control process: Managing change requests in a controlled way

5.11.2 Impact assessment: Assessing the impact of a requested change

5.11.3 Configuration management: Ensuring continuous accuracy of configuration items

5.11.1 Change control process

Managing change requests in a controlled way

The change control process provides a logical, controlled mechanism for identifying, evaluating and actioning requests to change the project's baseline (Figure 5.11.1). Change is an inevitable part of any project, so to maximise the chances of success, project professionals should define a clear process for managing changes, regardless of the project's approach or chosen life cycle.

Managing change requests in this way enables the project sponsor and other stakeholders to understand the implications of changes on the forecast outcomes of the project. They should:

- enable decisions to be made in line with the strategic objectives and appetite for risk
- manage impacts on other projects, programmes, portfolios or BAU activity

Requests to change a project may arise from many different sources, including issues discovered during project delivery, new stakeholder requirements or changes in the project's external environment.

Change requests are often raised through the project manager, who should implement the following process to control the change effectively:

- **Log change request:** A change register (or log) is used to record all requested changes.
- **Initial evaluation:** The change is reviewed to consider whether it is worth evaluating in detail. If not, it will be rejected at this stage.
- **Detailed evaluation:** An impact assessment is completed for the change, including how it may alter the project's benefits, scope, quality, schedule and costs. This is a detailed process requiring data analysis and inputs from various stakeholders. Change impact assessments are covered in greater detail in topic 5.11.2.
- **Decision:** Based on the detailed evaluation, a number of options and/or a recommendation are presented to the project sponsor. The project sponsor is accountable for ensuring a decision is made to approve, reject or defer the change. The decision is then communicated to the project team.
- **Update plans:** If the change is approved, plans are updated to reflect the change. Given the integrated nature of projects, updates will also need to be reflected in other processes, such as scope definition (5.2.1), solutions development (5.2.4) and configuration management (5.11.3).
- **Implement:** The necessary actions are taken to implement the change, and the impacts are monitored through to completion.

In scenarios where a change is implemented without formal approval, the project professional should manage the process retrospectively. Rather than being seen as unnecessary bureaucracy, change management should be seen as the best way to ensure the impacts of the change are fully understood.

In certain circumstances, projects may implement a change freeze, where no change requests are considered. This is usually implemented around exceptional events in the project's environment (e.g. Christmas, summer holidays) so as not to create unnecessary risks or issues.

It is important to differentiate change control from the wider discipline of change management. Change management is a holistic approach to organisational transformation, whereas change control is a process to manage changes to an individual project's baseline.

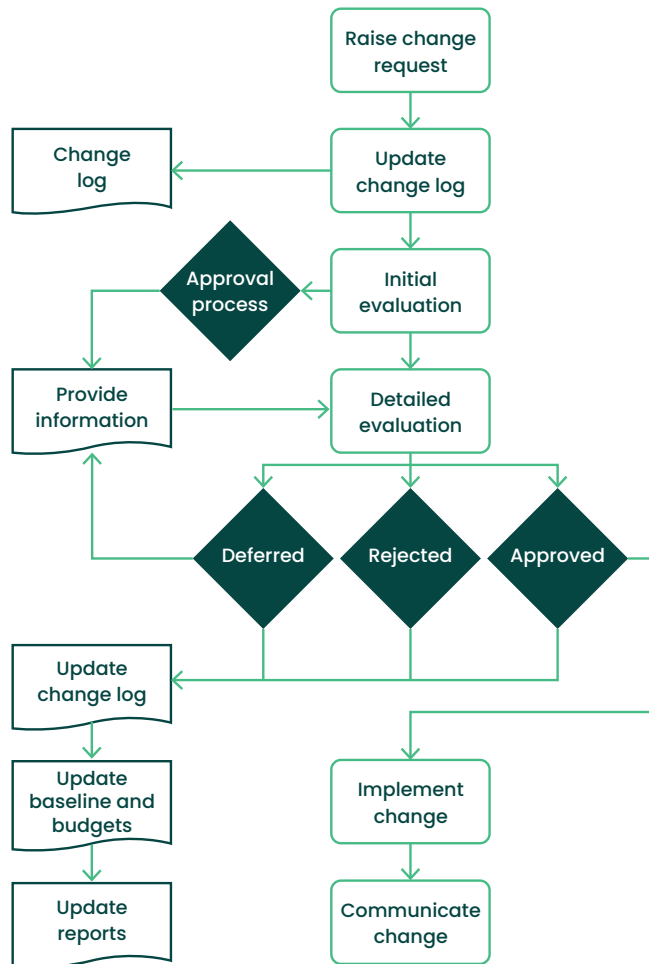


Figure 5.11.1 A change control process

Recommended reading

- APM's guide *Project Controls in the 21st Century* (2025) covers the change control process and the role the change control board.
- *The Project Workout: The Ultimate Guide to Directing and Managing Business-Led Projects* (2019) dedicates a chapter to the consideration of change control, with a particular emphasis on controlling change, accountabilities for change decisions, the change control process and advice regarding the change request form.

5.11.2 Impact assessment

Assessing the impact of a requested change

While all the steps in the change control process are important, quickly and effectively completing a detailed evaluation of a requested change is crucial for success. An impact assessment is a structured approach to assessing the implications of a change on the project's baseline. The impact assessment underpins the detailed evaluation, ensuring the project sponsor has all the information they need to make an informed decision about the change.

Project professionals and stakeholders will need to make time to assess the change, so it's wise for the project manager to determine the effort required before commencing. To do this, they should consider three key characteristics of the requested change:

- **The size and scale:** How big is the change? Does it touch every area of the project or just specific parts of the scope?
- **The complexity:** How complex is the change? Is it likely to be technically demanding or will the solution be basic?
- **The feasibility:** Considering the capabilities of the project and the organisation, is the change feasible?

Once the context of the change is defined, the project manager can coordinate the work required to assess the impact. This typically begins with an information-gathering exercise, using the knowledge of subject-matter experts, and internal and external data sources, to understand the specifics of the change requested. An example might be understanding the best technologies for a software solution.

From here, the team should assess the work and effort required to make the change happen. While the solution isn't defined in detail at this stage, subject-matter experts will need to make working assumptions on how the change would be implemented within the project.

When the work and effort have been determined, project professionals should assess and quantify the impact of the change on the project's baseline, specifically focusing on the following:

- **Scope:** Would the change fundamentally alter the project's scope?
- **Objectives and success criteria:** Would the outcomes and definition of success change?
- **Benefits:** Would the change create new benefits, or impact existing benefits?
- **Time:** Would the schedule change, becoming either longer or shorter?
- **Cost:** Would the change drive greater costs, or would it reduce costs in other areas?
- **Resource:** Are new resources required, or could the current profile (see 5.6.2) manage the work?
- **Quality:** Would the change improve or reduce the quality of project deliverables?
- **Risks:** Would the change create or reduce risk across the project?

For most change requests, several options may be evaluated, each with its own level of impact on the project's baseline. Project professionals should work hard to present the options in a way that the project sponsor can understand, striking a balance between summarising complex information and providing enough detail to enable effective decision-making.

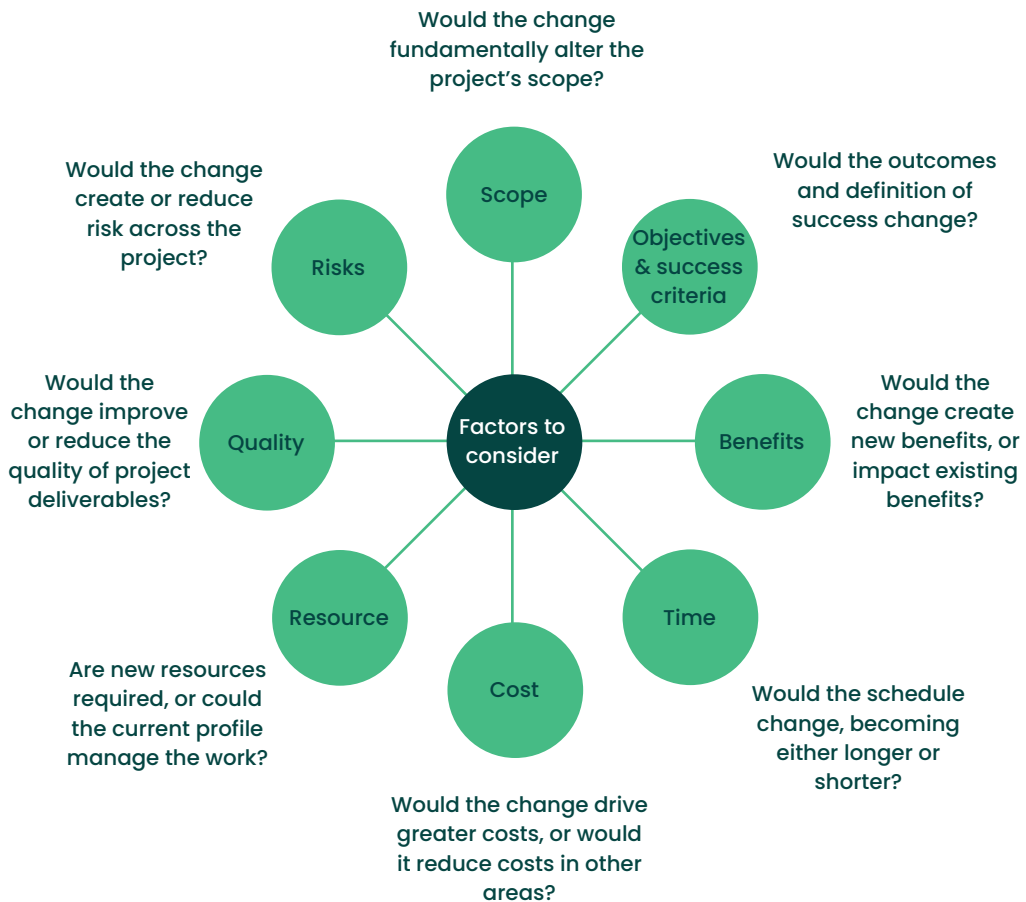


Figure 5.11.2 Factors to consider in impact assessment

Recommended reading

- *Impact Evaluation in Practice* (2016) provides an introduction to impact evaluation for policy makers and practitioners. It incorporates real-world examples to present practical guidelines for designing and implementing impact evaluations.

5.11.3 Configuration management

Ensuring continuous accuracy of configuration items

For highly technical projects, the change control process directly impacts the specification of the project's deliverables. When changes are agreed, they need to be clearly represented in the project's configuration, ensuring any changes to the functional and physical characteristics of the deliverables are documented and communicated to the entire team.

Configuration management describes the technical and administrative activities that manage the creation, maintenance, controlled change and quality control of the project's technical configuration (Figure 5.11.3). Done well, configuration management provides accuracy and alignment of the latest configuration, while providing clear traceability between versions.

At its simplest, configuration management oversees the version control of documents and information to ensure changes and knowledge are up to date (see also 3.5.5). But configuration management is more complex in projects where the design is multifaceted, combines multiple technical disciplines and includes a range of different assets. In some environments, it even takes the form of asset control, ensuring the configuration of all assets complies with organisational and regulatory standards.

Business information modelling (BIM) is related to configuration management, by creating and managing the digital representations of physical assets. Business information models are digital files that can be used to aid the visibility of complex configurations, while also enabling decision-making for a technical asset, such as a building or aircraft.

To effectively implement and control a project's configuration, project professionals should implement a configuration management process consisting of the following key steps:

- **Configuration management planning:** A configuration management plan describes any project-specific configuration procedures, how they're applied and who completes them; for example, the process for updating and publishing a construction project's technical designs.
- **Configuration identification:** Once a plan is in place, teams break down the project into configuration items, creating a unique number or reference for each item. This serves as an initial baseline and a single source of truth.
- **Configuration control:** When changes are approved through the change control process, configuration control ensures they are reflected in the latest configuration version. This also ensures that any interrelationships between configuration items are updated if required.
- **Configuration status accounting:** Throughout the project, reporting on the configuration's status is completed regularly, ensuring everyone has access to the most up-to-date information. It also enables traceability of configuration items throughout their development.
- **Configuration verification audits:** Audits are used when they are required to validate that deliverables conform to the documented configuration. Typically, a verification audit is completed at the end of a life cycle phase, when a deliverable is finished or at the point of transitioning the output into use.

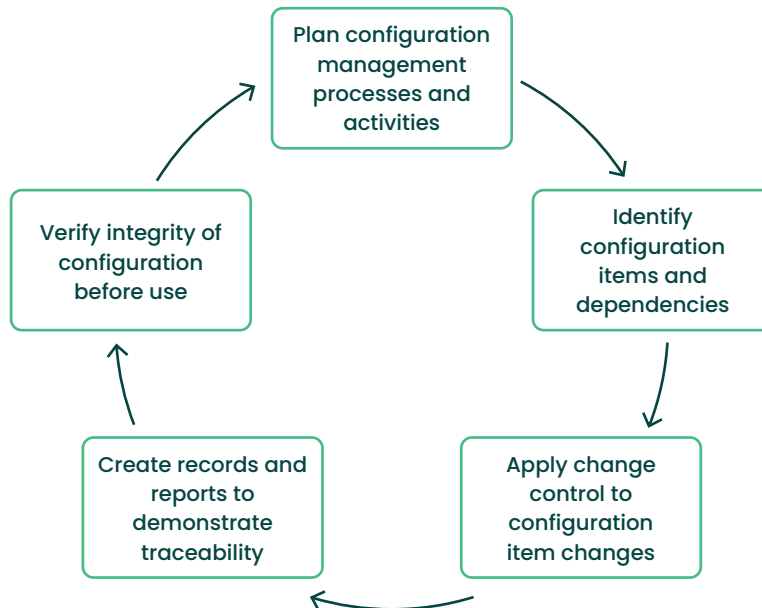


Figure 5.11.3 Essential activities to verify the configuration of an output

Recommended reading

- *Project Management* (2010) describes how configuration management is an essential project management tool to ensure that only the current specifications and designs are being used throughout the project, and how it links to change control and quality management.
- The APM Planning, Monitoring and Control Specific Interest Group's guide *Introduction to Project Control* (2010) demonstrates how configuration control fits into the wider suite of project controls.
- *BIM Handbook: A Guide to Building Information Modeling for Owners, Managers, Designers, Engineers and Contractors* (2018) is an established resource which aims to provide an in-depth understanding of BIM technologies, and of the business and organisational issues associated with its implementation. The 2018 edition incorporates coverage of BIM standards and a number of useful case studies.

6 Data analytics and AI in project management

The field of data analytics and AI is developing rapidly, as is their application in the field of project management. This chapter brings to the fore some key issues, challenges and questions project delivery professionals will face using data analytics and AI, together with some observations of the impact on the wider organisation. This chapter covers the following topics:

- The value of project data analytics and AI
- Data strategy – sources of data
- Data analysis techniques for project management
- Building a community
- Establishing and creating protocols, roles and responsibilities
- Building capabilities
- Applications of data analytics and AI on projects
- Ethics and safety
- Questions for the present and the future

Introduction

By their very nature, projects generate large volumes of data: design data, cost data, scheduling data, risk data, research data, data from pilots and prototypes, data about the project organisation, and so on. Many of the proprietary software applications that have become standard for projects automatically generate their own data. The challenge for projects historically has been that much of this data is unstructured, rarely integrated, and associated with different contractors and suppliers along the supply chain.

Data analytics (DA) is the process of organising and analysing this continual stream of project data and then using this data, sometimes combined with data from other projects, to make evidence-based decisions or predictions about what may happen in the future. Determining exactly what is collected, how it's analysed and how it's reported is usually defined by key performance indicators (KPIs) for tracking the performance of the project.

Artificial intelligence (AI) is the branch of computer science focused on creating systems that can perform tasks that typically previously required human intelligence. These tasks include learning from data, recognising patterns, solving problems, making decisions and understanding natural language. AI systems achieve this by using algorithms and models that can adapt and improve over time, often without explicit programming for each specific task (APM Project Data Advisory Group, 2023).

The 2024 Infrastructure and Projects Authority report *Data analytics and AI in government project delivery* highlights the benefits of leveraging both data and AI to help deliver better project outcomes, improve project effectiveness, value and efficiency, and support the development of the project delivery profession.

It underlines how leveraging these technologies will help to:

- ensure goal setting and business cases are more realistic and evidence-based
- enable better-informed project selection and greater opportunity for the use of options to provide more flexibility to projects following initiation
- improve lessons learned within and between projects to build capability
- ensure more relevant controls and evidence-based decision-making, leading to reduced duplication and rework
- enable the creation of new specialist roles and new future-focused career pathways

This section of the *Body of Knowledge* draws on existing research which APM has commissioned in data analytics and AI, and is written primarily with two audiences in mind: firstly, project professionals, who are not necessarily data specialists but who need to understand how analytics and AI can be applied, and secondly, data analysts and controls specialists who need to communicate the nature and value of their work to a diverse set of stakeholders.

The chapter opens with an overview of the value of data analytics and AI to provide some context, before diving into the strategy that underpins their use. We then cover the core techniques associated with both analytics and AI. The role of the delivery team is crucial in all of this, and so there is an explanation of the different roles and responsibilities for using these techniques and advice on building capability. The chapter explains some of the most common current applications and provides commentary on ethical and safety issues.

Developments in the use of data and AI to support project delivery are rapidly advancing, so this chapter will not refer to or recommend any specific analytic or AI tools but will focus more on underlying principles and techniques.

The value of project data analytics and AI

Project data analytics and AI provide significant value when delivering projects by enhancing efficiency, accuracy and decision-making throughout the project life cycle. Key benefits include the following:

- **Improved decision-making:** Data analytics and AI can analyse historical project data to predict potential risks, delays and cost overruns, allowing project managers to make proactive decisions. AI tools can monitor project performance in real time, offering actionable insights that help to optimise resource allocation, scheduling and budgeting.
- **Enhanced efficiency:** AI can automate repetitive tasks such as scheduling, progress tracking and reporting, freeing up time for project teams to focus on more strategic activities. AI algorithms can also optimise resource allocation by analysing availability, skill sets and workload, ensuring that resources are used effectively.
- **Risk identification and mitigation:** AI can identify patterns and trends in data that may indicate potential risks. It can also recommend mitigation strategies based on historical data and predictive models. AI tools can simulate different project scenarios, helping teams to assess the impact of various risks and uncertainties, and to develop contingency plans.

- **Enhanced collaboration and communication:** AI-driven analytics provide clear, data-backed insights that improve communication among stakeholders by reducing ambiguities and enhancing transparency. AI can enhance collaboration by integrating various project management tools, facilitating communication and ensuring that all team members have access to up-to-date information.
- **Cost and time savings:** AI can optimise project schedules by analysing constraints and dependencies, helping to reduce delays and avoid unnecessary costs. In addition, by providing accurate forecasts and identifying cost-saving opportunities, AI can help in keeping the project within budget.
- **Quality improvement:** AI can analyse data from ongoing projects to detect quality issues early, ensuring that corrective actions are taken before defects impact the final deliverable. AI systems learn from project data and outcomes, providing insights that can be applied to improve processes and methodologies in future projects.

By leveraging project data analytics and AI, organisations can deliver projects more successfully, with greater efficiency, reduced risks and better alignment with strategic goals. This ultimately leads to higher project success rates and more value creation for the organisation.

Any application of these techniques needs to start with a clear use case. Table 6.1 offers an overview of some of the ways in which projects, programmes and portfolios can leverage the value of data analytics and AI.

Table 6.1 Sources of value in data analytics and AI

Sources of value	Example applications
Operational value at the project level, through real-time, faster and evidence-based applications	• Data visualisation and stakeholder communication • Learning and development, and lessons learned • Planning and scheduling • Risk and opportunity management • Scope and requirements definition • Scope and risk deviance management • Software testing and systems simulations
Strategic value at the programme, portfolio and organisation levels, through enhanced capability	• Analysis of historical performance • Delivery team support with virtual project assistants, signposting help and guiding • Evidence-based decision-making • Progress monitoring, report preparation and distribution, compliance and assurance activities • Risk and performance comparisons across projects and programmes • Targeted portfolio prioritisation with less risk of human bias

Leveraging the changes to both process and behaviour that these techniques will drive requires data-savvy users who are able to adapt and build skills such as data visualisation and interpretation, along with strategic, creative and lateral thinking.

The genie is clearly out of the bottle. The use of AI is now in the mainstream and applications are accelerating across all aspects of business and home life. Realising the value is entirely dependent on an organisation's ability to navigate a minefield of practical and ethical considerations at a speed that may be uncomfortable or unfamiliar.

Data strategy – sources of data

The starting point for any project is to establish a data strategy to ensure that data is used effectively across the project. The data strategy establishes the foundations for data-driven decision-making and ensures that the collection, management and use of data are aligned with the project's objectives.

Project analytics, by necessity, relies on data and Table 6.2 shows some examples of the type of data that is typically available, and what data is required to enable a coherent and balanced analysis.

Table 6.2 Sources and forms of data

Internal data	For example: project management plans, risk management or stakeholder management plans, and ongoing project reports on progress, status and risk
External data	For example: commercially available research or statistics, and data from partner organisations
Quantitative data	For example: numerical data involving frequencies, values, ratings or scores – everything that lends itself to statistical or mathematical comparison. This may be harvested from an agreed set of sources
Qualitative data	For example: narratives and descriptions which may describe directions, trends and levels. They may not be reduced to statistical reporting but can include tools such as sentiment reporting to analyse stakeholder communications on project status, for example. This is likely to require more proactive methods for collection, for example: surveys, focus groups or benchmarking

Bear in mind that data analytics may combine data types from different sources. Whatever the sources and the collection methods, these should be documented in a collection plan as part of the overall data strategy. This will provide the project baseline for each data source and describe the method and frequency of collection.

Prior to any analysis, data typically needs to be processed and cleaned. This involves:

- determining the data's reusability to eliminate any sources that the project does not have legitimate access to
- identifying missing data and finding alternative sources to fill the gaps
- removing duplicate data to prevent redundant entries undermining the validity of the analysis

- standardising features to ensure that quantitative data is expressed in a common format
- framing the data to establish a basis for removing any outliers – data sets that are at either extreme of the range and might skew results
- validating any data that may be uncertain, through comparison with other sources

The process of building data sets, designing data capture techniques and validating data needs to be transparent and should involve subject-matter experts, data scientists and interested stakeholders. The importance of social learning in this process should not be underestimated, as it will not only help assure the quality of the data that is analysed but also underpin the understanding of those who, ultimately, will be making decisions based on the results. It will also inform them of the level of confidence they can apply to the information they are receiving. Data analytics are built on evidence, but that doesn't mean they are ever absolute.

Moving from an unplanned to a planned data environment places an emphasis on data standards (UK Government, 2021). These are fundamental to enabling reliable insights, interoperability, data exchange and data comparison, reducing the current administrative burden of manual data formatting, accelerating the adoption of data analytics, and ensuring compliance with General Data Protection Regulation (GDPR) and other data regulations.

Data analysis techniques for project management

Once a curated data set is available, data analytics can be applied to help inform those responsible for making decisions on what is happening in the project now and why. It is also possible to use data analytics to anticipate what may happen in the future, providing decision makers with an opportunity to make early decisions.

Data analytics can be categorised into different forms based on the type of analysis and the purpose it serves (*Project Data Analytics*, APM, 2020). Below are the five main forms of data analytics.

Descriptive analytics

This helps us to understand what has happened in the past.

It involves summarising historical data to identify trends, patterns and insights. It provides a clear picture of past performance by answering questions like 'What happened?' and 'What is happening now?'

Applications include accessing and searching lessons-learned repositories and bid analysis.

Diagnostic analytics

This helps us to understand why something happened.

It digs deeper into the data to identify the root causes of events or trends identified by descriptive analytics. It answers questions like 'Why did this happen?' by exploring relationships and correlations within the data.

Applications include root cause analysis and delay analysis.

Predictive analytics

This helps us to forecast future outcomes based on historical data.

It uses statistical models and machine learning algorithms to analyse past data and make predictions about future events. It answers questions like 'What is likely to happen?'

Applications include project assurance, simulations and optioneering.

Prescriptive analytics

This helps us to recommend actions based on data analysis.

Prescriptive analytics goes beyond predicting future outcomes by suggesting specific actions to achieve desired results or mitigate risks. It answers questions like 'What should we do?' by providing decision options.

Applications include weighting and ranking bids, and resource optimisation.

Cognitive analytics

This emerging form of analytics helps us to mimic human thought processes when analysing data.

It uses AI technologies such as natural language processing, machine learning and deep learning to analyse unstructured data (like text, images and video). It aims to simulate human reasoning and understanding to derive insights.

Each form of data analytics serves a unique purpose, from understanding past performance to predicting future outcomes and prescribing actions. Organisations can use a combination of these forms to gain comprehensive insights and make data-driven decisions.

It may be tempting to focus analysis on quantitative data because it is easier to validate and may be seen to be more scientific. However, no analysis, whether historical or forward-looking, should ignore qualitative data. Interviews, surveys and recorded observations all help make sense of why something is happening (or not happening). Decisions will often require a change of specific behaviours among team members or stakeholders, which qualitative data can inform and then measure.

Building a community

Successful application of analytics certainly requires data, but it also requires a delivery community that is informed and motivated. It is necessary to identify those stakeholders who will be drawing on the analysis to help guide their behaviour and inform their decisions.

Consider the following:

- Who is already familiar with data analysis and the use of evidence to inform decision-making, and who might be the champions for extending the use of data analytics?
- What is the level of their appetite? Where they are most struggling due to lack of data?

- How can initial pilots explore this demand and use readily available usable data?
- What will the application of data analytics allow them to do that they cannot do currently?
- What will a future data-enabled project team look and feel like? How will it work?

Establish and communicate protocols, roles and responsibilities

While every project team will be familiar with the techniques and the language associated with project delivery, part of the process of adopting data analytics and AI requires them to learn a new set of skills and a new vocabulary to go with them.

How the data analysis is represented (the data visualisation) will also play an important role in assuring the value of this activity. Decision makers, who may not necessarily be data-literate, need to be able to quickly read and recognise what is being presented to them, so that they can understand the implications.

Building capability and capacity will involve both specialists and generalists across the project delivery function (see Figure 6.1). Each organisation will approach this differently, but all need to answer some key questions:

- Is project data analytics core to a role or an enabling activity?
- Do we need data practitioners or translators (i.e. those who use analytical tools and techniques to crunch data and those who can translate the results into new processes) to integrate the data insights into business-as-usual behaviours?
- How are existing roles affected by a data-led approach?

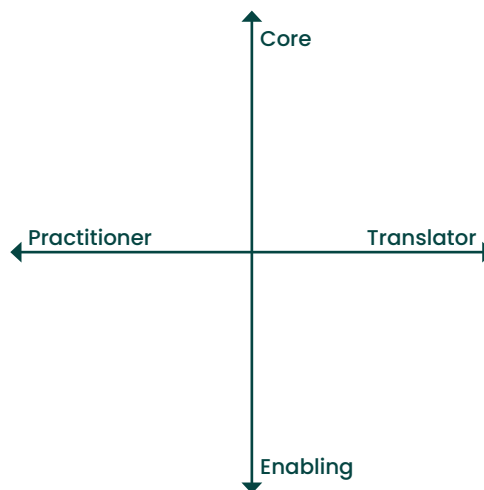


Figure 6.1 Data-related roles and skills across the delivery function

Build capabilities

The UK Government's *Digital and Data Profession Capability Framework* (2024) provides a useful guide to 45 common data and analytics roles, along with the nature (and level) of the skills associated with each.

Courses and programmes to learn, practise and develop these skills range from free (usually online) short courses and tutorials, to paid courses (in-person, hybrid and virtual) and certification programmes.

Apprenticeships are also available at every level, from Level 2 (equivalent to GCSE) right up to Level 7 (equivalent to master's).

Applications of data analytics and AI on projects

Project delivery professionals have been analysing project data for many years, with techniques such as earned value management (EVM) and risk management, often led by the project controls function. Today, AI enables data capture and comparison that is more accurate, faster and less time consuming. It also allows for an exponentially larger range and variety of data to be processed and for a much richer set of applications.

AI systems to support project delivery

In project delivery, several forms of AI can be highly beneficial, each offering different capabilities to enhance various aspects of project management. Below are some of the key forms of AI suitable for project delivery.

Machine learning (ML) can analyse historical project data to predict future outcomes, such as project timelines, costs and potential risks. By learning from past projects, ML models can provide forecasts and identify patterns that help in proactive decision-making.

Large language models (LLMs) can process and understand human language, which in turn enables us to interact with the data by asking questions. Essentially, they perform the role of virtual, digital, librarians. When combined with natural language processing, they allow computers not only to understand the vocabulary and grammar of human language but also to interpret the values, emotions, context and intent of written or spoken language. This can be used to analyse any human communication, and to express information that carries emotion and sentiment. It's worth noting that LLMs are not just text-to-text but are multimodal, in that they can work across different mediums such as text, audio and video.

Robotic process automation (RPA) allows an organisation to accelerate and improve the quality and consistency of regular processes. This can be used to improve data quality by using pattern recognition to prompt a user who is providing information. Examples include sending out an automated request or reminder to an individual who needs to authorise a decision, and streamlining the stages in a process according to the responses of the users.

AI-powered decision support systems combine ML, LLMs, natural language programming and process automation to create powerful dashboards that provide insight, prompts, options and recommendations to guide decision-making. These have the potential to operate at any level within a project, from helping to organise a kanban board right up to making strategic decisions within a portfolio.

Generative AI involves creating new and original data or content. Unlike traditional AI models that rely on large data sets and algorithms to classify or predict outcomes, generative AI models are designed to learn the underlying patterns and structure of the data and generate new outputs that mimic human creativity (Alan Turing Institute, 2023).

Collectively, these tools are moving towards being able to process information and communication in the way we do as humans, by adapting to the make-up of a given project team. This means, for example, that predictive analytics might be used in the future to suggest a course of action which reflects not just the real-time state of the project but also the capabilities and capacities of the delivery team. By implication, the technology should help to improve the basis of our decision-making and, for example, a team's understanding of a project's relative complexity and how well equipped they are for working in that context.

Ethics and safety

Any technology that impacts human development raises ethical issues. Given that one of the stated goals of AI is to remove human agency from (some) decision-making and replace it with non-human agents, organisations and industries should not be venturing down this path without the aid of an ethical framework and guardrails to ensure checks and balances.

The UK Government Digital Service document, *Artificial Intelligence Playbook for the UK Government*, includes advice on using AI safely and responsibly:

1. Transparency and explainability

AI should never be used deliberately or unconsciously to mislead, which means that governance processes should be clear about the basis for AI use, and how it is involved in decision-making. Make sure you always signpost whenever AI has been used to process data or create content.

2. Accountability and responsibility

The potential absence of human agency in decision-making should never leave people without recourse to holding organisations to account. This requires clear chains of human responsibility throughout the AI life cycle; that is, AI systems showing clear provenance, which can be audited, as well as up-front statements about the legal obligations and liability associated with any use of what has been generated.

3. Fairness, bias and discrimination

Automation bias occurs when people ascribe unreasonable authority to information, content or decisions that are generated by AI, simply because it is machine generated rather than human generated. AI and analytical systems are designed by humans, which means they are subject to the bias of their creators. Consequently, it is necessary to ensure AI outputs don't amplify social, demographic or cultural disparities, that they comply with human rights laws and that tests mitigate biases in AI systems, as part of their validation.

4. Information quality and misinformation

AI and analytics are dependent on the quality and veracity of the data on which they are working: 'rubbish in/rubbish out'. Verify and cross-reference AI-generated information with trusted sources and make sure there are easy and safe routes for users to challenge what is being generated.

5. Human involvement

AI should be used to enhance and improve what we can do as humans *but never to replace it entirely*. Appropriate and active human oversight in AI processes needs to be maintained and, in the case of high-risk or high-impact decisions, humans should always be the final arbiters.

6. Sustainability

It can be easy to overlook the environmental impact of software and digital systems. The World Economic Forum (2024) highlights just how hungry for energy these invisible systems actually are, with the energy required to run AI tasks already accelerating. It is predicted that, by 2028, AI could be using more power than the country of Iceland used in 2021.

This places an onus on organisations to consider and report the environmental impact of their AI systems alongside their other sources of energy usage.

The adoption and development of analytics and AI in large projects needs to capture the hearts as well as the minds of the project profession, which underlines the importance of robust governance that complies with statutory, regulatory and ethical standards.

Requirements for the present and directions for the future

AI is accelerating at a speed that makes future predictions largely meaningless, so, rather than anticipating emerging developments, we will conclude this chapter with four sets of needs statements that can help frame organisations' approaches to data analytics and AI. We will finish with three sets of questions we believe need to be kept in mind as we implement AI in our projects and wider organisations.

The need for purpose

Over the years, the project profession has learned the consequences of conflating the appetite for data with the ability to process and assimilate the lessons it offers. Many organisations have become buried in information at the expense of insight. AI is not boundless, and the inappropriate or unfocused use of AI will divert us from progress and accelerate our consumption of scarce resources.

The need for simplicity

The 2010s and early 2020s focused on the carbon associated with operations: the heating and cooling systems. This focus has shifted. Operational carbon is still a consideration, but we are now adopting approaches to tackle embodied carbon, including circular economy methods that favour alternatives to 'demolish and rebuild', and the use of recycled materials or those that sequester carbon, such as timber or hemp. The design of buildings, as a consequence, has become more complex, involving a greater range of parameters and systems. AI can provide a way of rethinking design and construction processes, such as these, so that they are more in tune with this shift, rather than applying the old processes to new materials (Major Projects Association, 2024).

The need to walk before we can run

Organisations need to address the fundamentals of data and data sharing: the quality, availability, consistency and comparability of data, and the requisite skills and capability associated with AI. Do our projects have a data strategy? Do we know how data is curated on our projects in the end-to-end supply chain? How does our organisation bring in new entrants with the right skills and how does it upskill the existing workforce?

The need to understand progress

For many organisations, auditing past decisions is limited to a focus on the critical moments when a project went wrong. How can organisations identify the everyday decisions associated with project delivery, the things that went right? How can we benchmark these decisions now and measure the progress of the efficiency and effectiveness of data- and AI-driven decision-making?

Questions to ask

As we move forward in our understanding and in our experience of AI, there are three important areas where questions need to be continually asked of our use of AI in data analytics.

Governance and oversight

Firstly, there will be a need for strong governance and oversight of all uses of AI within organisations. What is currently being managed by AI? How is this currently happening? How transparent are the AI processes and decisions to stakeholders? Are there mechanisms that allow stakeholders to understand and question AI decisions? Are the benefits and risks of using AI fully understood? Where else could AI be used? What are the situations where AI shouldn't be used? What are the mechanisms for closing AI projects or applications? How frequently are AI policies and governance structures reviewed and updated to keep pace with technological advancements? And, finally, is the system of governance and oversight fit for purpose?

Data

The benefits of AI come from its application to data sets. This means that there is an immediate need to understand the source of this data and the quality of the data being used by the applications. This raises two important questions. First, how useful is past project data for the current use of AI? Second, is it possible, ethical, legal and desirable to share data to improve the results coming from your AI?

People

In the future, people are going to be working in environments alongside AI. In the short term this will create a degree of uncertainty, which may unsettle people and need to be managed. But as we progress, we need to develop a clear understanding of how jobs and skill sets will need to change if this is to work effectively. This will impact on the skills and capabilities requirements of leaders too. But inevitably, the use of AI will result in a loss or deterioration of in-house human capabilities. While AI offers numerous and substantial benefits in project management, we must also consider what might be obscured in the

process. Relying on AI for project tasks can lead to significant efficiency gains; however, the opaque nature of AI's information processing – often described as a 'black box' – limits our understanding of the foundational logic behind its decisions. This obscurity introduces a subtle yet significant risk: the methodologies that underpin project analysis and decision-making may become less transparent, leading to outcomes that are both unintended and challenging to predict. In the context of rapid AI adoption, a key question emerges: how do we effectively mitigate these emergent risks? How do we ensure people work with AI, not against it? What does the organisation want AI to do and what are the domains and responsibilities of individuals, managers and directors?

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Glossary

This glossary is made up of terms used in the eighth edition of the *APM Body of Knowledge* only. Definitions are provided where terms used are unique to the profession, or have a unique meaning in the profession.

Acceptance criteria The requirements and essential conditions that have to be achieved before a deliverable is accepted.

Activity (1) A task, job, operation or process consuming time and possibly other resources. (2) The smallest self-contained unit of work in a project.

Adoption The optional additional phase in a linear life cycle that facilitates the use of project outputs to enable the acceptance and use of benefits.

Agile A family of development methodologies where requirements and solutions are developed iteratively and incrementally throughout the life cycle.

Analogous estimating An estimating technique based on comparison with, and factoring from, the cost of similar, previous work. Also known as comparative estimating.

Analytical estimating An estimating technique that uses detailed specifications to estimate time and cost for each product or activity. Also known as bottom-up estimating.

Assurance The process of providing confidence to stakeholders that projects, programmes and portfolios will achieve their objectives for beneficial change.

Baseline The reference levels against which a project, programme or portfolio is monitored and controlled.

Benchmarking The process of comparing your organisation's performance, processes or practices with industry standards or best practices to identify areas for improvement.

Benefit A positive and measurable impact of change.

Benefits management The identification, definition, planning, tracking and realisation of benefits.

Benefits realisation The practice of ensuring that benefits are derived from outputs and outcomes.

Bottom-up estimating An estimating technique that uses detailed specifications to estimate time and cost for each product or activity. Bottom-up methods include analytical estimating and Delphi technique.

Breakdown structure A hierarchical structure by which project elements are decomposed. Examples include cost breakdown structure (CBS), organisational breakdown structure (OBS), product breakdown structure (PBS) and work breakdown structure (WBS).

Budgeting The process of estimating and allocating financial resources for a project or organisational activities.

Buffer A term used in critical chain for the centralised management of schedule contingencies.

Business as usual An organisation's normal day-to-day operations. Also referred to as steady-state.

Business case Provides justification for undertaking a project, programme or portfolio. It evaluates the benefit, cost and risk of alternative options, and provides a rationale for the preferred solution.

Business information modelling (BIM)

Involves the generation and management of digital representations of physical and functional characteristics of buildings and places. Business information models are digital files (often, but not always, in proprietary formats and containing proprietary data) which can be extracted, exchanged or networked to support decision-making regarding a building or other built asset. Related to configuration management.

Change control The process through which all requests to change the approved baseline of a project, programme or portfolio are captured, evaluated and then approved, rejected or deferred.

Change freeze A point after which no further changes to scope will be considered.

Change log A record of all changes proposed, approved and implemented during a project, including their status and impact.

Change management The overarching approach taken in an organisation to move from the current to a future desirable state using a coordinated and structured approach in collaboration with stakeholders.

Change register (or log) A record of all proposed changes to scope.

Change request A request to obtain formal approval for changes to the approved baseline.

Closure The formal end point of a project, programme or portfolio – either because planned work has been completed or because it has been terminated early.

Communication The process of exchanging information and confirming there is shared understanding.

Communication plan A document outlining how information will be shared with stakeholders during a project.

Communities of practice A type of learning network used within and between organisations to maintain, develop and share knowledge.

Comparative estimating An estimating technique based on the comparison with, and factoring from, the cost of similar, previous work. Also known as analogous estimating.

Complexity Relates to the degree of interaction of all the elements that make up a project, programme or portfolio, and is dependent on such factors as the level of uncertainty, interaction between stakeholders and degree of innovation.

Compliance Adhering to laws, regulations and standards relevant to the organisation or project.

Concept The first phase in a linear life cycle. It develops an initial idea through initial studies and high-level requirements management and assessment of viability, including an outline business case.

Configuration The functional and physical characteristics of a product, as defined in its specification and achieved through deploying project management plans.

Configuration management Encompasses the technical and administrative activities concerned with the creation, maintenance, controlled change and quality control of the scope of work.

Conflict resolution The process of identifying and addressing differences that if left unmanaged would affect successful completion of objectives.

Context A collective term for the societal and/or organisational setting of a project, programme or portfolio. Also known as environment.

Contingency Provision of additional time or money to deal with the occurrence of risks in case they occur. See also risk budget and management reserve.

Continuing professional development

(CPD) The requirement for any professional to continually develop their competence.

Continuous improvement

The ongoing effort to enhance products, services or processes through incremental improvements.

Contract

An agreement made between two or more parties that creates legally binding obligations between them. The contract sets out those obligations and the actions that can be taken if they are not met.

Cost–benefit analysis

A systematic process of comparing the projected costs and benefits of a project to determine its overall value and feasibility.

Cost control

Managing and monitoring expenses to ensure a project remains within its budget.

Cost of capital

A term used in investment appraisal to reflect the percentage return an investment must deliver to satisfy lenders. Value is created only when the return is greater than the cost of capital. See also weighted average cost of capital (WACC).

Cost planning

The estimation of costs, the setting of an agreed budget, and management of actual and forecast costs against that budget.

Critical chain

A resource-based approach to scheduling, useful when time is critical and derived from the critical path, that protects critical chains of activities with buffers.

Critical path

A sequence of activities through a precedence network from start to finish, the sum of whose durations determines the overall duration.

Critical path analysis (CPA)

An activity-based scheduling technique that determines the overall duration of the identified work based on estimates and logical dependencies. The method of determining the critical path.

Decision bias

Psychological biases affecting individuals and groups when making risk-based decisions.

Decision gate

A point between phases of the life cycle that is used to review and confirm the viability of the work in line with the business case. Alternatively called stage gates or gates.

Deliverable

A product, set of products or package of work that will be delivered to, and formally accepted by, a stakeholder. Used interchangeably with 'output' and 'product'.

Delphi technique

The generation of an estimate through individual expert judgement followed by facilitated team consensus.

Dependency

A relationship between activities in a network diagram.

Earned value

A measure of progress that expresses costs committed and work achieved in the same units.

Earned value management

A project control process based on a structured approach to planning, cost collection and performance measurement. It facilitates the integration of project scope, time and cost objectives, and the establishment of a baseline plan of performance measurement.

Environment

A collective term for the societal and/or organisational setting of a project, programme or portfolio. Also known as Context.

Escalation

The process of drawing issues to the attention of a higher level of management.

Estimate

A forecast of the probable time or cost of completing work.

Estimating The use of a range of tools and techniques to produce forecasts of the probable time or cost of completing work.

Event-driven Control actions or reports that are triggered by a specific event.

Extended life cycle A life cycle approach that adds an adoption phase to a linear or iterative life cycle, with the purpose of ensuring that the accountability and governance of the investment stays with the change teams until change is fully embedded. It provides the missing connection to benefits realisation in a linear life cycle, and facilitates cooperation and knowledge sharing between change and business-as-usual teams.

Facilitation An approach to working with groups in a collaborative way to create energy and make it easy for the group to solve problems.

Feasibility study An analysis to assess the practicality and potential success of a proposed project or solution.

Feedback loop A system for gathering feedback on performance or outcomes and using it to make improvements.

Fixed cost A resource and associated cost that is not influenced by volume of business or quantity, for example a one-off capital cost.

Float The flexibility with which an activity may be rescheduled. There are various types of float, such as total float and free float.

Forecast A prediction of a defined future state, typically related to the duration and out-turn cost of a project or programme.

Funding The means by which the money required to undertake a project, programme or portfolio is secured and then made available as required.

Gantt chart A graphical representation of activity against time.

Governance The framework of authority and accountability that defines and controls the outputs, outcomes and benefits from projects, programmes and portfolios. The mechanism whereby the investing organisation exerts financial and technical control over the deployment of the work and the realisation of value.

Governance board A body that provides sponsorship to a project, programme or portfolio. The board will represent financial, provider and user interests. Members of a governance board oversee deployment and make decisions through the chosen life cycle. Alternatively called steering committee, steering group, project board, programme board, etc.

Handover The point in the transition phase of a linear life cycle where deliverables are commissioned and handed over to the permanent organisation to adopt.

Host organisation The organisation that provides the strategic direction of the project, programme or portfolio and is the primary investor and recipient of benefits. Used interchangeably with 'investing organisation' and 'client organisation'.

Hybrid life cycle A pragmatic approach to achieving beneficial change that combines a linear life cycle for some phases or activities with an iterative life cycle for others.

Impact assessment An assessment of the merits of pursuing a particular course of action or of the potential impact of a requested change.

Influencing The act of affecting the behaviours and actions of others.

Information management The collection, storage, curation, dissemination, archiving and destruction of documents, images, drawings and other sources of information.

Integrated assurance The coordination of assurance activities where there are a number of assurance providers. It can follow a three lines of defence model from corporate governance.

Integrated planning The application of management processes that bring together the planning of benefits, success criteria, scope, quality, time, resources, cost, risk, communications, etc to create the project management plan.

Internal rate of return (IRR) Used to determine the profitability of a potential investment. It is the discount rate that makes the net present value zero.

Investment appraisal The analysis done to consider the profitability of an investment over the life of an asset alongside considerations of affordability and strategic fit. An input to the investment decision.

Investment decision The decision made by the sponsor and governance board that justifies the investment in a project, programme or portfolio. Investment decisions rely on robust investment appraisal.

Issue A problem that is now breaching, or is about to breach, tolerances of delegated work on a project or programme. Issues require support from the sponsor to agree a resolution.

Issue management The process by which issues can be identified and addressed to remove the threats that they pose.

Iteration Repeating a process or activity to refine and improve results over time.

Iterative life cycle A life cycle that repeats one or more of the phases of a project or programme before proceeding to the next one, with the objective of managing uncertainty of scope by allowing objectives to evolve as learning and discovery takes place.

Key performance indicators (KPIs) Measures of success that can be used throughout the project to ensure that it is progressing towards a successful conclusion.

Knowledge management The holistic, cross-functional discipline and set of practices concerned with the way organisations create and use knowledge to improve outcomes.

Leadership The ability to establish vision and direction, to influence and align others towards a common purpose, and to empower and inspire people to achieve success.

Lessons learned Documented experiences that can be used to improve the future management of projects, programmes and portfolios.

Life cycle A framework comprising a set of distinct high-level stages required to transform an idea or concept into reality in an orderly and efficient manner. Life cycles offer a systematic and organised way to undertake project-based work and can be viewed as the structure underpinning deployment.

Linear life cycle A life cycle that aims to complete a project within a single pass through a set of distinct phases that are completed serially and span from the development of the initial concept to the deployment of an ultimate output, outcome or benefits.

Management reserve A sum of money that is part of overall cost contingency to cover the cost impact of unidentified risks, and potentially some already identified very low-probability, very high-impact risks. See also risk budget and contingency.

Maturity model An approach to understand the current capabilities, processes and behaviours deployed in the management of projects and to identify a structured path to increase the predictability of success.

Milestone A key event selected for its importance in the schedule, commonly associated with tangible acceptance of deliverables.

Minimum viable product A product with just enough features to satisfy early users and to provide feedback for future product development.

Monte Carlo simulation A technique often used in estimating the overall risk for a project, programme or portfolio that enables the combined effect of estimating uncertainty and specific risk events to be predicted.

MoSCoW prioritisation A technique for prioritising project requirements into 'must-have', 'should-have', 'could-have' and 'won't-have' categories.

Net present value (NPV) The difference between the present value of cash inflow and the present value of cash outflow over a period of time. It is the monetary value used to judge the value of an investment at a particular discount rate.

Network diagram A model of activities and their dependencies used in scheduling. Also known as a precedence network.

Objectives A generic term for predetermined results towards which effort is directed. Objectives may be defined in terms of outputs, outcomes and/or benefits.

Opportunity A positive risk event that, if it occurs, will have an upside or beneficial effect on the achievement of one or more objectives.

Optioneering An approach to exploring multiple options to optimally satisfy stakeholders' needs, requiring creativity and lateral thinking.

Organisational culture The unwritten rules that influence individual and group behaviour and attitudes. Applicable at multiple levels of organisation, including national culture or project culture.

Outcome The changed circumstances or behaviour that results from the use of an output and leads to realisation of benefits.

Output The tangible or intangible product typically delivered by a project. Used interchangeably with 'deliverable' and 'product'.

Parametric estimating An estimating technique that uses a statistical relationship between historical data and other variables to calculate an estimate.

Phase One of the major subdivisions of a life cycle.

Planned value The cost profile of a resource-optimised schedule used as the baseline to monitor actual spend and earned value. Alternatively called the budgeted cost of work scheduled (BCWS).

Portfolio A collection of projects and/or programmes used to structure and manage investments at an organisational or functional level to optimise strategic benefits or operational efficiency.

Portfolio management The selection, prioritisation and control of an organisation's projects and programmes in line with its strategic objectives and capacity to deliver.

Precedence network A model of activities and their dependencies used in scheduling. Also known as a network diagram.

Procurement strategy The high-level approach for securing the goods and services required from external suppliers to satisfy project, programme and portfolio needs. See also strategic sourcing.

Product A tangible or intangible component of a project's output. Used interchangeably with 'deliverable' and 'output'.

Product life cycle A life cycle approach that adds operation and termination phases to a linear life cycle to reflect the whole life of an asset. Enabling a full asset life cycle perspective encourages engagement with long-term future implications of project-related actions.

Professionalism The application of expert and specialised knowledge within a specific field and the acceptance of standards relating to that profession.

Programme A unique, transient strategic endeavour undertaken to achieve beneficial change and incorporating a group of related projects and business-as-usual (steady-state) activities.

Programme management The coordinated management of projects and business-as-usual (steady-state) activities to achieve beneficial change.

Project A unique, transient endeavour undertaken to bring about change and to achieve planned objectives.

Project-based working A collective term for project, programme and portfolio management. Used interchangeably with 'management of projects'.

Project charter A document that sets out the working relationships and agreed behaviours within a project team.

Project close-out The process of finalising all project matters, carrying out final project reviews, archiving project information and redeploying the remaining project team. See also handover.

Project controls Mechanisms to monitor and manage project scope, schedule, cost and quality.

Project leadership Guiding and motivating a team to achieve project objectives successfully.

Project management The application of processes, methods, knowledge, skills and experience to achieve specific objectives for change.

Project (programme or portfolio) management office (PMO) An organisational structure that provides support for projects, programmes and/or portfolios.

Project management plan (PMP) The output of the process of integrated planning for a project or programme.

Project professional A person in a role associated with the management of projects, programmes or portfolios.

Quality The fitness for purpose of the outputs of a process or the process itself, or their degree of conformance to requirements.

Quality assurance The process of evaluating overall project performance on a regular basis to provide confidence that the project will satisfy the relevant quality standards.

Quality control Inspection, measurement and testing to verify that the project outputs meet the acceptance criteria defined during quality planning.

Quality planning The process of taking the defined scope and specifying the acceptance criteria that will be used to validate that the outputs are fit for purpose to the sponsor.

Requirements The stakeholders' wants and needs, clearly defined with acceptance criteria.

Requirements management The process of capturing, assessing and justifying stakeholders' wants and needs.

Resource allocation The process by which labour and non-labour resources are attributed to activities.

Resource capacity planning Determining the resources needed to meet project demands, and balancing workloads.

Resource levelling An approach used during resource optimisation that delays activities such that resource usage is kept below specified limits. Also known as resource-limited scheduling.

Resource management The acquisition and deployment of the internal and external resources required to deliver the project, programme or portfolio.

Resource optimisation A collective term that refers to the methods for ensuring that labour and non-labour resources are matched to the schedule. See also resource levelling and resource smoothing.

Resource smoothing An approach used as part of resource optimisation that involves utilising float, or increasing or decreasing the resources required for specific activities, such that any peaks and troughs of resource usage are smoothed out, avoiding extension of the duration where possible. Also known as time-limited resource scheduling.

Resources All the labour and non-labour items required to undertake the scope of work to the required quality.

Responsibility assignment matrix (RAM) A diagram or chart showing assigned responsibilities for elements of work. It is created by combining the work breakdown structure with the organisational breakdown structure.

Retrospective A meeting held at the end of a project or iteration to reflect on successes and identify areas for improvement.

Return on investment (ROI) An expression of the value of an investment in change based on the gain in benefit relative to the cost.

Risk The potential of a situation or event to impact on the achievement of specific objectives.

Risk analysis An assessment and synthesis of estimating uncertainty and/or specific risk events to gain an understanding of their individual significance and/or their combined impact on objectives.

Risk appetite How much risk investors are willing to tolerate in achieving their objectives. Expressed as risk thresholds or tolerances.

Risk attitude The perception-driven choice of a person or group about an individual risk, or overall riskiness of a project, programme or portfolio.

Risk budget A sum of money that is part of overall cost contingency to cover the cost impact of identified risks. See also management reserve and contingency.

Risk event An uncertain event or set of circumstances that would, if it occurred, have an effect on the achievement of one or more objectives.

Risk management A process that allows individual risk events and overall risk to be understood and managed proactively, optimising success by minimising threats and maximising opportunities.

Risk owner The individual or group best placed to assess and manage a risk.

Risk register A document listing identified risk events and their corresponding planned responses. Used interchangeably with 'risk log' and 'risk repository'.

Risk response An action or set of actions to reduce the probability or impact of a threat, or to increase the probability or impact of an opportunity.

Rolling wave A process whereby short-term work is planned in detail and longer-term work is planned in outline only.

Root cause analysis Identifying the underlying cause of a problem to prevent recurrence.

Scenario planning A method used to anticipate potential future scenarios that is useful in preparing to deal with emergent change.

Schedule A timetable showing the forecast start and finish dates for activities or events within a project, programme or portfolio.

Scope The totality of the outputs, outcomes and benefits, and the work required to produce them.

Scope management The process whereby outputs, outcomes and benefits are identified, defined and controlled.

Share A risk management response to an opportunity that increases its probability and/or impact by sharing the risk with a third party.

Social system The network of relationships between people (actors) involved in the project, programme or portfolio and how the influences between actors work as a whole.

Sponsor A critical role as part of the governance board of any project, programme or portfolio. The sponsor is accountable for ensuring that the work is governed effectively and delivers the objectives that meet identified needs.

Sprint A regular repeatable work cycle in agile development. Also known as an iteration.

Stakeholder An individual or group that has an interest or role in the project, programme or portfolio, or is impacted by it.

Stakeholder engagement The systematic identification, analysis, planning and implementation of actions designed to influence stakeholders.

Strategic alignment Ensuring that project goals are consistent with the organisation's strategic objectives.

Strategic intent The aspirational plans, overarching purpose or intended direction of travel needed to reach an organisational vision.

Strategic sourcing An analysis of the buying strengths and weaknesses of an organisation that enables procurement strategies to maximise buying advantages and respond to risks of supply disruption.

Success criteria The satisfaction of stakeholder needs for the deployment of a project. Note that this is a different performance measure to benefits, which are focused on the strategic intent and delivering beneficial change.

Success factors Management practices that, when implemented, will increase the likelihood of success of a project, programme or portfolio. The degree to which these practices are established and embedded within an organisation indicates its level of maturity.

Sustainability An approach to business that balances the environmental, social, economic and administrative aspects of project-based working to meet the current needs of stakeholders without compromising or overburdening future generations.

Systems thinking Understanding and analysing complex systems as interconnected components rather than isolated parts.

Talent management The ability to attract, motivate and retain high-quality people to deliver the strategic goals and objectives of the organisation.

Team A group of people working in collaboration or by cooperation towards a common goal.

Temporary organisation (team) A project, programme or portfolio team brought together specifically to implement project-based work. Used to differentiate the organisational structure for project-based work from the permanent organisation.

Threat A negative risk event; a risk event that, if it occurs, will have a downside or detrimental effect on one or more objectives.

Three-point estimate An estimate in which optimistic best case, pessimistic worst case and most likely values are given.

Timebox A term used in iterative life cycle approaches for an iteration with a fixed end date that is not allowed to change, thereby adjusting the scope and quality to deliver on time and to cost.

Tolerance A level of delegated permission to vary performance from specified parameters.

Top-down estimating A high-level estimate based on top level-project assumptions without the need to understand all of the project detail. Top-down methods include Parametric and Analogous estimating.

Tranche A subdivision of the deployment phase of a programme designed to enable an incremental approach to development of outputs, outcomes and benefits.

Transformation Fundamental changes to processes, technology or culture to achieve significant improvements.

Transition The fourth phase in a linear cycle, where results are handed over, commissioned and accepted by the sponsor, culminating in formal closure.

Triple constraint A way of describing the fundamental trade-off between time, cost and quality in delivering the scope of a project. Also known as the Barnes Triangle, sometimes called the Iron Triangle.

User story An informal, simple-language description of one or more features of a system or tool. User stories are often written from the perspective of an end user or user of a system.

Users The group of people who are intended to work with deliverables to enable beneficial change to be realised.

Value A standard, principle or quality considered worthwhile or desirable. In value management terms, value is defined as the ratio of 'satisfaction of requirements' over 'use of resources'.

Value management A structured approach to defining what value means to the organisation. It is a framework that allows needs, problems or opportunities to be defined and then enables review of whether they can be improved to determine the optimal approach and solution.

Value proposition The unique benefits or value that a product or service delivers to customers.

Variable cost A cost that changes with use or quantity.

Virtual team A team where the people are separated by geography and potentially time zone.

VUCA conditions (volatility, uncertainty, complexity and ambiguity) An organisational context where there is inherent uncertainty that makes it difficult to predict and plan with great accuracy.

Weighted average cost of capital (WACC) The minimum average return that an organisation must earn on an existing asset base to satisfy its capital providers (creditors, owners, etc). See also cost of capital.

Whole-life costs The fixed and variable capital and operational costs required to develop, use and terminate a product or asset.

Work breakdown structure (WBS) Defines the total work to be undertaken and provides a structure for all control systems. It allows a project or programme to be divided by level into discrete groups for programming, cost planning and control purposes.

Work package A discrete element of project scope at the lowest level of each branch of the work breakdown structure. Collectively, the work packages specify all the work and products included in the project.

Workplace stress The adverse reaction that people have to excessive pressure or other types of demand placed on them.

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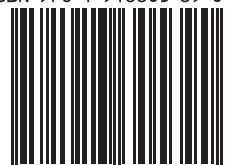
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